

CLIMATE VULNERABILITY ASSESSMENT FOR UTTARAKHAND

Comprehensive Climate Change Impact Assessment

May 2026



Submitted to



Submitted by



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Executive Summary

Uttarakhand's mountainous terrain, complex hydro-meteorological conditions, extensive river systems, fragile ecosystems and dependence on climate-sensitive livelihoods and infrastructure make the State particularly exposed to climate variability and climate-induced hazards. Changes in temperature and precipitation, together with the increasing variability and intensity of climate extremes, have implications for agriculture and horticulture, water resources, forests and biodiversity, energy, critical infrastructure, health, tourism and vulnerable communities.

This report presents the **Hazard Mapping and Sectoral Impact Assessment for Uttarakhand**, undertaken as part of the assessment of vulnerability to climate-induced hazards. The report establishes a spatial understanding of key climate-related hazards and examines their potential implications for major sectors of the State. It brings together downscaled climate projections, extreme climate indices, hazard datasets and sector-specific exposure information to identify areas and systems that may experience significant climate-related impacts.

The climate assessment is based on downscaled projections developed by **IIT Roorkee** and considers two future emissions pathways, **SSP1-2.6 and SSP5-8.5**, across the **2041–2070 and 2071–2100** periods. The analysis considers changes in temperature and precipitation characteristics, including **average temperature, maximum and minimum temperature, TX90p (warm days), TN90p (warm nights), annual precipitation, rainfall variability, RX1day (maximum 1-day rainfall) and consecutive dry days (CDD)**.

The assessment provides the climatic and hazard evidence required to understand how changes in climate conditions may translate into impacts across Uttarakhand.

Climate Extremes

The projected climate information indicates changes in both average climatic conditions and climate extremes across Uttarakhand. Temperature-related indicators provide evidence of changes in the thermal environment, including changes in maximum and minimum temperatures and the frequency of unusually warm days and nights.

The assessment of **Tmean, Tmax and Tmin** is particularly relevant for understanding changes in the frequency of average temperature, maximum and minimum relative to the historical climatological baseline. Changes in nighttime temperatures are important because warmer nights can reduce opportunities for overnight cooling and contribute to cumulative heat stress.

Precipitation changes are assessed through annual precipitation and rainfall variability. These indicators capture different dimensions of the precipitation regime. Changes in annual rainfall do not necessarily translate directly into increased or decreased water availability because

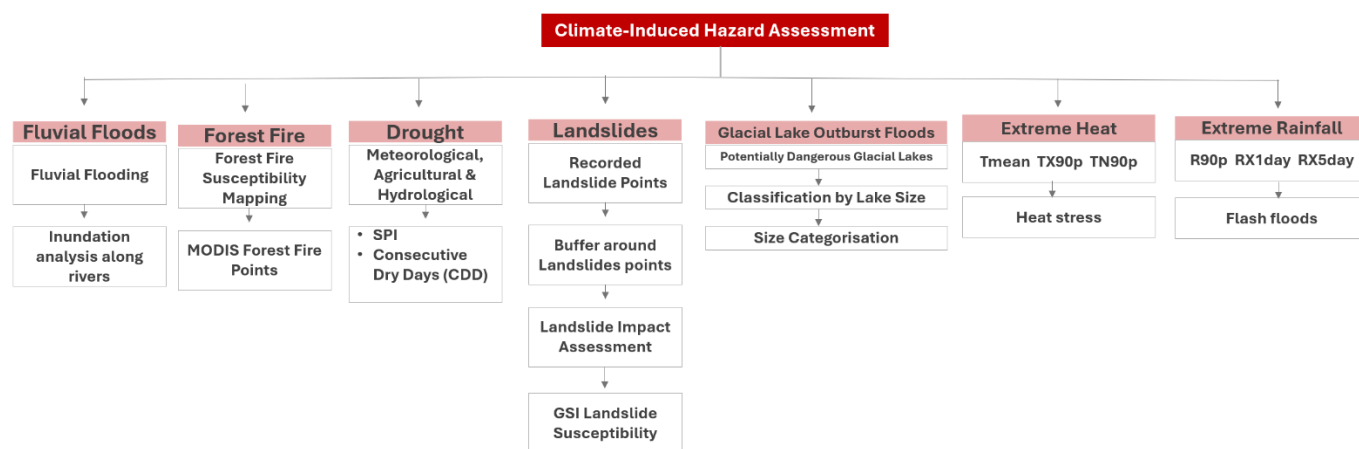
greater variability, intense rainfall events and longer dry spells can occur within the same climatic regime.

The combination of **extreme rainfall and prolonged dry periods** is therefore an important feature of future climate conditions for Uttarakhand, with implications for both water-related hazards and climate-sensitive sectors.

Hazard Mapping

The hazard mapping component translates the climate and environmental information into spatial representations of key hazards relevant to Uttarakhand.

The assessment considers **fluvial floods, drought, landslides, extreme heat and glacial lake-related hazards**, using appropriate datasets and methods for each hazard.



The mapped lake locations provide an important basis for understanding potential downstream exposure to GLOF-related impacts, particularly for settlements, roads, bridges, hydropower infrastructure and other assets located downstream of potentially dangerous glacial lakes.

Lake size provides an indication of the scale of water storage; however, lake size alone does not determine GLOF hazard, and other characteristics of individual glacial lakes and their downstream pathways are also relevant to detailed risk assessment.

The individual hazard layers demonstrate that climate-related hazards are not distributed uniformly across Uttarakhand. Different parts of the State may experience different combinations of hazards depending on their climatic conditions, topography, river systems and environmental characteristics.

The multi-hazard assessment therefore provides a basis for identifying locations where **multiple hazards may overlap or where exposure to one hazard may compound the consequences of another**.

This is particularly relevant in Uttarakhand because settlements, infrastructure and economic activities are frequently concentrated within narrow valleys and along mountain transport corridors. Consequently, the spatial coincidence of hazards with critical infrastructure can have consequences beyond the immediate hazard footprint.

Sectoral Impact Assessment

The hazard assessment is complemented by a sectoral impact assessment examining how identified climate hazards and projected climatic changes may affect major sectors of Uttarakhand.

The assessment considers:

- Agriculture
- Water resources & WASH
- Forests and biodiversity
- Energy
- Roads
- Health
- Tourism

The assessment focuses on the **hazard-exposure-impact pathway**, identifying how climate stressors and hazards may affect the functioning of sectoral systems.

S. No.	Sector	Key Climate Stressors / Hazards	Potential Impacts	Key Cross-Sectoral Implications
1	Agriculture	Rising temperatures; changes in precipitation; rainfall variability; prolonged dry periods; extreme rainfall	Changes in crop suitability and growing conditions; increased water stress; production uncertainty; impacts on chilling requirements and temperate fruit production.	Reduced farm productivity and income; livelihood impacts, particularly for household's dependent on climate-sensitive agriculture
2	Water Resources	Changes in precipitation and temperature; prolonged dry periods; extreme rainfall	Changes in water availability; increased runoff and erosion; flood-related impacts; stress on mountain water systems	Implications for agriculture, drinking water, ecosystems, livelihoods and energy systems
3	Forests & Biodiversity	Rising temperatures; changing precipitation; climate-induced disturbances	Changes in ecosystem functioning, species distribution, forest health and habitat suitability; increased sensitivity of vulnerable ecosystems	Impacts on ecosystem services, water regulation, soil conservation, biodiversity and communities dependent on natural resources
4	Energy	Floods; landslides; extreme rainfall; changes in hydrological conditions	Physical impacts on generation, transmission and distribution infrastructure; changes in water-dependent energy generation; disruption of energy services	Impacts on health facilities, water supply, communications and economic activities
5	Roads & Linear infrastructure	Landslides; fluvial floods; extreme rainfall	Physical damage to infrastructure; disruption of road connectivity; loss of functionality and service continuity	Reduced access to healthcare, markets, education, emergency services and tourism destinations

6	Health	Rising temperatures; warm days and nights; extreme events; disruption of water, transport and essential services	Increased heat-related health risks; impacts on climate-sensitive health outcomes; disruption to access and delivery of healthcare services	Increased pressure on health systems and vulnerable populations; impacts compounded by disruption of other essential services
7	Tourism	Landslides; floods; extreme rainfall; changing temperature and precipitation; environmental changes	Reduced destination accessibility ; disruption of transport connectivity; damage to tourism infrastructure; changes in tourism seasonality and suitability	Impacts on tourism-dependent businesses, workers and local livelihoods

These pathways have the potential for **cascading and interconnected impacts**, particularly where sectors depend on shared infrastructure, water resources, transportation networks and ecosystem services.

These interdependencies are particularly important for Uttarakhand, where the physical geography results in strong linkages between settlements, infrastructure, river systems, ecosystems and economic activities.

The Hazard Mapping and Sectoral Impact Assessment provides a consolidated understanding of the **climate hazards affecting Uttarakhand and their potential implications for the State's key social, economic, environmental and infrastructural systems**.

The assessment demonstrates the importance of considering both gradual climatic changes and extreme events, as well as the interactions between hazards and exposed systems. The results highlight that climate impacts in Uttarakhand are likely to be **spatially differentiated, sector-specific and interconnected**, with the potential for direct, compound and cascading consequences.

The outputs of this report provide the evidence base for the subsequent stages of the vulnerability and risk assessment by identifying **where hazards occur, which sectors and systems are exposed, and how climate stressors may translate into sectoral impacts**.

This provides the foundation for the subsequent assessment of **vulnerability and climate risk** and, ultimately, for identifying priorities for climate-resilient development in Uttarakhand.

1. Introduction

The magnitude of the climate crisis has intensified and its impacts are being experienced across the world. To address the ongoing crisis, countries around the globe have initiated climate action. In August 2021, the Intergovernmental Panel on Climate Change (IPCC) Working Group (WG) I released the Sixth Assessment Report (AR6), which stated that many of the climate changes that we are experiencing are largely unprecedented and are irreversible in nature (IPCC, 2021).

Furthermore, it adds that unless there are dedicated efforts to curb emissions, it will be inevitable to limit global warming to 1.5 degree or even 2 degrees in the coming decades.

These changes, however, may vary from one region to the other. The WG 2 AR6, released in February 2022, indicates the extent to which impacts are likely to be felt across different sectors including extremes like slow and sudden onset of events, along with an understanding on the regional implications of these changes (IPCC, 2022).

These changes in the climate system manifest themselves at a regional scale and have enormous consequences at the local scale. The impacts are likely to be felt across regions and sectors, with a social and economic footprint.

India has witnessed an increase of 0.7 °C in average temperature from 1901-2018. According to the Ministry of Earth Sciences (MoES, 2023), worst case scenario projections indicate an increase in temperature around 4.4 °C by the end of 21st century (as per 1976-2005 average).

The pace of climate change in the country has accelerated in recent years, with states reporting rising temperatures, increasing variability in rainfall leading to droughts and floods, unusual cyclonic activity along the coastline, melting of glaciers, thereby impacting the country's natural resource base. The excessive dependence of the country on climate-sensitive sectors (such as agriculture) exacerbates its susceptibility to climatic events.

Over the years the country has witnessed extreme rainfall, increased instances of forest fires and disasters like cyclones. In terms of geographical area, a large proportion is highly prone to extremes and disasters.

Estimates reflect that approximately 68% of cultivable land is vulnerable to droughts, while 58.6% is prone to earthquakes and seismic activity. Additionally, 12% of the land remains prone to floods, 5700 km of coastline is likely to be affected by cyclonic activity and 15% of land may be impacted by landslides. With an overwhelming share of the country's population still dependent on natural resources, such changes in climatic parameters have huge implications on the economy including loss of lives and livelihoods.

Such high susceptibility to changes in the climate system warrant urgent climate action across the country, with a focus on climate mitigation and adaptation actions at the centre of the development process.

The country has responded by initiating a slew of responses to limit the impact of climate change and respond effectively. India has been ranked seven in terms of climate risk as per the latest Global Climate Risk Index, 2021 owing to the high levels of exposure and vulnerability to extreme events.

Since, The Himalayan region is widely recognized as one of the most climate-sensitive and hazard-prone regions globally. Uttarakhand, situated in the central Himalayas, exemplifies this vulnerability due to its complex topography, dynamic geological conditions, and increasing exposure to climate variability.

The state has experienced several high-impact disasters over the past two decades, including the 2013 Kedarnath floods, the 2016 forest fires, and the 2021 Chamoli disaster, highlighting the urgent need for a comprehensive understanding of climate-induced risks.

Climate change is altering the fundamental drivers of natural hazards in Uttarakhand. Rising temperatures are accelerating glacier melt and changing snow dynamics, while increasing variability in precipitation is contributing to both flood and drought conditions.

These changes are interacting with non-climatic factors such as land-use change, deforestation, unplanned urbanization, and infrastructure development, thereby exacerbating the overall risk landscape.

Recognizing these challenges, this study aims to undertake a detailed climate change impact assessment across key sectors in Uttarakhand.

The assessment leverages downscaled climate projections, historical climate data, and sectoral information to evaluate the implications of climate variability and change. It forms a foundational component of a broader effort to develop a Climate Vulnerability Atlas at the block level, supporting evidence-based planning and decision-making.

The objectives of this report are threefold. First, it provides a comprehensive overview of the climatic trends and projected changes in Uttarakhand. Second, it establishes the baseline characteristics of the state, including its geographical, socio-economic, and environmental context.

Third, it sets the stage for sectoral impact assessments by identifying key climate drivers and their potential implications.

The assessment adopts an integrated approach, consistent with the Intergovernmental Panel on Climate Change (IPCC) risk framework, which conceptualizes risk as a function of hazard, exposure, and vulnerability.

By situating climate impacts within this framework, the report aims to provide a robust analytical foundation for subsequent vulnerability and risk analyses.

2. State Profile Uttarakhand

The state of Uttarakhand was created in November 2000 as the 27th state of the country. Formerly a part of Uttar Pradesh (called Uttaranchal then), the state is located at the foothills of the mighty Himalayas and shares international borders with China and Nepal. Neighbouring Indian states include Himachal Pradesh in the northwest and Uttar Pradesh in the south. Uttarakhand covers a geographical area of 53483 sq. km. The topography of the state is largely hilly with steep hills and large areas covered under snow. The topography also consists of snow-clad peaks, glaciers, deep canyons, streams, and lakes.



Map 1: Uttarakhand District Map

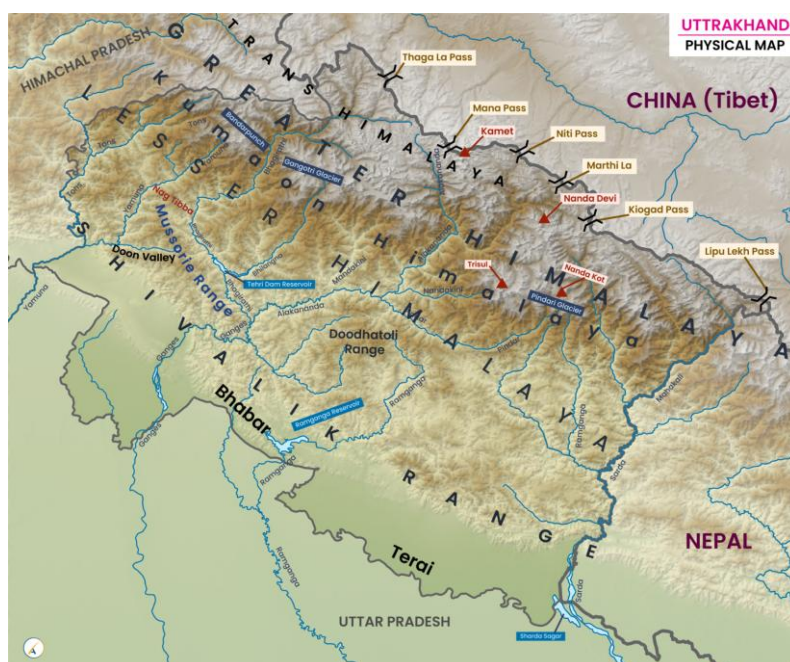
The state is home to some of the highest mountains in the world including Nanda Devi (7817 m), which is the second highest peak in India. Uttarakhand is often divided into multiple physiographic zones, running parallel to each other from northwest to southeast. The northern zone, popularly known as the Himadri, includes segments of the Zaskar and the Great Himalayan ranges, with elevations ranging in the range of 3000 to 7600 metres above mean sea level. Adjacent to the Great Himalayas are the Lesser Himalayas, popularly known as Himachal, with elevations between about 2000 and 3000 metres above mean sea level. To the south of Himachal is a stretch of the Shiwalik range. The entire area containing Himadri, Himachal, and Shiwaliks is broadly known as Kumaon Himalayas. The southern edge of the Shiwalik range merges with a narrow bed of gravel and alluvium known as Bhabar, which interfaces to the southeast with the marshy terrain known as Tarai.⁸ The state is broadly categorized into the Garhwal and Kumaon region and has 13 administrative divisions. The districts Dehradun, Haridwar, Chamoli, Rudrapur, Tehri Garhwal, Uttarkashi and Pauri

Garhwal, constitute the Garhwal Division. On the other hand, Kumaon division comprises Almora, Nainital, Pithoragarh, Udham Singh Nagar, Bageshwar and Champawat. Dehradun is the capital of the state and is located in the Dun valley, on the watershed of Ganga and Yamuna rivers. Uttarakhand is a mountainous state located in northern India, forming part of the central Himalayan region. It shares international borders with China (Tibet Autonomous Region) to the north and Nepal to the east, while being bounded by Himachal Pradesh to the west and Uttar Pradesh to the south. The state spans a geographical area of approximately 53,483 square kilometers and is characterized by a wide altitudinal range, extending from the Gangetic plains at around 300 meters above sea level to peaks exceeding 7,000 meters in the Greater Himalayas (Government of Uttarakhand, 2024). The state is divided into 13 districts and 95 development blocks, with significant spatial heterogeneity in terms of topography, climate, and land use. The major river systems, including the Ganga and Yamuna, originate in the glaciers of Uttarakhand, making it a critical hydrological region for northern India. The presence of numerous glaciers, glacial lakes, and snow-fed rivers underscores the importance of cryospheric processes in shaping the state's climate and water resources.

Uttarakhand's geographical setting makes it highly susceptible to natural hazards. Steep slopes, active tectonics, and fragile geological formations contribute to landslide susceptibility, while intense rainfall events and glacial dynamics increase the risk of floods and GLOFs. The spatial distribution of settlements and infrastructure across hazard-prone areas further amplifies the risk.

2.1 Physiographic conditions

The physiography of Uttarakhand is broadly divided into three distinct regions: the Greater Himalayas, the Lesser Himalayas, and the Shivalik ranges, along with the adjoining Terai and Bhabar plains. Each of these regions exhibits unique geological, geomorphological, and climatic characteristics.



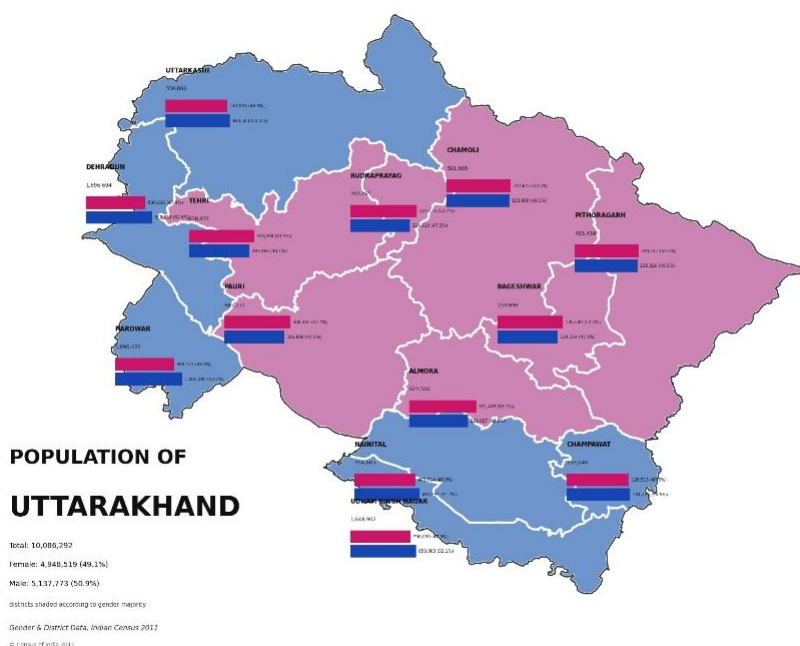
Map 2: Uttarakhand Physical Map

The Greater Himalayas, located in the northern part of the state, are characterized by high-altitude snow-covered peaks, extensive glaciation, and rugged terrain. This region is the source of major river systems and is highly sensitive to climate-induced changes such as glacier retreat and snow variability. The Lesser Himalayas form the middle zone and are characterized by steep slopes, deep valleys, and dense forest cover. This region is particularly prone to landslides and slope instability due to a combination of geological fragility and intense rainfall. The Shivalik ranges and the Terai-Bhabar plains constitute the southern part of the state. These areas are relatively lower in elevation and are characterized by alluvial soils, dense forests, and significant agricultural activity. However, they are also prone to flooding, waterlogging, and heat stress.

The geological composition of Uttarakhand is marked by young fold mountains, active fault lines, and highly weathered rock formations. These characteristics, combined with high seismicity, make the region one of the most hazard-prone areas in India (Geological Survey of India, 2020).

2.2 Demographic and social profile

The state is divided into 2 administrative divisions, with 13 districts; 102 tehsils; 95 development blocks; 670 nyaay panchayats; and 7950-gram panchayats. There are 16,793 census villages of which 15,745 are inhabited and 1048 uninhabited (2011 census). The population of Uttarakhand was about 10 million in 2011, accounting for about 0.83% of India's population (Statistical Diary, Government of Uttarakhand, 2013). Around 66% of the population resides in rural areas. Between 2001 and 2011, the population of the state grew by 18.81%, higher than the national average of 17.6%. Within the state the population in plains grew at a much faster rate (over 32%) as compared to hills (12.7%).



Map 3: Demographic Map of Uttarakhand (Census 2011)

The population growth rate in some hill districts was observed to be negative: –1.73 % in Almora and –1.51 % in Garhwal (Pauri), while districts such as Dehradun, Udham Singh Nagar, Nainital and Haridwar saw a high rate of decadal increase in population – highlighting the trend of migration from the hills. The data also points towards

an outflux of people from many hill-regions of the state; as in the case of Almora and Garhwal (Pauri) districts.

Where an absolute decline of 17,868 persons in the population between 2001 and 2011 was recorded. Further, in the last 10 years, a total of 3,83,726 persons from 6338-gram panchayats have migrated on a semi-permanent basis. Additionally, between 2001 and 2011, as many as 1,18,981 permanent migrations were reported from 3946-gram panchayats. Thus, the data indicates that there are more semi-permanent migrants than permanent migrants in all districts of the state.

In terms of the Human Development Index (HDI), the state was ranked 18 on the HDI 2018. The state saw an improvement over the period, from 0.531 in , 2011-12 to 0.718 in 2017 - higher than the all-India HDI value (0.64) and just above the medium HDI score (0.70) as per United Nations Development Programme (UNDP) 2018. Then top three districts of Dehradun (0.765), Haridwar (0.733), and Udham Singh Nagar (0.717) are the plain districts.

Among the remaining hill districts, Rudrapur (0.626), Champawat (0.620) and Tehri Garhwal (0.611) are placed at the bottom. This gap between hill and plain districts is mostly marginal in terms health and education facilities, however, there is considerable difference in their per capita GSDP.

Lastly, in terms of the Multi-dimensional Poverty Index (MPI), the state had an overall score of 11.90. The MPI presents a comprehensive understanding of health, education and standard of living. Hill districts of Chamoli, Champawat, Pithoragarh, Tehri Garhwal and Uttarkashi reported the highest MPI, thereby indicating poor performance on these aspects. The plain districts – Haridwar, Udham Singh Nagar and Dehradun - have performed better on the MPI, depicting lower deprivation in education, health, and standard of living.

Therefore, Women play a critical role in natural resource management and household economies yet often face disproportionate burdens during climate-induced stresses. Similarly, children, elderly populations, and marginalized groups are particularly vulnerable to climate impacts due to limited adaptive capacity.

Access to healthcare, education, and basic services varies significantly across regions, with remote mountainous areas often facing significant challenges. These disparities contribute to differential vulnerability across the state.

2.3 Economic profile

Uttarakhand has shown impressive growth after its formation in 2000. In 2011-12, per capita income (per capita Net State Domestic Product or NSDP at factor cost at 2004–05 prices) in Uttarakhand stood at INR 50,303, against the highest of INR 1,12,626 in Delhi and the lowest of INR 13,226 in Bihar. Between 2001–02, 2012–13 and

2018–19, the state's Gross State Domestic Product (GSDP) grew at a compound annual growth rate of 10.85%, to INR 2.37 trillion.

According to the 2011 census, the majority of Uttarakhand's working population, about 51% (from 60% in 2001), was dependent on agriculture for livelihood. This contrasts sharply with the fact that agriculture contributed only 12% to the total output of the state in 2010-11. Data on land holdings reflects that small and marginal land holdings have increased over the decade – from an already high 55% of the total area of operational holdings in 2001 – to 64% in 2011. This increase was due to fragmentation of the medium and large land holdings which, together with limited irrigation facilities especially in hilly areas, could have contributed to the low share of agricultural in the GSDP.

Trends indicate that the growth rate of GDSP (at constant prices) in the state declined to 4.3% in 2019–20, as compared to 5.8% in the previous year. Sector-wise contribution indicates that in 2019-20, agriculture, manufacturing, and services sectors contributed 9%, 52%, and 39%, respectively to the state economy. The financial year 2019-20 reflects a decline in the growth of manufacturing and services sectors as compared to 2018–19. Lastly, the per capita GSDP of Uttarakhand in 2019-20 (at constant prices) was INR 1,78,050, which is 3.1% higher than that of 2018-19

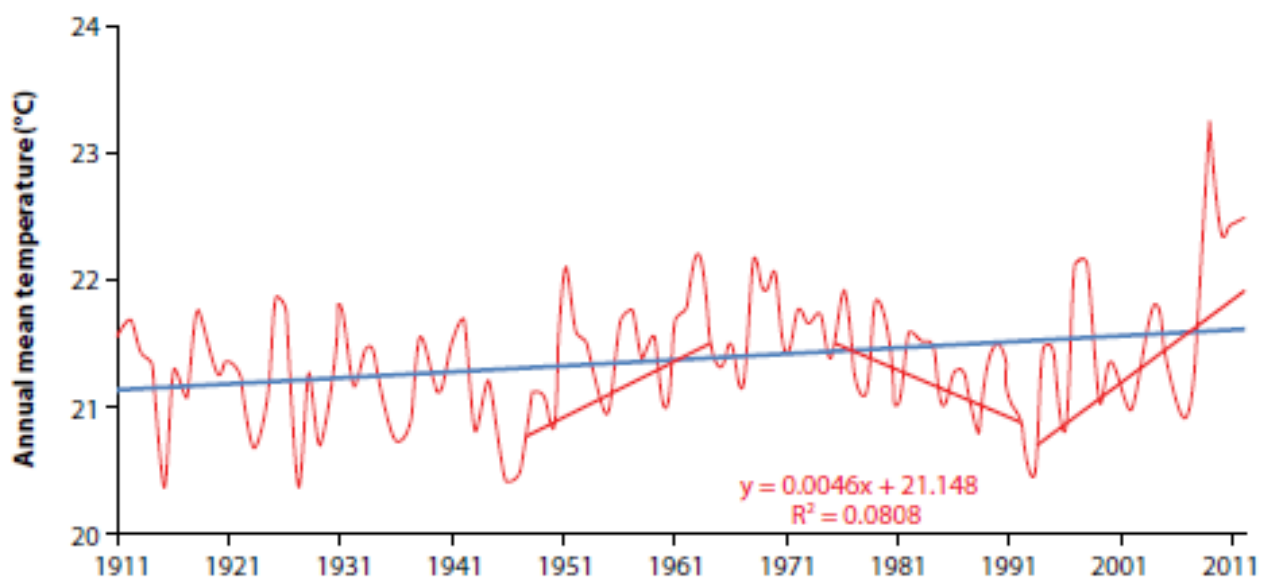
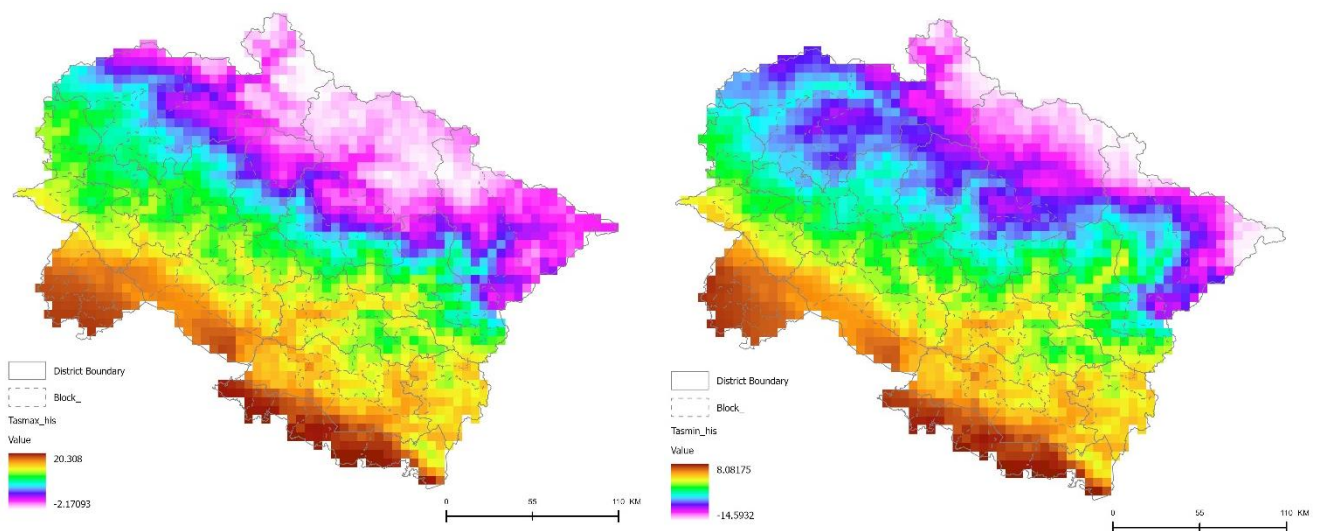
The state also has significant hydropower potential, contributing to its energy sector. However, hydropower infrastructure is exposed to risks from glacial hazards, sedimentation, and extreme hydrological events.

According to the Economic Survey of Uttarakhand (2024-25), the state's economy has shown steady growth, but regional disparities persist. The dependence on climate-sensitive sectors underscores the importance of integrating climate resilience into economic planning and development strategies.

3. Climate Profile

3.1 Observed climate and variability

Temperature: Changes in the temperature of a region is one of the important reasons behind the increasing occurrence of natural disasters. In the last hundred-year period (1911-2011), the state of Uttarakhand has shown a warming annual mean temperature trend (Mishra, 2017; Yadav, et al. 2014). There has been a clear increase in mean annual temperature in the last decade (Figure 4.1). High elevation districts like Uttarkashi, Chamoli, Rudraprayag and Pithoragarh noticed higher increase in temperature as compared to the low-lying districts (Mishra, 2017). The annual mean maximum temperature and minimum temperature for all the districts of Uttarakhand (1950–2013) has a decreasing and increasing trend respectively. (Pranuthi, et al. 2014; Gosain et al. 2016).



An increasing trend of 0.42 °C/63 years for the annual minimum temperature and decreasing trend of -0.25°C/63 years for the annual maximum temperature has been inferred. These

changes are not statistically significant, but in the seasonal trend analysis significant variations were noted.

The seasonal variation of mean maximum and minimum temperature of Uttarakhand (over 1951–2013). It is evident that the southern districts of the state have a comparatively higher minimum temperature than the northern ones.

It can also be noticed that the variability of minimum temperature is higher than that of the maximum temperature.

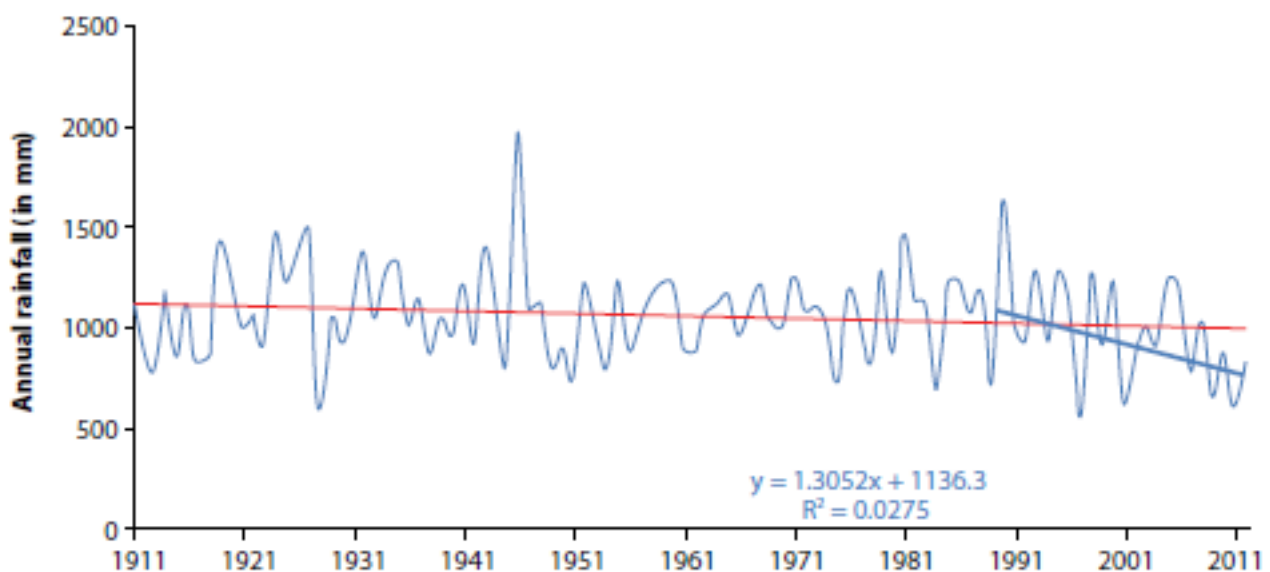
Rainfall:

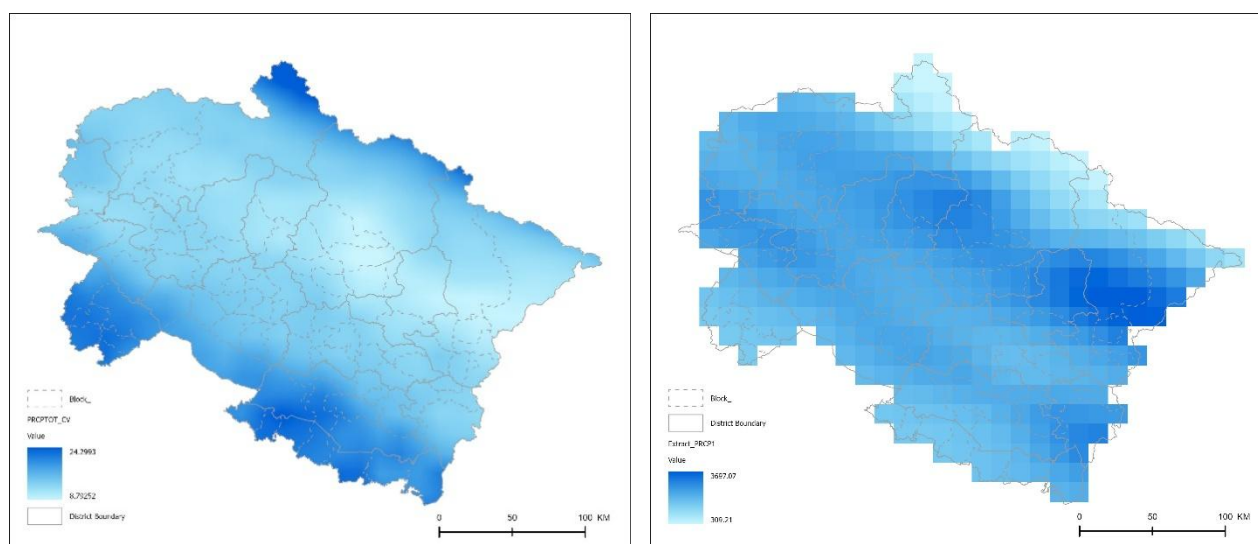
Change in the rainfall pattern is inevitable in a changing climate scenario. Higher variations in rainfall will lead to meteorological extremes (like floods or droughts) which in turn adversely affect livelihood. In the state of Uttarakhand, the precipitation can be rain as well as snow at some locations in the winters.

The rainfall in Uttarakhand is monsoon and orographic in case of extreme events. The annual rainfall lies between 948 mm and 2986 mm – of which 60-85% is received in the monsoon season, i.e., June - September (Basistha, et al. 2007). The eastern districts of the state receive majority of the monsoon rainfall, i.e., >15 mm/day (Banerjee, et al 2019).

The rainfall pattern of Uttarakhand during 1911–2011. Over this period, the trend did not vary much except for the years after 1990s – where a decreasing trend is prominent.

This declining rainfall has put a huge pressure on the water resources of the region. Figure 4.5 depicts the monthly rainfall changes in the different districts of Uttarakhand from 1911 to 2011 (Mishra 2017).





Map 5: Rainfall Variability (left) and Mean annual rainfall (right) map of Uttarakhand

Changes in Climate Extremes: Changes Observed

Temperature extremes. Long-term station records for the northwest Himalaya show winter (December–February) warming outpacing summer warming, a trend of roughly 0.05–0.09°C per year in winter minimum temperature at high-altitude stations against a considerably weaker summer trend (Bhutiya et al., 2007), consistent with the broader Hindu Kush Himalaya finding that high-elevation warming is occurring faster than the global and regional lowland average, an “elevation-dependent warming” signal now treated as an established feature of the HKH region rather than a local anomaly (Krishnan et al., 2019, *Hindu Kush Himalaya Assessment*; IPCC AR6 WGI, Chapter 10). For Uttarakhand specifically, this shows up as a measurable reduction in the number of cold days and cold nights per year and a corresponding increase in warm days and warm nights, changes that matter disproportionately for the state’s high-altitude blocks, where the same warming shifts the rain–snow threshold elevation upward, converts a growing share of winter precipitation from snow to rain, and feeds directly into the glacier mass-balance and spring-recharge decline documented elsewhere in this report (Section 5.1, Section 9.4.1 water-sector findings).

Precipitation extremes. The more consequential observed shift is not in total rainfall, which shows no strong long-term trend across most of Uttarakhand, but in how that rainfall now arrives. IMD and IIT Roorkee’s own downscaled hazard profile both show a growing share of the state’s annual rainfall concentrated into fewer, more intense rainfall days, consistent with the all-India and HKH-wide finding that a decline or stagnation in total monsoon rainfall has coincided with a rise in the frequency of extreme (>100mm/day) rainfall events (Krishnan et al., 2019; Roxy et al., 2017, for the wider Indian monsoon context). This is the direct precipitation mechanism behind the state’s rising cloudburst, flash-flood and landslide-triggering-rainfall frequency documented in Section 4: the same, or even a declining, seasonal rainfall total is now more capable of triggering a hazard event because it arrives in short, intense bursts rather than distributed evenly across the monsoon. The state’s own recorded-disaster pattern is consistent with this: recorded hazard events concentrate

overwhelmingly in the June–September window, and within it, August specifically, with events increasingly attributed in local reporting to short-duration, high-intensity cloudburst cells rather than sustained monsoon rainfall.

Seasonal Shifts: Changes Observed

Three seasonal-timing shifts recur across the observed record and the IIT Roorkee downscaled profile used in this assessment:

A less reliable, more erratic monsoon onset and withdrawal. Monsoon onset and withdrawal dates over the wider Indian Himalayan region have become measurably more variable year-to-year over the past two to three decades, with a tendency toward delayed onset and, in some recent years, an extended withdrawal period, lengthening the effective monsoon season in some years while compressing it in others (Krishnan et al., 2019; IMD long-period monsoon records). For Uttarakhand’s predominantly rainfed hill agriculture, already identified in this assessment’s Agriculture chapter as its principal source of climate sensitivity, this onset variability is at least as consequential as any trend in total seasonal rainfall, since sowing decisions are timed to onset, not to the seasonal total.

A shrinking, more erratic winter-precipitation (western disturbance) season. A materially different seasonal driver, winter precipitation delivered by western disturbances, shows a distinct and, for Uttarakhand’s high-altitude blocks, arguably more consequential shift: observed weakening and increased year-to-year variability in western-disturbance-driven winter snowfall across the northwest Himalaya (Dimri et al., 2015; Krishnan et al., 2019), producing winters that are, in a growing share of years, both warmer and drier of snow than the historical baseline. Because winter snowpack is the slow-release source that sustains the spring and stream discharge documented as failing in this assessment’s Water Resources chapter (Section 9.4.1), a shift in winter-precipitation reliability compounds, rather than sits separately from, the spring-depletion mechanism already identified as a driver of village abandonment in Uttarakhand’s hill districts.

Lengthening pre-monsoon dry spells and a compressed effective growing season. Taken together, later and more erratic monsoon onset and a drier, less reliable winter season are lengthening the pre-monsoon dry spell separating the two, a period of moisture stress that falls directly on the rabi-to-kharif transition in the state’s cropping calendar and on the late-winter/early-summer period when spring discharge is already at its seasonal low. No single Uttarakhand-specific study quantifying this compound dry-spell lengthening in growing-season-day terms could be confirmed in the course of this research; this sits alongside the other sector-specific evidence gaps flagged throughout this assessment as a priority for the state’s own agrometeorological monitoring network to close, since it directly affects both the AGVI’s rainfed-dependence sensitivity indicator and the WRVI’s spring-discharge indicator without currently being measured as a distinct seasonal-timing variable in either.

4. Hazard Mapping

Uttarakhand's complex Himalayan terrain, strong altitudinal gradients, monsoon-driven rainfall, glaciated landscapes, extensive forest cover and concentration of settlements and infrastructure along valleys make the State highly exposed to a diverse range of natural and hydro-meteorological hazards.

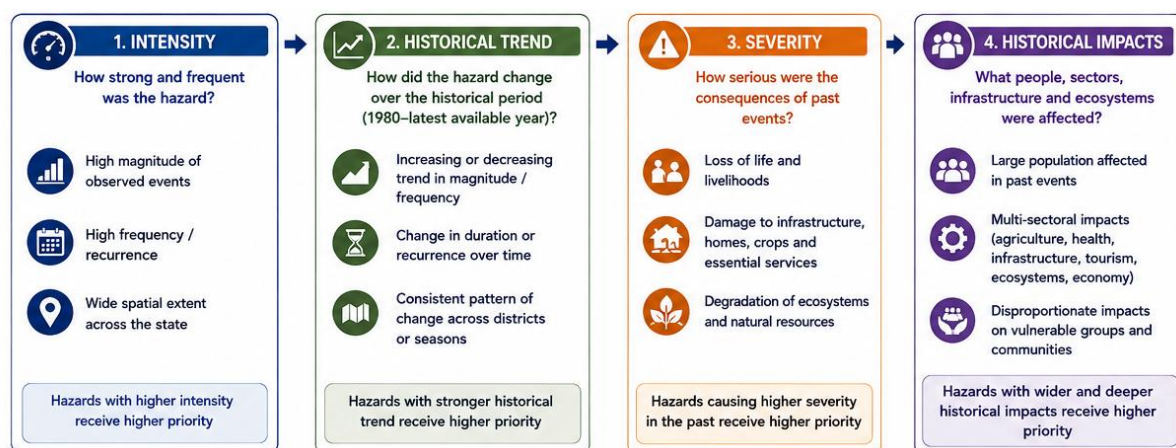
The hazard mapping exercise under this assessment adopts a **historical evidence-based approach**, with hazard characteristics analysed from **1980 to the latest year for which reliable data are available**. The objective is to establish the spatial and temporal characteristics of hazards that have actually occurred across Uttarakhand and to identify those hazards that have demonstrated sufficient intensity, recurrence, severity and sectoral impacts to warrant detailed mapping.

The approach deliberately distinguishes **historical hazard assessment** from future climate projection. The identification and prioritisation of hazards presented in this chapter are therefore based on **observed and documented evidence from the past**, rather than on assumptions regarding future changes in climate.

The historical period provides a sufficiently long record to examine recurring hazard behaviour, extreme events, changes in frequency or magnitude, and the consequences of hazards across people, livelihoods, infrastructure, ecosystems and economic activities.

Hazard Priority

The initial hazard universe was developed from the major climate-sensitive and hydro-meteorological hazards known to affect Uttarakhand. Candidate hazards were subsequently screened using four dimensions:



1. **Intensity:** How strong or extreme has the hazard been?
2. **Trend:** How has the occurrence, frequency, magnitude or spatial extent of the hazard changed over the historical period?
3. **Severity:** How serious have the consequences of individual events been?
4. **Impacts:** How widespread and multi-sectoral have the observed impacts been?

The four dimensions provide the basis for distinguishing hazards that are merely present in the State from hazards that have demonstrated sufficient historical significance to be considered priority hazards for spatial mapping and vulnerability assessment.

Historical Data Period

The assessment uses **1980 as the common historical reference point**, wherever adequate data are available, while recognising that individual datasets may have different temporal coverage.

Type of event	Date	District	River	Location	Casualties/ damage	References
Landslide	1803	Chamoli	Alaknanda	Birahi	73 humans	Rautela and Pande (2005)
Landslide	1880, 19 Sep	Nainital	-	Sher Danda	151 humans	Rautela and Pande (2005)
Lake burst	1894, 25 Aug	Chamoli	Alaknanda	Birahi	Property lost	Rautela and Pande (2005)
Lake outburst flood	1894	Chamoli	Alaknanda	Gohna Tal	-	Blanford (1894)
Landslide	1898, 17 Aug	Nainital	-	Kakarighat	29 humans	Middlemiss (1898)
Flood	1951	Pauri	Nayar	Satpuli	>50 humans	Niyaz and Kumar (1989)
Cloudburst	1970	Chamoli	Alaknanda	Belakuchi	Property lost	Rana et al. (2013)
Rainfall induced landslide	1970, 20 July	Chamoli	Alaknanda	Belakuchi	70 humans	Niyaz and Kumar (1989)
Cloudburst	1971, 13 July	Pithoragarh	-	Debatia village	12 humans, 35 animals	Khanduri et al.
Flood	1976, 1 July	Chamoli	Nandakini	Nandprayag	>5 humans	Niyaz and Kumar (1989)
Cloudburst	1978, 4 Aug	Uttarkashi	Bhagirathi	Upper Bhagirathi	25 humans	Prasad and Rawat
Cloudburst	1979, 17 June	Chamoli	Alaknanda	Suklot	3 humans, 70 animals	Joshi and Kumar
Landslide	1979, 17 Aug	Chamoli	Alaknanda	Kumna village	39 humans, property lost	Khanduri and Kumar
Landslide	1980, 23 June	Uttarkashi	Bhagirathi	Kandyoga Gad	45 humans	Agarwal and Chak (1989)
Landslide	1980, 9 Sep	Uttarkashi	Bhagirathi	Mandakhal	15 humans	Agarwal and Chak (1989)
Rainfall induced cloudburst	1982, 31 July	Pauri	Alaknanda	Karmi village	3 humans, 80 animals	Joshi and Kumar (2006)
Cloudburst	1983, 22 July	Bageshwar	-	-	150 humans, 20 animals	Joshi and Kumar (2006)

Rainfall induced landslide	1990, 9 July	Tehri	Bhagirathi	Neketh	100 humans, 10 houses	Joshi and Kumar (2006)
Cloudburst	1991, 16 Aug	Chamoli	Alaknanda	Gopeswar, Dronagiri	63 humans, 38 houses	Asthana (2014)
Cloudburst	1992, 2 Sep	Chamoli	Alaknanda	Ghat	14 humans, 31 animals	Asthana (2014)
Cloudburst	1993, 3 July	Chamoli	-	Pabni, Musdiyari	25 animals	Joshi and Kumar
Cloudburst	1995, 13 Aug	Pauri	Alaknanda	Baniyara	6 humans	Sah (2007)
Cloudburst	1998, 17 Jul	Pithoragarh	-	Berinag	5 humans, 85 animals	Joshi and Kumar
Rainfall induced landslide	1998, 11–19 Aug	Rudraprayag	Mandakini	Rambara	92 houses	Joshi et al. (2000)
Cloudburst	2000, July	Uttarkashi	Bhagirathi	Phitargad	Property lost	Bhakuni (2015)
Landslide	2001, 11 Aug	Rudraprayag	Mandakini	Budha Kedar	-	Sah et al. (2003)
Cloudburst	2001, 16 July	Rudraprayag	Mandakini	Para Bugyal	21 humans	Joshi and Kumar (2006)
Cloudburst	2001, 17 Aug	Rudraprayag	Mandakini	Guna village	28 animals	Semwal (2014)
Cloudburst	2001, 13 Aug	Tehri	Bhilangna	-	-	Semwal (2014)
Cloudburst landslide	2002, 10 Aug	Chamoli	Alaknanda	Budha Kedar	38 humans	SDC (2003)
Cloudburst landslide	2002, 29 Aug	Uttarkashi	Bhagirathi	Syanachatti	26 animals	SDC (2003)
Rainfall induced landslide	2003, 5 Aug	Uttarkashi	Bhagirathi	Paravat	207 animals	Asthana
Landslide	2004, July	Chamoli	Alaknanda	Lam Bagar	16 humans	Khanduri and Pande
Landslide	2004, July	Uttarkashi	Yamuna	Kaldi	6 humans	Kumar and Pande
Cloudburst	2005, 18 June	Chamoli	Alaknanda	Govindghat	90 humans	Rautela and Pande (2005)
Cloudburst	2005, 30 June	Chamoli	Alaknanda	Hanuman Chatti	11 humans	Rautela and Pande (2005)
Rainfall induced cloudburst	2005, 18 Aug	Rudraprayag	Mandakini	Bhatwari village	8 humans, 192 animals	Thakur (2000)
Cloudburst	2006, 26 July	Rudraprayag	-	Jadoli, Byoli, Chaur	19 animals	SDC
Cloudburst	2007, Sep	Pithoragarh	-	Baram	5 humans, 9 animals	Sawal et al. (2007)

Rainfall induced landslide	2009, 18 Sep	Pithoragarh	Jakula	Kimy village	43 humans	Sarkar and Sati (2010)
Rainfall induced landslide	2010, 1–2 Sep	Uttarkashi	Bhagirathi	Bhatwari	167 humans	GSI (2014)
Landslide	2010, 18 Aug	Bageshwar	-	Sangari village	18 children	Parkash (2015)
Rainfall induced landslide	2010, 18–21 Sep	Rudraprayag	Ganga	Madhyamaheshwar	220 humans	Semwal (2011)
Flood and debris flow	2010, 25 Aug	Uttarkashi	Bhagirathi	Dabrani Glacier	Extensive damage	Kumar et al. (2014)
Flash flood	2012, 3 Aug	Uttarkashi	Bhagirathi	Assiganga	48 humans, livestock loss	Rana et al. (2012)
Cloudburst flash flood	2012, 15 Sep	Rudraprayag	Mandakini	Okhimath	591 livestock	Gupta et al. (2013)
Cloudburst flash flood	2013, 16–17 June	Chamoli	Alaknanda	Kedarnath	>4000 humans	Bahuguna et al. (2013)
Cloudburst flash flood	2013, 16–17 June	Uttarkashi	Bhagirathi	Gangotri Glacier	Infrastructure loss	Arora et al. (2016)
Cloudburst	2016, 23 May	Tehri	-	Koti and Silwara	100 animals, 120 houses	Dimri et al. (2017)
Cloudburst	2017, 8 May	Pithoragarh	Ramganga	Dhanari, Talla	8 animals, property lost	Chaudhary et al.
Cloudburst	2017, 26 May	Pithoragarh	-	Mangri and Malpa	51 animals	Mehrotra et al. (2018)
Flash flood	2021, 7 Feb	Chamoli	Rishiganga–Dhauliganga	Raini village	206 missing, 70+ deaths	Sati et al. (2021)
Extreme Rainfall / Flash Flood	Oct-21	Nainital, Almora, Champawat	Gaula, Kosi, Sarda	Nainital town, Ramnagar, Haldwani	72 deaths; widespread inundation; 6,000+ displaced	Mishra, V. et al. (2022)
Urban Land Subsidence	Jan-23	Chamoli	Alaknanda	Joshimath town	868+ cracked structures; 250+ families displaced	Sati. et al. (2023)
Cloudburst / Flash Flood	Aug-20	Pithoragarh	Kali / Saryu	Dharchula, Munakot	16 deaths; 10 missing; 200+ houses damaged	Sati. et al. (2021)
Extreme Monsoon Flooding	Monsoon 2020	Multiple districts	Alaknanda, Kosi, Ramganga	State-wide	200+ deaths (season); climate change signal in precipitation	Bharti, V. et al. (2022)

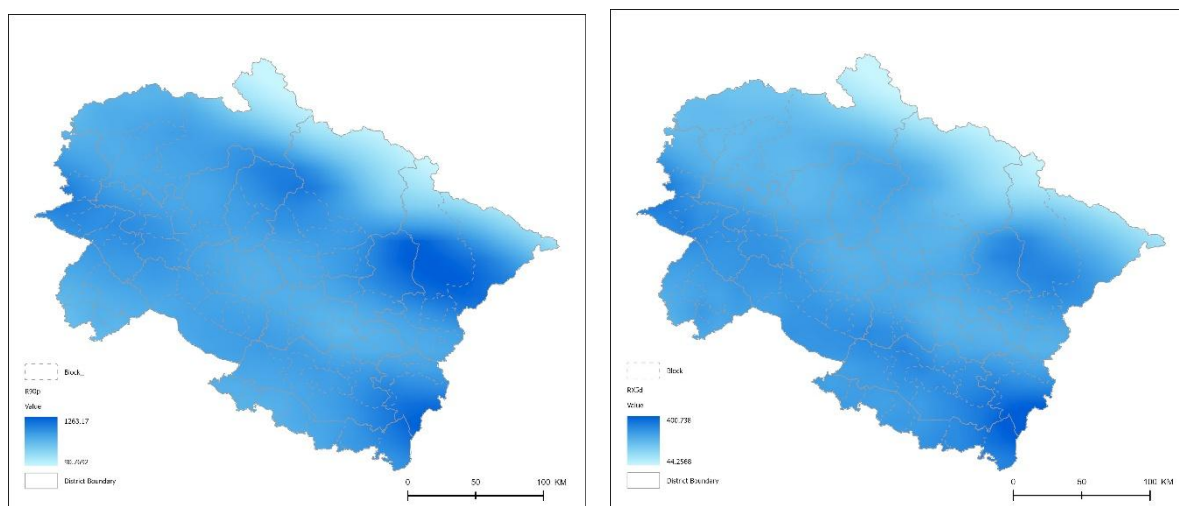
Earthquake-triggered Landslides	2020–2022	Rudraprayag, Chamoli	Mandakini, Alaknanda	Kedarnath & Badrinath corridors		Martha, T.R. et al. (2021)
Forest Fire	Mar-May 2022	Nainital, Pauri, Almora, Tehri	—	Kumaon & Garhwal forests	~3,000 ha burnt; biodiversity loss; climate drivers documented	Sharma, R. et al. (2023)
Monsoon rainfall landslide	2023, 26 Jun- 12 Jul	Multiple (Uttarakhand affected)	—	Mandi, Solan, Hamirpur districts and Uttarakhand region	62 humans	EM-DAT (2023)
Extreme monsoon rainfall / flooding	2023, 25 Jun- 7 Sep	Multiple districts	—	Uttarakhand and multiple Indian states	1529 humans; 10,250,000 affected	EM-DAT (2023)
Landslide / heavy rainfall event	2023, 23-24 Aug	Multiple (Uttarakhand & Himachal)	—	Himalayan region	10 humans; 605 affected	EM-DAT (2023)
Landslide	2025, 28 Feb	Chamoli	—	Chamoli district	8 humans killed; 46 injured; 219 affected	EM-DAT (2025)
Flood and landslide	2025, 20–26 Jun	Multiple districts	—	Rajasthan, Gujarat, Odisha, Uttarakhand region	27 humans; 33,584 affected	EM-DAT (2025)
Monsoon rainfall landslide	2025, 20 Jun - 7 Jul	Multiple districts (Uttarakhand affected)	—	Himalayan region	140 humans; 74 injured; 80,074 affected	EM-DAT (2025)
GLOF	2025, 5 Aug	Uttarkashi	Bhagirathi basin	Dharali village	105 humans	EM-DAT (2025)
Lightning / landslide event	2025, 15-17 Sep	Dehradun, Champawat, Haridwar, Pauri Garhwal, Bageshwar, Pithoragarh, Nainital, Udham Singh Nagar	—	State-wide Uttarakhand	33 humans; 900 affected	EM-DAT (2025)

Hazard Profile

Extreme Rainfall

Intensity. Two complementary layers characterise rainfall intensity at the block level. The maximum multi-day rainfall accumulation layer (RXGd) ranges from 44.3 mm to 400.7 mm across the state, with the highest values concentrated in the southern Bhabar-Tarai belt and in a distinct pocket along the state's central-eastern corridor. R95p, the standard extreme-precipitation index measuring total annual rainfall contributed by very-wet days (days above the local 90th-percentile threshold), ranges from 30.8 mm to 1,263.2 mm, an over 40-fold spread across the state that is itself evidence of how unevenly extreme-rainfall risk is distributed: the same south and central-east blocks that score highest on RXGd also carry the highest R95p values, confirming that the state's most intense single rainfall events and its heaviest cumulative extreme-rainfall burden concentrate in the same geography.

Trend. Consistent with the observed shift toward fewer, more intense rainfall days documented in Section 5.1a, the extreme-rainfall indices show that a rising share of the state's annual rainfall is now delivered through the very-wet-day tail of the distribution (R90p) rather than through steady, moderate rainfall, a trend most pronounced in the blocks already scoring highest on both layers.



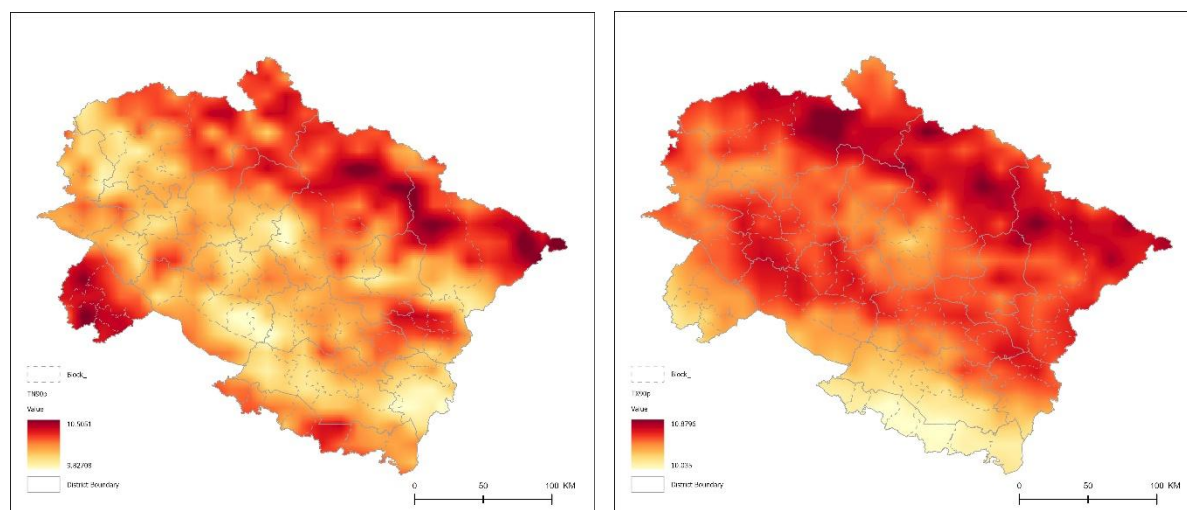
Map: RX90p(left) and RX5d(right) spatial map of Uttarakhand (IMD 1990-2020)

Severity. Severity is highest where high R90p values intersect with steep terrain and dense drainage, since extreme rainfall in these blocks translates directly into the flash-flood and landslide-triggering rainfall rather than being absorbed by the landscape.

Impacts. Extreme rainfall is the direct upstream trigger for the flood, landslide and GLOF hazards profiled below, and is the principal driver of the state's recorded-disaster concentration in the June–September monsoon window

Future Projections. Under SSP5-8.5, regional projections carried through this assessment indicate a 6-16 per cent increase in monsoon precipitation by end-century, with the extreme tail of the distribution (captured by R90p) expected to grow faster than the seasonal mean, intensifying rather than relocating the current high-R90p hotspots.

Extreme Temperature'



Map 6: TN90p (left) and TX90p (right) map of Uttarakhand (IMDAA 1990-2020)

Intensity: The warm-days indicator mapped for this assessment (TX90p, the frequency of days with maximum temperature above the local 90th-percentile threshold) ranges from 10.0 to 10.9 across the state on the trend metric used here, with the highest values concentrated along the state's northern and eastern high-elevation margins and in a distinct pocket in the southwest, indicating that the warm-days trend is strengthening fastest not only in the already-warm plains but also at the margins of the high-Himalayan belt, consistent with the elevation-dependent warming signal

Trend. The warm-days trend is positive across almost the entire state, with no block showing a declining trend in the frequency of hot days, reinforcing the observed warming.

Severity. Severity concentrates in the Bhabar-Tarai plains and valley towns (Dehradun, Haridwar, Udham Singh Nagar), where warm-days frequency compounds with already-

high baseline temperature (20–24.9°C annual-mean band) to produce the state’s most acute heat-stress exposure.

Impacts. Heat-stress impacts concentrate on outdoor and agricultural workers, the elderly, and infants in the plains and valley towns; heatwave conditions also elevate electricity demand for cooling at exactly the time grid reliability is most tested

Future Projections. The 2.3–5.9°C end-century warming range under SSP5-8.5 (Section 5.2) implies a substantial further increase in warm-days frequency across all districts, with the plains districts crossing progressively more heat-stress thresholds per year.

Drought

Intensity. Consecutive Dry Days (CDD), the layer referenced as Map 9 in this chapter, ranges from 21.2 to 70.8 days across the state, with the longest dry spells concentrated in the southern plains and a distinct secondary concentration in the northeastern corner of the state; the state’s interior and northern-central blocks show comparatively shorter dry spells. The block-wise drought-risk classification confirms this pattern, with Very High risk concentrated in the northwestern and northeastern corners of the state and scattered Moderate-to-High pockets through the central-southern belt.

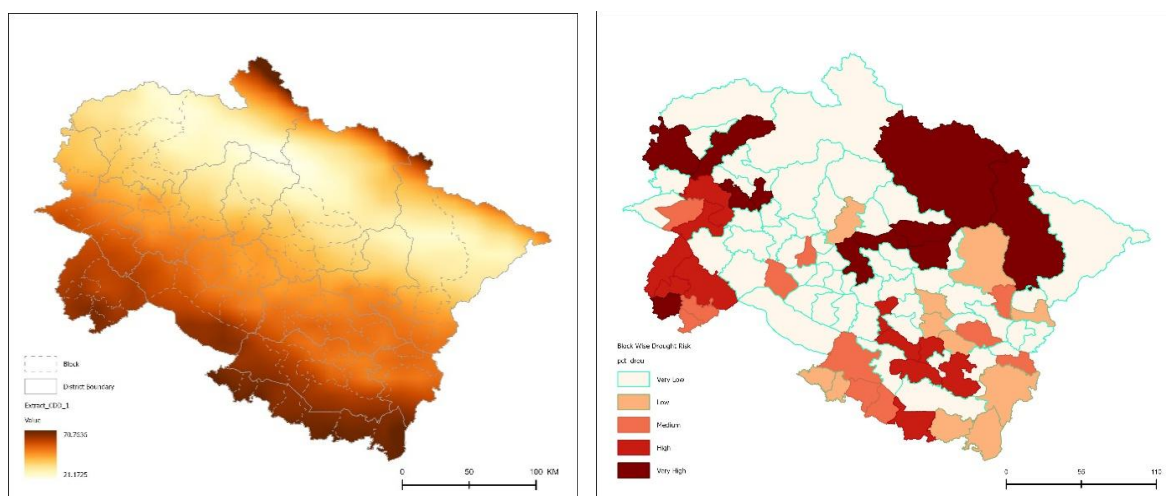
Trend. The CDD pattern is consistent with the lengthening pre-monsoon dry-spell trend discussed in Section 5.1b, and with the drought hazard’s classification as driven by IMD’s long-period (1901–2025) Drought Normalised Vulnerability Index.

Severity. Severity is highest where long dry spells intersect with high rainfed-agriculture dependence and low irrigation coverage, precisely the AGVI sensitivity mechanism, and where they intersect with the spring-dependence pattern, since consecutive dry days are the direct climatic driver of the spring-discharge decline already identified as a migration driver in Uttarakhand’s hill villages.

Impacts. Drought damages rainfed crop yield, dries mountain springs, and reduces fodder and livestock water availability; it is the hazard most directly linked, via the CDD-spring-discharge pathway, to the village-abandonment pattern documented in this assessment’s Water and Agriculture chapters.

Future Projections. Regional projections of greater year-to-year monsoon variability imply that dry-spell length is likely to become more variable rather than uniformly

longer or shorter, meaning the state's drought-risk classification should be revisited each time the underlying CDD baseline is updated rather than treated as static.



Consecutive Dry Days, mean

Block-wise Drought Risk classification

Landslide

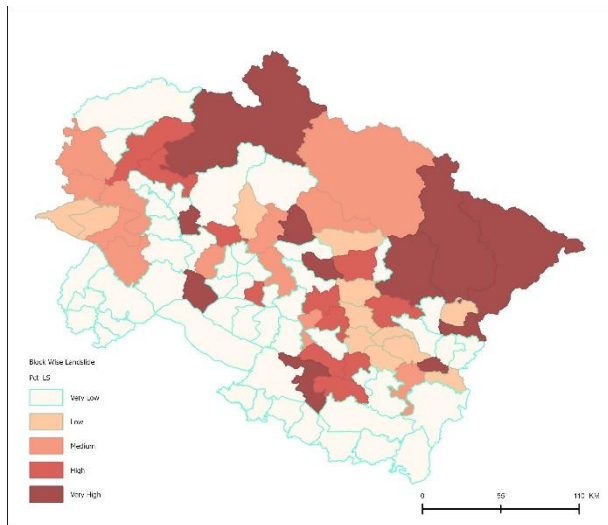
Intensity and Severity. The block-wise landslide-risk classification shows Very High risk concentrated in a broad band across the state's northern districts and in a large contiguous area in the east, consistent with the Char Dham corridor and Pithoragarh trekking-belt landslide hotspots and in the Roads GEDSI addendum (81 per cent of landslide occurrences on the Char Dham corridor confined within a 100-metre buffer of the road itself).

Trend. Landslide frequency on the state's road corridors has been rising over the observed record, a trend this assessment links directly to the combination of extreme-rainfall intensification and construction-driven slope destabilisation on the same corridors.

Impacts. Landslides damage roads, settlements and hydropower assets, sever the single-access routes that isolate resident hill communities and, on the evidence reviewed in the Roads GEDSI addendum, disproportionately affect the migrant construction workforce building and widening the same corridors.

Future Projections. Projected landslide risk increases of up to 90 per cent in certain districts during monsoon months by the latter half of the century indicate an

intensification, rather than a spatial shift, of the Very High risk zones already mapped below.



Block-wise Landslide Risk classification

Flood (Fluvial/Riverine)

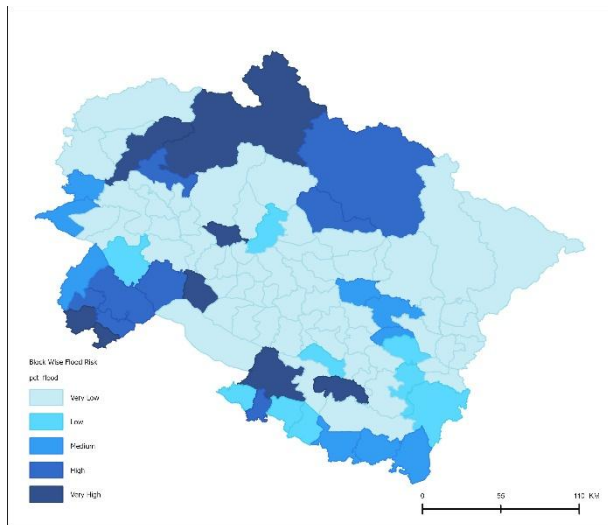
Intensity and Severity. The block-wise flood-risk classification shows High to Very High risk concentrated in a cluster of northern blocks and a distinct cluster in the west, with additional scattered High-risk blocks in the southern plains, consistent with the three flood mechanisms: flash flooding in steep northern catchments, riverine flooding along the west's river valleys, and urban pluvial flooding risk in the expanding southern towns.

Trend. Flood risk in the state's steep northern catchments tracks the extreme-rainfall intensification trend directly while flood risk in the southern plains is compounded by rapid urbanisation increasing impervious surface cover and pluvial flood risk independent of any change in rainfall itself.

Impacts. Flooding damages settlements, agricultural land and water-supply infrastructure, and is a leading cause of the state's recorded-disaster events in the monsoon window.

Future Projections. The same 6–16 per cent projected increase in monsoon precipitation is expected to elevate flood risk fastest in the already-High-risk northern

and western clusters, while continued urbanisation is expected to independently raise pluvial flood risk in the southern plains regardless of the rainfall trend.



Block-wise Flood Risk classification

Forest Fire

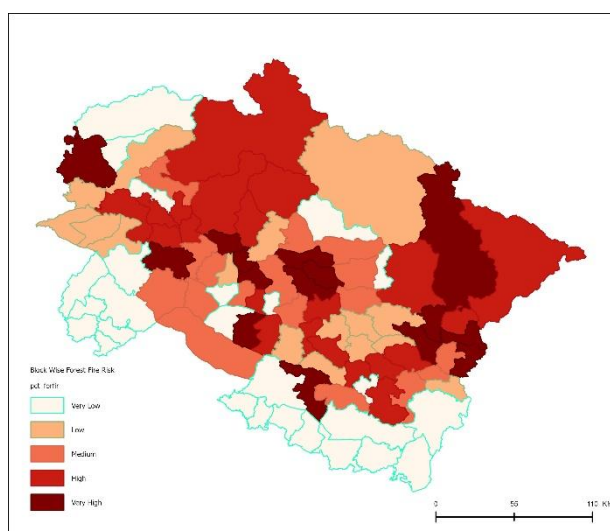
Intensity and Severity. The block-wise forest-fire-risk classification shows High to Very High risk spread across a wide central-to-eastern belt of the state, with additional Very High pockets in the northwest and south, a substantially wider geographic footprint than the landslide and flood hazards above. This is consistent with the 11,256 forest-fire incidents recorded across the Almora-Bageshwar-Nainital mid-elevation belt between November 2023 and June 2024 alone.

Trend. Forest-fire risk tracks the Fire Weather Index and dry-spell-length trend directly; the lengthening pre-monsoon dry spell documented in Section 5.1b is the principal

mechanism connecting the state's broader climate-extremes trend to its forest-fire risk.

Impacts. Forest fire damages the state's pine forest cover degrades downstream spring recharge and soil stability, and displaces wildlife into human settlements, compounding the human-wildlife-conflict hazard (Section 4.2.2, hazard 14).

Future Projections. Continued warming and drier pre-monsoon periods are expected to widen the Very High forest-fire-risk footprint beyond its current mid-elevation concentration, particularly where it already borders the High-risk zones mapped below.



Block-wise Forest Fire Risk classification

High-risk districts

Applying this index to the hazard record compiled for this assessment, four districts recur consistently at the top of the Multi-Hazard Index, and consistently as the same four districts identified independently through the Historical Disaster: **Chamoli, Uttarkashi, Rudrapur** and **Pithoragarh**. This is not a coincidence of index construction: these four districts are the only ones in the state where extreme rainfall, landslide susceptibility, GLOF exposure and avalanche risk all score in the highest AHP-weighted tier simultaneously, precisely the “High Himalayan Multi-Hazard Cluster” identified independently from the sector vulnerability indices. **Bageshwar** and **Almora** form a secondary tier, driven less by glacial and flash-flood hazard than by the forest-fire and drought hazard scores. The Bhabar-Tarai districts (**Haridwar, Udham Singh Nagar**) and Dehradun instead score highest on a narrower hazard set, riverine/urban flood, heatwave and fog, consistent with their distinct, lowland hazard profile rather than the high-Himalayan multi-hazard signature.

Most impacted sectors and groups

Overlaying the Multi-Hazard Index onto the sector and social vulnerability findings elsewhere in this report identifies three groupings of impact that recur across the high-risk districts identified above:

- **Natural resource sectors:** Water, Biodiversity and Forest. Water resources are affected through both ends of the hazard spectrum at once, GLOF and flash-flood destruction of water-supply infrastructure at one extreme, and drought- and consecutive-dry-day-driven spring depletion at the other. Forest and biodiversity assets carry the compounding fire, drought and human-wildlife-conflict hazard load documented for the mid-elevation belt, with direct downstream consequences for water-sector spring recharge and soil stability.
- **Economic sectors:** Energy, Agriculture, urban planning/critical infrastructure, and Tourism. Energy infrastructure concentrates in exactly the GLOF- and landslide-exposed river valleys carrying the highest Multi-Hazard Index scores, agriculture carries the compounding exposure of extreme rainfall, drought, hailstorm and cold-wave hazard against a rainfed, fragmented-landholding base; critical infrastructure, roads, bridges, transmission lines, sits directly within the landslide and flash-flood hazard footprint mapped; and tourism's seasonal, corridor-concentrated economic base is exposed to the same landslide and extreme-weather hazards that threaten the roads it depends on.
- **Community dimensions:** Gender and poverty, and Health. The hazard record compiled in this chapter is, on its own, hazard-neutral: it does not distinguish who is exposed within an affected district. Read alongside the Social Vulnerability Index proposed, however, the same four high-risk districts identified here, Chamoli, Uttarkashi, Rudraprayag and Pithoragarh, are also among the state's highest out-migration and most feminised-residency districts, meaning the population most exposed to this chapter's compounding hazard load is disproportionately women, elderly residents and children left in place as working-age men migrate for work. Health-system adaptive capacity is lowest in precisely these same districts driven by long travel times to functional health facilities rather than hazard exposure alone, so that a hazard event in the state's highest Multi-Hazard Index districts also strikes where emergency health response is weakest, a compounding pattern this report treats as a single finding rather than three coincidental ones.

5. Sectoral Impact Assessment

This assessment adopts the IPCC AR6 risk framing, in which climate risk emerges from the interaction of a climate-related hazard with the exposure and vulnerability of the system at risk. Vulnerability is decomposed into sensitivity (the degree to which the system is affected when exposed) and adaptive capacity (the ability to cope, respond and recover):

$$\text{Risk} = f(\text{Hazard} \times \text{Exposure} \times \text{Vulnerability})$$

$$\text{Whereas Vulnerability} = f(\text{Sensitivity, Adaptive Capacity})$$

5.1 Components of the Framework

Component	Definition adopted	Operationalisation
Hazard	A climate-related physical event or trend that may damage crops, orchards, livestock or aquaculture, or disrupt production	Erratic/extreme monsoon rainfall, temperature rise & chilling deficit, extreme weather (flood/cloudburst/hailstorm/landslide), drought, Section 7.4
Exposure	Presence of farmland, orchards, livestock and aquaculture assets in places that could be adversely affected	GIS overlay of hazard layers with cultivated area, orchard belts, grazing routes and fishery sites, at block level
Sensitivity	Degree to which the sector is affected by climate variability once exposed	Rain-fed share, chilling-dependent perennial-crop share, HWC intensity, out-migration, gendered labour burden, Section 7.6
Adaptive capacity	Ability to cope with, respond to and recover from harm	Crop insurance penetration, irrigation coverage, cold-chain/market access, veterinary service coverage, non-farm income diversification

Risk framework components and their operationalisation.

5.2 Methodological Steps

The methodology is designed to be consistent across sectors, transparent in its evidence base, and directly usable as the foundation for the vulnerability indicator frameworks and adaptation priorities developed later in the assessment.

5.2.1 Purpose and Scope

The purpose of the climate impact pathway analysis is threefold: to provide a common, replicable structure for identifying the mechanisms of climate risk across sectors; to make the causal logic of each risk explicit, so that the points at which adaptation can intervene become visible; and to build the analytical bridge between climate information and adaptation planning. Rather than cataloguing hazards in isolation, the pathway approach treats each impact as a sequence, and each link in that sequence as a candidate entry point for action. The analysis covers the principal climate-sensitive sectors assessed in the CRVA — agriculture and allied livelihoods, water resources, tourism, energy, roads and transport, health, and forests — across the ninety-five blocks of the thirteen districts of Uttarakhand. Its temporal scope spans baseline (observed) conditions and mid-century projections under the SSP2-4.5 and SSP5-8.5 scenarios. The pathway analysis is qualitative to semi-quantitative in character: it establishes the causal structure and direction of risk, which the subsequent indicator-based vulnerability assessment then quantifies wherever data permit.

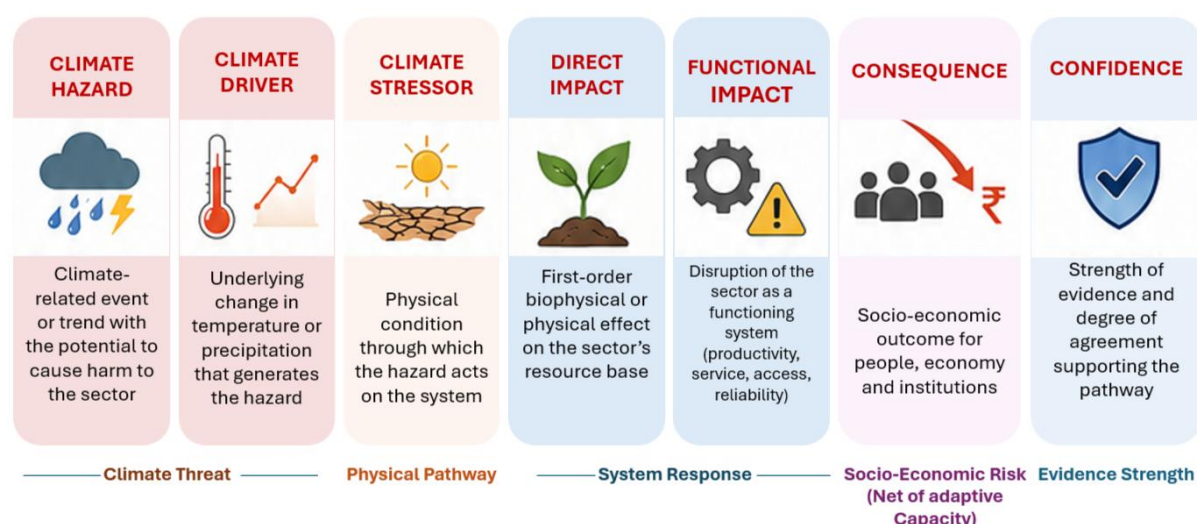
5.2.2 Conceptual Foundation

The methodology is rooted in the risk framing of the IPCC Sixth Assessment Report, in which risk arises from the interaction of hazard, exposure and vulnerability, the last of these comprising sensitivity and adaptive capacity (IPCC, 2022). This risk concept was introduced in the Fifth Assessment Report (IPCC, 2014), superseding the earlier vulnerability framing, and it is now the standard basis for climate risk and adaptation assessment. The framework is operationalised through the “impact chain” concept aligned with the AR5 risk concept in its Risk Supplement (Zebisch et al., 2021). This lineage is directly pertinent: the present assessment is delivered under the GIZ CONSULTATION programme, and the impact-chain method is GIZ’s own standardised framework for climate vulnerability and risk assessment, applied in more than twenty countries. An impact chain is a participatory analytical model that traces the climatic and non-climatic factors driving risk in a defined system and serves as the backbone for an indicator-based vulnerability assessment that combines data, models and expert judgement (Zebisch et al., 2021). The climate impact pathway used here is a sector-tailored, linear expression of this impact chain.

5.2.3 The Climate Impact Pathway Structure

Each pathway is expressed as a single causal chain, read from left to right across seven elements. The first two: climate hazard and climate driver, define the climatic threat. The climate stressor is the physical condition through which that threat acts on the

system. The direct impact and functional impact describe how the threat is realised given the exposure and sensitivity of the sector: the direct impact is the first-order biophysical or physical effect, while the functional impact is the disruption of the sector as a working system. The consequence is the resulting socio-economic outcome, which reflects the residual risk once the sector's (typically limited) adaptive capacity is taken into account. A confidence rating accompanies each chain. Table 7.1 defines each element and maps it to the corresponding concept in the IPCC frameworks.



Pathway Element	What it captures	Correspondence to the risk concept
Climate Hazard	The climate-related event or trend with the potential to cause harm to the sector	Hazard- entry point of the impact chain
Climate Driver	The underlying change in the mean or extreme behaviour of temperature or precipitation that generates the hazard	Climatic signal of the hazard
Climate Stressor	The physical condition through which the hazard acts on the system (e.g. heat stress, reduced recharge, erosion)	Hazard–exposure interface; intermediate impact in the chain

Pathway Element	What it captures	Correspondence to the risk concept
Direct Impact	The first-order biophysical or physical effect on the sector's resource base	Impact realised given exposure and sensitivity; intermediate impact
Functional Impact	The disruption of the sector as a functioning system (productivity, service reliability, access)	Higher-order impact reflecting sensitivity
Consequences	The socio-economic outcome for people, the economy and institutions	Risk / end-of-chain outcome, net of adaptive capacity
Confidence	The strength of evidence and the degree of agreement supporting the chain	IPCC calibrated language of confidence

Elements of the climate impact pathway and their correspondence to the IPCC AR6 and GIZ risk concepts

The pathways are deliberately drawn as linear chains for clarity and comparability across sectors. In reality, impacts interact through feedbacks and cascades; the compound-hazard pathway included in each sector table captures the most important of these interactions, and the approach can be extended to non-linear “impact webs” where cross-sectoral feedbacks are material (Zebisch et al., 2021).

5.2.4 Analytical Workflow

The pathways were developed through a seven-step process that combines climate information, secondary literature and expert judgement, consistent with the step-wise, mixed-methods approach of the Vulnerability Sourcebook (Fritzsche et al., 2014). The process is summarised in Table 7.2.

Its participatory elements in particular the sector-expert consultation and the CONSULTATION stakeholder workshop are integral rather than incidental: co-developing the chains with those who hold local and sectoral knowledge both improves their accuracy and builds the shared understanding on which adaptation planning depends.

Step	Activity	Principal output
1	Scoping and sector definition: define the systems of interest, select the sectors	Sector scope and spatial framework
2	Hazard and driver identification, compile observed climate trends to identify relevant hazards and drivers	Sector hazard profile
3	Impact chain construction trace each hazard from driver to stressor, direct impact, functional impact and consequence, sector by sector	Draft climate impact pathways
4	Evidence assembly ground each link in peer-reviewed and institutional literature, government statistics and regional studies	Referenced evidence base
5	Expert and stakeholder validation review and refine the pathways through sector-expert consultation	Validated pathways
6	Confidence assessment assigns a confidence rating to each pathway based on evidence and agreement	Confidence-rated pathway tables
7	Linkage to indicators and adaptation translate the pathways into vulnerability indicators and adaptation entry points	Indicator frameworks and adaptation priorities

Analytical workflow for constructing the climate impact pathways

5.2.5 Evidence Base and Data Sources

Each link in every pathway is grounded in an explicit evidence base drawn from four sources. Climate data provide the hazard and driver characterisation: observed meteorological records establish baseline trends, and downscaled CMIP6 projections prepared for the risk assessment under the SSP2-4.5 and SSP5-8.5 scenarios inform the forward-looking elements.

Scientific literature, peer-reviewed studies on Himalayan and Uttarakhand-specific climate impacts, together with the IPCC assessments, supports the causal links. Institutional and government sources, including NITI Aayog, ICIMOD, state government statistics and the District Statistical Digests, provide sectoral and contextual data.

Finally, expert and stakeholder input, gathered through consultation and the consultation workshop, supplies local knowledge and validation in keeping with the participatory ethos of the impact-chain method.

The assessment applies strict provenance discipline. Statistics are attributed to their source, and where a robust site-specific figure is unavailable, the direction and mechanism of change are stated qualitatively while specific numbers are flagged for insertion from the project's downscaled dataset rather than estimated.

This keeps the pathways defensible and their evidential status transparent.

5.2.6 Assumptions and Limitations

The approach carries several limitations that should be read alongside its results. It is a qualitative causal model rather than a quantitative risk-probability model; the magnitude of risk is addressed subsequently in the indicator-based vulnerability assessment.

Its linear representation simplifies a reality of feedbacks and cascades, a simplification only partly relaxed by the compound-hazard pathway and extendable, in principle, to impact webs (Zebisch et al., 2021).

Its forward-looking elements depend on the SSP2-4.5 and SSP5-8.5 projections and their downscaling assumptions, and deep uncertainty remains, particularly for precipitation extremes. Block-level and sector-specific data are uneven across the state, so some links rest on regional analogues rather than local measurement, a limitation the confidence ratings are designed to signal.

And the participatory elements, while essential, introduce expert judgement; transparent sourcing and structured validation mitigate, but do not eliminate, its subjectivity.

5.2.7 From Pathways to Vulnerability Indicators and Adaptation

The pathways are not an end in themselves; they are the analytical backbone for the remainder of the risk. Each functional impact and consequence points to measurable dimensions of exposure, sensitivity and adaptive capacity, which are operationalised in the sector vulnerability indicator frameworks that underpin the assessment's vulnerability indices.

The links that recur across multiple pathways identify priority entry points, which are carried forward into the adaptation options and the block-level adaptation matrix. In this way the methodology closes the loop from climate information, through impact analysis, to prioritised and actionable adaptation, the full adaptation planning cycle for which the Vulnerability Sourcebook approach was designed (Fritzsche et al., 2014; Zebisch et al., 2021).

6. Agriculture

Uttarakhand is situated at the foothills of the Western Himalayan range in the northern part of the country, has a unique agricultural landscape due to its diverse agro-climatic zones. The agriculture sector, providing livelihood to nearly 70% of the state population and contributing about 8.45% to state GDP. The total land **area of the state is 53.48 lakh ha**, out of which **13% (i.e., 6.96 lakh ha) is under cultivation**. of this, about **3.45 lakh hectares** are irrigated.

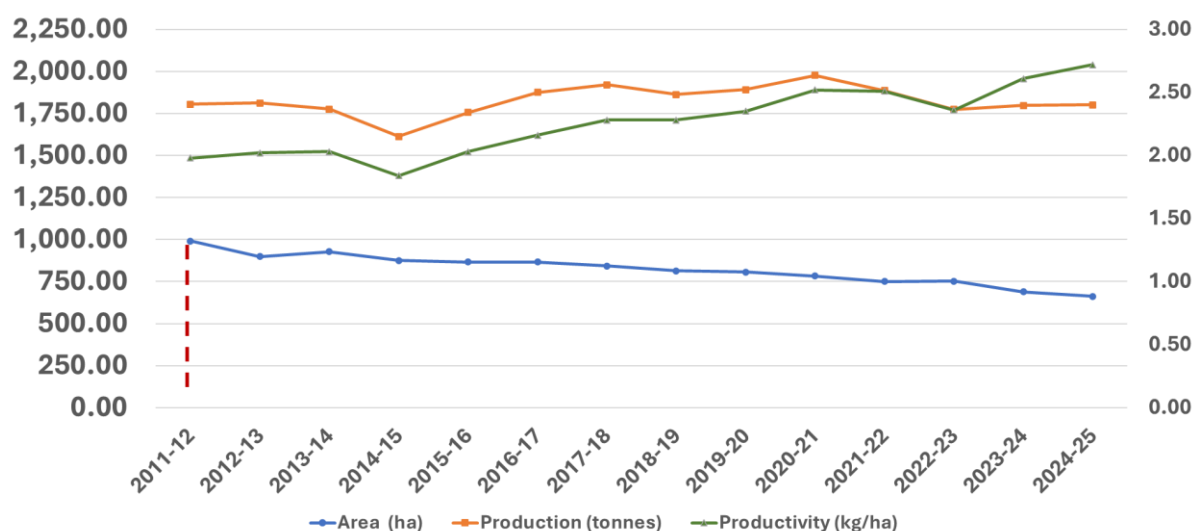


Figure 6.1: Cultivated area, production and productivity, Uttarakhand, 2011-12 to 2024-25 (Department of Agriculture, Government of Uttarakhand)

6.1 Agricultural Profile and Farming Systems

Agriculture is the foundational sub-sector, with a net sown area of approximately 661,853 hectares, of which over 55 per cent is rain-fed. Traditional crops, paddy, wheat, finger millet (mandua), barnyard millet (jhangora) and amaranth, dominate hill cultivation, while the Terai supports commercial sugarcane, wheat and rice. Land under traditional food grains has declined by 27 per cent over the last decade as climate stress, out-migration and declining economic returns erode the viability of hill farming.

Agricultural systems in Uttarakhand exhibit marked heterogeneity across the hill and plain regions owing to differences in topography, agro-climatic conditions, resource availability, and production systems. The Tarai and Bhabar plains are characterized by relatively high agricultural productivity, greater irrigation coverage, and market-oriented farming systems dominated by irrigated crops such as rice, wheat, and sugarcane. In contrast, the hill regions are predominantly characterized by rainfed agriculture, small and fragmented landholdings, and diversified farming systems comprising traditional cereals, millets, pulses, horticultural crops, livestock, and agroforestry.

These systems are largely dependent on monsoonal precipitation and spring-fed water sources, rendering agricultural production intrinsically sensitive to climatic variability.

Agro-climatic zone	Irrigation	Annual Rainfall (in mm)	Districts that fall in the zone	Principal crops produced
ZONE A (UP TO 1000 M) 	Tarai irrigated	1400	US Nagar, Haridwar	Rice, wheat, sugarcane, lentils, chickpea, rapeseed, mustard, mango, litchi, guava, peach, plums
	Bhabar irrigated	1400	Nainital, Dehradun, Pauri Garhwal	Rice, wheat, sugarcane, rapeseed, mustard, potato, lentils, mango, guava, litchi
	Irrigated lower hills (600-1000 M)	2000-2400	Champawat, Pauri Garhwal, Dehradun, Nainital, Tehri Garhwal	Rice, wheat, onion, chilly, peas, potato, radish, pulses, cauliflower, oilseeds, soyabean, mango, guava, plums, litchi
	Rain-fed lower hills (600-1000 M)	2000-2400	Champawat, Pauri Garhwal, Dehradun, Nainital, Tehri Garhwal, Bageshwar	Finger millet, maize, rice, wheat, pulses, mango, guava, plum, peaches
ZONE B (1000-1500 M) 	Mid hills South aspect	1200-1300	Pithoragarh, Almora, Chamoli, Bageshwar	Rice, finger millet, wheat, potato, tomato, peas, cole crops, pulses, peach and plums
ZONE C (1500-2400 M) 	High hills	1200-2500	Pithoragarh, Almora, Chamoli, Bageshwar	Amaranth, finger millet, french-beans, cole crops, potato, peas, peach, plum, pear, apple, stone fruit
ZONE D (>2400 M) 	Very high hills	1300	Pithoragarh, Chamoli, Uttarkashi	Amaranth, buckwheat, peas, cole crops, apple and potato.

Figure 6.2: Source: Draft revised Uttarakhand's State Action Plan on Climate Change

The spatial distribution of irrigation infrastructure is highly uneven across the state, with the majority of irrigated agricultural land concentrated in the plains, while irrigation coverage in the hill districts remains comparatively limited. Consequently, farming systems in the hill regions exhibit greater sensitivity to fluctuations in precipitation, prolonged dry spells, and changes in seasonal water availability. In addition, the biophysical characteristics of the Himalayan landscape, including steep slopes, shallow soils, and fragmented agricultural holdings, constrain mechanization and limit the scope for irrigation expansion, thereby influencing the adaptive capacity of farming communities. . Irrigation in the state is mostly confined to the plain regions, as out of the 3.45 lakh hectares that is irrigated, 85.83% of it lies in the plain regions. The primary crops that are grown in the plains include water intensive ones, such as rice, sugarcane, etc. Since the percentage of irrigated area in the hills is significantly lower, the crops grown in the hilly areas are less water intensive.

This includes a few staple crops and horticulture products, such as fruits, vegetables, flowers, spices, medicinal plants and herbs, etc.

Projected changes in temperature and precipitation regimes are expected to amplify existing vulnerabilities within these contrasting production systems. Rising temperatures are likely to increase crop evapotranspiration and irrigation water requirements, while greater variability in monsoon rainfall and the increasing frequency and intensity of extreme weather events may affect crop establishment, growth, and yield stability. In the plains, irrigated cropping systems may experience increased pressure on water resources, alongside heightened risks associated with heat stress, waterlogging, and flood-induced crop damage. In the hill regions, where agricultural production is predominantly rainfed, projected declines in the reliability of precipitation and spring discharge, coupled with increasing temperatures, are expected to adversely

affect crop productivity, horticultural systems, soil moisture availability, and the suitability of existing cropping patterns. These changes may also contribute to shifts in pest and disease dynamics and exacerbate soil erosion on vulnerable slopes.

The Government of Uttarakhand continues to strengthen the agriculture sector through centrally sponsored and state sector schemes, including **PM-KISAN, PM Fasal Bima Yojana, Kisan Credit Card, Soil Health and Fertility Scheme, Paramparagat Krishi Vikas Yojana (PKVY), Sub-Mission on Agricultural Mechanization (SMAM), Rainfed Area Development Programme, Chief Minister Apple Development Scheme, and the Uttarakhand State Millet Mission**. These initiatives aim to improve farm productivity, promote sustainable agriculture, encourage mechanization and organic farming, enhance soil health, strengthen climate-resilient crops such as millets, and improve farmers' incomes through diversification and value addition.

Due to which, foodgrain production has remained relatively stable, fluctuating between 1.61 and 1.98 million metric tonnes. Uttarakhand has progressively shifted from **low-value crops towards high-value agriculture** through targeted policy interventions. Productivity increased from 1.98 tonnes per hectare in 2011-12 to 2.72 tonnes per hectare in 2024-25, representing an increase of approximately 37%. The consistent improvement reflects the adoption of high-yielding seed varieties, improved agricultural technologies, better crop management practices, and enhanced farmer capacity-building programmes.

The state has also witnessed notable growth in horticulture and allied sectors. Fruit and vegetable production increased from 16.47 lakh metric tonnes at the time of state formation to 17.97 lakh metric tonnes in 2023–24. Fruit productivity has increased from approximately 5 metric tonnes per hectare to around 9 metric tonnes per hectare, supported by high-density orchard development and improved cultivation practices. The Apple Mission, Kiwi Policy, and promotion of aromatic crops, mushroom cultivation, and beekeeping have further strengthened agricultural diversification and enhanced farmers' income opportunities.

The Government of Uttarakhand has introduced several initiatives, including the **Millet Policy, Kiwi Policy, Apple Mission, Scheme for Upgradation of Post-Harvest Apple Logistics (SUPAHL), Chief Minister Apple Development Scheme**, and programmes promoting organic farming, agricultural mechanization, soil health management, and crop diversification. These interventions span the entire agricultural value chain from quality seed and planting material to post-harvest infrastructure, processing, and market linkages.

6.1.1 Sericulture

Sericulture, mulberry silkworm (*Bombyx mori*) cultivation for raw silk, supports approximately 25,000 farm families, concentrated in the Doon Valley, Almora, Nainital and Pauri Garhwal districts. Uttarakhand is one of North India's principal mulberry silk-producing states, integrated into cottage-industry value chains administered through the Uttarakhand Silk Federation and the Central Silk Board's regional office (Central Silk Board, 2016; Sengupta et al., 2020).

6.1.2 Coldwater Fisheries

Coldwater fisheries are both ecologically and economically significant. Himalayan rivers support populations of snow trout (*Schizothorax* spp.), mahseer (*Tor* spp.) and introduced rainbow trout (*Oncorhynchus mykiss*), the last anchoring a growing aquaculture economy in spring-fed and high-altitude stream environments. The fishery is a critical protein source for riparian communities where alternative animal protein is limited.

6.1.3 Animal Husbandry and Livestock

Animal husbandry and livestock provide 35-40 per cent of rural household income in the hills, with cattle, buffalo, sheep, goat and poultry as the principal species. The state's 10.8 million livestock units (Census, 2019) depend on a seasonal grazing system in which van panchayat forests, agricultural crop residues and high-altitude bugyals (alpine meadows) constitute the primary feed base. Transhumance, seasonal herd movement between valley floors and alpine pastures, remains defining practice in Chamoli, Uttarkashi, Pithoragarh and Bageshwar, linking lowland winter shelter to high-altitude summer grazing across elevation gradients of 2,000–4,000 m.

6.2 Defining Drivers and Their Climate Implication

Agriculture and allied sectors in Uttarakhand are intrinsically climate-sensitive because production depends directly on temperature, precipitation, snow dynamics, hydrological processes, and seasonal climate variability. Unlike most economic sectors, climate functions not only as an external driver of risk but also as a fundamental production input governing crop growth, horticultural productivity, livestock performance, fisheries, and sericulture. Consequently, changes in climatic conditions influence agricultural systems through multiple interacting biophysical, ecological, and socio-economic pathways.

The climatic sensitivity of the sector is further amplified by Uttarakhand's physiographic setting. The state lies within the transition zone between the winter precipitation regime associated with Western Disturbances and the summer Indian Summer Monsoon, making agricultural production responsive to changes in both climate systems. At the same time, the state's exceptional altitudinal gradient—from approximately 210 m in the Terai to more than 7,800 m in the Greater Himalaya—creates distinct agro-climatic zones, each characterised by narrow thermal and moisture regimes that determine crop suitability, livestock production systems, and allied livelihood activities. Progressive warming is therefore expected to modify the spatial distribution of agro-climatic suitability rather than simply increasing temperatures uniformly across the landscape.

Crop production forms the foundation of the agricultural economy and remains predominantly rainfed, with more than 65% of the net sown area dependent on rainfall. Agricultural productivity is therefore highly sensitive to the timing, distribution, and intensity of precipitation rather than seasonal rainfall totals alone. Variability in

monsoon onset, prolonged intra-seasonal dry spells, and increasing rainfall intensity affect sowing, flowering, grain filling, and harvesting, while changes in winter precipitation influence soil moisture availability for rabi crops. Concurrently, increasing temperatures accelerate crop phenology, shorten grain-filling periods, increase evapotranspiration, and alter crop suitability across elevations, particularly for climate-sensitive cereals, pulses, and vegetables.

Temperate horticulture exhibits a distinct climatic dependence because winter chilling requirements govern dormancy, bud break, flowering, and fruit development. Progressive winter warming reduces the accumulation of chilling hours required by temperate fruit species, thereby affecting flowering synchronisation, fruit set, yield, and fruit quality. This sensitivity is reinforced by the long productive lifespan of perennial orchards, where investment decisions remain fixed over several decades, limiting the capacity for rapid adaptation under changing climatic conditions. In addition to long-term warming trends, hailstorms, late frosts, and untimely precipitation pose recurrent risks during flowering and fruit development.

Livestock systems are sensitive to climate through both feed availability and animal physiology. Alpine grazing systems depend on seasonal snow accumulation and snowmelt to sustain high-altitude pastures, while lowland livestock production is increasingly exposed to heat stress, water scarcity, and declining fodder availability. Rising temperatures reduce animal productivity through lower feed intake, reduced reproductive performance, declining milk yield, and increased disease susceptibility. Climate-induced changes in forest productivity, grassland condition, and wildfire occurrence further influence fodder availability for livestock-dependent households.

Cold-water fisheries represent one of the most climate-sensitive allied sectors because aquatic species occupy narrow thermal and hydrological niches. Increasing air temperatures, altered snowmelt contributions, changing river discharge regimes, and declining dissolved oxygen concentrations affect habitat suitability, breeding success, and species distribution. In addition, extreme rainfall, flash floods, sediment transport, and channel instability increase physical risks to hatcheries, raceways, and aquaculture infrastructure located within river valleys.

Sericulture is similarly dependent on stable microclimatic conditions throughout the silkworm life cycle. Temperature and relative humidity regulate larval development, cocoon formation, and silk quality, while climatic variability influences mulberry phenology and the synchronisation between host plant development and silkworm rearing cycles. Warmer and more humid conditions also favour the proliferation of fungal and protozoan pathogens, increasing production risks and reducing cocoon productivity.

Collectively, these defining characteristics demonstrate that climate change affects agriculture and allied sectors through multiple, interconnected pathways involving crop

physiology, hydrology, ecosystem processes, livestock health, and production systems. The interaction of progressive climatic change with existing structural constraints—including fragmented landholdings, high dependence on rainfed agriculture, limited irrigation, and mountainous terrain—further increases the sensitivity of agricultural livelihoods to future climate variability and extremes. These characteristics provide the basis for identifying climate impact pathways and subsequent vulnerability assessment presented in the following sections.

Defining Feature	Description	Relevance for Climate Sensitivity
Predominantly rainfed agricultural system	A large proportion of cultivated land (>65%) depends on direct rainfall, with limited irrigation coverage, particularly across the mid- and higher Himalayan regions.	Agricultural productivity is highly sensitive to changes in rainfall amount, distribution, monsoon onset, intra-seasonal dry spells, and increasing drought frequency, with limited capacity to buffer climatic variability.
Strong dependence on seasonal climate regimes	Cropping systems are synchronized with the southwest monsoon for kharif cultivation and with winter precipitation associated with Western Disturbances for rabi production.	Shifts in monsoon timing, changes in winter precipitation, and increasing climate variability directly influence sowing, crop establishment, growth duration, and harvest timing.
Elevation-dependent agro-climatic production systems	Agricultural production is distributed across diverse agro-climatic zones ranging from the Terai plains to high-altitude Himalayan valleys, supporting distinct crop, horticulture, livestock, and allied production systems.	Rising temperatures modify agro-climatic suitability, alter crop phenology, shift cultivation zones, and change the spatial distribution of suitable production systems across elevations.
Climate-sensitive horticultural systems	Temperate fruit production, including apple, pear, peach and plum, depends on winter chilling accumulation, distinct seasonal temperature regimes, and favourable spring climatic conditions.	Increasing winter temperatures reduce chilling accumulation, while late frosts, hailstorms, and untimely precipitation affect flowering, fruit set, productivity, and fruit quality.
Integrated crop–livestock production systems	Farming systems combine crop cultivation with livestock rearing, relying on crop residues, natural grazing lands, forest fodder, and common property resources for animal husbandry.	Climate-induced changes in fodder availability, pasture productivity, forest condition, heat stress, and water availability simultaneously affect crop and livestock productivity, amplifying livelihood vulnerability.
Dependence on mountain ecosystem services	Agricultural production relies on ecosystem services including snowmelt-fed streams, springs, fertile mountain soils, forests,	Climate-induced changes in hydrology, snow dynamics, soil moisture, biodiversity, pollination services, and forest ecosystems influence

	pollinators, and natural nutrient cycling.	agricultural productivity and long-term system resilience.
Smallholder and fragmented landholdings	Agriculture is dominated by small and marginal farmers cultivating fragmented terraced fields with limited mechanisation and financial resources.	Small-scale production systems have lower adaptive capacity to absorb climatic shocks, adopt new technologies, diversify livelihoods, or invest in climate-resilient infrastructure.
High dependence on climate-sensitive allied sectors	Rural livelihoods depend not only on crop production but also on horticulture, livestock, fisheries, sericulture, and other climate-dependent allied activities.	Climate impacts propagate across multiple interconnected livelihood systems, increasing cumulative economic losses and reducing household resilience to climate variability and extreme events.
Mountain terrain and fragile production landscapes	Agricultural lands are located on steep slopes and narrow valleys characterised by fragile soils and high geomorphic instability.	Extreme rainfall, landslides, soil erosion, debris flows, and flash floods damage cultivated land, irrigation channels, terraces, and rural infrastructure, reducing long-term agricultural productivity.

6.3 Key Climate Hazards for Agriculture sector

The Himalayan Mountain ecosystems despite being rich in biodiversity and traditional knowledge, are highly susceptible to natural hazards and climate change. Four hazards dominate climate risk to the agriculture and allied sectors in Uttarakhand. For each, this section sets out the physical mechanism, the sub-sectors and geographies affected, and the documented historical record. Together the record establishes that climate risk to the sector is not prospective but an ongoing, measurable loss stream, with two signatures: rare, catastrophic extreme-weather events that destroy standing crops and orchards outright, and a chronic, slow-onset erosion of rainfall reliability and winter chilling that degrades production every season without triggering a disaster declaration.

6.3.1 Hazard

Hazard	Mechanism of damage	Stage of production cycle affected	Sub-sectors affected	Principal districts
Erratic and intensifying monsoon rainfall	Deficit of low-intensity, infiltration-dominant rain at sowing and germination; high-intensity pulses causing waterlogging, terrace scour, soil loss and lodging	Sowing; germination; vegetative; grain-fill	Rain-fed cereals and millets; horticulture	All hill districts; acute in Uttarkashi, Chamoli, Pauri, Almora
Temperature rise and chilling deficit	Insufficient winter chill accumulation for dormancy break in temperate fruit; accelerated and desynchronised phenology;	Dormancy; flowering; fruit set; livestock productivity	Horticulture; livestock; sericulture	Almora, Chamoli, Nainital, Pauri, Uttarkashi (orchard belts)

	heat stress on livestock and sericulture			
Extreme weather (cloudburst, flash flood, hailstorm, landslide)	Destruction of standing crops and orchard infrastructure within hours; terrace and irrigation-channel failure; debris and silt deposition; irrigation-water contamination	Any stage; catastrophic at flowering and fruit development	All sub-sectors	Uttarkashi, Chamoli, Rudraprayag, Pithoragarh, Bageshwar
Drought and groundwater stress	Soil-moisture deficit reducing rain-fed yield in the hills; compensatory groundwater over-abstraction in the Tarai	Whole season; inter-annual carry-over	Agriculture; livestock (pasture)	Statewide hill drought; Haridwar and U.S. Nagar (groundwater)
Spring discharge decline and irrigation-source failure	Reduction and seasonal shift in spring (naula, dhara) and gadhera discharge on which gravity-fed guhl irrigation and livestock watering depend	Rabi irrigation; lean-season watering	Agriculture; livestock	All hill districts; acute in mid-altitude spring-dependent blocks
Forest fire, pasture degradation and fodder deficit	Loss of grazing and lopped-fodder biomass; destruction of NTFP resource base; post-fire loss of ground cover raising runoff and slope instability above terraces	Livestock nutrition (pre-monsoon lean period); post-fire monsoon slope risk	Livestock; NTFP-based livelihoods; agriculture (terrace stability)	Pauri, Almora, Nainital, Tehri, Champawat
Climate-amplified biotic stress and human-wildlife conflict	Altitudinal expansion of pest and pathogen ranges and reduced overwintering mortality; shifting wild-herbivore movement raising crop-raiding pressure at the forest-farm interface	Vegetative; pre-harvest; harvest	Agriculture; horticulture	Statewide; conflict acute in forest-fringe blocks

Hazard register for the Uttarakhand agriculture and allied sectors.

1. Erratic and Intensifying Monsoon Rainfall

Rainfall is the primary determinant of agricultural viability in Uttarakhand's rain-fed hill economy. Over 55 per cent of the cultivated area has no irrigation access, so crop outcomes are contingent almost entirely on monsoon onset, intra-seasonal distribution and cessation (ICAR, 2023). The hazard is therefore inadequately represented by change in seasonal rainfall total alone; what governs rain-fed yield is the joint distribution of rainfall intensity, timing and dry-spell length.

Since 2000, total annual monsoon precipitation over the State has declined by approximately 12 per cent, while the frequency of extreme-rainfall events exceeding 100 mm/day has increased by approximately 35 per cent (IMD, 2023). The two findings together, and neither alone, define the hazard: the same or a smaller annual water volume is being delivered in fewer, more violent pulses.

For a rain-fed terrace system this is the least favourable possible reorganisation of a given rainfall total.

It reduces the number of days on which rainfall is infiltration-dominant and recharges the root zone, and increases the number of days on which rainfall intensity exceeds the infiltration capacity of a cultivated terrace and is converted directly to runoff, carrying topsoil and applied nutrients with it.

Pathway to agriculture-sector damage

Failure pathway	What is lost	Why recovery is slow
Delayed or failed monsoon onset	The sowing window for kharif rain-fed cereals and millets	A single growing season at altitude affords no realistic re-sowing opportunity
Extended intra-seasonal dry spell at tillering or grain-fill	Yield per unit area rather than area sown -a loss invisible in area statistics	The loss registers only at harvest, after the full input cost has been incurred
High-intensity rainfall pulse on a cultivated terrace	Topsoil, applied nutrients and terrace risers	Terrace reconstruction is manual and labour-intensive; out-migration has depleted the available labour pool
Waterlogging in valley-floor and Tarai fields	Root-zone oxygen; crop lodging and quality downgrade	Field drainage infrastructure is largely absent on hill terraces
Pulse-driven siltation of guhl irrigation channels	Conveyance capacity for the following rabi season	Desilting is a community labour obligation now weakly enforced in depopulated villages

Rainfall-reorganisation failure pathways and their recovery horizons.

Documented impact

In 2023, 44,882 hectares of farmland were lost to extreme weather — floods, landslides, cloudbursts and hailstorms (Climate Trends, 2024).

Barnyard millet (jhangora) production declined 20 per cent, from 75,929 tonnes (2012–13) to 60,569 tonnes (2021–22); finger millet (mandua) production fell 27 per cent, from 157,792 to 115,125 tonnes over the same period (Department of Agriculture, GoU, 2022, in Carbon Impacts, 2023). Barley production declined 11 per cent, from 28,222 to 25,256 tonnes (2012–2022), compounding food-security risk in subsistence hill communities for whom millets and coarse grains are the staple.

2. Temperature Rise, Chilling Deficit and Phenological Disruption

Uttarakhand's mean temperature has risen at approximately 0.02°C per year between 1970 and 2022, accumulating some 1.5°C of warming since 1970, with higher elevations experiencing amplified rates through elevation-dependent warming (Climate

Trends, 2024; Shekhar et al., 2023). For temperate agriculture the most consequential expression of this trend is the progressive reduction in winter chilling accumulation.

Temperate deciduous fruit species, apple, pear, walnut, apricot, plum and cherry require a cultivar-specific accumulation of exposure below a physiological threshold in order to break endodormancy. Insufficient accumulation produces delayed, protracted and irregular bud break, poor flower quality, low and erratic fruit set, and increased alternate bearing.

Because the chilling requirement is fixed at the point of planting and cannot be adjusted in a standing orchard, warming does not reduce yield smoothly and proportionately across all orchards.

It removes the viability of specific cultivars at specific altitudes in a threshold-like manner. The spatial expression of this process is the upslope migration of the viable cultivation band.

Documented impact

Apple cultivation has been forced upward from approximately 6,000 ft to 6,500–7,000 ft; orange and apricot have migrated from approximately 4,500 ft to 6,000 ft (CarbonImpacts, 2023). Initial establishment cost for apple at the higher altitude has approximately doubled, driven by transport and labour costs.

Red spider mite infestation in apple is expanding under warmer, drier conditions; fruit sunburn and cracking are increasing in apple, apricot and cherry at high altitude (ResearchGate, 2020 — primary citation to be substituted before submission).

Warmer winters have accelerated snowmelt, reducing the snow-covered area that moderates temperature extremes and insulates high-altitude orchards against frost injury (Climate Trends, 2024).

The third of these findings carries a counter-intuitive implication that should be stated explicitly: reduced snow cover increases, rather than reduces, frost-injury exposure.

The insulating snowpack is lost while radiative frost nights persist, so warming at the seasonal mean can coexist with rising frost damage at the plant.

Hazard metrics built on mean winter temperature will not detect this.

The orchard record: a disinvestment signal

Apple production fell from 123,228 tonnes (2013) to 64,881 tonnes (2023), a decline of 47 per cent, with apple area contracting by 55 per cent over the same period.

Statewide fruit area contracted 54 per cent, from 177,324 to 81,693 hectares between 2016–17 and 2022–23, with fruit yield declining 44 per cent.

An area contraction of this magnitude over seven years is not a yield-response signal; it is a disinvestment signal. Growers are removing perennial capital rather than accepting reduced returns from it. This is the economically rational response to a chilling regime expected to continue shifting, and it is the clearest available indicator that a significant part of the horticulture sub-sector has already moved from incremental adaptation into transformation or exit. Upslope relocation, the principal autonomous adaptation observed, is bounded: above the current band, land availability, slope, soil depth, road access and land tenure all constrain replacement. Where the viable band migrates beyond the cultivable envelope, the response ceases to be adaptation and becomes withdrawal from the sub-sector.

Livestock and sericulture

Heat stress reduces dry-matter intake, milk yield, reproductive efficiency and conception rates in cattle and buffalo, with exposure concentrated in the Tarai–Bhabar belt and the lower valleys. Sericulture is sensitive to temperature and humidity during the rearing period, and to the phenology of host-plant leaf flush.

3. Extreme Weather Events: Cloudbursts, Flash Floods, Hailstorms and Landslides

Extreme weather is the most sudden and, per event, the most economically destructive climate hazard affecting the agricultural system. Cloudbursts, flash floods, hailstorms and landslides can destroy entire standing crops within hours, damage perennial orchard infrastructure irreparably, contaminate irrigation water, and initiate multi-season soil erosion on agricultural terraces. The State Disaster Management Authority documents a 240 per cent increase in hydro-meteorological disasters between 2000 and 2022, with cloudburst incidents rising from 11 events (2000–2010) to 83 events (2011–2023).

Two characteristics distinguish this hazard class in the agricultural context. First, the destructive agent is not only water but also ice (hail) and mass movement (landslide), which act through mechanical rather than hydrological pathways.

Second, the damage function is strongly conditioned by phenological timing: a hailstorm in March over a dormant orchard is a minor event, whereas the same storm in the April–June fruit-development window is a total-loss event.

Sub-hazards and their agricultural damage functions

Sub-hazard	Damage mechanism	Phenological window of maximum loss	Recovery horizon
Hailstorm	Mechanical injury to fruit skin, foliage and bearing shoots; total loss of marketable grade even where biological yield survives	April–June (fruit development in apple and stone fruit)	One season for annuals; two to three seasons where tree structure is damaged

Cloudburst	Terrace scour and riser failure; crop burial; destruction of irrigation channels	Monsoon, coinciding with kharif vegetative stage	Two to five seasons where the terrace and soil profile are lost
Flash flood	Removal of the soil profile in valley-floor fields; silt and debris deposition; contamination of irrigation water	Monsoon	Multi-season; governed by soil rehabilitation, not replanting
Landslide	Permanent loss of cultivable land; severance of access to fields and to markets	Monsoon and post-monsoon	Effectively permanent for land lost
Unseasonal rain and frost at harvest	Grain sprouting and quality downgrade; harvest and seed-stock loss	Rabi harvest (March–April)	One season, with carry-over effect through degraded seed stock

Extreme-weather sub-hazards, damage functions and recovery horizons.

Documented impact

- In 2023, extreme weather destroyed 44,882 hectares of farmland, approximately 5.6 per cent of the State's total cultivated area (Climate Trends, 2024; Department of Agriculture, 2023).
- Hailstorms during the April–June fruit-development window are a primary driver of commercial loss for apple and stone-fruit growers, with 75–100 per cent crop loss documented in badly affected orchards from single events.
- Post-flood soil contamination with debris, silt and pathogen-laden water causes multi-season productivity loss beyond the immediate crop damage; upstream industrial effluent and road-construction spoil compound irrigation-water quality risk.

4. Drought and Groundwater Stress

Drought exposure in Uttarakhand is bimodal, and the two modes require different treatment. In the hills, drought is a soil-moisture and spring-discharge phenomenon: because irrigation infrastructure is largely absent, a meteorological deficit translates directly and immediately into agricultural drought, with no intervening storage layer to buffer it. In the Tarai, drought is a hydrological and groundwater phenomenon: irrigation exists, so the deficit is transferred to the aquifer rather than to the crop.

Hill drought is a yield and food-security hazard borne by subsistence households in the season it occurs; plains drought is a resource-depletion hazard borne by the aquifer and deferred to future users. A single drought-exposure metric conceals this distinction, which is why the composite index reports the two separately and interprets them differently by terrain.

Documented impact

- Ten of the thirteen districts experienced drought conditions during 2007–09 (Singh, 2023).

- Crop yields in drought years fall by 30–40 per cent relative to average-rainfall conditions.
- In the Tarai plains, declining surface-water reliability is driving compensatory groundwater over-extraction: annual draft rose 66 per cent, from 0.99 BCM (2013) to 1.64 BCM (2017), approaching stage-of-development thresholds that signal long-term aquifer degradation (CGWB, 2017).

The Tarai groundwater trajectory is the clearest documented instance of maladaptation in the sector: a response that sustains production in the short term while degrading the resource base on which production depends over the longer term. It is recorded here as a hazard consequence rather than an adaptation outcome and is carried forward into the recommendations as a constraint on any strategy that relies on expanded groundwater irrigation.

5. Spring Discharge Decline and Irrigation-Source Failure

Hill irrigation in Uttarakhand is neither canal-based nor tubewell-based. It depends on springs (naula, dhara) and small perennial streams (gadhera), conveyed by gravity through unlined community channels (guhl). Springs are recharged by diffuse infiltration in mid-altitude recharge zones, with discharge governed by recharge volume, subsurface residence time and land cover. The hazard arises from the interaction of three drivers already established above: the reduction in infiltration-dominant rainfall; reduced snow accumulation and earlier melt, which erodes the natural seasonal reservoir; and land-cover and slope modification, which shortens residence time and converts recharge to runoff. The consequence for agriculture is a reduction in lean-season discharge together with a seasonal shift in availability away from the rabi irrigation window. Because guhl systems are gravity-fed and unstored, a reduction in discharge translates directly and proportionately into a reduction in assured command area; there is no storage element to absorb variability. The same reduction simultaneously constrains livestock watering during the pre-monsoon lean period.

6. Forest Fire, Pasture Degradation and Fodder Deficit

Uttarakhand's mid-altitude belt is dominated by chir pine (*Pinus roxburghii*), whose resinous and highly flammable needle litter creates a near-ideal surface fuel bed through the dry pre-monsoon months. Fire is normally assessed as a forestry or an energy-infrastructure hazard. Its agriculture and allied sector pathway is distinct and is assessed here through four mechanisms.

- **Fodder and grazing loss.** Fire removes standing grazing biomass and the lopped tree fodder on which hill livestock nutrition substantially depends, and it does so in the pre-monsoon months — precisely the lean period of maximum fodder scarcity. The temporal coincidence of peak fire risk and peak fodder deficit is the defining feature of this pathway.

- **NTPF resource destruction.** Fire destroys the non-timber forest produce base — fuelwood, fibre, medicinal and aromatic plants, and resin — that supplements smallholder income and is disproportionately important to women's livelihoods and to the poorest households.
- **Post-fire slope destabilisation.** Loss of ground cover and litter increases runoff coefficients and reduces slope cohesion. Fire in the pre-monsoon of a given year therefore contributes to terrace damage and landslide loss in the monsoon of the same year, creating an intra-annual compound hazard sequence that neither hazard, assessed alone, would reveal.
- **Direct livestock exposure.** Smoke, heat and the loss of shade and water points during fire events impose acute stress on grazing animals and on transhumant herds moving through affected corridors.

Documented impact:

The 2024 fire season recorded over 1,100 major fire incidents across the State, with projections indicating that days of severe fire risk in the State's dry forests could increase by approximately 60 per cent under continued warming.

Alpine pasture and transhumant systems

Earlier snowmelt and shifting flowering phenology in the alpine meadows (bugyal) alter both the timing and the nutritional quality of the grazing window on which transhumant pastoral systems depend. Where the meadow green-up advances relative to the customary migration calendar, herds arrive after peak forage quality has passed. This is a distributionally significant hazard because it falls on communities with few alternative livelihood options and limited representation in agricultural planning.

7. Climate-Amplified Biotic Stress and Human–Wildlife Conflict

Two biotic hazards are climate-conditioned rather than climate-caused, and both act at the pre-harvest stage, where the full input cost has already been incurred and the economic loss per unit of biological damage is therefore at its maximum.

The first is the altitudinal expansion of insect pests and pathogens. Warming permits range expansion into altitudes previously protected by winter cold, shortens generation times, and reduces overwintering mortality, so both the range and the within-season pressure of a given pest increase together. The documented expansion of red spider mite in apple under warmer and drier conditions is the best-evidenced State-level instance (ResearchGate, 2020).

The second is human–wildlife conflict. Changes in forest fodder availability, fruiting phenology and water-point distribution alter the movement and foraging behaviour of wild herbivores, primates and wild boar, increasing crop-raiding pressure at the forest–farm interface. The climatic contribution is indirect but material, and it interacts with the abandonment dynamic in a way that is self-reinforcing: crop raiding is a

documented proximate driver of land abandonment, and abandoned land adjoining cultivated plots increases cover and approach routes for wildlife, raising raiding pressure on the plots that remain. Individual abandonment decisions thereby aggregate into village-scale withdrawal from cultivation.

Documented impact

- Between 2008 and 2022 the State recorded 1,47,512 permanent out-migrants, with the number of fully depopulated “ghost” villages rising from 1,034 in 2011 to 1,792 in 2022.
- In November 2025, the Pradhan Mantri Fasal Bima Yojana was revised to include wild-animal crop damage as an insurable risk, with Uttarakhand identified as a priority State, an institutional recognition that crop loss to wildlife has reached a scale requiring formal risk-transfer treatment.

Compound and Cascading Hazard Interactions

The seven hazards in this register are not statistically independent, and the composite vulnerability index must be constructed accordingly. Four interactions are material.

- **Shared driver, distinct expression.** Extreme rainfall (Hazard 3) is the proximate trigger for the landslide sub-hazard, and is itself the tail of the rainfall distribution described in Hazard 1. These must not be summed as independent contributions to a composite index.
- **Warming as a common cause.** Temperature rise (Hazard 2) drives spring decline through reduced snow storage (Hazard 5), fire risk through fuel dryness and season length (Hazard 6), and pest range expansion (Hazard 7). Blocks scoring high on warming exposure will therefore score high across several hazards for a single underlying reason.
- **Intra-annual sequencing.** Pre-monsoon fire in a given year raises the probability of monsoon terrace loss and landslide in the same year. The compound sequence carries a higher expected loss than the sum of the two hazards assessed independently.
- **Cross-sector dependency.** For perishable horticultural produce, road severance during the harvest window converts a physically undamaged crop into a total economic loss.
- The February 2021 Chamoli event illustrates the wider pattern: beyond direct damage, it disrupted irrigation and water supply in the affected valleys and severed market access.

Agricultural climate risk in Uttarakhand cannot be assessed independently of the roads and water sector chapters, and the recommendations treat market-access continuity as an agricultural adaptation measure.

Chronology of Documented Climate-Hazard Impacts

Year	Event	Documented agriculture-sector consequence	Lesson carried into the framework
2007–09	Multi-year hill drought	10 of 13 districts affected; groundwater draft begins rising sharply thereafter	Slow-onset drought is as consequential for the sector as any single extreme event
2012–13 to 2021–22	Sustained millet and coarse-grain decline	Mandua –27%; jhangora –20%; barley –11%	Chronic yield erosion, not one bad season, defines the baseline against which risk is measured
2013 (June)	Kedarnath disaster - extreme rainfall with moraine-dammed lake breach	Loss of cultivated terraces, irrigation channels and market access across the Mandakini and Alaknanda valleys	Compound hazards exceed the design basis of individually appraised protective works
2013–2023	Apple production collapse	Apple production 123,228 t → 64,881 t (–47%); apple area –55%	Chilling-deficit impact compounds over the orchard investment cycle, not the crop season
2013–2017	Tarai groundwater draft rises	Annual draft 0.99 → 1.64 BCM (+66%)	Adaptation in one sub-region can constitute maladaptation at the State scale
2016–17 to 2022–23	Statewide horticulture contraction	Fruit area –54% (177,324 → 81,693 ha); fruit yield –44%	Horticulture is the most visibly climate-impacted sub-sector, and the contraction is a disinvestment signal
2021 (Feb)	Chamoli rock-ice avalanche and flash flood	Irrigation and water-supply disruption feeding into agriculture in the affected valleys; market access severed	Cross-sector cascades matter for agricultural water security and for perishable-produce logistics
2021 (Oct)	Extreme post-monsoon rainfall, Kumaon	Damage to standing kharif crops at or near harvest	The hazard season is lengthening beyond the conventional monsoon window, into the harvest period
2023	Statewide extreme-weather season	44,882 ha of farmland destroyed (~5.6% of cultivated area)	A single adverse monsoon can equal several years of chronic decline
2024	Forest fire season — >1,100 major incidents	Grazing and fodder biomass loss in the pre-monsoon lean period; post-fire slope instability above terraces	Fire is an agricultural and livestock hazard, not solely a forestry one

2008–2022	Sustained rural out-migration	1,47,512 permanent migrants; 1,792 ghost villages by 2022, from 1,034 in 2011	Climate stress and out-migration are mutually reinforcing; labour loss is itself an adaptation constraint
2025 (Nov)	PMFBY revision	Wild-animal crop damage added as an insurable risk; Uttarakhand named a priority State	Institutional recognition of human–wildlife conflict as a climate-amplified agricultural hazard

Chronology of documented climate-hazard impacts on the Uttarakhand agriculture and allied sectors.

6.4 Exposure Assessment

Exposure is assessed by intersecting hazard-susceptibility layers with the geo-referenced agricultural asset base. Four hazards are carried into the spatial overlay, rainfall erraticism/drought, chilling deficit (temperature), extreme weather (flood/cloudburst/hailstorm/landslide), and human-wildlife-crop conflict intensity, because susceptibility layers or documented incidence for these are available at a resolution consistent with block-level reporting. Groundwater stress is treated as a plains-specific sub-hazard within the drought layer.

Exposure Metrics

Asset class	Exposure metric	Unit reported
Rain-fed cropland	Area within high/very-high drought & rainfall-erraticism classes ÷ total rain-fed area	% of area exposed
Cultivated land (all)	Area within high/very-high extreme-weather classes ÷ total cultivated area	% of area exposed

Exposure metrics by asset class. “Exposed” means intersecting the high or very-high class of the hazard layer.⁴

Classification Bands

Class	Range (% of asset exposed)	Score	Interpretation
Very Low	0 – 10 %	1	Marginal exposure; standard practice adequate
Low	>10 – 25 %	2	Localised exposure; targeted protection of identified farms/orchards

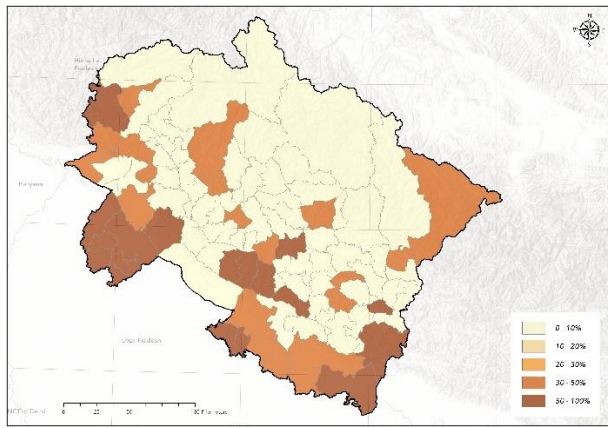
Class	Range (% of asset exposed)	Score	Interpretation
Moderate	>25 – 50 %	3	Material exposure; block-level risk treatment required
High	>50 – 75 %	4	Systemic exposure; varietal adaptation, insurance and infrastructure priority
Very High	>75 – 100 %	5	Critical exposure; the production system is effectively co-located with the hazard

Exposure classification bands and the score used in the composite.

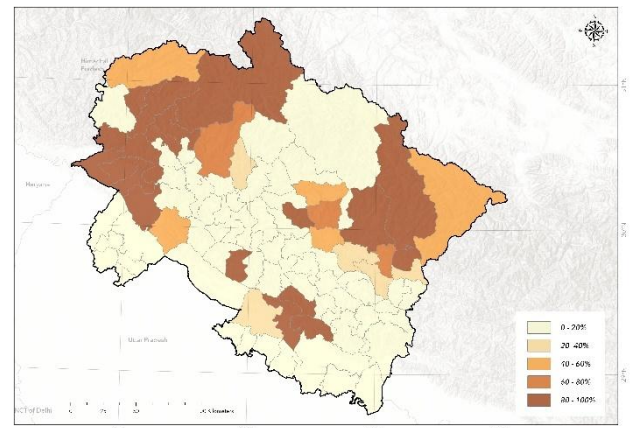
Hazard-Wise Exposure: Distribution of Blocks

This section assesses how climate-induced hazards translate into impacts on the agriculture sector through identifiable cause-effect pathways. The analysis focuses on the dominant climate hazards affecting Uttarakhand's agricultural land - floods, landslides, droughts, extreme rainfall, and forest fire and examines, for each hazard, the exposure of agricultural land, farming communities, irrigation infrastructure, and agricultural production systems.

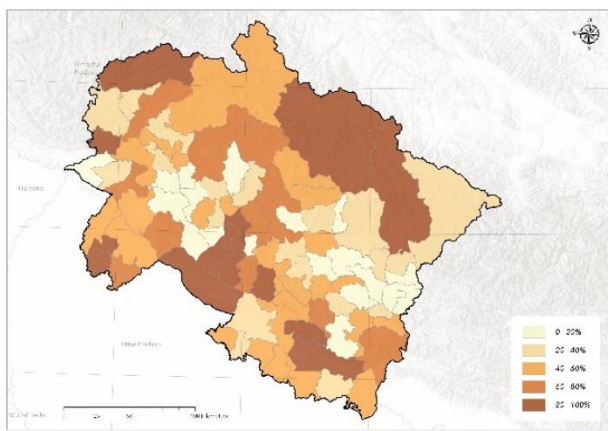
Given the dependence of agriculture on climatic conditions, exposure is evaluated not only in terms of physical damage to cropland but also in terms of disruption to agricultural operations, input accessibility, water availability, and seasonal crop productivity. Agricultural land represents the primary production asset within the sector and forms the first point of climate impact. Unlike discrete infrastructure assets, agricultural landscapes are spatially extensive and directly exposed to climatic variability. Because crop production depends on seasonal rainfall, soil moisture, temperature regimes, and stable terrain, even moderate changes in climatic conditions can significantly influence productivity across large areas. Consequently, exposure of agricultural land reflects both the direct susceptibility of cultivated areas to climate hazards and the broader implications for food production and rural livelihoods.



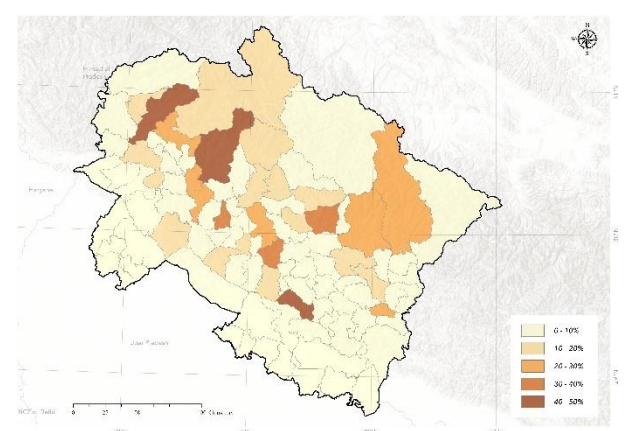
Cropland exposed to floods



Cropland exposed to extreme rainfall



Cropland exposed to Drought



Cropland exposed to Landslide

Composite Exposure Formula

The block composite is the weighted mean of the four hazard-specific exposure scores. Weights reflect the relative contribution of each hazard to documented agriculture-sector loss in the State, with extreme weather weighted highest because it is the most acute destroyer of standing crops and orchard infrastructure, followed closely by chilling deficit given its irreversible, multi-year impact on horticulture:

$$E(\text{block}) = 0.25 \cdot S(\text{extreme rainfall}) + 0.25 \cdot S(\text{floods}) + 0.25 \cdot S(\text{drought}) + 0.25 \cdot S(\text{Landslide})$$

where S is the 1–5 class score from Table 7.8. Composite scores are then banded: <2.0 Very Low; 2.0–2.69 Low; 2.70–3.39 Moderate; 3.40–4.19 High; ≥4.20 Very High, identical banding to the Energy and Water Sector CIAs so that composite classes are comparable across sectors.

Block-Wise Exposure of Agricultural Systems

District	Block	Extreme Rainfall	Floods	Drought	Landslide	Composite exposure
Uttarkashi	Bhatwari	Very High	High	High	High	Very High

District	Block	Extreme Rainfall	Floods	Drought	Landslide	Composite exposure
	Dunda	High	High	Moderate	High	High
	Chinyalisaur	High	High	Moderate	High	High
	Naugaon	High	High	Moderate	High	High
	Purola	Moderate	Moderate	Moderate	High	Moderate
	Mori	High	Very High	High	Moderate	High
Chamoli	Dasholi	Very High	Very High	High	Moderate	Very High
	Joshimath	Very High	Very High	Very High	Low	Very High
	Ghat	Very High	High	High	Moderate	High
	Karnaprayag	High	High	High	High	High
	Dewal	High	High	Moderate	High	High
	Tharali	High	High	High	High	High
	Narayanbagar	High	High	Moderate	High	High
	Pokhri	High	Moderate	Moderate	High	High
	Gairsain	Moderate	Moderate	Moderate	High	Moderate
Rudraprayag	Ukhimath	Very High	High	High	Moderate	High
	Agastyamuni	High	High	High	Moderate	High
	Jakholi	High	Moderate	High	High	High
Tehri Garhwal	Chamba	High	Moderate	Moderate	High	High
	Jaunpur	High	Moderate	Moderate	High	High
	Kirtinagar	High	Moderate	Moderate	High	High
	Narendranagar	High	Low	Moderate	Very High	High
	Pratapnagar	High	High	High	High	High
	Devprayag	High	Moderate	Moderate	High	High
	Jakhnidhar	High	Moderate	Moderate	High	High
	Bhilangana	Very High	High	High	Moderate	High
	Thauldhar	High	Moderate	Moderate	High	High

District	Block	Extreme Rainfall	Floods	Drought	Landslide	Composite exposure
Dehradun	Chakrata	High	High	Moderate	Very High	High
	Kalsi	High	Moderate	Moderate	Very High	High
	Vikasnagar	Moderate	Low	Moderate	Moderate	Moderate
	Sahaspur	Low	Very Low	Moderate	Moderate	Low
	Raipur	Moderate	Low	High	Moderate	Moderate
	Doiwala	Low	Very Low	High	Low	Low
Pauri Garhwal	Kot	High	Moderate	Moderate	Very High	High
	Kaljikhali	High	Moderate	Moderate	Very High	High
	Pauri	High	Moderate	Moderate	Very High	High
	Pabau	High	Moderate	Moderate	Very High	High
	Thalisain	High	Moderate	Moderate	Very High	High
	Bironkhal	High	Moderate	Moderate	Very High	High
	Dwarikhali	High	Low	Moderate	Very High	High
	Dugadda	High	Low	Moderate	High	Moderate
	Jaharikhal	High	Moderate	Moderate	Very High	High
	Rikhnikhali	Moderate	Low	Moderate	High	Moderate
	Yamkeshwar	High	Low	Moderate	Very High	High
	Eleshwar	Moderate	Low	Moderate	High	Moderate
	Khirsu	Moderate	Moderate	Moderate	High	Moderate
	Nainidanda	High	Moderate	Moderate	Very High	High
	Pokhra	High	Moderate	Moderate	High	High
Nainital	Betalghat	High	Moderate	High	Very High	High
	Bhimtal	High	High	Very High	Very High	Very High
	Dhari	High	Moderate	High	Very High	High
	Kotabag	Moderate	Low	High	Very High	High
	Okhalkanda	High	Moderate	High	Very High	High

District	Block	Extreme Rainfall	Floods	Drought	Landslide	Composite exposure
	Ramgarh	High	High	High	Very High	Very High
	Ramnagar	Low	Low	High	Very High	Moderate
	Haldwani	Low	Very Low	Very High	High	Moderate
Almora	Hawalbagh	High	Very High	Moderate	Very High	Very High
	Takula	High	High	Moderate	Very High	High
	Lamgara	High	High	Moderate	Very High	High
	Dhauladevi	High	High	Moderate	Very High	High
	Bhaisiyachhana	High	High	Moderate	Very High	High
	Bhikiyasain	High	High	Moderate	Very High	High
	Sult	High	High	Moderate	Very High	High
	Chaukhutia	High	High	Moderate	Very High	High
	Dwarahat	High	High	Moderate	Very High	High
	Syaldey	Moderate	Moderate	Moderate	High	Moderate
	Tarikhhet	High	High	Moderate	Very High	High
Bageshwar	Bageshwar	High	High	High	High	High
	Kapkot	Very High	Very High	High	Moderate	Very High
	Garur	High	High	High	High	High
Pithoragarh	Pithoragarh	High	High	High	High	High
	Munakot	High	High	High	High	High
	Gangolihat	High	Moderate	Moderate	Very High	High
	Berinag	High	High	High	Very High	Very High
	Didihat	High	High	High	High	High
	Kanalichhina	High	High	High	High	High
	Dharchula	Very High	High	Very High	Low	High
	Munsyari	Very High	Very High	Very High	Low	Very High
Champawat	Champawat	High	Moderate	Moderate	Very High	High

District	Block	Extreme Rainfall	Floods	Drought	Landslide	Composite exposure
	Lohaghat	High	Moderate	Moderate	Very High	High
	Pati	High	High	High	Very High	Very High
	Barakot	Moderate	Moderate	High	High	High
Haridwar	Bahadrabad	Moderate	Very Low	Low	Low	Low
	Roorkee	Moderate	Very Low	Low	Very Low	Very Low
	Narsan	Low	Very Low	Low	Very Low	Very Low
	Bhagwanpur	Low	Very Low	Low	Low	Very Low
	Laksar	High	Very Low	Low	Very Low	Low
	Khanpur	Moderate	Very Low	Low	Very Low	Very Low
Udham Singh Nagar	Kashipur	Moderate	Very Low	Low	Low	Low
	Jaspur	Moderate	Very Low	Low	Low	Low
	Bazpur	Moderate	Very Low	Low	Low	Low
	Gadarpur	High	Very Low	Moderate	Low	Low
	Rudrapur	Moderate	Very Low	Moderate	Very Low	Low
	Sitarganj	High	Very Low	Moderate	Low	Low
	Khatima	High	Very Low	Moderate	Low	Low

Priority Blocks

The Very High composite blocks cluster in the high-altitude orchard belts and the districts already documented as suffering the steepest horticultural decline, Almora, Chamoli, Pithoragarh, combining acute chilling deficit, extreme-weather exposure and, in several cases, severe human- wildlife-crop conflict.

Block	District	Exposure driver	Agricultural systems at stake
Joshimath	Chamoli	Full hazard stack: extreme weather, chilling deficit, rainfall/drought	Terraced cereal plots; high-altitude orchards
Munsyari	Pithoragarh	Extreme weather, chilling deficit, rainfall, remote, glacier-fed	Subsistence cereals; transhumant grazing
Dharchula	Pithoragarh	Extreme weather, rainfall/drought; border-area isolation	Terraced agriculture; livestock corridors
Kapkot	Bageshwar	Extreme weather, chilling deficit; Pindari catchment	High-altitude orchards; transhumance route
Hawalbagh	Almora	Chilling deficit (Very High) with severe HWC	Apple/pear orchards; documented 84% fruit-yield decline district
Bhimtal	Nainital	Chilling deficit, rainfall/drought, HWC all Very High/High	Orchard belt; tourism-adjacent farmland
Dasholi	Chamoli	Extreme weather (Very High), chilling deficit (Very High)	Alaknanda-corridor terraces and orchards

Table 6.10 Illustrative Very-High composite-exposure blocks and their agricultural systems at stake.

Asset-Class Findings

Rain-fed cereal and millet systems

Exposure to rainfall erraticism and drought is highest in the mid-altitude hill blocks where over 65% of area is rain-fed and no irrigation buffer exists. Compounding this, the same blocks carry the highest extreme-weather exposure, so the two hazards reinforce rather than offset each other, the “paradox” of less total rain delivered in more destructive pulses.

Horticultural orchards

Chilling-deficit exposure is overwhelmingly concentrated in Almora, Chamoli, Nainital, Bageshwar and Pithoragarh, the traditional apple and stone-fruit belts, consistent with the documented 47% collapse in state apple production (2013–2023) and the 84% fruit-yield decline recorded in Almora district specifically. This is the sub-sector where composite exposure translates most directly into irreversible, multi-year economic loss given 5–15-year orchard investment cycles.

Livestock and transhumance routes

HWC intensity and extreme-weather exposure jointly bind on the transhumance corridors of Chamoli, Uttarkashi, Pithoragarh and Bageshwar, and separately on the Tarai dairy belt through heat-stress days. Pauri Garhwal and Almora blocks combine Very High HWC with High-to-Moderate other hazards, the documented pattern of farm abandonment enabling wildlife encroachment that then further suppresses remaining cultivation.

Population and Livelihood Exposure

Asset exposure is only half the picture: the consequence of a hazard depends on how many farm households and how much livelihood income depend on the affected land. For each block the assessment therefore also records the farm-household population dependent on exposed rain-fed or orchard land, the share of household income derived from the exposed sub-sector, out-migration rate (a proxy for eroding labour available to maintain terraces and guard crops against wildlife), and block accessibility during hazard events (repair/relief-access proxy). The distinction matters: high-altitude blocks such as Munsyari and Dharchula have very high production-system exposure but small, already-declining populations, while mid-hill blocks in Almora and Pauri Garhwal combine high exposure with the state's largest remaining farm populations and the highest out-migration pressure, the combination that should draw the most adaptation investment on a livelihood-continuity criterion.

District wise Priority

District	Flood	Extreme rainfall	Drought	Landslide
Uttarkashi	Low in the northern high-altitude blocks; a high-exposure cluster in the west	Among the highest in the State; upper classes dominant	Upper-middle to high	Contains some of the State's highest-exposure blocks
Chamoli	Generally low; scattered mid-valley exposure	High in the northern blocks	Moderate to high	Mostly low with localised peaks
Rudraprayag	Low to moderate	Moderate	Moderate to high	Localised high
Tehri Garhwal	Scattered moderate to high	Low to moderate	Moderate to high	Localised high in one or more blocks
Dehradun	Prominent high-exposure cluster in the western hill blocks	High in the west; low along the valley and Tarai fringe	Moderate	Low to moderate
Pauri Garhwal	Scattered moderate to high	Predominantly low	Moderate to high	Localised; at least one high-exposure block

Nainital	Moderate to high in the southern blocks	Low	Moderate	Low to moderate
Almora	Scattered	Low	Moderate	Low with localised moderate
Bageshwar	Moderate	Moderate to high	High	Moderate
Pithoragarh	Moderate in the eastern blocks	Among the highest in the State	Among the highest in the State	Moderate band across the east
Champawat	Moderate	Low to moderate	Moderate to high	Low to moderate
Haridwar	Moderate to high in parts of the southern belt — materially above the superseded rating	Very low	Low to moderate	Negligible
Udham Singh Nagar	High across parts of the southern and south-eastern belt	Very low	Low to moderate	Negligible

6.5 Climate Variability Impacting the Agriculture and Allied Sectors

This section addresses sensitivity: given exposure, how strongly does climate variability translate into loss of agricultural production and livelihood? For Uttarakhand the answer is that sensitivity is unusually high, because almost every link in the agricultural chain, resource base, production, post-harvest handling and household labour, is climate-dependent. The section first sets out the observed variability, then traces sensitivity pathways by sub-sector, and finally identifies where variability is already binding.

Observed Climate Variability and Trends

Climate variable	Observed change	Consequence for agriculture and allied sectors
Total monsoon rainfall	Decline of ~12% since 2000 (IMD, 2023)	Less percolating rain for rain-fed germination
Extreme rainfall frequency	Increase of ~35% since 2000 (IMD, 2023)	Terrace erosion, waterlogging, standing-crop destruction
Hydro-met disaster frequency	+240% between 2000 and 2022 (SDMA, 2023)	44,882 ha farmland destroyed in 2023 alone
Air temperature	+1.5°C since 1970; elevation-dependent amplification	Chilling-hour deficit; upward migration of crop and orchard belts
Winter chilling hours (apple/pear/walnut belts)	Progressive reduction at lower elevations	Orchard belt forced upward 500–1,000+ ft; commercial unviability below threshold altitude

Climate variable	Observed change	Consequence for agriculture and allied sectors
Cloudburst incidence	11 events (2000–2010) → 83 events (2011–2023)	Primary trigger for agricultural landslide and flash-flood damage
Drought frequency	10 of 13 districts affected, 2007–09 (Singh, 2023)	Yield falls 30–40% in drought years; groundwater over-extraction in Tarai
Out-migration	1,47,512 permanent migrants (2008–2022); 1,792 ghost villages	Terrace abandonment, labour scarcity, wildlife-conflict amplification

Table 7.11 Observed climate variability in Uttarakhand and its agriculture-sector consequence.

Agricultural Chain Transmission Matrix

The matrix traces each climate variable through every stage of the agricultural chain. Reading down a column shows what a single stage is exposed to; reading across a row shows how a single variable propagates.

Climate variable	Resource base	Production	Post-harvest & market	Household & community
Rising temperature	Chilling-hour loss; upward belt migration	Poor flowering/fruit-set; heat stress on livestock & silkworms	—	Adaptation cost (higher-altitude relocation)
Erratic monsoon rainfall	Recharge & soil-moisture disruption	Germination failure or waterlogging; yield variance	Quality loss (pre-harvest sprouting, fungal infection)	Income volatility
Extreme rainfall/cloudburst	Terrace/soil erosion	Direct crop/orchard destruction	Access loss to markets & cold storage	Multi-season recovery burden
Drought	Groundwater over-extraction	30–40% yield loss	—	Distress migration
Hailstorm	—	75–100% single-event crop loss (fruit)	Total consignment loss	Acute income shock

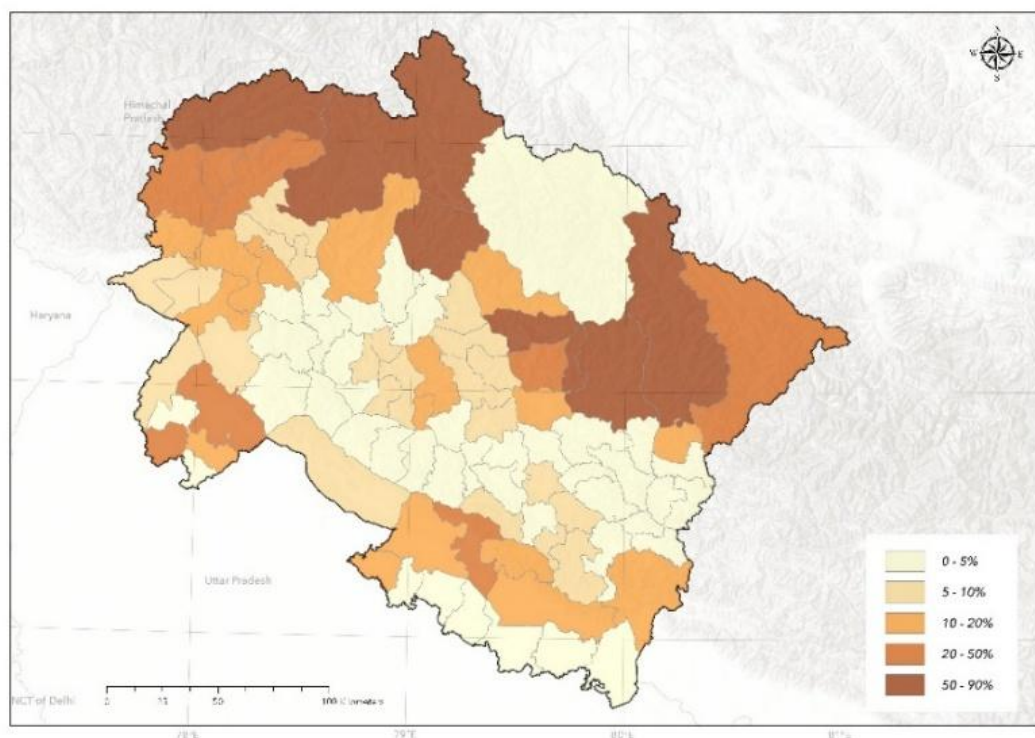
Climate variable	Resource base	Production	Post-harvest & market	Household & community
Warming winters (snow retreat)	Loss of frost/insulation buffer for orchards	Frost injury; pest range expansion (red spider mite)	—	—
Climate-driven wildlife range shift	Forest-edge advance into fallow land	Crop raiding (16–24% productivity loss in conflict zones)	—	Farm abandonment; crop-guarding burden on women

Transmission of climate variability through the agricultural chain. Blank cells indicate no material pathway.

Climate Impact Pathways

Agriculture: Crop Yield Decline and Landscape Transformation

Uttarakhand's hill cropping system is characterised by diversified subsistence cultivation, mandua, jhangora, maize, rice, wheat, gahat (horse gram) and minor millets intercropped on terraced slopes entirely dependent on monsoon precipitation.



The Tarai plains support more commercial, partially irrigated wheat-rice rotations.

Traditional millets possess inherent drought tolerance that has historically buffered erratic rainfall, but climate change is beginning to breach even these tolerances: the

critical vulnerability is the increasingly erratic intra-seasonal distribution of rainfall, longer dry spells during germination and vegetative growth, followed by intense precipitation causing waterlogging and physical crop damage.

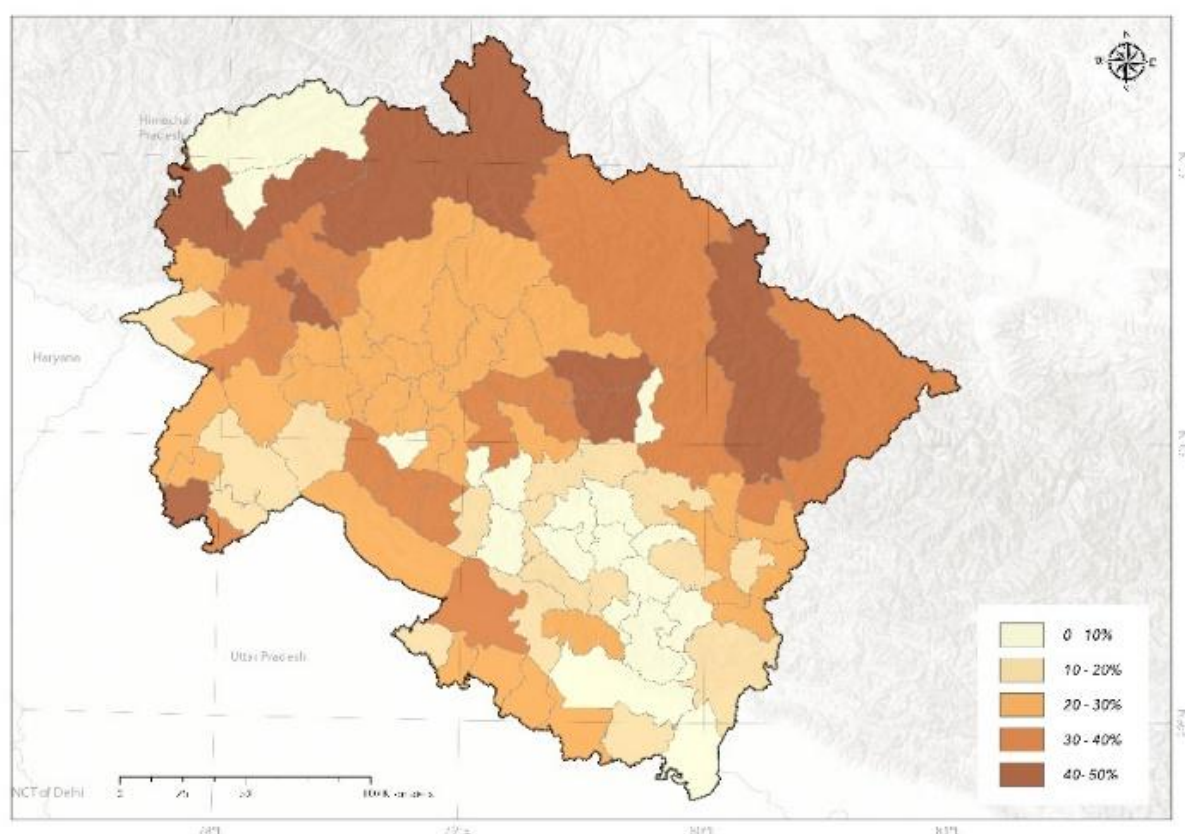
Climate warming has allowed maize and kharif cultivation to advance upward by 200–400 m, while simultaneously degrading the traditional millet belts below through changed rainfall seasonality (Thadani, 2015).

Irrigation Infrastructure Exposure to Disruption

Irrigation infrastructure constitutes a critical enabling system for agricultural production, particularly as rainfall variability increases.

Canal networks, diversion structures, lift irrigation systems, water harvesting structures, and community irrigation channels provide essential water supply during dry periods and enhance agricultural resilience.

However, these assets are highly sensitive to climate-induced physical damage and hydrological variability, making irrigation infrastructure an important component of agricultural exposure.



The analysis indicates that irrigation systems across Uttarakhand face increasing exposure to multiple climate hazards. Extreme rainfall and floods can damage canals, diversion weirs, pump houses, and distribution networks through erosion, sediment deposition, and structural failure. In mountainous districts, landslides frequently

damage gravity-fed irrigation channels and obstruct water conveyance systems, reducing irrigation reliability during the cropping season. Drought conditions further increase exposure by reducing streamflow, spring discharge, and water storage availability, thereby limiting irrigation supplies precisely when agricultural demand is highest.

Forest fires contribute indirectly by degrading upstream catchments, reducing watershed health, and accelerating soil erosion, which increases sediment loads within irrigation systems. The combined effects of these hazards reduce irrigation efficiency, increase maintenance requirements, and constrain agricultural productivity, particularly in blocks dependent on localized surface water sources and traditional irrigation networks.

Tarai Plains Agriculture

Haridwar and Udham Singh Nagar support the state's most productive, commercially oriented agriculture, irrigated wheat (a record 3,244 kg/ha in 2024), paddy, sugarcane, oilseeds and vegetables.

This zone, while more resilient than hill agriculture due to irrigation access, faces climate risks of a different character: heat stress during the March–April wheat grain-filling period, waterlogging and flood damage from intensified monsoon, and groundwater depletion from rising irrigation pressure.

Unseasonal March–April rains cause quality loss in wheat through pre-harvest sprouting and fungal infection even without measurable yield loss, and irrigated rice yield is projected to decrease across all agro-climatic zones under both SSP2-4.5 and SSP5-8.5, by up to 20% by the 2090s in comparable Himalayan-foothill systems (Singh et al., 2024).

Horticulture: Orchard Decline and Altitude Migration

Horticulture is one of the most visibly and severely climate-impacted sub-sectors. Apple production has fallen from 123,228 tonnes (2013) to 64,881 tonnes (2023), a nearly 50 per cent decline in a decade. The broader contraction is equally stark: total fruit cultivation area fell 54 per cent between 2016–17 and 2022–23 (177,324 to 81,693 ha), and total fruit yield fell 44 per cent (662,847 to 369,447 MT).

Apple area fell from **25,201.58 ha (2016–17) to 11,327.33 ha (2022–23)**, a 55% decline, with a corresponding 30% decline in yield per remaining hectare; the Kumaon apple belt has shifted upward as lower altitudes no longer accumulate sufficient chilling.

Fruit production in **Almora fell 84%** (2016–17 to 2022–23), the steepest decline of any district; **Chamoli saw a 53% yield reduction** despite only 13% area contraction, indicating severe per-hectare productivity collapse.

Pear, apricot, plum and walnut, all highly chilling-dependent, show maximum production decline; lemon yield fell 58%, while mango and litchi fell only 20–24%, showing mixed tropical-crop resilience relative to temperate types.

Farmers are adopting low-chilling cultivars (the “Anna” apple variety, requiring only 200–300 chilling hours) and shifting to kiwi, pomegranate, dragon fruit and guava, adaptations that carry socioeconomic and ecological risk from opening new, less-stable terrain at higher altitude.

Vegetables, Spices and Off-Season Cultivation

High-altitude districts have historically supplied off-season vegetables to plains markets during May–September, when plains heat prohibits production. This comparative climatic advantage is eroding as Himalayan temperatures rise: the altitude threshold for competitive off-season production is increasing, reducing the aggregate production area, while warming at lower elevations extends the plains vegetable season and narrows the market window for Himalayan-origin produce.

Animal Husbandry: Pasture Degradation and Heat Stress

Livestock constitutes a critical secondary livelihood for over 70% of rural households. The sector is tightly integrated with forest and pasture systems: Van Gujjar and Bhotiya pastoral communities practise transhumance between Terai winter pastures and alpine (bugyal) summer grazing, a system directly threatened by warming, reduced snowpack and altered vegetation at both ends of the altitudinal range.

Warming in the Terai/Bhabar belt is increasing Temperature-Humidity Index stress days for crossbred dairy cattle: above THI 72, milk production falls measurably; above 78, sharp declines in productivity, conception rate and animal welfare follow (Cartwright et al., 2023).

- Pasture quality and availability are declining in climate-stressed areas; warming-induced upward migration of vegetation zones is shrinking the altitude range of productive alpine bugyals (Thadani, 2015).
- Reduced black-bear hibernation duration and depth is forcing bears to forage at lower altitudes during traditional hibernation periods, increasing bear-livestock conflict and compound livestock mortality alongside leopard predation (WII, 2019).
- Loss of traditional transhumance routes to infrastructure development and climate-altered snow conditions is disrupting pastoral mobility, forcing sedentarisation on Van Gujjars and Bhotiyas and reducing herd sizes (Negi et al., 2017).

Climate hazard	Climate driver	Climate stressor	Direct impact	Functional impact	Consequences	Confidence
Rising temperature and chilling deficit	Increasing mean and winter minimum temperatures with elevation-dependent amplification (+1.5°C since 1970)	Accumulated winter chilling falls below cultivar requirement; accelerated and desynchronised phenology	Delayed, protracted and irregular bud break; poor flower quality and low fruit set in apple, pear, walnut, apricot, plum and cherry	Loss of commercial viability of established cultivars below a threshold altitude; upward migration of the viable orchard belt	Apple production –47% (2013–2023); apple area –55%; near-doubling of establishment cost at higher altitude; stranded perennial capital	High
Erratic and delayed monsoon	Shifting onset and cessation; longer intra-seasonal dry spells; ~12% decline in total monsoon rainfall since 2000	Soil-moisture deficit at germination and tillering; reduction in infiltration-dominant rainfall	Germination failure; reduced tillering and grain-fill in rain-fed millets, maize, wheat and paddy	Rising inter-annual yield variance on the >55% of cultivated area without irrigation access	Mandua –27%, jhangora –20%, barley –11% (2012–13 to 2021–22); income volatility; erosion of subsistence food security	High
Extreme rainfall and cloudburst	~35% increase in events exceeding 100 mm/day; cloudburst incidence 11 → 83 events	Runoff exceeding terrace infiltration capacity; slope saturation and mass movement	Terrace scour and riser failure; crop burial; destruction of guhl channels; landslide loss of cultivable land	Permanent reduction in net cultivable area; multi-season loss of soil productive capacity	44,882 ha destroyed in 2023 (~5.6% of cultivated area); multi-season recovery burden; abandonment of damaged terraces	High
Hailstorm	Increasing convective instability in the pre-monsoon and early monsoon period	Mechanical impact energy delivered during the fruit-development window	Injury to fruit skin, foliage and bearing shoots; destruction of marketable grade	Loss of the entire marketable consignment even where biological yield partially survives	75–100% crop loss in badly affected orchards from single events; acute and largely uninsured income shock	High
Drought and soil-	Rainfall deficit combined with	Prolonged root-zone moisture	30–40% yield reduction in drought	Loss of a full production season	10 of 13 districts affected 2007–09; Tarai groundwater	High

moisture deficit	higher evaporative demand under warming	deficit; declining surface-water reliability	years; pasture and fodder biomass loss	with no irrigation buffer in the hills; transfer of the deficit to the aquifer in the Tarai	draft +66% (0.99 → 1.64 BCM); 4–6 additional hours per day of water collection borne by women	
Spring discharge decline and glacier retreat	Reduced recharge, declining snowfall fraction and sustained negative glacier mass balance	Falling lean-season discharge in springs, gadhera and glacier-fed canals; seasonal shift away from the rabi window	Contraction of the command area of gravity-fed gul and nali irrigation; ~12,000 springs dried statewide	Progressive reversion of assured-irrigated land to rain-fed status, compounding every rainfall-driven pathway above	Land abandonment where no alternative source exists; peak-water transition around 2050 in glacier-dependent commands in Chamoli, Uttarkashi, Pithoragarh and Tehri Garhwal	Medium–High
Warming winters and snow retreat	Rising winter minima; declining snow-covered area and duration	Loss of the insulating snowpack while radiative frost nights persist; reduced overwintering pest mortality	Frost injury to orchards despite rising seasonal means; altitudinal expansion of red spider mite and other pests	Increased crop-protection cost and input dependence; yield and quality loss at high altitude	Rising cost of production at precisely the altitudes to which cultivation is being forced to migrate	Medium–High
Forest fire and pasture degradation	Warmer, drier pre-monsoon conditions and an extended fire season	Loss of grazing biomass, lopped fodder and NTFP resource in the lean pre-monsoon period; post-fire ground-cover loss	Fodder deficit at the point of maximum seasonal scarcity; increased runoff and reduced slope cohesion above terraces	Reduced livestock productivity and distress sale; elevated terrace-failure probability in the monsoon of the same year	>1,100 major incidents in 2024; severe-fire-risk days projected to rise ~60%; compound fire–landslide loss sequence within a single year	High (fire); Medium (fodder link)
Heat and humidity stress on livestock and sericulture	Rising Temperature–Humidity Index in the Tarai–Bhabar belt and lower valleys; warmer	THI exceeding the 72 and 78 thresholds; September–October rearing temperatures	Measurable milk-yield decline above THI 72; sharp productivity, conception-rate and welfare decline above THI 78; reduced cocoon shell weight	Loss of the commercially critical autumn silk crop; reduced reliability of dairy income; phenological mismatch between	Erosion of the secondary livelihood on which over 70% of rural households depend; elevated muscardine and pebrine outbreak risk capable of causing complete batch loss	Medium–High

	autumn rearing season	persistently above the 26°C optimum	and elevated larval mortality	mulberry leaf flush and rearing cycle		
Climate-driven wildlife range shift	Altered forest fodder and fruiting phenology; reduced black-bear hibernation duration; forest-edge advance into abandoned land	Increased crop-raiding pressure from wild boar, rhesus macaque, nilgai, porcupine and bear; increased livestock predation	16–24% productivity loss for maize, wheat and paddy in conflict zones; livestock mortality from bear and leopard predation	12–17% reduction in net cropped area relative to non-conflict areas; diversion of household labour to crop guarding	Over 50% of erstwhile cultivated land lying barren in some Pauri Garhwal and Almora villages; guarding burden falling on women and the elderly; PMFBY revised in 2025 to insure the risk	Medium–High
Compound hazards (chronic decline + episodic destruction + labour scarcity)	Convergence of slow-onset productivity decline, episodic asset destruction and demographic depletion	Simultaneous loss of yield, of the productive asset and of the labour required to reconstruct it	Terraces damaged in one season are not reconstructed in the next; orchards removed are not replanted	Transition from recoverable loss to permanent abandonment — the sensitivity threshold beyond which the system does not return	1,792 ghost villages by 2022 (from 1,034 in 2011); unmaintained terraces raise landslide and erosion risk on slopes formerly regulated by active cultivation	High

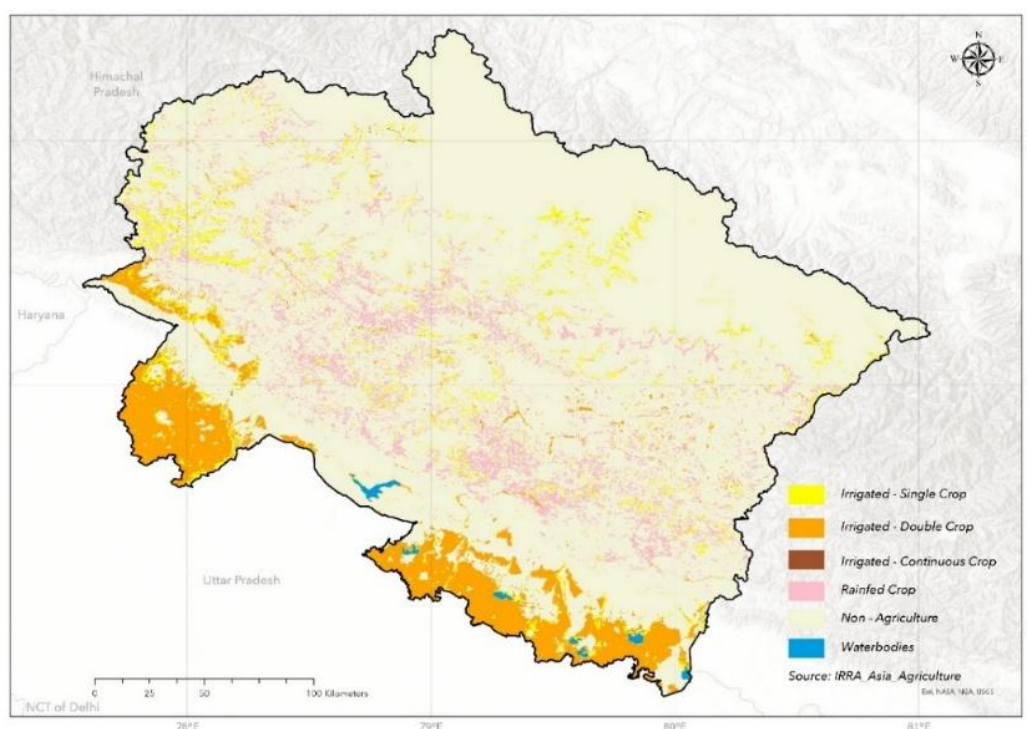
Human–Wildlife–Crop Conflict: A Climate-Amplified Agricultural Hazard

The interaction between climate-driven wildlife range expansion and progressive hill-farm abandonment has created a severe, escalating human-wildlife-crop conflict crisis that is simultaneously a driver and a consequence of agricultural abandonment. As villages empty and terraces lie fallow, forest edges advance into former agricultural buffers, opening new wildlife access corridors into remaining cultivated areas.

- Wild boar, rhesus macaque, neel gai, porcupine and bears are the primary crop-raiding species; in some Pauri Garhwal and Almora villages, **over 50% of erstwhile cultivated land** lies barren, partly attributable to the impossibility of defending crops against persistent animal pressure (NIRD, 2015).
- Crop productivity of maize, wheat and paddy in conflict zones is reduced **16–24%**, and net cropped area declines 12–17% relative to non-conflict areas (Kumar Thakur et al., 2022).
- The Government of India's revised PMFBY (November 2025) has, for the first time, included **wild-animal crop damage** as an insurable risk, explicitly naming Uttarakhand a priority state, official acknowledgement of the problem's severity.

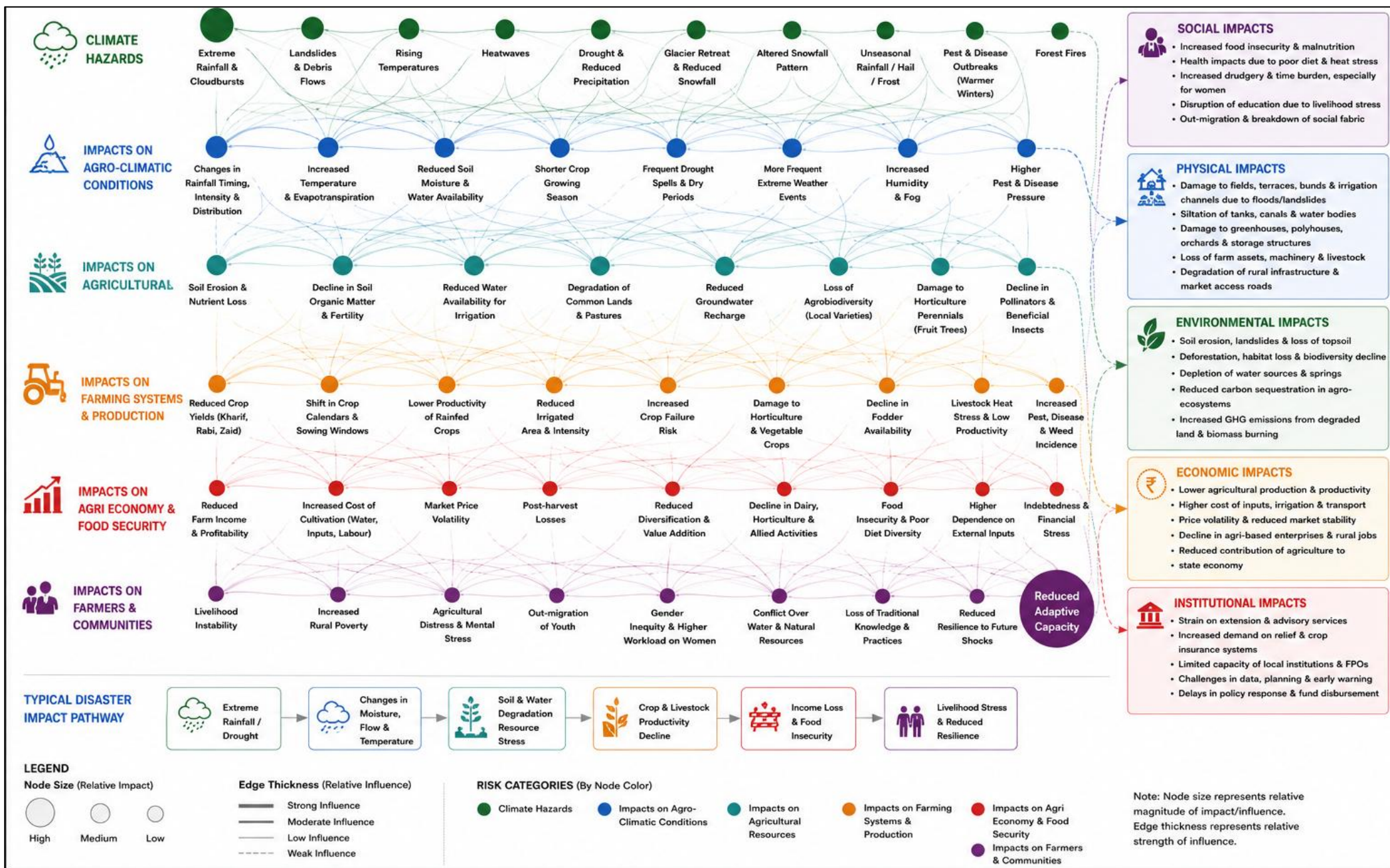
Agricultural Water Security

Over 75% of the state's total water demand is consumed by agriculture (CGWB, 2013), making agricultural water security the sector's most fundamental climate-sensitive dependency. Approximately 12,000 springs have dried up statewide, directly reducing the spring-fed irrigation channels (guls and nalis) that sustain terraced hill agriculture; spring failure forces land abandonment where no alternative source exists. Glacier-fed irrigation systems face a peak-water transition around 2050, after which sustained flow reductions will reduce the command area of glacier-dependent canals in Chamoli, Uttarkashi, Pithoragarh and Tehri Garhwal.



Farming Communities: Migration, Land Abandonment and Food Sovereignty

Farming communities are at the frontline of every climate impact in this chapter, but in Uttarakhand the relationship between climate stress and community vulnerability is mediated by a compound out-migration dynamic. Between 2008 and 2022, **1,47,512 people** migrated permanently from Uttarakhand's villages, and 1,792 villages were rendered completely uninhabited by 2022 (up from 1,034 in 2011), 734 new ghost villages in seven years. Districts with the highest out-migration, Almora, Tehri, Pauri Garhwal, Chamoli, are precisely the districts with the highest climate vulnerability, highest HWC pressure and most severe horticulture decline; this geographic co-location reflects climate stress as a structural driver of out-migration alongside economic and service-access factors. As villages hollow out, the remaining agricultural labour force is disproportionately women, the elderly and children, who also bear the primary burden of crop-guarding against wildlife and, per PSI (2022), spend 4–6 additional hours per day collecting water during drought. Farm abandonment also has ecological consequences: unmaintained terraces lose structural integrity, increasing landslide and erosion risk on slopes once hydrologically regulated by active agriculture.



6.6 Potential Future Climate Change

Because temperate horticulture and rain-fed cereal systems depend directly on winter chilling and monsoon reliability respectively, the projected change in temperature and rainfall extremes is the single most consequential projection for the sector. This section sets out the far-future (2071–2100, relative to the 1995–2014 baseline) change in the variables that govern agricultural production, under a low-emissions (SSP1-2.6) and a high-emissions (SSP5-8.5) scenario.

Variable (far future, 2071–2100)	SSP1-2.6	SSP5-8.5	Mechanism and agriculture-sector relevance
Mean annual temperature ($\Delta^{\circ}\text{C}$)	(), indicative +1.0 to +1.8 $^{\circ}\text{C}$	(), indicative +3.5 to +5.5 $^{\circ}\text{C}$	Drives chilling-hour loss, heat stress and phenological disruption across all five sub-sectors; EDW amplifies at 3,000+ m
Chilling hours (below 7.2 $^{\circ}\text{C}$, Oct–Feb)	(), ~15–20% reduction at current apple-belt altitudes	(), commercial apple cultivation eliminated below ~2,800 m	Direct threat to temperate horticulture; belt contracts upward 300–500 vertical m by 2050s
Extreme-rainfall day frequency	(), moderate increase (~22 events/yr)	(), large increase (~27 events/yr)	Terrace erosion; farmland destruction; landslide-triggered crop loss
Wheat yield (hill, rain-fed, temperature effect only)	(), ~5–8% reduction	(), ~10–15% reduction	Grain-filling period shortened; heat-sensitive varieties most affected
Millet (mandua/jhangora) yield trajectory	(), decline continues, some altitude compensation	(), significant yield stress in traditional zones	Erratic intra-seasonal rainfall distribution is the binding constraint, not total rainfall
Agricultural land lost per monsoon season	(), 50,000–60,000 ha/yr risk	(), 70,000–90,000 ha/yr potential	Non-linear amplification from terrace degradation and extreme-event frequency
Fruit horticulture production area (state total)	(), further contraction without	(), temperate fruit at <2,200 m	Adaptation window (varietal shift, altitude migration) narrows rapidly

Variable (far future, 2071–2100)	SSP1-2.6	SSP5-8.5	Mechanism and agriculture-sector relevance
	cultivar adaptation	commercially unviable	
THI >72 stress days (Teraï dairy cattle)	(), moderate increase from current 3–4 months/yr	(), 5–6 months/yr by 2050	Milk yield, conception rate and feed-conversion efficiency decline

Table 7.14 Projected far-future change in agriculture-relevant variables and their significance for the sector (synthesised from CMIP6 ensemble outputs, ICIMOD 2019, Krishnan et al. 2020, Dubey et al. 2022, and Jain et al. 2025).

Sectoral Projections Beyond Crops

For sericulture, any warming sustained above 28°C during October rearing cycles will systematically suppress the commercially critical autumn cocoon crop without climate-controlled rearing infrastructure not currently in use at farm scale. For coldwater fisheries, continued warming at 0.2–0.47°C per decade will progressively eliminate suitable thermal habitat for snow trout at lower elevations, driving range contraction upward and increasing local extinction risk in the most accessible, historically most productive river reaches. For livestock, the projected national mean temperature increase by 2050 will extend physiological heat stress beyond current adaptations, particularly for exotic and crossbred dairy breeds with narrower thermal tolerance than indigenous breeds.

Adaptation Instruments: Insurance and Infrastructure

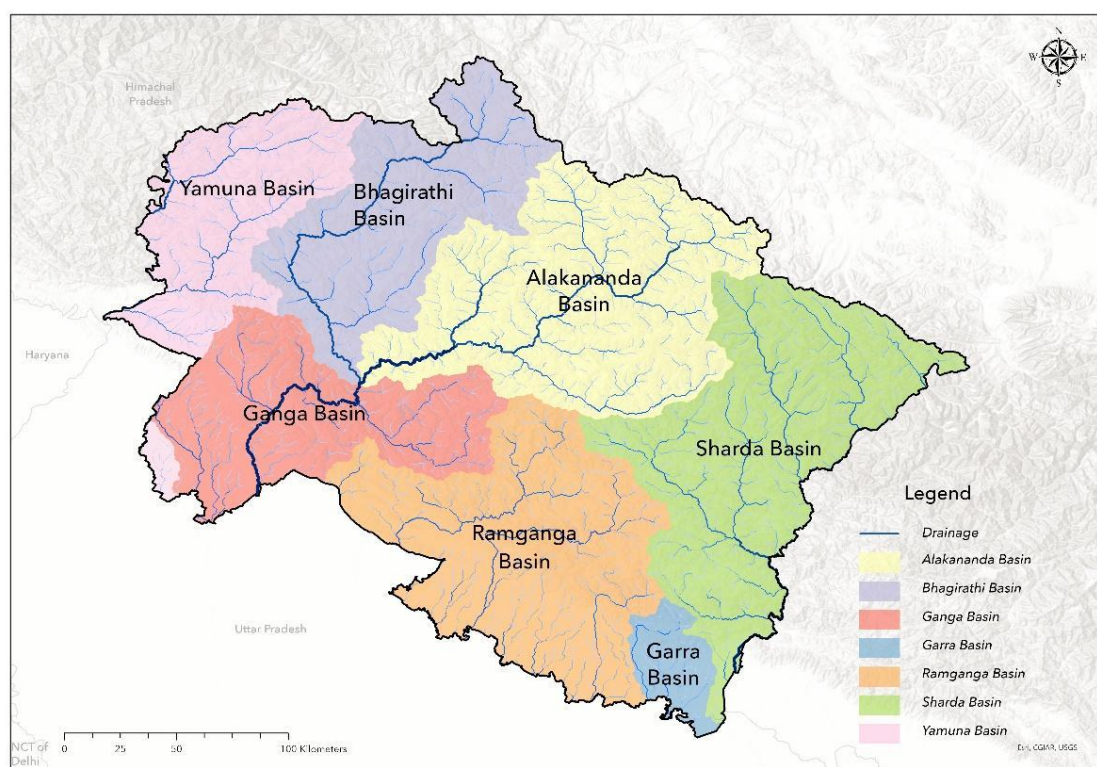
The primary economic-loss equation for crop agriculture used across this assessment is:

$$\text{Economic Loss (crop)} = \text{Agricultural Revenue} \times \text{Weighted Average Yield Reduction}$$

where Agricultural Revenue is the sum across crops of (farmgate price × production volume), and Weighted Average Yield Reduction is the area-weighted average of crop-specific yield-loss rates. Micro-insurance is a structurally important adaptation instrument given fragmented landholdings (average 0.68 ha) and limited financial reserves; PMFBY enrolment in hill districts remains constrained by the mismatch between its unit-area yield assessment and the high spatial variability of losses in mountainous terrain, though the November 2025 revision adding wild-animal crop damage as an insurable risk is a material improvement. Priority infrastructure investments include terrace strengthening and drainage improvement (with MGNREGA-funded convergence), anti-hailstorm netting for orchards (60–80% damage reduction demonstrated elsewhere), and cold storage combined with last-mile road connectivity to reduce the 15–30% post-harvest losses currently recorded in horticulture.

7. Water & WASH

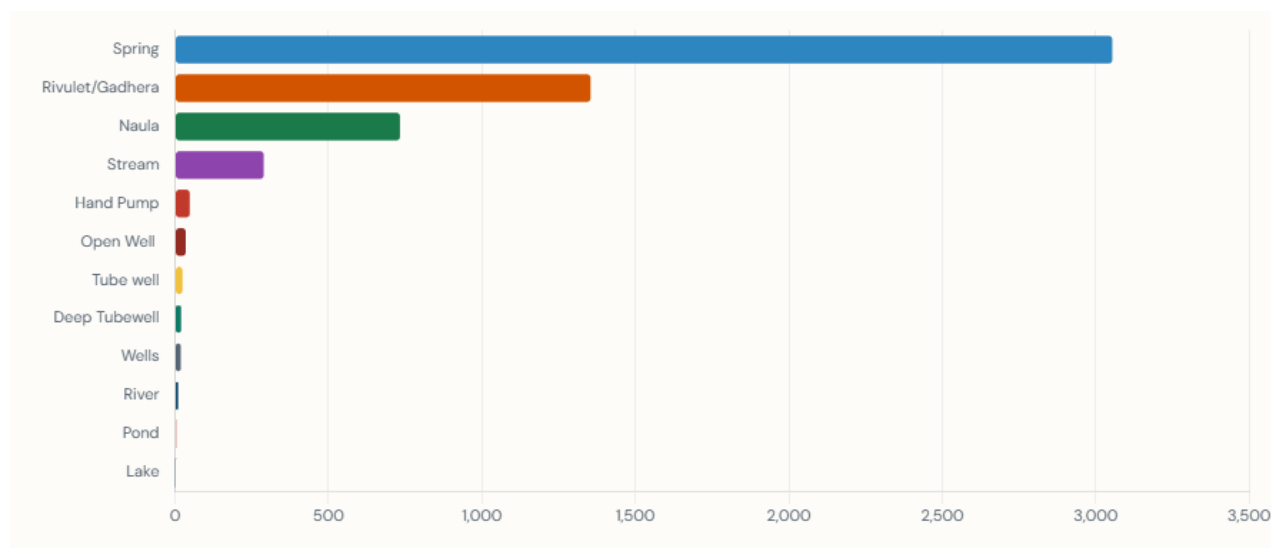
Uttarakhand, known for its mountainous landscapes, is adorned with many different freshwater reserves. These resources not only fulfil the water need of the different communities of the state but also of India as a whole. The water reserves of the state include 17 rivers, 31 lakes, and 8 major river catchments comprising 8 watersheds, 116 sub-watersheds, and 1110 micro-watersheds and most of the northern parts of the state are part of Greater Himalayan Ranges and are covered by high Himalayan peaks and glaciers. The seven key rivers of the state are Yamuna, Bhagirathi, Alaknanda, Ganga, Ramganga, and Kaliganga. Uttarakhand being a water-rich state also is a site of 8 huge man-made reservoirs in the districts of Tehri Garhwal and Udham Singh Nagar. These reservoirs are spread over an area of 20,075 ha. Surface water is available only to people living in the vicinity of riverbanks, but for people living upstream and on top of the hills, streams and groundwater are the primary sources of water. In the plains, groundwater is extracted by human-made structures such as open wells, hand pumps, shallow and deep tube wells but in hills natural sources like springs are the main source. Groundwater in the hilly region occurs mostly in fissures/fractures and emerges as springs. These water resources are also very often observed to be fulfilling needs of many other sectors like agriculture, livestock, energy, tourism, and urban use. More than 75% of the total water demand is accounted by the agriculture sector, making it the greatest consumer of water in the state. Though the water resources of the state are in abundance but at the same time districts of Uttarakhand have reported issues of water scarcity both for drinking and for irrigation purposes. Changes in the climate may further strain and impact the future availability of utilizable water.



River basins of Uttarakhand

7.1 Water Resource Endowment

The state contains 968 glaciers covering 2,857 km², 17 major rivers, 31 lakes, and 8 major river catchments comprising 116 sub-watersheds and 1,110 micro-watersheds (State Specific Action Plan for Water Sector [SSAP], 2018). This endowment is overwhelmingly a flux rather than a stock: the sector's effective "reservoir" is the seasonal snowpack and the ~900+ glaciers that release meltwater through the dry months, neither of which the sector has to build or maintain, and both of which are being eroded by warming.



Distribution of principal water sources in Uttarakhand.

According to a report published by NITI Aayog, the state has been witnessing dependence on water resources lying outside the state and other neighbouring areas. Significant gaps have been observed in the demand and the supply of the water requirement of the state. To meet this demand, the citizens of the area were observed to have been widely using potable water, groundwater, natural fresh water resources, and surface water. In the state's urban areas, 78.4% of the households have access to piped water supply.

However, the number of households using tap water from treated sources is only 72.7%. It is evident that the population living in the urban areas of the state has greater access to tap water from treated sources as compared to the population living in the rural areas.

A gap has been observed in per capita availability of water in urban and rural areas of Uttarakhand. The per capita availability of water in urban areas of Uttarakhand is depicted in Table 8.11.

In urban areas, there are 17 local bodies where per capita availability of water is as per Government of India's (GoI) parameter. There are 36 local bodies where availability ranges between 100 and 135 litres per capita availability (LPCD). There are 39 ULBs wherein residents get less than 100 LPCD.

In rural areas, as of April 1, 2020, there are 19%, which is around 7271 habitations receiving 40 LPCD and 3187 habitations are partially receiving 40 LPCD. Out of 14,61,910 households, only 2,17,120 households are having functional household tap connection.

Supply and Distribution Backbone

Mountain water access in Uttarakhand is structurally distinct from any assumption underpinning national-level water programmes. Ninety per cent of the state's hill and high-hill population depends on natural springs locally called dhara, naula or umre ko pani, for drinking water (NITI Aayog, 2018; Garhwal Post, 2024).

These springs are fed by shallow groundwater emerging from fractured, fissured rock aquifers, and are acutely sensitive to changes in precipitation timing, winter snowpack, forest cover and temperature. Distribution is dominated by gravity-fed schemes drawing from these springs; only the plains districts rely primarily on pumped groundwater.

This makes the source, not the pipe, the binding constraint on service, a structural property with no equivalent in plains water systems.

Institutional Structure

The sector is administered principally by the Uttarakhand Jal Sansthan and the Uttarakhand Peyjal Nigam (drinking-water supply and scheme construction), the Uttarakhand Jal Vidyut Nigam (water for hydropower), the Irrigation Department (canals and flood works), Jal Sansthan and Pollution Control Board (water quality).

Rural service delivery is now organised chiefly through the Jal Jeevan Mission (JJM) under Har Ghar Jal, with source protection and rejuvenation nominally coordinated with the NITI Aayog Spring Atlas framework and Van Panchayat catchment management.

A structural governance gap runs through this arrangement: JJM secures delivery infrastructure while the hydrological source on which delivery depends is secured by a separate, weaker instrument.

Demand Profile

Demand is dominated by agriculture, which consumes over 75 per cent of the state's total water demand (CGWB, 2013), followed by domestic drinking-water and WASH demand and a rapidly growing tourism/pilgrimage load.

The demand profile is strongly seasonal and increasingly mis-aligned with supply: peak agricultural and tourism demand falls in the pre-monsoon and summer months precisely when spring discharge and lean-season river flow are at their lowest, and when warming is advancing snowmelt so that runoff arrives before, not during, the period of peak demand.

90%	12,000	50%+	95.21%
Hill population dependent on natural springs (NITI Aayog, 2018)	Springs fully dried up of ~2,60,000 inventoried (UNDP / Garhwal Post, 2024)	Discharge reduction across 500 water sources (UNDP / Garhwal Post, 2024)	JJM tap coverage reported vs field reality of drying sources (JJM, 2023)

7.2 Defining Features and Their Climate Implication

The water sector in Uttarakhand is fundamentally regulated by the state's mountainous physiography, cryosphere-fed hydrology, monsoon-driven precipitation regime, and forested catchments. As the headwater region of the Ganga and Yamuna river systems, Uttarakhand plays a critical role in sustaining water availability both within the state and across the Indo-Gangetic plains. Water resources originate from a combination of glaciers, seasonal snowpack, springs, rivers, groundwater, and monsoon rainfall, making the sector intrinsically dependent on climatic processes governing precipitation, temperature, snow accumulation, glacier mass balance, evapotranspiration, and catchment hydrology. Consequently, projected changes in climate are expected to influence water availability, seasonal flow regimes, groundwater recharge, water quality, and the reliability of water supply systems through interconnected hydrological and geomorphological pathways.

Water availability across the Himalayan districts is predominantly sustained by natural springs, which constitute the principal source of domestic water supply for rural communities. Spring discharge is governed by precipitation, groundwater recharge, snowmelt, forest condition, and subsurface hydrogeological processes. Climate-induced changes in rainfall seasonality, increasing temperatures, declining snow accumulation, and changes in vegetation cover are expected to alter recharge dynamics, resulting in reduced spring discharge, increased seasonal variability, and declining reliability of water supplies, particularly during the pre-monsoon period.

The state's hydrological regime is strongly influenced by cryospheric processes. Hundreds of glaciers and extensive seasonal snow cover sustain river discharge during the dry season by providing delayed meltwater contributions to major Himalayan rivers. Climate warming is expected to alter glacier mass balance, reduce snow cover duration, and shift the timing of snowmelt, thereby modifying seasonal runoff patterns. While accelerated glacier melt may temporarily increase river discharge in some basins, continued glacier retreat is expected to reduce long-term meltwater availability, resulting in progressive changes to lean-season flows and overall water security.

Water supply systems in the Himalayan region are largely characterized by **run-of-catchment hydrology**, where communities depend directly on natural springs and gravity-fed water sources with relatively limited engineered storage capacity. Consequently, water availability closely follows seasonal recharge and streamflow conditions, leaving the sector with limited capacity to buffer climatic variability. Increasing interannual variability in precipitation and prolonged dry periods are therefore expected to translate directly into fluctuations in water availability and supply reliability.

Recent expansion of rural drinking water infrastructure through the **Jal Jeevan Mission (JJM)** has substantially improved household tap connectivity across the state. However, the long-term sustainability of these systems remains dependent on the health and productivity of the source springs and catchments that supply them. Climate-induced reductions in spring discharge or deterioration of source conditions may therefore constrain the reliability of piped water supply even where physical infrastructure coverage is high.

Water infrastructure in Uttarakhand including spring chambers, gravity-fed pipelines, intake structures, pumping stations, and treatment facilities is predominantly located within steep Himalayan terrain. These assets are exposed to climate-sensitive geomorphological processes such as landslides, flash floods, debris flows, cloudbursts, riverbank erosion, and glacial lake outburst floods (GLOFs). Increasing frequency and intensity of extreme hydro-meteorological events are expected to influence both the physical integrity and operational reliability of water supply infrastructure.

Climate change also has significant implications for water quality. Rising temperatures, changing hydrological regimes, intense rainfall events, and increasing flood frequency influence sediment transport, turbidity, microbial contamination, nutrient loading, and pollutant mobilization. Open spring sources, inadequate sanitation, tourism-related pressures, and flood-induced mixing of sewage with drinking water sources increase the vulnerability of both surface water and spring-fed systems to water quality deterioration, with direct implications for public health

Defining feature	Description	Implied climate sensitivity
Spring-dependent	~90% of hill population relies on natural springs	Supply is a direct function of recharge, snowpack and forest cover
Cryosphere-linked	968 glaciers / 2,857 km ² sustain lean-season base flow	Non-linear: transient melt gain followed by long-term decline (“peak water”)
Run-of-catchment, low storage	Little engineered storage in the hills	No buffer against flow variability; supply tracks instantaneous recharge/flow
Source–pipe mismatch	JJM secures pipes, not the source spring	Tap coverage overstates actual water security
Mountain-located assets	Spring chambers, gravity lines on steep, active slopes	Landslide, GLOF, cloudburst and flash-flood exposure
Contamination-prone	Open sources, flood-linked sewage, tourism load	Rising temperature and extreme rainfall degrade water quality
Downstream-critical	Feeds ~500 million people in the Indo-Gangetic plain	Source-side change propagates far beyond the state

Structural features of the water sector and the climate sensitivity each implies.

7.3 Key Climate Hazards for the Water Sector

Six hazards dominate climate risk to the water sector in Uttarakhand. For each, this section sets out the physical mechanism, the stage of the water chain at which it acts, the assets and geographies affected, and the documented historical record. Together the record establishes that climate risk to the sector is not prospective but an ongoing, measurable loss stream with two signatures: rare, catastrophic events that destroy water infrastructure outright, and a chronic, pervasive attrition of spring discharge that never triggers a disaster declaration yet accounts for the larger share of cumulative loss.

Hazard	Mechanism of damage	Water-chain stage affected	Principal districts / basins
Spring / source depletion	Reduced recharge from lower snowpack, oak-to-pine conversion, higher evapotranspiration; discharge falls or ceases	Source (slow-onset, chronic)	Bageshwar, Chamoli, Rudraprayag, Almora, Pauri, Tehri
GLOF	Debris-laden surge destroys intakes, spring chambers and valley schemes; forces communities onto uncharacterised sources	Source; capture; quality (total loss)	Chamoli, Uttarkashi, Pithoragarh, Bageshwar
Extreme rainfall / cloudburst	Flash flooding destroys capture works and gravity lines; triggers contamination	Capture; conveyance; quality	Uttarkashi, Chamoli, Rudraprayag, Pithoragarh, Nainital
Riverine & flash flooding	Inundation of valley/plains schemes; sewage overflow into sources; well & handpump contamination	Capture; distribution; quality	Alaknanda–Mandakini valleys; Haridwar, U.S. Nagar
Landslide & slope failure	Burial/rupture of spring chambers and gravity pipelines; severs access for repair	Capture; conveyance	All hill districts; acute in Rudraprayag, Chamoli, Tehri, Pauri
Temperature rise & quality loss	Faster pathogen growth; lower self-purification; enhanced glacial-melt contaminant flux	Quality (chronic)	Statewide; tourism corridors, plains

Hazard register for the Uttarakhand water sector.

Spring and Source Depletion

Spring failure is the most consequential slow-onset climate impact on Uttarakhand's water sector. Unlike glacial retreat, which operates over decades and is trackable, spring failure is occurring at the village level across thousands of locations simultaneously, without systematic

monitoring and without triggering formal disaster declarations. The causal chain is well established: reduced winter precipitation and declining snowpack provide less sub-surface recharge; the progressive replacement of moisture-retaining banj oak (*Quercus leucotrichophora*) with fire-prone chir pine (*Pinus roxburghii*) reduces soil infiltration capacity by 30–40 per cent (Singh et al., 1984); post-fire soil hydrophobicity further reduces infiltration; and warming-driven evapotranspiration reduces net recharge.

- The NITI Aayog (2018) Spring Atlas found that nearly half of the ~3 million springs across the Indian Himalayan Region are dying or significantly reduced. In Uttarakhand, ~12,000 of approximately 2,60,000 inventoried springs have fully dried up, and 500 water-supply sources have lost more than 50 per cent of their discharge (UNDP, in Garhwal Post, 2024).
- Of Uttarakhand's 39,202 villages, only 21,363 have drinking-water facilities; the remaining 17,839 face severe shortages (NRDWP, in Garhwal Post, 2024). The Migration Commission has explicitly identified water scarcity as a driver of permanent out-migration.
- Sharma (2026, *Frontiers in Water*) frames drying springs as a non-traditional security threat, undermining human security, demographic-territorial security in border areas, and hydrological security at the national scale.

Glacial Lake Outburst Floods (GLOF)

A GLOF occurs when a moraine- or ice-dammed proglacial lake fails, releasing a large volume of water in minutes; in the Uttarakhand Himalaya the failure is frequently compound, with a rock-ice avalanche generating a debris-laden surge down exactly the confined valleys in which valley water infrastructure sits. For the water sector the hazard is doubly severe: it destroys capture and conveyance works outright, and it forces surviving communities onto uncharacterised groundwater that can carry geogenic contaminants.

Documented impact: the Chamoli disaster, 7 February 2021. A 27-million-m³ rock-ice avalanche (Shugar et al., 2021, *Science*) destroyed 17 water-supply schemes in the Rishiganga and Dhauliganga valleys. Subsequent CGWB (2022) surveys detected arsenic at 3–8 times WHO limits in villages forced onto emergency groundwater, and AIIMS Dehradun (2023) documented new cases of arsenical keratosis and bladder cancer with no prior history. The SLRMC (2025) inventory identifies 25 Potentially Dangerous Glacial Lakes in the state, so the potential for future GLOF-triggered water-supply and contamination events is non-trivial and, under warming, growing.

Extreme Rainfall, Cloudbursts and Flash Flooding

Extreme hydrometeorological events are the most sudden and structurally destructive climate impact on water supply. Cloudbursts and flash floods destroy spring-capture chambers, break gravity-line pipelines, inundate wells and handpumps, and contaminate aquifers with pathogens and heavy metals, leaving communities on unsafe surface water for weeks while recovery is blocked by road closures. The State Disaster Management Authority (2023) documents a 240 per cent increase in hydro-meteorological disasters between 2000 and 2022, and extreme-rainfall events (≥ 100 mm/day) have risen ~35 per cent since 2000 (IMD, 2023).

The 2021 Nainital floods overwhelmed sewage systems, contaminating 73 springs with *E. coli* and causing 3,200 cases of acute diarrhoea within two weeks; paediatric Hepatitis E in Chamoli rose 30 per cent during the monsoon (State Health Report, 2022).

The 2013 Kedarnath disaster simultaneously destroyed water-supply infrastructure across three districts, with recovery timelines measured in months to years, during which communities are systematically exposed to waterborne disease from contaminated surface sources.

Landslides and Slope Failure

Landslides in Uttarakhand are overwhelmingly rainfall-triggered, occurring where intense or prolonged precipitation raises pore-water pressure in already-marginal slopes of young, fractured, tectonically active lithology. For the water sector they bury or rupture spring chambers and gravity pipelines — the very assets on which hill supply depends — and they sever the roads on which repair crews and materials travel, so that restoration time is frequently governed by road clearance rather than by the water repair itself. This co-location of pipeline and hazard makes landslides a chronic, every-monsoon source of water-service loss.

Temperature Rise and Water-Quality Degradation

Rising temperature acts directly on water quality: it increases pathogen proliferation in spring and surface water, reduces the self-purification capacity of rivers, and — combined with more intense rainfall — sweeps surface contamination (open-defecation residue, agricultural runoff, construction waste) into recharge zones. It also enhances glacial melt, increasing the flux of emerging contaminants; microplastics have been detected in 91 per cent of Bhagirathi River samples during glacial-melt periods (Talukdar et al., 2023). Tourism-driven sewage — 58 million domestic arrivals in 2023 — is now among the primary drivers of spring-water-quality degradation in hill towns (Kothari et al., 2023).

Chronology of Past Impacts on the Water Sector

Year	Event	Documented water-sector consequence	Lesson carried into the framework
2013	Kedarnath disaster (extreme rainfall + moraine-lake breach)	Water-supply infrastructure destroyed across three districts; multi-year recovery	Compound hazards exceed the design basis of individually appraised schemes
2021 (Feb)	Rishiganga–Chamoli rock-ice avalanche / GLOF	17 water schemes destroyed; post-event arsenic 3–8× WHO limits; arsenical keratosis & bladder cancer documented	GLOF converts a source loss into a chronic contamination and health crisis
2021	Nainital monsoon floods	73 springs contaminated with <i>E. coli</i> ; 3,200 acute diarrhoea cases in two weeks	Flood-borne contamination is a predictable, recurrent water hazard

Year	Event	Documented water-sector consequence	Lesson carried into the framework
2019	Pithoragarh drought	80% of handpumps ran dry; 57% rise in waterborne conjunctivitis from reuse of contaminated water	Slow-onset drought produces measurable health and gendered burdens
2000–2022	Rising hydro-meteorological disasters	240% increase in disaster frequency; recurrent destruction of hill water schemes	Chronic attrition, not a single event, sets the baseline loss rate
2000–present	Accelerating spring failure	~12,000 springs dried; 500 sources –50% discharge; out-migration driver	The dominant loss is invisible to event-based hazard mapping

Chronology of documented climate-hazard impacts on the Uttarakhand water sector.

7.4 Exposure Assessment

Exposure is assessed by intersecting hazard-susceptibility layers with the geo-referenced water-asset inventory. Four hazards are carried into the spatial overlay — spring/source depletion, flood, extreme rainfall and landslide because susceptibility layers for these are available at a resolution consistent with block-level reporting. GLOF is treated separately as a basin-level siting constraint on valley schemes; temperature-driven quality loss is handled on the sensitivity side (Section 9.6).

Data Layers

Hazard layer	Source and basis	Classification	Resolution
Spring / source depletion	NITI Aayog Spring Atlas; district spring-dependence; discharge-decline & dried-source records; recharge (snowpack, oak-pine, ET)	5 classes (Very Low – Very High)	Block / habitation
Landslide susceptibility	GSI / BMTPC mapping refined with slope, lithology, land cover and rainfall-trigger thresholds	5 classes	30 m raster
Flood hazard	NRSC flood hazard atlas; historical inundation; river buffer and terrace analysis	5 classes	30 m / vector
Extreme rainfall	IMD gridded daily rainfall (0.25°); frequency of ≥95th/99th-percentile days; cloudburst record	5 classes	Downscaled to block

Hazard layer	Source and basis	Classification	Resolution
Water assets	Jal Sansthan / Peyjal Nigam spring & scheme inventory; JJM pipeline network; SBM-G sanitation inventory	Point / line vector	Asset-level

Data layers used in the exposure overlay.

Exposure Metrics

Asset class	Exposure metric	Unit reported
Springs & gravity schemes	Schemes drawing on high/very-high-depletion or hazard-exposed sources ÷ total schemes in the block	% of schemes exposed
Gravity pipelines	Pipeline length within high/very-high landslide & flood classes ÷ total length	% of length exposed
Sanitation (IHHL)	Household latrines in high slope-failure / flood classes ÷ total IHHLs	% of IHHLs exposed
Population served	Block population dependent on exposed sources ÷ total block population	% of population exposed

Exposure metrics by asset class. “Exposed” means intersecting the high or very-high class of the hazard layer.

Classification Bands

Class	Range (% of asset exposed)	Score	Interpretation
Very Low	0 – 10 %	1	Marginal exposure; standard practice adequate
Low	>10 – 25 %	2	Localised exposure; targeted source/asset protection
Moderate	>25 – 50 %	3	Material exposure; scheme-level risk treatment required
High	>50 – 75 %	4	Systemic exposure; source rejuvenation / redundancy required
Very High	>75 – 100 %	5	Critical exposure; the water system is effectively co-located with the hazard

Exposure classification bands and the score used in the composite.

Composite Exposure Formula

The block composite is the weighted mean of the four hazard-specific exposure scores. Weights reflect the relative contribution of each hazard to documented water-sector loss in the State, with spring/source depletion weighted highest because it is the dominant, chronic driver of water insecurity in the hills:

$$E(\text{block}) = 0.30 \cdot S(\text{spring}) + 0.25 \cdot S(\text{extreme rainfall}) + 0.20 \cdot S(\text{flood}) + 0.25 \cdot S(\text{landslide})$$

Spring Dependence: the Governing Exposure Variable

Because ~90 per cent of hill supply is spring-fed, the share of a district's schemes drawing on springs is the single most important determinant of water-source exposure. The gravity-spring dependence of the schemes surveyed is set out below; it is the empirical basis for the spring-depletion column of the block table.

District	Total schemes	Gravity spring	Spring dep. % (gravity-spring ÷ total schemes)	Gravity gadhera	Pumped/TW	Pipeline (km)
Bageshwar	119	106	89.1%	7	0	402
Chamoli	128	113	88.3%	5	0	557
Rudraprayag	92	79	85.9%	13	0	323
Dehradun	114	92	80.7%	4	13	444
Uttarkashi	99	73	73.7%	22	0	396
Tehri Garhwal	128	91	71.1%	32	1	438
Pauri Garhwal	193	130	67.4%	55	0	652
Pithoragarh	178	114	64%	55	1	632
Nainital	97	57	58.8%	34	4	278
Champawat	147	85	57.8%	52	2	484
Almora	170	72	42.4%	73	2	335
Haridwar	103	0	0%	0	85	210

Table 9.10 Drinking-water scheme profile and gravity-spring dependence by district (source: state scheme inventory). Note: the Gravity-spring, Gravity-gadhera and Pumped/TW columns are a partial breakdown and do not necessarily sum to Total schemes, as some schemes draw on other or mixed source types.

Block-Wise Exposure of Water Infrastructure

Status of the values below. The hazard classes in Table 9.11 are a physiography-based pre-classification, derived from terrain, lithology, forest type, river proximity, district spring-dependence and the documented event record. They are an analytical scaffold, not the output of the GIS overlay. Once the overlay described in Sections 9.5.1–9.5.2 is run, the computed exposure percentages replace these classes directly — the table structure, weights and banding do not change. The exceptions to watch are plains blocks whose supply is groundwater-based (spring-depletion low but flood exposure high) and mixed hill–valley blocks whose schemes are concentrated in one physiographic zone.

District	Block	Drought	Flood	Extreme rainfall	Landslide	Composite exposure
Uttarkashi	Bhatwari	Very High	High	Very High	Very High	Very High
	Dunda	High	Moderate	High	High	High
	Chinyalisaur	High	Moderate	High	High	High
	Naugaon	High	Moderate	High	High	High
	Purola	High	Moderate	Moderate	Moderate	Moderate
	Mori	Very High	Moderate	High	High	High
Chamoli	Dasholi	Very High	High	Very High	Very High	Very High
	Joshimath	Very High	Very High	Very High	Very High	Very High
	Ghat	Very High	Moderate	High	Very High	Very High
	Karnaprayag	High	High	High	High	High
	Dewal	High	Moderate	High	High	High
	Tharali	High	High	High	High	High
	Narayanbagar	High	Moderate	High	High	High
	Pokhri	Very High	Low	Moderate	High	High
	Gairsain	High	Low	Moderate	Moderate	Moderate
Rudraprayag	Ukhimath	Very High	High	Very High	Very High	Very High
	Agastyamuni	High	High	High	Very High	Very High
	Jakholi	High	Moderate	High	High	High
Tehri Garhwal	Chamba	High	Low	Moderate	High	Moderate
	Jaunpur	High	Low	Moderate	High	Moderate
	Kirtinagar	High	Moderate	Moderate	High	High
	Narendranagar	High	Moderate	Moderate	High	High
	Pratapnagar	Very High	Moderate	High	High	High
	Devprayag	Moderate	High	Moderate	High	High
	Jakhnidhar	High	Low	Moderate	High	Moderate

District	Block	Drought	Flood	Extreme rainfall	Landslide	Composite exposure
	Bhilangana	Very High	High	High	Very High	Very High
	Thauldhar	High	Low	Moderate	High	Moderate
Dehradun	Chakrata	Very High	Low	Moderate	High	High
	Kalsi	High	Moderate	Moderate	High	High
	Vikasnagar	Moderate	High	Moderate	Low	Moderate
	Sahaspur	Low	Moderate	Moderate	Very Low	Low
	Raipur	Moderate	High	High	Moderate	High
	Doiwala	Low	High	High	Low	Moderate
Pauri Garhwal	Kot	High	Low	Moderate	High	Moderate
	Kaljikhali	High	Low	Moderate	High	Moderate
	Pauri	High	Low	Moderate	High	Moderate
	Pabau	High	Low	Moderate	High	Moderate
	Thalisain	High	Low	Moderate	High	Moderate
	Bironkhal	High	Low	Moderate	High	Moderate
	Dwarikhali	High	Moderate	Moderate	High	High
	Dugadda	Moderate	Moderate	Moderate	High	Moderate
	Jaharikhali	High	Low	Moderate	High	Moderate
	Rikhnikhali	High	Low	Moderate	Moderate	Moderate
	Yamkeshwar	Moderate	Moderate	Moderate	High	Moderate
	Ekeshwar	High	Low	Moderate	Moderate	Moderate
	Khirsu	High	Low	Moderate	Moderate	Moderate
	Nainidanda	High	Low	Moderate	High	Moderate
	Pokhra	High	Low	Moderate	High	Moderate
Nainital	Betalghat	High	Moderate	High	High	High
	Bhimtal	High	Moderate	Very High	High	High

District	Block	Drought	Flood	Extreme rainfall	Landslide	Composite exposure
	Dhari	High	Low	High	High	High
	Kotabag	Moderate	High	High	Moderate	High
	Okhalkanda	High	Moderate	High	High	High
	Ramgarh	High	Low	High	High	High
	Ramnagar	Low	High	High	Low	Moderate
	Haldwani	Low	Very High	Very High	Low	Moderate
Almora	Hawalbagh	High	Low	Moderate	High	Moderate
	Takula	High	Low	Moderate	High	Moderate
	Lamgara	High	Low	Moderate	High	Moderate
	Dhauladevi	High	Low	Moderate	High	Moderate
	Bhaisiyachhana	High	Low	Moderate	High	Moderate
	Bhikiyasain	High	Moderate	Moderate	High	High
	Sult	High	Low	Moderate	High	Moderate
	Chaukhutia	High	Moderate	Moderate	High	High
	Dwarahat	High	Low	Moderate	High	Moderate
	Syaldey	Moderate	Low	Moderate	Moderate	Moderate
	Tarikhet	High	Low	Moderate	High	Moderate
Bageshwar	Bageshwar	High	High	High	High	High
	Kapkot	Very High	High	Very High	Very High	Very High
	Garur	High	Moderate	High	High	High
Pithoragarh	Pithoragarh	High	Moderate	High	High	High
	Munakot	High	Moderate	High	High	High
	Gangolihat	High	Low	Moderate	High	Moderate
	Berinag	High	Low	High	High	High
	Didihat	High	Moderate	High	High	High

District	Block	Drought	Flood	Extreme rainfall	Landslide	Composite exposure
	Kanalichhina	High	Moderate	High	High	High
	Dharchula	Very High	Very High	Very High	Very High	Very High
	Munsyari	Very High	High	Very High	Very High	Very High
Champawat	Champawat	High	Low	Moderate	High	Moderate
	Lohaghat	High	Low	Moderate	High	Moderate
	Pati	High	Moderate	High	High	High
	Barakot	Moderate	Moderate	High	High	High
Haridwar	Bahadrabad	Very Low	High	Moderate	Very Low	Low
	Roorkee	Very Low	High	Moderate	Very Low	Low
	Narsan	Very Low	Moderate	Moderate	Very Low	Very Low
	Bhagwanpur	Very Low	Moderate	Moderate	Very Low	Very Low
	Laksar	Very Low	Very High	Moderate	Very Low	Low
	Khanpur	Very Low	Very High	Moderate	Very Low	Low
Udham Singh Nagar	Kashipur	Very Low	High	Moderate	Very Low	Low
	Jaspur	Very Low	High	Moderate	Very Low	Low
	Bazpur	Very Low	High	Moderate	Very Low	Low
	Gadarpur	Very Low	Very High	High	Very Low	Low
	Rudrapur	Very Low	High	High	Very Low	Low
	Sitarganj	Very Low	Very High	High	Very Low	Low
	Khatima	Very Low	Very High	High	Very Low	Low

● Priority Blocks

The Very High composite blocks cluster in the high-Himalayan, glaciated and highly spring-dependent zones- the same districts the water sector already identifies as most stressed. They combine chronic spring depletion with acute cloudburst, flood and landslide exposure to valley schemes, and several also carry GLOF exposure treated separately as a basin-level constraint.

Block	District	Exposure driver	Water assets at stake
Bhatwari	Uttarkashi	Upper Bhagirathi; spring depletion, cloudburst, GLOF	Gangotri-corridor schemes; high-altitude spring sources

Block	District	Exposure driver	Water assets at stake
Dasholi	Chamoli	Alaknanda confluence; spring depletion, extreme rainfall	District gravity schemes; valley intakes
Joshimath	Chamoli	Full hazard stack: GLOF, flood, landslide, subsidence	Valley schemes; springs on subsiding slope
Ukhimath	Rudraprayag	Mandakini valley; 2013 footprint; spring depletion	Kedarnath-route schemes; spring sources
Kapkot	Bageshwar	Pindari catchment; highest district spring-dependence	Sarju-Pindar spring schemes; remote habitations
Dharchula	Pithoragarh	Kali-Dhauliganga; GLOF, flood, spring depletion	Border-area schemes with no redundancy
Munsyari	Pithoragarh	Goriganga; glacial-fed sources; remote	Isolated high-altitude spring schemes

Illustrative Very-High composite-exposure blocks and their water assets at stake.

Asset-Class Findings

Springs and gravity schemes. Source exposure varies systematically with district spring-dependence and elevation. The high-dependence districts- Bageshwar (89.1%), Chamoli (88.3%) and Rudraprayag (85.9%) have the largest share of their water systems exposed to spring depletion, and their high-altitude blocks combine this with cloudburst and landslide exposure to the gravity lines that convey the water. Almora, despite a lower headline dependence (42.4%), carries an acute documented spring-failure burden across most of its blocks. The plains district of Haridwar (0% spring dependence) is essentially unexposed to source depletion but highly exposed to flood.

Sanitation infrastructure and ODF sustainability. Sanitation exposure is a distinct dimension: individual household latrines (IHHLs) built to SBM-G standards do not incorporate seismic or slope-stability requirements, so a single cloudburst or landslide can revert ODF-certified villages to open defecation within hours. The ODF-gap and climate-risk profile by district is set out below.

District	Total HH	HH with toilet	ODF gap (HH)	ODF gap %	ODF risk level
Haridwar	219K	108K	111.0K	50.6%	Critical
Almora	133K	75K	58.4K	43.8%	High
Rudraprayag	55K	32K	23.2K	42.4%	High
Pauri Garhwal	120K	72K	48.1K	40%	High
Udham Singh Nagar	183K	112K	70.5K	38.5%	High
Pithoragarh	101K	66K	35.0K	34.8%	Medium
Tehri Garhwal	122K	81K	40.6K	33.4%	Medium
Uttarkashi	61K	41K	20.5K	33.4%	Medium
Champawat	50K	34K	16.4K	32.7%	Medium
Chamoli	69K	49K	20.1K	29.2%	Low
Bageshwar	55K	39K	15.7K	28.5%	Low
Dehradun	103K	74K	28.9K	28%	Low
Nainital	94K	68K	25.5K	27.2%	Low

Sanitation ODF-gap and climate-risk level by district (source: SBM-G / state sanitation inventory).

Population and Service Exposure

Asset exposure is only half the picture: the consequence of a failure depends on how many people and critical services depend on the affected source. The assessment therefore also records, for each block, the population served by exposed sources, the health and school facilities dependent on them, the mean restoration time after a hazard event, and block accessibility during hazards (a proxy for repair-crew access). The distinction is not academic: high-altitude blocks such as Dharchula and Munsyari have very high source exposure but small populations, while plains blocks such as Haldwani and Rudrapur have low spring exposure but very large populations on flood-exposed, groundwater-based supply — so a ranking on source exposure alone would misallocate investment on a service-continuity criterion.

1.5 Climate Variability Impacting the Water Sector

This section addresses sensitivity: given exposure, how strongly does climate variability translate into loss of water service? For Uttarakhand the answer is that sensitivity is unusually high, because almost every link in the water chain- source recharge, capture, quality, conveyance and demand is climate-dependent. The section first sets out the observed variability, then traces each sensitivity pathway from climate variable to water outcome and finally identifies where variability is already binding.

Observed Climate Variability and Trends

Climate variable	Observed change	Consequence for the water sector
Total monsoon rainfall	Broadly stable-to-declining annual total	Same or less water delivered in fewer, more violent pulses
Extreme rainfall frequency	Increase of ~35% since 2000 (IMD, 2023)	More scheme destruction, contamination and landslide damage
Hydro-met disaster frequency	+240% between 2000 and 2022 (SDMA, 2023)	Rising acute destruction of capture and conveyance works
Air temperature	Warming with elevation-dependent amplification	Faster pathogen growth; higher ET; accelerated glacier melt
Winter precipitation / snowfall	Declining snowfall fraction; rising snowline	Less sub-surface recharge; spring discharge falls
Glacier mass balance	Sustained negative balance across all 8 basins	Transient flow gain then permanent decline — “peak water”
Glacial lake formation	Number & size increasing; 25 PDGLs (SLRMC, 2025)	Rising base probability of GLOF across glaciated basins
Spring discharge	~12,000 dried; 500 sources -50% discharge	Chronic, village-scale erosion of the drinking-water base

Observed climate variability in Uttarakhand and its water-sector consequence.

Climate Variable → Water Chain Transmission Matrix

The matrix traces each climate variable through every stage of the water chain. Reading down a column shows what a single stage is exposed to; reading across a row shows how a single variable propagates.

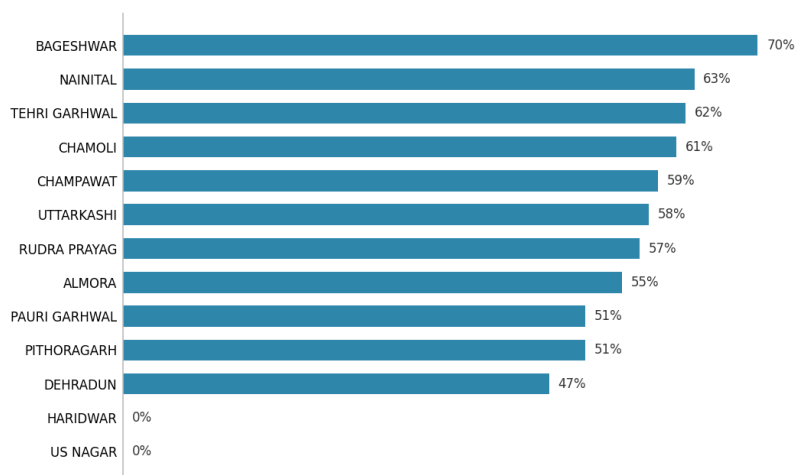
Climate variable	Source & recharge	Capture & storage	Conveyance & distribution	End use & quality
Rising temperature	Higher ET; earlier snowmelt; lower recharge	Faster evaporation from open storage	—	Faster pathogen growth; lower self-purification
Declining snowfall fraction	Loss of seasonal reservoir; lower spring/lean flow	Lower stored volume	—	Concentration of contaminants at low flow
Increased rainfall intensity	Runoff instead of recharge; springs flashier	Scour/burial of spring chambers & intakes	Pipeline breakage; landslide severance	Flood-borne pathogen contamination
Declining total rainfall	Lower annual recharge	Lower yields; source failure	—	—
Glacier retreat	Peak water then permanent base-flow decline	—	Reduced canal/scheme reliability	Altered melt-borne contaminant flux
GLOF / mass movement	Sediment renders sources unusable	Total destruction of capture works	Loss of conveyance & access	Geogenic contamination (arsenic)

Transmission of climate variability through the water chain. Blank cells indicate no material pathway.

Spring Systems: The Vanishing Mountain Lifeline

The spring system is simultaneously Uttarakhand’s most widespread, most critical and most climate-exposed drinking-water infrastructure. Unlike surface diversions or tube wells, springs are ecological phenomena: they exist where geology, hydrology, land cover and recharge align, and climate change is shifting all four parameters at once. The Moldhar village case (Down to Earth, 2025) illustrates the

community-level reality — over 80 households on a single spring that began to fail two years ago, with families now abandoning the village and the spring failure explicitly identified as a migration driver. Critically, the Jal Jeevan Mission’s reported 95.21% coverage counts physical tap installation, not water delivery: where the source spring is dry or reduced to a pre-monsoon trickle, the tap delivers intermittently or not at all.



District-wise spring dependence — share of drinking-water supply reliant on spring sources. This district-level supply-reliance measure is distinct from, and not directly comparable to, the scheme-count percentage in Table 9.10 (gravity-spring schemes ÷ total schemes). Haridwar shown as 0%, consistent with its 0% spring-dependence recorded elsewhere in this chapter.

Groundwater: Rising Extraction, Declining Quality

In the plains districts — Haridwar (≈76% of supply from groundwater) and Nainital (≈93% groundwater-dependent for city supply) — climate change is driving compensatory groundwater over-extraction as surface water declines. Annual groundwater draft rose 66% between 2013 and 2017 (0.99 to 1.64 BCM; CGWB, 2017), and Haridwar, Nainital and Udham Singh Nagar shifted from “critical” to “semi-critical” classification. Chemometric studies find Heavy Pollution Index values above threshold for iron and toxic elements in Dehradun urban groundwater (Nayak et al., 2024) and WHO-limit exceedances in Ramganga-basin groundwater (Gautam et al., 2024), confirming that intensifying extraction is mobilising geogenic and co-contaminating pollutants.

Water Quality: Contamination Pathways Under Climate Change

Water-quality degradation is a distinct and independently severe dimension operating through four documented pathways: (i) pathogen contamination from flood and sewage-overflow events, which produces predictable waterborne-disease clusters each monsoon; (ii) GLOF-released geogenic contamination, exemplified by post-2021-Chamoli arsenic at 3–8× WHO limits; (iii) tourism-driven sewage overwhelming hill-town treatment capacity and degrading source springs; and (iv) microplastic contamination, now present in 91% of Bhagirathi samples during melt periods. Each pathway is amplified by warming and by the reorganisation of rainfall into more intense pulses.

Sanitation, ODF Sustainability and the Gendered Water Burden

Two sensitivity pathways are frequently omitted from water assessments but are acute in Uttarakhand. First, ODF status is structurally fragile in mountain terrain: cloudbursts and landslides can destroy latrine superstructures across hundreds of households simultaneously, and monsoon road blockages (42 consecutive days of PHC inaccessibility in Uttarkashi in 2021) sever the supply chains needed to repair them. Second, the burden of water scarcity falls disproportionately on women: PSI (2022) documents 4–6 additional hours per day of water collection during drought, and UNFPA (2023) a 300% rise in gynaecological infections during drought months from inadequate water for menstrual hygiene, a direct, measurable, climate-induced public-health burden absent from standard monitoring.

Agricultural Water Security

Agriculture consumes over 75% of the state’s water, making agricultural water security the sector’s most quantitatively dominant dimension. The nexus operates through three pathways: glacier retreat reducing

canal base flows (with the ~2050 peak-water transition reducing command areas in Chamoli, Uttarkashi, Pithoragarh and Tehri); spring failure reducing the spring-fed gul and nali systems that sustain terraced hill agriculture; and groundwater over-extraction as surface water becomes unreliable. With over 55% of cultivated area rainfed, erratic intra-seasonal rainfall — less total rain in more violent events — is the primary climate driver of crop failure.

Institutional Response: The JJM Paradox and Governance Gaps

The Jal Jeevan Mission is the largest single investment in rural water supply in the state’s history, and its 95.21% coverage is a genuine achievement in physical installation. But its design limitation is now well documented: JJM secures delivery infrastructure without securing the source springs and streams feeding it. The Himachal parallel is instructive — ₹4,000 crore spent yet acute shortages persist in 2024 — and the problem is structural to mountain-terrain implementation, not state-specific. The state’s spring-rejuvenation programme targets 13,000 springs but had reached under 20% by 2023, and key adaptation instruments (National Adaptation Fund, NMHS) have received no allocation since 2022–23 (Down to Earth, 2025).

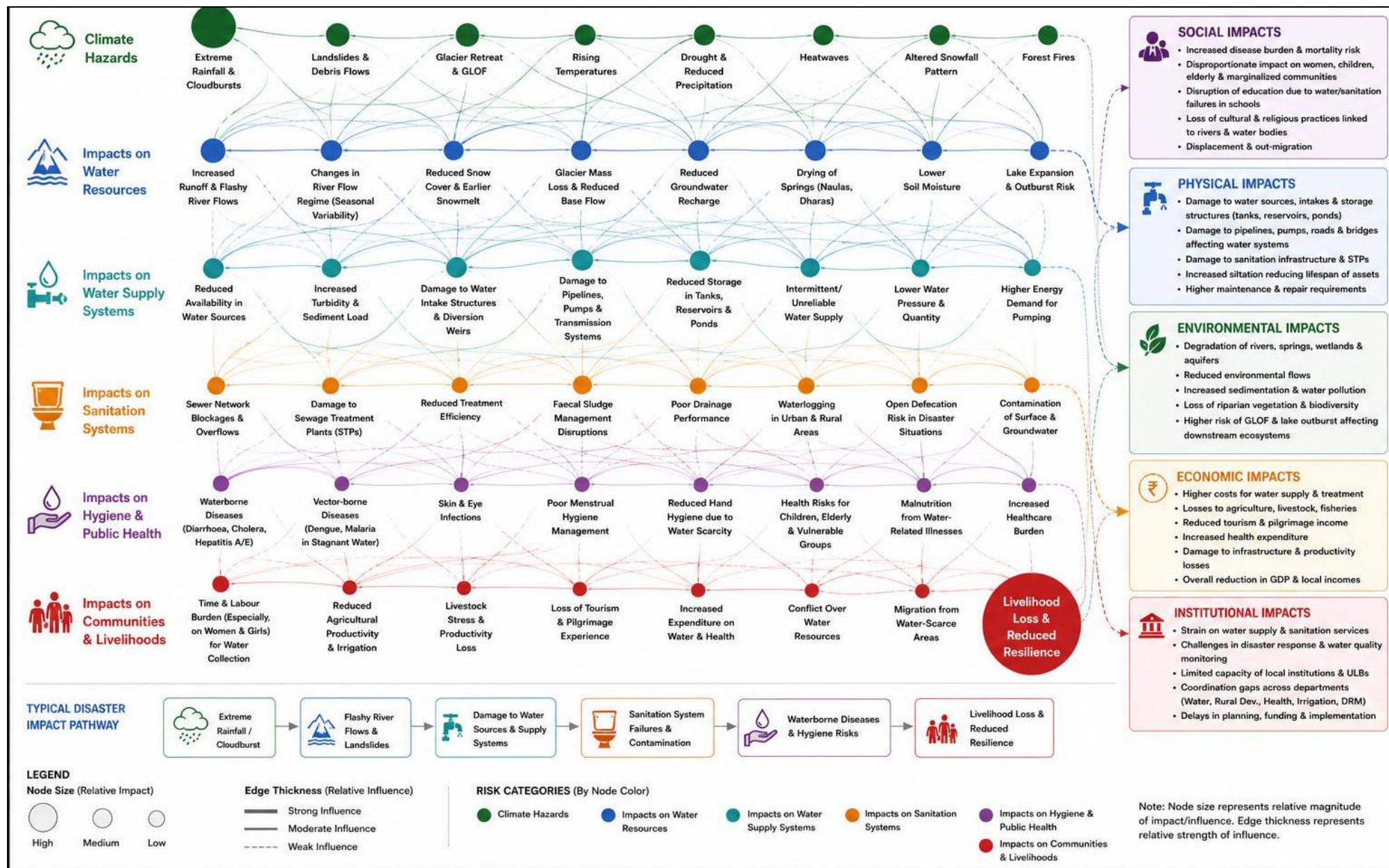
Cross-Sector and Cascading Sensitivity

Water impacts propagate outward and back. Energy → water: loss of electricity halts pumping and treatment, as in the 2021 Pithoragarh cloudburst and recurrent Almora–Haldwani outages that produced water crises as a second-order effect. Water → energy: competition for water among hydropower, irrigation, drinking supply and ecological flow intensifies as availability becomes more variable. Water → health & out-migration: spring failure drives both waterborne disease clusters and, over time, permanent out-migration that thins civilian presence in strategically sensitive border blocks. These couplings should be scored where possible rather than ignored or spuriously quantified.

Where Variability Already Binds

Sensitivity	Status today	Evidence
Spring discharge decline	Binding now, worsening	~12,000 springs dried; 500 sources –50%
Flood-borne contamination	Binding now, every monsoon	73 springs contaminated, 3,200 cases (Nainital 2021)
Scheme destruction by extreme rainfall	Binding now	17 schemes lost (Chamoli 2021); 240% disaster rise
Gendered water burden	Binding now	4–6 extra hours/day; +300% gynaecological infections
GLOF destruction & geogenic contamination	Binding now, low-frequency/catastrophic	2021 Chamoli; arsenic 3–8× WHO; 25 PDGLs
Peak-water reversal in glacier-fed basins	Not yet binding; will bind by ~2050	Sustained negative glacier mass balance

Binding status of water-sector sensitivities.



7.6 Potential Future Climate Change

Because the sector's effective reservoir is the snowpack and glaciers, the projected change in water its quantity, its phase (snow versus rain) and above all its seasonal timing is the single most consequential projection for the sector. This section sets out the far-future (2071–2100, relative to the 1995–2014 baseline) change in the water-related variables that govern the sector, under a low-emissions (SSP1-2.6) and a high-emissions (SSP5-8.5) scenario.

Variable (far future, 2071–2100)	SSP1-2.6	SSP5-8.5	Mechanism and water-sector relevance
Mean annual temperature ($\Delta^{\circ}\text{C}$)	() indicative +1.0 to +1.8 $^{\circ}\text{C}$	() indicative +3.5 to +5.5 $^{\circ}\text{C}$	Drives melt rate, snowline, ET and pathogen growth; high-altitude basins warm faster than the State mean
Mean annual precipitation ($\Delta\%$)	() indicative +5 to +10 %	() indicative +10 to +25 %	Total volume rises but is concentrated in high-intensity events, so recharge does not rise proportionally
Extreme-rainfall day frequency	() moderate increase	() large increase	More scheme destruction, contamination and landslide damage
Snowfall fraction of precipitation	() moderate decline	() steep decline	Erodes the seasonal reservoir; lower and earlier spring recharge
Snowline / zero-degree isotherm	() rises	() rises substantially	Reduces the recharge-contributing snowmelt area
Glacier volume change (HKH, to 2100)	() indicative –30 to –50 %	() indicative –65 to –80 %	Sets the long-term lean-season base-flow floor after peak water
Spring discharge (hill sources)	() continued decline	() severe decline	Source failure spreads to a majority of hill blocks
Seasonal shift in flow	() earlier spring peak	() pronounced earlier peak, deeper pre-monsoon trough	Water arrives before peak demand; lean-season deficit deepens
GLOF frequency & lake area	() elevated	() substantially elevated (+60–90% lake area)	Hazard itself grows; exposure layer is non-stationary
Waterborne-disease burden	() seasonal spikes	() large increase; concurrent outbreaks	Flood-borne contamination overwhelms patchy surveillance

Projected far-future change in water-related variables and their significance for the water sector.

2. Energy

Energy is widely regarded as the foundation of the modern economy and development of any region. Energy forms the essential elements for the livelihood of inhabitants and industrial needs, viz., cooking, lighting, space heating, cooling, mining, manufacturing, production, transport, etc. Currently, the major energy needs are met through fossil-based fuels (coal, oil, gas) and the renewable energy share is progressively increasing across all nations to alleviate the persisting climate risks. Uttarakhand is often referred to as a hydro-energy state because of its enormous hydropower potential. The state is surrounded by many important rivers like Ganga, Yamuna, Bhagirathi, Kali, Gori, Alaknanda, Kosi, Sharda, Ramganga, DhaultiGanga, Mandakini, Pindar, Tonnes, Saryu, Bhilangana, etc. A recent State assessment (2017-23) places Uttarakhand's exploitable hydropower potential from large projects (each above 25 MW) at about 13,481 MW across 55 projects, of which roughly 4,035 MW (19 projects) is already in operation and a further 1,264 MW is under construction; an additional ~2,000 MW of pumped-storage potential has been identified, of which the 1,000 MW (2×500 MW) Tehri pumped-storage plant is the first to be developed—500 MW was commissioned in 2025 and the balance 500 MW is scheduled for commissioning by the end of 2025-26 (State Profile on Hydro Development for Uttarakhand; CEA Resource Adequacy Plan for Transmission, October 2025). The State's gross theoretical potential is often cited at around 25,000 MW.

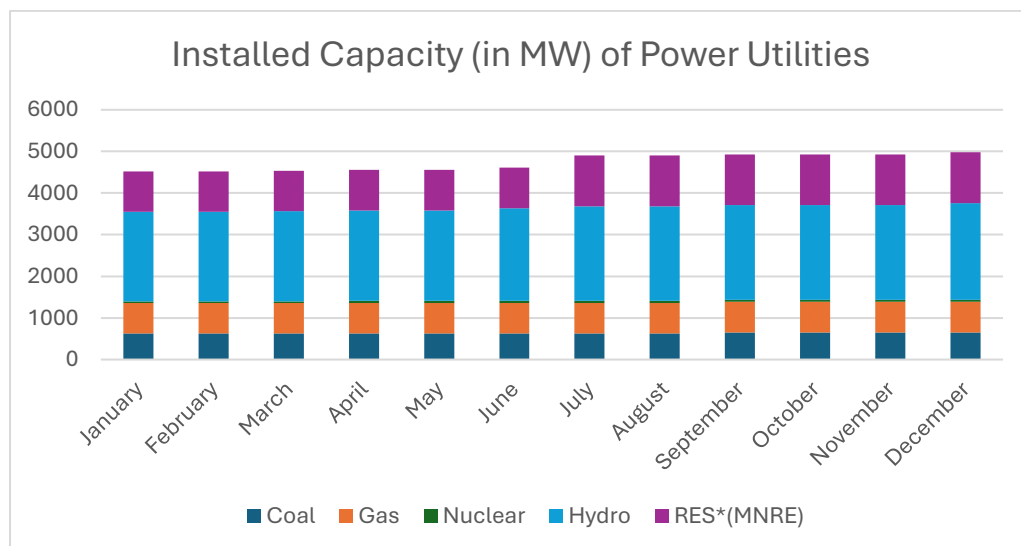


Figure 10.1a Installed capacity (in MW) of Uttarakhand's power utilities by source (monthly build-up over 2024-25; year-end total 4,979.92 MW)

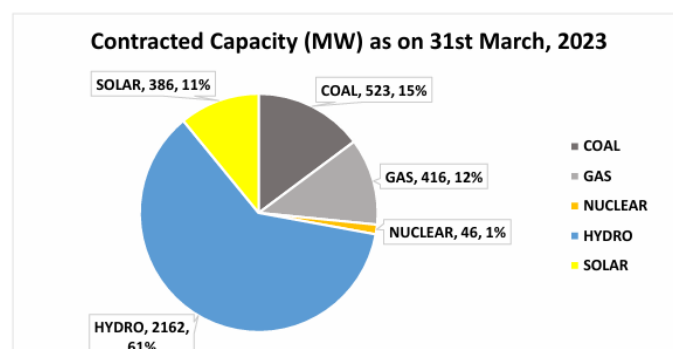
The distinction between an energy resource (a potential- a stock for fossil fuels, a flux for renewables) and energy supply (the converted, deliverable output) is central to how the State's exposure should be read. Because Uttarakhand's endowment is overwhelmingly a renewable flux governed by the hydrological and cryospheric cycle rather than a geological stock, it is intrinsically more climate-sensitive than a fossil-based system of comparable size (Schaeffer et al., 2012).

Generation Profile

As of December 2025, the total installed capacity for Uttarakhand is about 4,980 MW. Out of the total installed capacity, the share of non-fossil-fuel-based capacity is 72.1 %. Approximately 61 per cent of hydropower generated in the State is exported to neighbouring States under inter-State allocation arrangements, even as Uttarakhand continues to record peak-hour energy deficits during winter and the lean-flow season. The State's most recent Resource Adequacy Plan (CEA, October 2025) records a contracted capacity of 4,080 MW as of June 2025, of which about 76 per cent is non-fossil, and projects that

contracted capacity will need to rise to roughly 9,928 MW by 2035-36 to meet demand reliably. On the existing and planned contracts alone, that study estimates unserved energy climbing from about 2.4 BU in 2025-26 to about 9.1 BU, close to 30 per cent of the annual energy requirement—by 2035-36, underscoring how tightly the State’s adequacy margin is now bound to the climate-sensitive generation base described below.

Source	Installed Capacity (MW)	Share (%)
Coal	653.58	13.1
Gas	733.66	14.7
Nuclear	47.51	1.0
Hydropower	2,323.89	46.7
Other renewables (RES)	1,221.28	24.5
Total	4,979.92	100.0



Generation profile and the principal climate dependency of each segment.

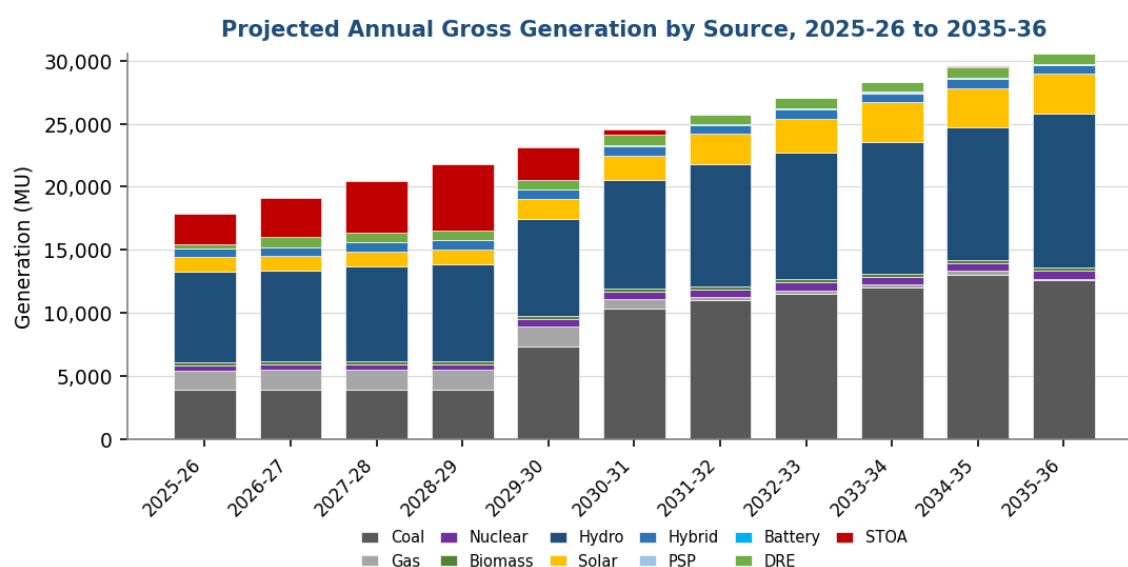


Figure 10.1b Projected annual gross generation by source, 2025-26 to 2035-36 (MU).

Source: CEA, Resource Adequacy Plan (Generation) for Uttarakhand, 2025-26 to 2035-36 (Version 3.0), October 2025 (Figure 14).

Institutional Structure

The sector is unbundled under the Electricity Act, 2003 into three State utilities: Uttarakhand Jal Vidyut Nigam Limited (UJVNL) for generation; Power Transmission Corporation of Uttarakhand Limited (PTCUL) for transmission; and Uttarakhand Power Corporation Limited (UPCL) for distribution. Central-sector entities: NTPC, THDC India Limited and SJVN Limited, operate large stations within the State, while UREDA anchors the off-grid and renewable programme.

Transmission and Distribution Backbone

Corridors at 132 kV, 220 kV and 400 kV traverse steep slopes, dense forest, narrow river valleys and high passes to connect upper-catchment stations with load centres in the Bhabar–Tarai plains and the Northern Regional Grid. Last-mile distribution to more than 16,000 inhabited villages depends heavily on overhead lines exposed to landslide, slope failure, snow loading, lightning and forest fire.

Two structural properties drive systemic risk. First, transmission lines are linear, geographically extended assets: a single 220 kV corridor may run hundreds of kilometres across multiple hazard zones, so the

probability that some segment intersects an active hazard at any moment is high. Second, in hill terrain the network is largely series-connected with minimal redundancy, so failure of one tower or span can isolate downstream substations and entire load pockets. Exposure of the transmission system is therefore simultaneously an asset risk and a network-level systemic risk.

At 220 kV and above, the existing intra-State transformation capacity is about 5,515 MVA and the intra-State transmission line length about 1,499 ckt-km. Under the CEA Resource Adequacy Plan for Transmission (October 2025), these are planned to rise to roughly 11,870 MVA and 1,930 ckt-km by 2029-30 and to about 14,000 MVA and 2,148 ckt-km by 2034-35, with new 400 kV substations at Roorkee, Khurpia Farm, SIDCUL (Haridwar) and Kotdwar and extensive reconductoring with HTLS conductors. Because the State meets much of its demand particularly in the low-hydro winter months—through imports over the inter-State network, this import dependence is itself climate-exposed: as of October 2025 Uttarakhand held an Available Transfer Capacity of 1,800 MW against an operationalised General Network Access of 1,402 MW, leaving limited headroom when a hazard event coincides with peak demand.

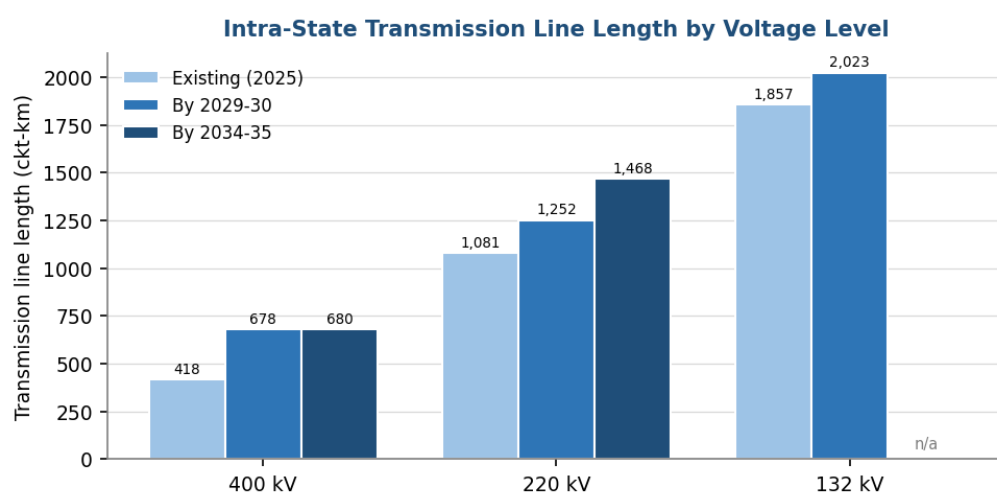


Figure 10.1c Intra-State transmission line length by voltage level (ckt-km).

Source: CEA, Resource Adequacy Plan for Transmission System of Uttarakhand up to 2034-35, October 2025 (Tables 1.2, 2.16, 3.8). 132 kV not consolidated for 2034-35.

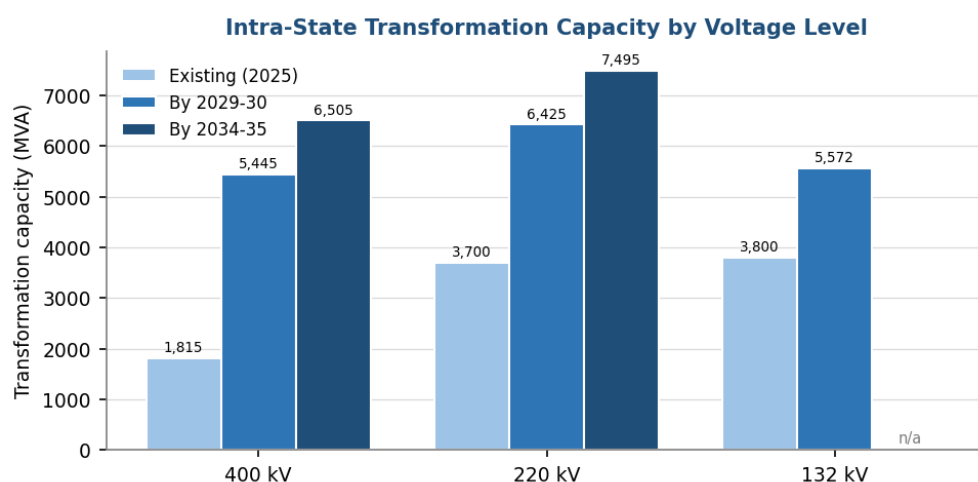


Figure 10.1d Intra-State transformation capacity by voltage level (MVA).

Source: CEA, Resource Adequacy Plan for Transmission System of Uttarakhand up to 2034-35, October 2025 (Tables 1.2, 2.16, 3.8). 132 kV not consolidated for 2034-35.

Demand Profile

Analysis of the 2024-25 hourly demand pattern shows that peak electricity demand in Uttarakhand now occurs during the non-solar hours of June, with demand during the non-solar hours of May and June

markedly higher than in the rest of the year, as seen in the month-wise average hourly demand profile for 2024-25 below. The daily solar and non-solar peaks for 2024-25, also shown below, confirm this seasonality: the daily peak falls in the non-solar (evening and night) hours through the summer months, whereas in the winter season the daily peak shifts into the solar (daytime) hours. This is a clear reorganisation from the earlier pattern—as recently as 2022-23 and 2023-24 the annual peak fell in the winter months of December and January—but the all-time peak of 2,863 MW in 2024-25 occurred in June 2024 during non-solar hours, and in 2025-26 (up to October 2025) a fresh all-time peak of 2,910 MW was again recorded in June, driven by summer cooling load, Char Dham tourism, rapid urbanisation and the emerging electric-vehicle and industrial load (CEA Resource Adequacy Plans for Generation and Transmission, October 2025). Winter nonetheless remains the season of greatest supply stress, because in-State hydro generation falls sharply in the lean-flow months and the State must draw more power over the inter-State network. Optimal utilisation of resources through short-term contracts such as banking or STOA/MTOA, as currently practised, remains one of the more effective ways of managing this seasonal variation and ensuring resource adequacy. Over the planning horizon the State’s energy requirement is projected to grow at a CAGR of about 4.9 per cent and peak demand at about 4.6 per cent between 2025-26 and 2035-36, with peak demand rising to roughly 3,917 MW by 2029-30 and about 4,707 MW by 2034-35 and the annual energy requirement approaching 30,850 MU by 2035-36; a Planning Reserve Margin of 13 per cent has been assessed for the State. In 2025-26 the State’s energy requirement reached 16,538 MU, with only 63 MU (0.4 per cent) unmet, while peak demand touched 2,910 MW and was fully met (Uttarakhand power supply-position report, 2025-26).

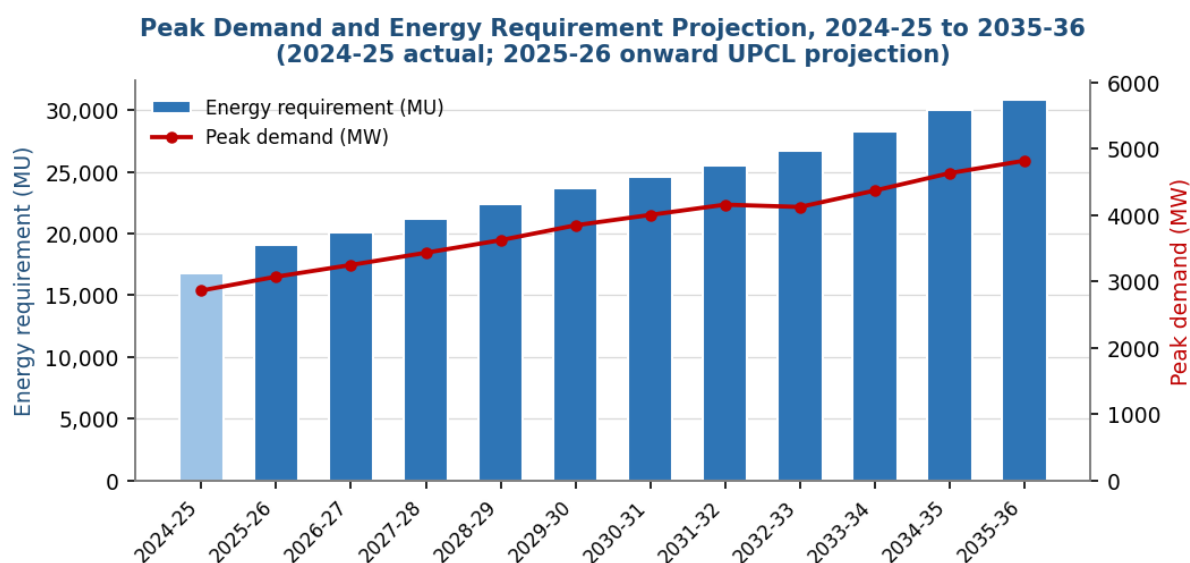
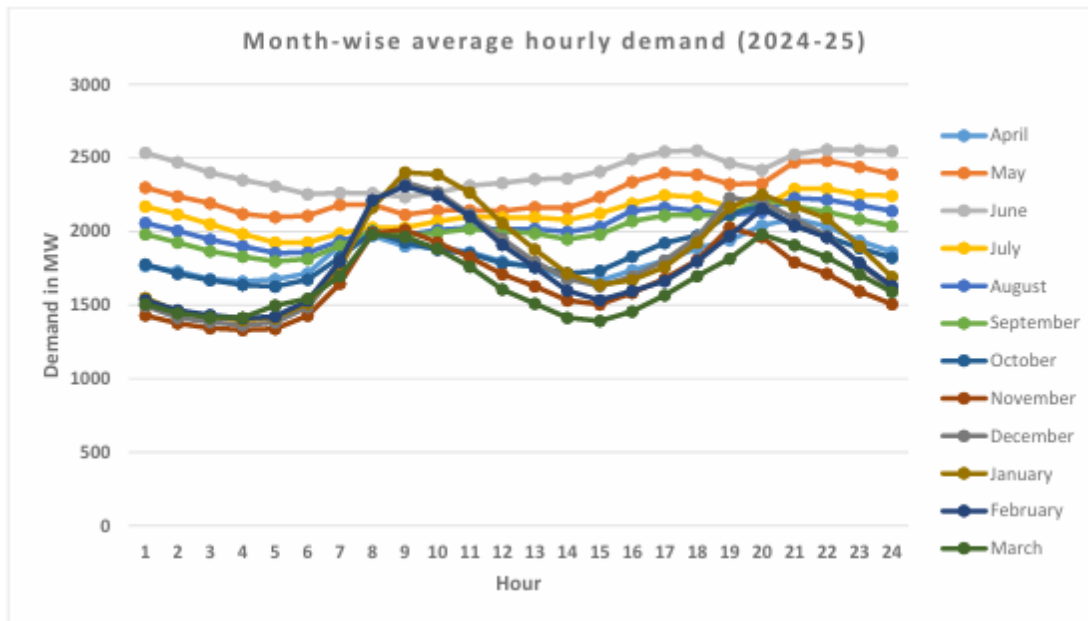
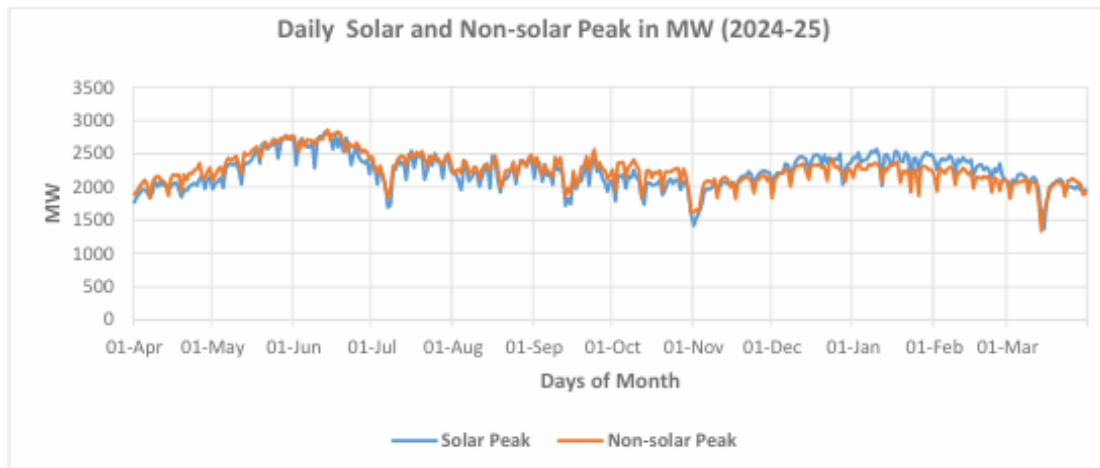


Figure 10.2e Peak demand and energy requirement projection, 2024-25 to 2035-36.

Source: CEA, Resource Adequacy Plan (Generation) for Uttarakhand, October 2025 (Table 7, UPCL projection); 2024-25 actual from Table 5.



Average Hourly Demand Variation (Month-wise) for 2024-25



Solar and non-solar Peak in MW for the year 2024-25

8.2 Defining Features and Their Climate Implication

The energy sector in Uttarakhand is intrinsically coupled with the regional climate system through hydrological, atmospheric, and geomorphological processes that govern electricity generation, transmission, distribution, and energy demand. Owing to the state's mountainous physiography and river-fed hydrology, the electricity system is dominated by hydropower, making energy production directly dependent on the magnitude, timing, and seasonal variability of river discharge, which is regulated by precipitation, snow accumulation, glacier melt, and catchment hydrological processes. Consequently, projected changes in temperature, precipitation regimes, cryospheric dynamics, and the frequency and intensity of extreme weather events are expected to influence multiple components of the energy system through interconnected physical pathways.

Hydropower generation is primarily controlled by basin-scale hydrological processes. Seasonal precipitation, snowpack accumulation, glacier mass balance, and the timing of snowmelt collectively determine streamflow regimes and reservoir inflows. Climate-induced alterations in these processes may modify both the seasonal availability and interannual variability of water resources, thereby

influencing hydropower generation potential, reservoir operations, and system flexibility. In glacier-fed catchments, long-term glacier retreat is expected to progressively alter meltwater contributions to river discharge, while increasing variability in monsoon rainfall may amplify fluctuations in runoff generation and sediment transport.

Electricity transmission and distribution infrastructure are closely linked to the climatic and geomorphological characteristics of the Himalayan environment. Transmission corridors traverse steep slopes, river valleys, and landslide-prone terrain, where infrastructure performance is influenced by intense precipitation, slope instability, flash floods, avalanches, debris flows, and glacial lake outburst floods (GLOFs). These hydro-meteorological and geomorphic processes influence the physical exposure of substations, transmission towers, distribution networks, and associated infrastructure throughout the state.

Climate also exerts a direct influence on electricity demand through its effects on thermal comfort and seasonal energy requirements. Rising temperatures are expected to modify cooling demand in the rapidly urbanizing foothill districts, while changes in winter temperature regimes may influence space-heating requirements in higher elevations. Increasing climatic variability may also alter the temporal distribution of electricity demand, resulting in changes to seasonal and diurnal load profiles.

Defining Feature	Description	Relevance for Climate Sensitivity
Hydropower-dominated energy system	Hydropower constitutes the predominant source of electricity generation within Uttarakhand, with generation concentrated in Himalayan River basins.	Electricity generation is intrinsically dependent on the timing, magnitude, and seasonal variability of river discharge, which is regulated by precipitation, snowmelt, glacier melt, and catchment hydrology.
Predominance of run-of-river hydropower	A substantial proportion of installed hydropower capacity comprises run-of-river schemes with limited live storage capacity.	Generation is highly sensitive to short-term and seasonal fluctuations in streamflow due to limited capacity to regulate hydrological variability.
Cryosphere-dependent River systems	Major Himalayan River basins are sustained by seasonal snowpack and glacier melt, particularly during the lean-flow period.	Changes in snow accumulation, glacier mass balance, snowmelt timing, and glacier retreat influence seasonal flow regimes and long-term water availability for hydropower generation.
Mountain-based energy infrastructure	Hydropower facilities, reservoirs, substations, and transmission infrastructure are located within steep, geologically active Himalayan terrain.	Energy infrastructure is exposed to climate-sensitive geomorphological processes, including landslides, flash floods, debris flows, glacial lake outburst floods (GLOFs), and slope instability.
Extensive transmission network	Electricity is transmitted through long-distance transmission corridors traversing mountainous terrain, with inter-state power evacuation forming an important component of the power system.	Transmission reliability is influenced by climate-induced disruptions affecting transport corridors, slope stability, river crossings, and associated infrastructure.
Forest-linked catchments and transmission corridors	Hydropower catchments and transmission networks intersect extensive forested landscapes that regulate watershed processes and ecosystem functions.	Climate-induced changes in forest condition, wildfire occurrence, vegetation dynamics, and catchment processes influence watershed stability, sediment dynamics, and the operational environment of energy infrastructure.

The energy sector is further influenced by climate-dependent ecosystem processes. Forested catchments regulate runoff generation, sediment transport, slope stability, and watershed hydrology that underpin hydropower operations. Climate-induced changes in forest composition, wildfire occurrence, vegetation cover, and catchment degradation may modify sediment yields, reservoir sedimentation rates, and watershed functioning, thereby affecting the long-term performance of hydropower assets.

8.3 Key Climate Hazards

Seven hazards dominate climate risk to the energy sector in Uttarakhand. For each, this section sets out the physical mechanism, the stage of the energy chain at which it acts, the assets and geographies affected, the documented historical record, and the residual data gap. Together the record establishes that climate risk to the sector is not a prospective concern but an ongoing, measurable loss stream: with two distinct signatures: rare, catastrophic, capital-destroying events and a chronic, high-frequency attrition of the transmission and distribution network that never reaches the national news but accounts for the larger share of cumulative economic loss.

Key Hazard

Hazard	Mechanism of damage	Energy-chain stage affected	Assets primarily affected	Principal districts / basins
GLOF	Instantaneous debris-laden surge destroys barrages, intakes, powerhouses and tunnels; deposits sediment that takes decades to clear	Resource; conversion (total asset loss)	Head-works, powerhouses, head-race tunnels of upper-basin projects	Chamoli, Uttarkashi, Pithoragarh, Bageshwar
Landslide and slope failure	Tower foundation loss; conductor snapping; burial of access roads preventing restoration	Transmission and distribution; conversion (penstocks)	Towers, poles, penstocks, approach roads	All hill districts; acute in Rudraprayag, Chamoli, Tehri, Pauri, Pithoragarh
Extreme rainfall / cloudburst	Flash flooding of valley-floor substations; scour of tower foundations; forced turbine shutdown on safety grounds; sediment pulse	Conversion; transmission; distribution	Substations, barrages, tailrace channels, feeders	Uttarkashi, Chamoli, Rudraprayag, Pithoragarh, Bageshwar, Nainital
Riverine and flash flooding	Inundation of valley-floor infrastructure; channel migration undermining structures	Conversion; transmission; distribution	Substations, switchyards, intakes, Tarai distribution assets	Alaknanda–Mandakini valleys; Haridwar and U.S. Nagar in the plains
Forest fire	Conductor and pole damage; flashover across corridors; aerosol loading suppresses solar irradiance	Transmission and distribution; solar conversion	Overhead corridors through chir-pine forest; solar installations	Pauri, Almora, Nainital, Tehri, Champawat
Temperature extremes	Transformer and substation derating in heat; conductor sag; snow and ice loading on lines; peak cooling load in the Tarai	Transmission; distribution; demand	Transformers, substations, conductors; demand side	Tarai belt (heat); high-altitude blocks (cold, snow loading)
Land subsidence (climate-conditioned)	Compromises stability of underground powerhouses and head-race tunnels; slow-onset, non-reversible	Conversion; siting	Underground works, tunnels, surface structures	Joshimath (Chamoli); other tunnelled reaches

Hazard register for the Uttarakhand energy sector

Glacial Lake Outburst Floods

A GLOF occurs when a moraine- or ice-dammed proglacial lake fails, releasing a large volume of water in minutes. In the Uttarakhand Himalaya the failure mechanism is frequently compound: a rock-ice avalanche detaches from an over-steepened, warming rock wall, impacts a lake or a glacier surface, and generates a debris-laden surge that entrains channel sediment as it descends. The resulting flow is not water but a

hyper-concentrated slurry, with an erosive and impact capacity orders of magnitude greater than the design flood of a hydropower barrage. Because such flows travel down exactly the confined, steep valleys in which run-of-river projects are sited, the hazard and the asset base are geometrically co-located.

Why this hazard is uniquely severe for the energy sector

- The failure mode is total, not partial: intakes, barrages and powerhouses are destroyed rather than damaged, so there is no repair pathway, only reconstruction.
- The loss extends beyond the asset to the site: post-event sediment and debris aggradation can render a site unusable and impose clearance liabilities that exceed the capital value of the project.
- The event has no useful precedent in the operational record, so the design basis cannot be derived from historical frequency, the central methodological difficulty identified in the climate-impact literature.
- Fatality risk is concentrated among project workers in tunnels and powerhouses, converting an asset risk into a life-safety risk.

Documented impact: the Chamoli disaster, 7 February 2021

A rock-ice avalanche in Chamoli district triggered a flash flood along the Rishi Ganga, Dhauliganga and Alaknanda valleys. The 13.2 MW Rishi Ganga Power Project was entirely obliterated. NTPC's 520 MW Tapovan Vishnugad Hydropower Plant, among the largest under construction in India, sustained catastrophic damage, with tunnels inundated and construction infrastructure destroyed (ICIMOD, 2021). Over 200 persons were killed or reported missing, many of them project workers trapped in the head-race tunnel.

The financial accounting measures the sector's material exposure. Climate Risk Horizons (2024) estimated the direct financial impact at approximately ₹1,625 crore. Debris removal alone is projected to cost ₹3,400 crore: the volume of displaced material, approximately 174.6 million cubic metres, would require an estimated 133 years to clear at the pace of a major urban demolition project (Down To Earth, 2024). A single event thus generates a liability far exceeding the capital value of the destroyed projects — a decisive argument for treating GLOF exposure as a siting constraint rather than an insurable operating risk.

Climate Risk Horizons (2024) identifies 15 hydropower projects with a combined investment of approximately ₹70,000 crore situated in high-risk zones. Of the eight major river basins in the State, the Joshimath–Srinagar basin is classified as most sensitive to hydropower project risk, followed by Tehri–Uttarkashi and Pithoragarh–Bageshwar. The number and size of glacial lakes forming at the termini of retreating glaciers continues to increase, raising the base probability of future events across all these basins (Mongabay India, 2023). The implication for this assessment is that GLOF exposure cannot be treated as a static layer: the hazard itself is growing under the same warming that drives the other hazards in this register.

Landslides and Slope Failure

Landslides in Uttarakhand are overwhelmingly rainfall-triggered, occurring where intense or prolonged precipitation raises pore-water pressure in already-marginal slopes. The State's geology compounds the trigger: young, tectonically active, highly fractured lithologies with steep relief and thin, poorly cohesive colluvial cover. Research in Sustainable and Resilient Infrastructure (2024) confirms that these formations substantially increase landslide incidence following extreme rainfall. Anthropogenic slope-cutting for roads, and the spoil generated by tunnelling and hydropower construction, further destabilise the very slopes on which the transmission network is founded: so the sector partly manufactures its own hazard.

Pathway to energy-sector damage

Failure pathway	What is lost	Why restoration is slow
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Tower foundation on colluvial slope loses support	Tower collapse; conductor severed; corridor de-energised	New foundation must be sited on stable ground; may require re-routing and fresh right-of-way
Debris flow across a corridor span	Conductor snapping; insulator and hardware loss	Debris must be cleared before crews can reach the span
Slide burying the approach road	No asset loss, but the asset becomes unreachable	Restoration time is governed by road clearance, not by the electrical repair itself
Slope failure at an intake or penstock	Generation halted; penstock rupture	Repair requires dewatering and stable working platform
Distribution pole loss on hill slopes	Village-level supply cut	Dispersed, low-priority assets are restored last

Documented impact

Every major monsoon season produces documented loss of transmission capacity. In Almora and Haldwani, torrential rain has repeatedly destroyed both transmission lines and local distribution networks, triggering prolonged outages that compound the water crisis as pumping stations and treatment plants lose power. Uttarakhand's 66/33/11 kV grid is predominantly overhead and routed through deeply incised valleys and along slopes of high landslide susceptibility, so the network is, in effect, aligned with the hazard.

Recognising this structural vulnerability, the Asian Development Bank approved a USD 200 million loan in December 2023 for the Uttarakhand Climate Resilient Power System Development Project. A central element is the replacement of exposed overhead conductors with underground cable in the most hazard-prone corridors; the ADB assessment explicitly notes the superior resilience of underground infrastructure to high winds, falling trees and flying debris — the primary failure modes of the current overhead network. The investment is, in effect, an acknowledgement that the existing T&D grid was not designed for the climate the State now experiences.

Extreme Rainfall and Cloudbursts

A cloudburst, conventionally, rainfall exceeding 100 mm in an hour over a small area, delivers a runoff volume that small Himalayan catchments cannot route without overtopping. The energy-sector consequence is threefold: direct inundation of valley-floor infrastructure; scour of foundations by high-velocity flow; and a sediment pulse that forces generating plant offline even where no structure is damaged. Extreme rainfall is also the proximate trigger for the landslide hazard above, so the two hazards are not independent and must not be double-counted in the composite index.

Documented impact

The 2021 Pithoragarh cloudburst, which delivered 147 mm of rainfall in two hours, destroyed four primary health centres and disrupted electricity supply to 17 villages. The event is instructive not for its magnitude but for its cascade: loss of a single distribution node removed power from health facilities at the moment of greatest demand for them. Analogous events propagate through transmission nodes that have no redundancy or alternative routing.

The trend that matters

Total monsoon rainfall in Uttarakhand has declined by approximately 12 per cent since 2000, while the frequency of extreme rainfall events has increased by approximately 35 per cent (IMD, 2023). The two findings together, not either alone, define the hazard: the same or less annual water is being delivered in fewer, more violent pulses. For a run-of-river system this is the worst possible reorganisation of the same rainfall total — fewer optimal-flow days for generation, and more days on which flow exceeds the safe operating envelope.

8. Riverine and Flash Flooding

Flood exposure in the State is bimodal. In the hills, flooding is flashy, sediment-laden and channelised: it undermines barrages, intakes and tailrace channels, and inundates valley-floor switchyards sited on terraces that historical flood records classified as safe. In the Bhabar–Tarai plains, Haridwar, Udham Singh Nagar and the southern blocks of Nainital and Dehradun, flooding is areal and prolonged: it inundates distribution networks, substations and consumer premises across wide areas, with far larger populations affected but with lower per-asset destructive force. The two require different treatment. Hill flooding is a capital-destruction hazard; plains flooding is a service-continuity and consumer-safety hazard. A single flood exposure metric conceals this distinction, which is why Section 10.4 reports flood exposure separately from extreme rainfall and interprets the two differently by terrain.

Channel migration is a distinct and under-assessed sub-hazard: rivers that shift their thalweg after a high-magnitude flood can leave a substation that was 200 m from the bank sitting on the bank. Because substation siting decisions are effectively permanent, this converts a transient event into a permanent change in exposure.

Forest Fire

Uttarakhand’s mid-altitude belt is dominated by chir pine (*Pinus roxburghii*), whose resinous, highly flammable needle litter creates a near-ideal surface fuel bed through the dry pre-monsoon months. Fire affects the energy sector through two independent pathways that are usually assessed separately and should not be:

- The network pathway. Fire beneath and around overhead corridors damages wooden and composite poles, anneals and sags conductors, causes flashover across the ionised hot-air column between phases, and destroys the right-of-way. Because corridors in the mid-hills are routed through precisely the chir-pine belt where fire risk is highest, the correlation between corridor alignment and fire susceptibility is strongly positive.
- The generation pathway. Fire aerosols absorb and scatter incoming solar radiation, suppressing irradiance at the surface. A 2022 study found that aerosols from the 2021 Uttarakhand fire season, which affected large tracts of Garhwal and Kumaon forest, caused a documented loss of solar energy output valued at approximately ₹78 lakh at affected installations.

Why the dual pathway compounds risk

The State’s stated strategy is to scale solar capacity in order to diversify away from climate-stressed hydropower. Fire undermines both terms of that strategy simultaneously: it damages the network that would carry the diversified generation, and it suppresses the resource on which the diversification depends. The 2024 fire season recorded over 1,100 major fire incidents, and projections indicate that days of severe fire risk in the State’s dry forests could increase by approximately 60 per cent under continued warming. Fire is, on the evidence, the fastest-growing hazard in this register, and the one for which the sector’s current risk treatment is weakest.

Temperature Extremes

Extreme	Effect on the energy system	Where it binds in Uttarakhand
Heat	Transformer and switchgear derating; accelerated insulation ageing; conductor thermal sag reducing line ratings precisely when load peaks; reduced thermal-plant and PV efficiency	Tarai–Bhabar belt; Haridwar, U.S. Nagar, Haldwani, Rudrapur, Dehradun valley
Cold, snow and ice loading	Mechanical loading on conductors and towers; ice accretion causing galloping and outage; access closure for maintenance crews	High-altitude blocks: Bhatwari, Mori, Joshimath, Dharchula, Munsyari, Kapkot, Chakrata
Warm winters (absence of cold)	Reduced snow accumulation — the erosion of the natural seasonal reservoir; earlier and lower lean-season flow	All snow-fed upper catchments

Temperature extremes and where they bind. Note that the third row the absence of cold is the most consequential for a hydro-dominant system, yet it is not an "extreme event" at all and is invisible to event-based hazard mapping.

Land Subsidence

The 2023 Joshimath subsidence event raised acute concerns about the long-term stability of underground powerhouses and head-race tunnels in the area. Subsidence is a slow-onset, effectively irreversible hazard whose relationship to climate is indirect but real: enhanced meltwater and altered groundwater regimes change pore pressures in the loose paraglacial material on which much of the settlement and infrastructure sits. It is included in this register because it acts on precisely the asset class — tunnelled works — that is least amenable to retrofit, and because it is invisible to hazard frameworks built exclusively around extreme events.

9. Chronology of Past Impacts on the Energy Sector

Year	Event	Documented energy-sector consequence	Lesson carried into the framework
2010	Widespread monsoon landslides and cloudbursts (Kumaon and Garhwal)	Extensive damage to distribution networks and access roads; prolonged village-level outages	Chronic monsoon attrition, not a single catastrophic event, sets the baseline loss rate
2012	Assi Ganga flash flood, Uttarkashi	Small hydro projects on the Assi Ganga damaged; intake and channel works lost	Small hydro in steep tributary catchments carries disproportionate exposure per MW
2013	Kedarnath disaster-extreme rainfall combined with moraine-dammed lake breach	Damaged or destroyed the 99 MW Singoli-Bhatwari and 76 MW Phata-Byung projects and significant components of the Vishnuprayag and Srinagar projects	Compound hazards exceed the design basis of individually appraised projects
2021 (Feb)	Rishi Ganga–Tapovan rock-ice avalanche and flash flood, Chamoli	13.2 MW Rishi Ganga destroyed; 520 MW Tapovan Vishnugad severely damaged; >200 fatalities; ~₹1,625 crore direct loss; ~174.6 Mm ³ debris	No-precedent events cannot be assessed from the historical record alone
2021 (Oct)	Extreme post-monsoon rainfall, Kumaon	Landslide and flood damage to transmission and distribution assets; multi-day outages across Nainital district	Hazard season is lengthening beyond the conventional monsoon window
2021	Pithoragarh cloudburst-147 mm in two hours	Supply disrupted to 17 villages; four primary health centres lost power	Cascading failure into health and water services must be scored, not just asset loss
2021	Forest fire season, Garhwal and Kumaon	Aerosol loading suppressed irradiance; ~₹78 lakh solar generation loss at affected installations	Fire attacks the diversification strategy itself
2023 (Jan)	Joshimath land subsidence	Acute concern over stability of underground powerhouses and head-race tunnels	Slow-onset hazards need separate treatment from extremes
2023 (Dec)	ADB approves USD 200 million Climate Resilient Power System Development Project	Formal recognition that the overhead T&D network is not fit for the current climate	Adaptation finance is now available; the constraint is prioritisation, which this index supplies
2024	Forest fire season->1,100 major incidents	Corridor damage across the chir-pine belt; suppression of solar output	Fire is the fastest-growing hazard in the register
2010–present	Recurrent monsoon flash floods and cloudbursts	Cumulative damage to barrages, intakes, penstocks, tailrace channels, towers and substations across Uttarkashi, Chamoli, Rudraprayag, Pithoragarh and Bageshwar	The cumulative chronic loss exceeds the headline catastrophic loss

Chronology of documented climate-hazard impacts on the Uttarakhand energy sector

8.4 Exposure Assessment

Exposure is assessed by intersecting hazard-susceptibility layers with the geo-referenced energy asset inventory. Four hazards are carried into the spatial overlay, landslide, flood, extreme rainfall and forest fire, because susceptibility layers for these are available at a resolution consistent with block-level reporting.

GLOF is treated separately as a basin-level siting constraint, temperature extremes are handled on the sensitivity side (Section 10.6); subsidence is treated as a site-specific constraint.

Data Layers

Hazard layer	Source and basis	Classification	Resolution
Landslide susceptibility	GSI / BMTPC landslide susceptibility mapping, refined with slope, lithology, land cover and rainfall-trigger thresholds	5 classes (Very Low – Very High)	30 m raster
Flood hazard	NRSC flood hazard atlas; historical inundation extents; river buffer and terrace analysis in hill reaches	5 classes	30 m raster / vector extents
Extreme rainfall	IMD gridded daily rainfall (0.25°); frequency of days exceeding the 95th and 99th percentile; cloudburst incidence record	5 classes	0.25° downscaled to block
Forest fire risk	FSI forest fire susceptibility; chir-pine distribution; fire incidence record (VIIRS/MODIS hotspots)	5 classes	250 m – 1 km
Energy assets	PTCUL transmission network; UPCL distribution and substation inventory; UJVNL and IPP generation inventory; UREDA solar	Point and line vector	Asset-level

Data layers used in the exposure overlay.

Exposure Metrics

Asset class	Exposure metric	Unit reported
Transmission lines (132/220/400 kV)	Length within high and very-high hazard classes ÷ total line length in the block	% of line length exposed
Distribution network (66/33/11 kV)	Feeder length within high and very-high classes ÷ total feeder length in the block	% of feeder length exposed
Substations	Substations within high and very-high classes ÷ total substations in the block	% of substations exposed
Hydropower stations	Stations (and MW) within high and very-high classes ÷ total in the block	% of stations and % of MW exposed
Solar installations	Installed capacity within high fire-risk classes ÷ total installed capacity in the block	% of MW exposed

Exposure metrics by asset class. "Exposed" means intersecting the high or very-high class of the hazard layer.

Classification Bands

Class	Range (% of asset exposed)	Interpretation
Very Low	0 – 10 %	Marginal exposure; standard operating practice adequate
Low	>10 – 25 %	Localised exposure; targeted hardening of identified assets
Moderate	>25 – 50 %	Material exposure; corridor-level risk treatment required
High	>50 – 75 %	Systemic exposure; re-routing, undergrounding or redundancy required
Very High	>75 – 100 %	Critical exposure; the asset base is effectively co-located with the hazard

Exposure classification bands and the score used in the composite.

Composite Exposure Formula

The block composite is the weighted mean of the four hazard-specific exposure scores. Weights reflect the relative contribution of each hazard to documented energy-sector damage in the State (Section 10.4.9), with landslide weighted highest because it is both the most frequent cause of network failure and the principal control on restoration time:

$$E(\text{block}) = 0.30 \cdot S(\text{landslide}) + 0.20 \cdot S(\text{flood}) + 0.25 \cdot S(\text{extreme rainfall}) + 0.25 \cdot S(\text{forest fire})$$

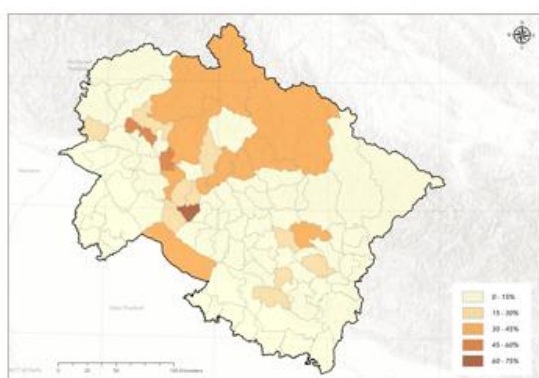
where S is the 1-5 class score from Table 10.12. Composite scores are then banded: <2.0 Very Low; 2.0–2.69 Low; 2.70–3.39 Moderate; 3.40–4.19 High; ≥4.20 Very High.

Hazard-Wise Exposure: Distribution of Blocks

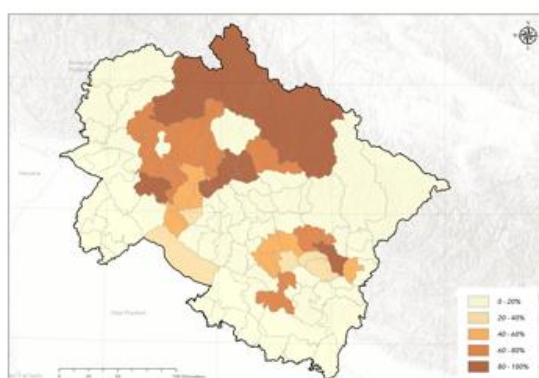
This section assesses how climate-induced hazards translate into impacts on the energy sector through identifiable cause-effect pathways. The analysis focuses on four dominant hazards in the Uttarakhand context: landslides, floods, extreme rainfall, and forest fires; examines, for each, the exposure of generation and transmission assets, the population affected, and the resulting accessibility disruption.

Transmission & Distribution exposure to disruption

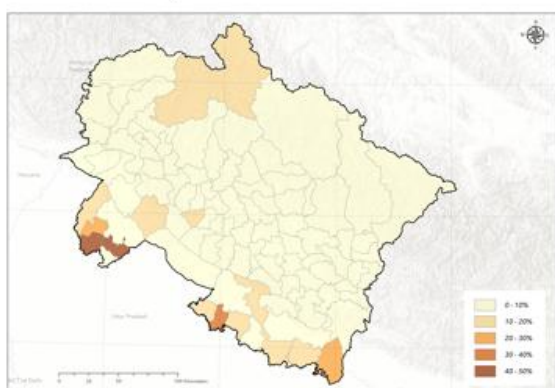
Transmission lines occupy a uniquely critical position in the hazard pathway for two reasons. First, they are linear, geographically extended assets, a single 220 kV corridor can run for hundreds of kilometres across multiple hazard zones, so the probability that some segment intersects an active hazard at any given time is high. Second, they are series-connected with little inherent redundancy in hill terrain, so failure of a single tower or span can isolate downstream substations and entire load pockets. The exposure of transmission lines is therefore both a direct asset risk and a network-level systemic risk.



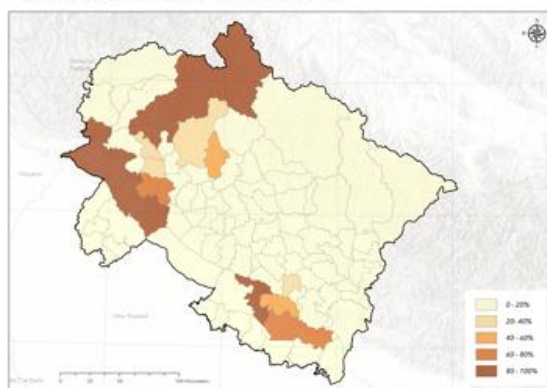
Transmission line exposed to Landslide



Transmission line exposed to Forest Fire



Transmission line exposed to Floods



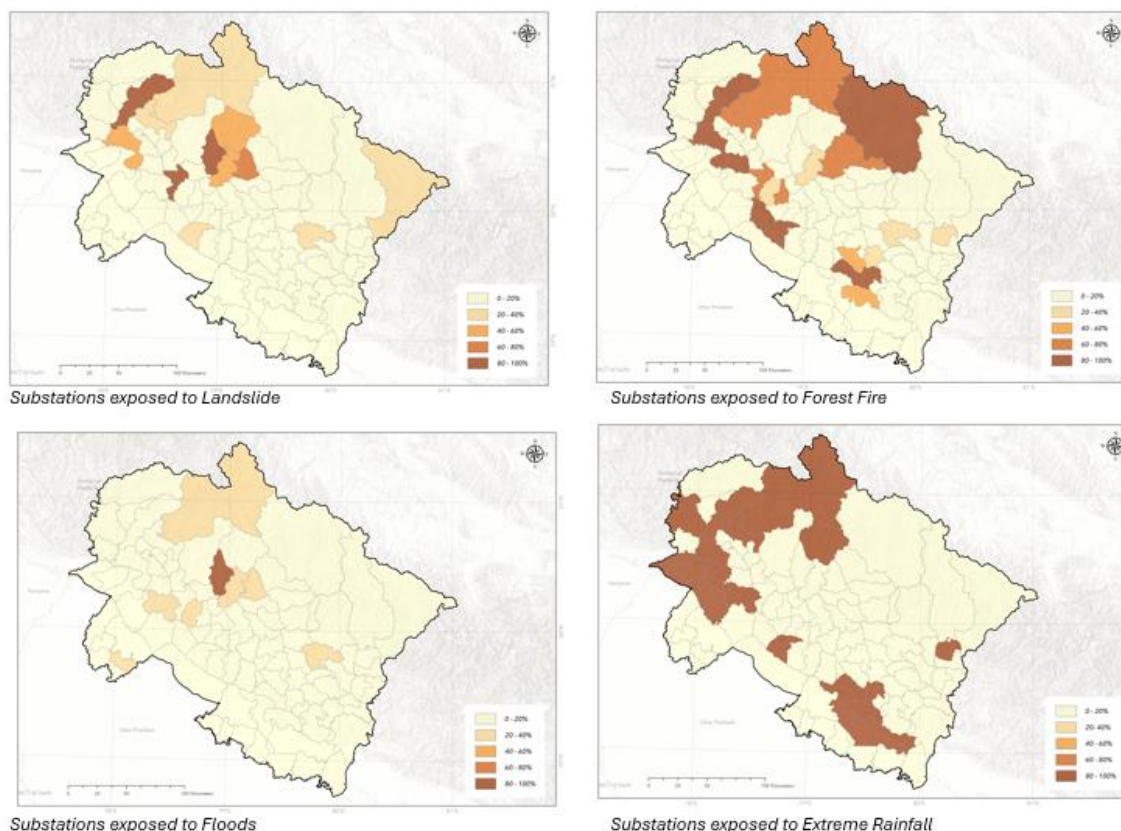
Transmission line exposed to Extreme Rainfall

The analysis indicates that transmission line infrastructure in Uttarakhand is increasingly exposed to multiple climate hazards including landslides, forest fires, floods, and extreme rainfall events. Exposure patterns vary spatially, with mountainous districts showing greater vulnerability due to rugged terrain, slope instability, and changing climatic conditions.

Forest fire exposure appears to affect a substantial proportion of transmission corridors, particularly in central and northern districts, increasing the risk of damage to transmission towers, conductor failures, and service interruptions. Landslides and extreme rainfall further threaten infrastructure stability through slope

failures, erosion, and damage to access routes required for maintenance activities. Pauri, Nainital, Almora, Dehradun, and U.S. Nagar as areas with relatively higher exposed transmission line lengths.

Substations exposure to disruption



10. Among the assessed hazards, forest fire and extreme rainfall exposure affect a larger proportion of substations, particularly across the northern and western districts. Substations located in mountainous areas are increasingly vulnerable to forest fire events, which can damage transformers, transmission equipment, and supporting infrastructure. Extreme rainfall and associated slope instability further increase risks through flooding, erosion, and landslide-triggered damage.

11.

12. Landslide exposure is concentrated in selected hill districts where unstable slopes and intense rainfall create conditions for infrastructure disruption. Flood exposure appears comparatively localized but remains critical because even limited impacts on key substations can trigger widespread power supply interruptions.

13. Block-Wise Exposure of Energy Infrastructure

District	Block	Landslide	Flood	Extreme rainfall	Forest fire	Composite exposure
Uttarkashi	Bhatwari	Very High	High	Very High	Moderate	Very High
	Dunda	High	Moderate	High	High	High
	Chinyalisaur	High	Moderate	High	High	High
	Naugaon	High	Moderate	High	High	High
	Purola	Moderate	Moderate	Moderate	High	Moderate
	Mori	High	Moderate	High	Moderate	High

District	Block	Landslide	Flood	Extreme rainfall	Forest fire	Composite exposure
Chamoli	Dasholi	Very High	High	Very High	Moderate	Very High
	Joshimath	Very High	Very High	Very High	Low	Very High
	Ghat	Very High	Moderate	High	Moderate	High
	Karnaprayag	High	High	High	Moderate	High
	Dewal	High	Moderate	High	Moderate	High
	Tharali	High	High	High	Moderate	High
	Narayanbagar	High	Moderate	High	Moderate	High
	Pokhri	High	Low	Moderate	High	Moderate
	Gairsain	Moderate	Low	Moderate	High	Moderate
Rudraprayag	Ukhimath	Very High	High	Very High	Moderate	Very High
	Agastyamuni	Very High	High	High	Moderate	High
	Jakholi	High	Moderate	High	High	High
Tehri Garhwal	Chamba	High	Low	Moderate	High	Moderate
	Jaunpur	High	Low	Moderate	High	Moderate
	Kirtinagar	High	Moderate	Moderate	High	High
	Narendranagar	High	Moderate	Moderate	Very High	High
	Pratapnagar	High	Moderate	High	High	High
	Devprayag	High	High	Moderate	High	High
	Jakhnidhar	High	Low	Moderate	High	Moderate
	Bhilangana	Very High	High	High	Moderate	High
	Thauldhar	High	Low	Moderate	High	Moderate
Dehradun	Chakrata	High	Low	Moderate	Very High	High
	Kalsi	High	Moderate	Moderate	Very High	High
	Vikasnagar	Low	High	Moderate	High	Moderate
	Sahaspur	Very Low	Moderate	Moderate	High	Low
	Raipur	Moderate	High	High	High	High
	Doiwala	Low	High	High	High	High
Pauri Garhwal	Kot	High	Low	Moderate	Very High	High
	Kaljikhali	High	Low	Moderate	Very High	High
	Pauri	High	Low	Moderate	Very High	High
	Pabau	High	Low	Moderate	Very High	High
	Thalisain	High	Low	Moderate	Very High	High
	Bironkhal	High	Low	Moderate	Very High	High
	Dwarikhali	High	Low	Moderate	Very High	High
	Dugadda	High	Moderate	Moderate	Very High	High
	Jaiharikhali	High	Low	Moderate	Very High	High
	Rikhnikhali	Moderate	Low	Moderate	Very High	Moderate
	Yamkeshwar	High	Moderate	Moderate	Very High	High

District	Block	Landslide	Flood	Extreme rainfall	Forest fire	Composite exposure
	Ekeshwar	Moderate	Low	Moderate	High	Moderate
	Khirsu	Moderate	Low	Moderate	High	Moderate
	Nainidanda	High	Low	Moderate	Very High	High
	Pokhra	High	Low	Moderate	High	Moderate
Haridwar	Bahadabad	Very Low	High	Moderate	Low	Low
	Roorkee	Very Low	High	Moderate	Very Low	Low
	Narsan	Very Low	Moderate	Moderate	Very Low	Very Low
	Bhagwanpur	Very Low	Moderate	Moderate	Low	Low
	Laksar	Very Low	Very High	Moderate	Very Low	Low
	Khanpur	Very Low	Very High	Moderate	Very Low	Low
Nainital	Betalghat	High	Moderate	High	Very High	High
	Bhimtal	High	Moderate	Very High	Very High	Very High
	Dhari	High	Low	High	Very High	High
	Kotabag	Moderate	High	High	Very High	High
	Okhalkanda	High	Moderate	High	Very High	High
	Ramgarh	High	Low	High	Very High	High
	Ramnagar	Low	High	High	Very High	High
	Haldwani	Low	Very High	Very High	High	High
Almora	Hawalbagh	High	Low	Moderate	Very High	High
	Takula	High	Low	Moderate	Very High	High
	Lamgara	High	Low	Moderate	Very High	High
	Dhauladevi	High	Low	Moderate	Very High	High
	Bhaisiyachhana	High	Low	Moderate	Very High	High
	Bhikiyasain	High	Moderate	Moderate	Very High	High
	Salt	High	Low	Moderate	Very High	High
	Chaukhutiya	High	Moderate	Moderate	Very High	High
	Dwarahat	High	Low	Moderate	Very High	High
	Syaldey	Moderate	Low	Moderate	High	Moderate
	Tarikhet	High	Low	Moderate	Very High	High
Bageshwar	Bageshwar	High	High	High	High	High
	Kapkot	Very High	High	Very High	Moderate	Very High
	Garur	High	Moderate	High	High	High
Pithoragarh	Pithoragarh	High	Moderate	High	High	High
	Munakot	High	Moderate	High	High	High
	Gangolihat	High	Low	Moderate	Very High	High
	Berinag	High	Low	High	Very High	High
	Didihat	High	Moderate	High	High	High
	Kanalichhina	High	Moderate	High	High	High

District	Block	Landslide	Flood	Extreme rainfall	Forest fire	Composite exposure
	Dharchula	Very High	Very High	Very High	Low	Very High
	Munsyari	Very High	High	Very High	Low	High
Champawat	Champawat	High	Low	Moderate	Very High	High
	Lohaghat	High	Low	Moderate	Very High	High
	Pati	High	Moderate	High	Very High	High
	Barakot	High	Moderate	High	High	High
Udham Singh Nagar	Kashipur	Very Low	High	Moderate	Low	Low
	Jaspur	Very Low	High	Moderate	Low	Low
	Bazpur	Very Low	High	Moderate	Low	Low
	Gadarpur	Very Low	Very High	High	Very Low	Low
	Rudrapur	Very Low	High	High	Very Low	Low
	Sitarganj	Very Low	Very High	High	Low	Moderate
	Khatima	Very Low	Very High	High	Low	Moderate

Block-wise exposure of energy infrastructure by hazard (95 blocks, 13 districts). Composite score computed as $0.30 \cdot \text{landslide} + 0.20 \cdot \text{flood} + 0.25 \cdot \text{extreme rainfall} + 0.25 \cdot \text{forest fire}$, on the 1–5 class scores of Table 10.12.

Priority Blocks

Seven blocks return a Very High composite exposure. They do not form a single group, and the difference between them is the most policy-relevant result in this section:

Block	District	Exposure driver	Energy assets at stake
Bhatwari	Uttarkashi	Upper Bhagirathi; landslide, extreme rainfall, GLOF	Maneri Bhali complex; Assi Ganga small hydro; upper Bhagirathi corridor
Dasholi	Chamoli	Alaknanda confluence zone; landslide and extreme rainfall	Alaknanda corridor transmission; district substations
Ukhimath	Rudraprayag	Mandakini valley; 2013 event footprint	Singoli-Bhatwari and Phata-Byung catchment; Kedarnath-route distribution
Bhimtal	Nainital	Very high fire and extreme-rainfall exposure in the chir-pine belt	Kumaon distribution network; tourism-node load; Bhimtal–Haldwani corridor
Kapkot	Bageshwar	Pindari glacier catchment; landslide; extreme rainfall	Sarju–Pindar small hydro; remote feeders
Joshimath	Chamoli	The full hazard stack: GLOF, landslide, flood and subsidence	Tapovan Vishnugad (520 MW); Rishi Ganga site; head-race tunnels
Dharchula	Pithoragarh	GLOF, landslide, extreme rainfall, flood on the Kali–Dhauliganga	Dhauliganga HEP and transmission; border-area distribution with no redundancy

Blocks returning Very High composite exposure (scores per Table 10.14).

Asset-Class Findings

Transmission lines

Transmission-line exposure varies systematically with terrain. Mountainous districts show markedly higher exposure owing to slope instability and the routing of corridors along ridgelines and valley walls. Forest fire affects a substantial proportion of corridors in the central and northern districts, raising the risk of tower damage, conductor failure and prolonged interruption. Landslide and extreme rainfall compound this by threatening not only the assets but the access routes required for restoration — a critical distinction, since restoration time in hill terrain is frequently governed by road access rather than by the repair itself. Pauri Garhwal, Nainital, Almora, Dehradun and Udham Singh Nagar carry the highest absolute lengths of exposed

line; the first three are exposed principally through terrain-driven hazards, the last two combine long line lengths with flood and extreme-rainfall exposure in the plains and foothills.

Substations

Among the assessed hazards, forest fire and extreme rainfall affect the largest proportion of substations, particularly across the northern and western districts. Substations in mountainous areas face fire risk to transformers, switchgear and supporting infrastructure, while extreme rainfall and associated slope instability add flooding, erosion and landslide-triggered damage. Landslide exposure is concentrated in the hill districts where unstable slopes coincide with intense rainfall. Flood exposure is comparatively localised, but remains critical, because a limited impact on a single key substation can trigger widespread supply interruption across the whole load pocket it serves.

Population and Service Exposure

Asset exposure is only half the picture: the consequence of a failure depends on how many people and how many critical services depend on the affected asset. For each block the assessment therefore also records:

Variable	Definition	Source
Population served by exposed assets	Block population within the service area of substations/feeders classed as exposed	UPCL consumer data + Census/projected population
Critical facilities dependent	Health, water-supply, telecom and education facilities on exposed feeders	Line departments; Economic Statistics document
Mean restoration time	Average hours/days to restore supply after a hazard event, by block	UPCL/PTCUL outage records, last 5–10 years
Accessibility during hazard	Share of block road network exposed to landslide/flood (proxy for repair-crew access)	PWD road network × hazard layer

Population and service exposure variables. These carry directly into the sensitivity and adaptive-capacity dimensions of the index.

The distinction between asset exposure and service exposure is not academic. Dharchula and Munsyari have very high asset exposure and small populations; Haldwani and Rudrapur have low landslide exposure and very large populations on flood-exposed feeders. A ranking on asset exposure alone would send investment to the former and none to the latter, which would be indefensible on a service-continuity criterion.

8.5 Climate Variability Impacting the Energy Sector

This section addresses sensitivity: given exposure, how strongly does climate variability translate into loss of energy service? For Uttarakhand the answer is that sensitivity is unusually high, because almost every link in the value chain- resource availability, generation, transfer and demand is climate-dependent. The section first sets out the observed variability, then traces each sensitivity pathway from climate variable to energy outcome and finally identifies where variability is already binding today.

Observed Climate Variability and Trends

Climate variable	Observed change	Consequence for the energy sector
Total monsoon rainfall	Decline of approximately 12% since 2000 (IMD, 2023)	Fewer optimal-flow days for run-of-river generation; lower assured annual energy
Extreme rainfall frequency	Increase of approximately 35% (IMD, 2023)	More forced turbine shutdowns; more landslide-driven T&D failure; larger sediment pulses
Rainfall concentration	Same or less annual total, delivered in fewer, more intense pulses	The flow-duration curve steepens — the single most damaging reorganisation of rainfall for a storage-less system
Air temperature	Warming, with elevation-dependent amplification at altitude	Accelerated snow and glacier melt; higher cooling demand in the Tarai; transformer derating
Winter precipitation and snowfall	Declining snowfall fraction; rising snowline	Erosion of the natural seasonal reservoir that sustains lean-season base flow

Glacier mass balance	Sustained negative mass balance across the ~900 glaciers of the State	Transient flow gain, then permanent decline — the "peak water" trajectory
Glacial lake formation	Number and size of proglacial lakes increasing (Mongabay India, 2023)	Rising base probability of GLOF across all eight major basins
Forest-fire season severity	>1,100 major incidents in 2024; severe-fire-risk days projected to rise ~60%	Corridor damage; aerosol suppression of solar irradiance
Seasonality of hazard	Damaging events now recorded outside the conventional monsoon window (e.g. Oct 2021 Kumaon)	Maintenance and restoration planning built around a June–September hazard season is mis-calibrated

Observed climate variability in Uttarakhand and its energy-sector consequence.

Climate Variable → Energy Chain Transmission Matrix

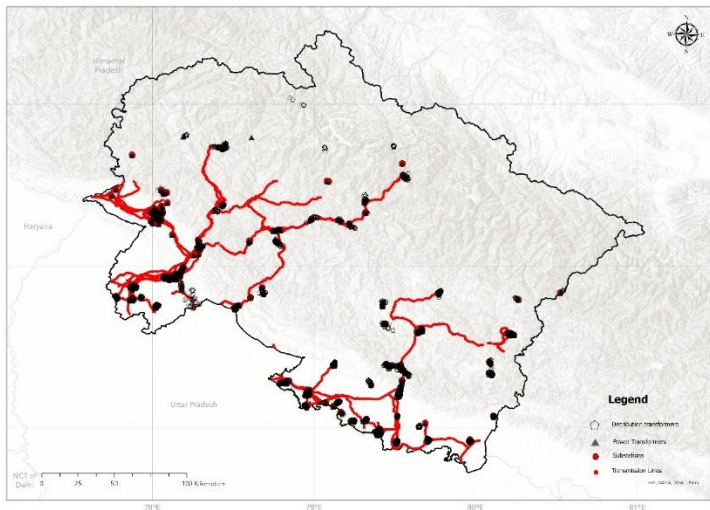
The matrix below traces each climate variable through every stage of the energy chain, which is the discipline the framework in Section 10.3 imposes. Reading down a column shows what a single stage is exposed to; reading across a row shows how a single variable propagates.

Climate variable	Resource	Conversion / generation	Transmission & distribution	Demand
Rising air temperature	Accelerated melt; earlier runoff; higher evapotranspiration	PV module derating; thermal-cycle efficiency loss; gas-turbine output loss	Transformer derating; conductor sag reducing line rating at peak load	Cooling load up in the Tarai; heating load down in the hills; irrigation pumping up
Declining snowfall fraction	Loss of the natural seasonal reservoir; lower spring and lean-season flow	Reduced firm power; lower plant availability Dec–May	Reduced snow/ice loading (a rare positive)	Marginal reduction in hill heating load
Increased rainfall intensity	Runoff becomes episodic; less usable, more spilled	Forced shutdowns; sediment pulses; turbine abrasion	Landslide-triggered tower and pole loss; foundation scour	Outage-driven demand suppression (not a benefit)
Declining total monsoon rainfall	Lower annual runoff volume	Lower annual energy; lower plant load factor	—	—
Glacier retreat	Peak water then permanent decline in base flow	Long-run erosion of firm power; stranded-asset risk	—	—
Forest-fire severity	Loss of fuelwood biomass resource for rural households	Aerosol suppression of solar irradiance	Conductor and pole damage; corridor flashover	Loss of supply; fallback to biomass and diesel
Extreme wind and storm	—	Turbine and structure damage	Conductor galloping; tree fall on lines	—
GLOF / mass movement	Sediment aggradation renders sites unusable	Total destruction of head-works and powerhouses	Loss of corridor and access	Long-duration regional outage

Transmission of climate variability through the energy chain. Blank cells indicate no material pathway.

Transmission Line Damage

While catastrophic project-level failures attract national media attention, the more pervasive and economically costly impact of climate change on the power sector is the chronic destruction of transmission and distribution (T&D) infrastructure. Landslides, cloudbursts, and flash floods routinely sever high-voltage transmission towers, snap aerial conductors, and destroy substations located in hazard-prone valleys, cutting power supply to communities for days to weeks at a stretch.



Uttarakhand's 66/33/11 kV grid is predominantly composed of overhead lines routed through deeply incised valleys and along hill slopes where landslide susceptibility is high. Research documented in **Sustainable and Resilient Infrastructure** (Tandfonline, 2024) confirms that the geologically young, tectonically active, and fragile rock formations of the Himalayan state increase landslide incidence substantially following extreme rainfall. Every major monsoon season results in documented loss of transmission capacity: in Almora

and Haldwani, torrential rains have repeatedly destroyed both transmission lines and local distribution networks, triggering prolonged outages that compound the water crisis as pump stations and treatment plants lose power (Headlines India, 2024). The 2021 Pithoragarh cloudburst, which delivered 147 mm of rainfall in two hours, destroyed four primary health centres and disrupted electricity supply to 17 villages; analogous events cascade through transmission nodes that have no redundancy or alternative routing.

Recognising this structural vulnerability, the Asian Development Bank approved a USD 200 million loan in December 2023 for the Uttarakhand Climate Resilient Power System Development Project (ADB, 2023). A central element of the project is the replacement of exposed overhead conductors with underground cable systems in the most hazard-prone corridors. The ADB assessment explicitly notes that underground infrastructure demonstrates superior resilience to high winds, falling trees, and flying debris associated with extreme weather events, the primary failure modes of the current overhead network. This investment acknowledges, in effect, that the existing T&D grid is not designed for the climate that the state now experiences.

The transmission consequences also extend to the national grid. Because Uttarakhand feeds surplus hydropower into the Northern Regional Grid via inter-state transmission links, damage to key substations and transmission corridors in the state propagates supply shortfalls to Uttar Pradesh, Haryana, and Delhi during peak summer and post-monsoon demand periods, amplifying regional energy insecurity.

Climate Hazard	Climate Driver	Climate Stressor	Direct Impact	Functional Impact	Consequences	Confidence
Rising Temperature	Increasing average and summer temperatures	Higher cooling demand and thermal stress on electrical equipment	Reduced efficiency of transformers, substations and transmission equipment	Increased peak electricity demand and reduced system efficiency	Higher operational costs, increased outage risk and pressure on grid reliability	High
Changing Rainfall Pattern	Erratic monsoon onset, prolonged dry spells and uneven rainfall distribution	Altered river discharge and seasonal water availability	Reduced dependable generation from run-of-river hydropower projects during lean seasons	Lower reliability of electricity generation and seasonal supply variability	Reduced energy security and increased dependence on external power sources	High
Drought / Reduced Lean-season Flows	Rainfall deficit, reduced snow accumulation and declining baseflows	Reduced river discharge available for hydropower generation	Declining hydropower output and turbine operating efficiency	Reduced availability of renewable electricity during peak demand periods	Power shortages, higher generation costs and reduced energy resilience	High
Extreme Rainfall	High-intensity precipitation and cloudburst events	High sediment load, debris flow and sudden inflow surges	Damage to hydropower intakes, desilting chambers and powerhouse infrastructure	Temporary shutdown of hydropower facilities and reduced generation capacity	Service interruptions, infrastructure repair costs and generation losses	High
Landslides & Slope Instability	Intense rainfall and unstable mountain slopes	Damage to transmission towers, distribution lines and access roads	Failure of transmission corridors and restricted maintenance access	Reduced reliability of power transmission and delayed restoration	Prolonged outages in remote communities and increased maintenance costs	High
Glacier Retreat & Changing Snowmelt	Rising temperatures and declining glacier mass	Altered seasonal river flow and increasing hydrological variability	Reduced long-term hydropower potential and unpredictable inflows	Reduced reliability of hydropower planning and reservoir operations	Long-term decline in hydropower generation potential and investment uncertainty	Medium–High
High Sediment Load	Glacier melt, erosion and intense rainfall	Increased sediment transport in rivers	Turbine abrasion, accelerated equipment wear and reduced plant efficiency	Increased maintenance frequency and forced shutdowns	Higher operating costs and reduced asset lifespan	High
Forest Fires	Warmer, drier pre-monsoon conditions	Fire damage to transmission and distribution infra	Damage to poles, conductors and communication systems	Disruption of electricity transmission and field operations	Localized power outages and increased restoration costs	Medium–High
Compound Climate Hazards (Heat + Extreme Rainfall + Landslides)	Increasing frequency and convergence of multiple climate extremes	Simultaneous disruption of hydropower generation, transmission and distribution systems	Cascading failures across the electricity network and emergency response constraints	Reduced reliability and resilience of the power system	Widespread outages, economic disruption and reduced resilience of critical services	High

Cryospheric Dependence

The State hosts roughly 900 glaciers, including the Gangotri, Satopanth, Bhagirathi-Kharak, Pindari and Milam systems. These act as natural seasonal regulators, releasing meltwater that sustains base flow through the dry months, and as such they are functionally a reservoir that the sector did not have to build and does not maintain. Snowpack performs the same function on an annual cycle: it stores winter precipitation and releases it in spring, exactly when reservoir-less run-of-river plant would otherwise be starved.

Warming attacks this storage in two ways. It reduces the fraction of precipitation falling as snow, so less is stored; and it advances the timing of melt, so what is stored is released earlier. The result is a spring flow peak that arrives before demand does, and a deeper trough in the pre-monsoon months when demand driven by rising Tarai cooling load is climbing. If existing reservoirs are not designed to manage earlier increased flows, energy is simply wasted through spillover.

Over the longer run the sensitivity is non-linear. Accelerated melt produces a transient increase in flow the sector's revenue may even improve followed by permanent decline as ice volume is exhausted. This is the "peak water" trajectory, and it is the single most consequential assumption failure in the sector's planning basis, because conventional plant design is calibrated to a stationary historical hydrology and therefore cannot represent a resource that first grows and then shrinks. A project appraised on the flows of its first two decades may be financed against a hydrology that will not exist in its later operating life.

Sediment and Turbidity

Climate-driven changes in rainfall intensity, slope stability and glacial debris production directly raise sediment loads in Himalayan rivers. Sediment is a critical operational variable for hydropower, and one that rarely appears in climate-impact assessments because it is treated as an engineering problem rather than a climate one. High silt concentration causes:

- Abrasive wear of turbine runners, guide vanes and seals, particularly where the sediment contains a high quartz fraction which Himalayan sediment does;
- Shortened overhaul intervals and longer outage windows, reducing the annual plant availability factor;
- Forced generation shutdown during peak-monsoon high-turbidity episodes, precisely when flow is greatest so the plant is unable to convert the water it is designed for;
- Reservoir and desilting-basin sedimentation, progressively reducing live storage and hence the buffering capacity that would otherwise offset flow variability.

The sensitivity is acute for projects on the Alaknanda, Mandakini and Dhauliganga, where catchments are tectonically active and prone to mass movement. Its importance to this assessment is that it is a chronic revenue-erosion mechanism operating continuously, even in years with no headline disaster: it does not appear in disaster loss statistics, and is therefore systematically absent from the sector's risk accounting. Post-GLOF sediment aggradation converts this chronic mechanism into a step change, the 174.6 million cubic metres mobilised in 2021 will influence sediment loads on the Alaknanda for decades.

Network and Infrastructure

T&D assets are exposed to almost the full hazard spectrum: overhead lines on steep slopes are vulnerable to rainfall-triggered landslides; tower foundations on debris and colluvial slopes are susceptible to scour and toe erosion during flash floods; valley-floor substations are at risk from channel migration and inundation; and forest-fire incursion, intensified by drier winters and warmer pre-monsoon months, threatens corridors through pine-dominated forest. Higher temperatures independently reduce the current-

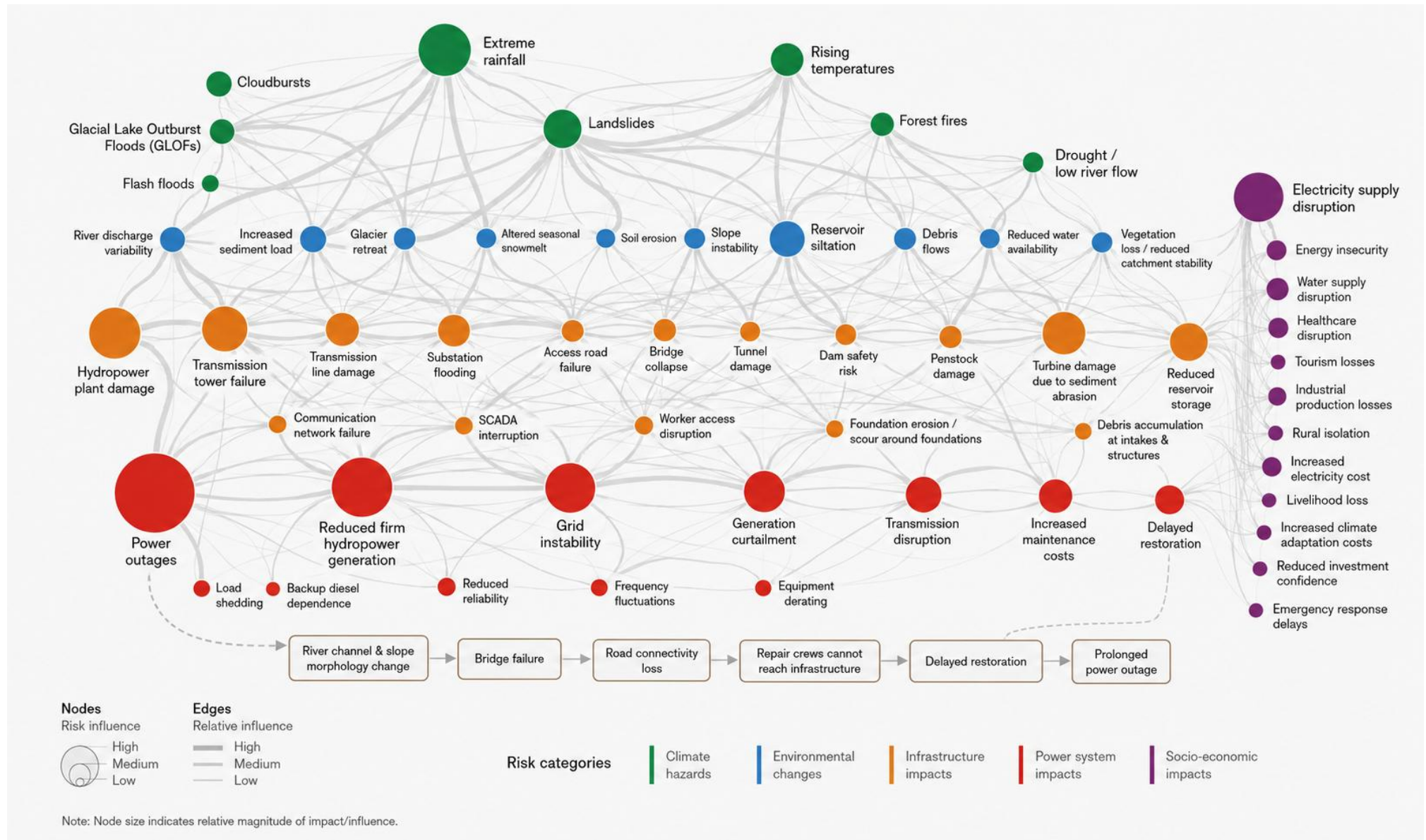
carrying capacity of conductors and the rating of transformers, an effect that bites hardest at the summer peak, when the network is most loaded.

What converts this asset-level sensitivity into service-level sensitivity is the absence of redundancy. In hill terrain the network is largely radial and series-connected: there is no alternate path, so a single failure de-energises everything downstream of it. Restoration time is then governed not by the electrical repair but by physical access, which the same hazard has typically severed. This is why outage duration in the hills is measured in days rather than hours, and why the ADB programme's emphasis on undergrounding is well targeted, it addresses the failure mode, not merely the asset.

14. Cross-Sector and Cascading Sensitivity

Climate impacts on the energy sector propagate outward, and impacts on other sectors propagate back. Three couplings are material in Uttarakhand:

- Energy → water. Loss of electricity halts pumping and treatment. The 2021 Pithoragarh cloudburst and the recurrent Almora and Haldwani outages both produced water crises as a second-order effect of power loss, at the moment when clean water was most needed.
- Water → energy. Competition for water among hydropower, irrigation, drinking supply and ecological flow intensifies as availability becomes more variable. In a drought year, hydropower is not the priority claimant, so the generation loss exceeds the hydrological loss.
- Energy → health, telecom, tourism. Health facilities, telecom towers and the tourism economy on which many hill blocks depend are all electricity-dependent, and all lose service in the same event that cuts the feeder. The vulnerability index therefore scores critical-facility dependence explicitly. The methodological point, drawn directly from the review literature, is that *ceteris paribus* analysis of a single sector has the advantage of reducing uncertainty and isolating direct impacts, but it cannot represent these couplings. They should be acknowledged qualitatively and scored where possible, rather than either ignored or spuriously quantified.



8.6 Potential Future Climate Change

Because more than 85 per cent of in-State generation is hydropower, the projected change in water, in its quantity, its phase (snow versus rain) and above all its seasonal timing, is the single most consequential projection for the sector. This section sets out the far-future (2071–2100, relative to the 1950–2014 baseline) change in the water-related variables that govern generation, under a low-emissions scenario (SSP1-2.6) and a high-emissions scenario (SSP5-8.5).

Projected Change in Water-Related Variables

Variable (far future, 2071–2100)	SSP1-2.6	SSP5-8.5	Mechanism and energy-sector relevance
Mean annual temperature (Δ°C)	() — indicative +1.0 to +1.8 °C	() — indicative +3.5 to +5.5 °C [State ensemble reports hottest-day (TXx) change of +3.9 to +4.8 °C for SSP5-8.5; confirm mean-annual value from the projections chapter]	Drives melt rate, snowline elevation, evapotranspiration and cooling demand; elevation-dependent warming means high-altitude basins warm faster than the State mean
Mean annual precipitation (Δ%)	+17 to +24 % (State ensemble, district means)	+56 to +63 % (State ensemble, district means)	Total volume rises, but the increase is concentrated in high-intensity events, so it does not translate proportionally into usable generation
Extreme-rainfall day frequency	() — moderate increase	() — large increase	More forced shutdowns, larger sediment pulses, more landslide-driven T&D failure
Snowfall fraction of precipitation	() — moderate decline	() — steep decline	Erodes the natural seasonal reservoir; lower and earlier lean-season flow
Snowline / zero-degree isotherm	() — rises	() — rises substantially	Reduces glacier accumulation area and the catchment contributing snowmelt
Glacier volume change (HKH, to 2100)	() — indicative –30 to –50 %	() — indicative –65 to –80 %	Sets the long-term lean-season base-flow floor after peak water is passed
Annual streamflow (basin mean)	() — stable to modest increase	() — higher annual volume, high interannual variance	Annual totals mislead: what matters is the flow-duration curve, not the mean
Seasonal shift in flow	() — earlier spring peak	() — pronounced earlier peak; deeper pre-monsoon trough	The critical result: generation shifts away from the period of peak demand
Lean-season (Dec–May) flow	() — modest decline	() — significant decline post peak-water	Directly reduces firm power — the binding constraint on a hydro-dominant system
GLOF frequency and lake area	() — elevated	() — substantially elevated	Hazard itself grows; exposure layer is non-stationary

Projected far-future change in water-related variables and their significance for the energy sector.

3. Health

Climate change is an evolving issue that has a profound impact on the health of individuals. The Intergovernmental Panel on Climate Change's (IPCC) recent '1.5°C Health Report' provides evidence about the global implications of climate change on human health. It further warns that temperature rise from 1.5°C to 2°C will have disastrous consequences on human health—manifesting as increase in mortalities and morbidities driven by climate related disasters, malnourishment driven by food insecurity, and increased incidence of diseases and poverty (Ebi, et al, 2018). It has been predicted that climate change in addition to causing sizable losses to the infrastructure will also have huge economic implications on the health sector, with the direct costs amounting between USD 2-4 billion per year by 2030 (WHO 2018).

The quality and delivery of health care services are predominantly governed by the adequacy of the health infrastructure and availability of trained human resources in any state. The state of Uttarakhand secured the 15th position and attained a score of 52.5 as per the NITI Aayog's Health Index, 2026. The state performed well in terms of intermediate outcomes, such as full immunization coverage, institutional deliveries, notified cases of tuberculosis, treatment success rate of new micro biologically confirmed tuberculosis, and share of people living with HIV (PLHIV) on anti-retroviral therapy (ART). Additionally, Uttarakhand also indicates progress in terms of health monitoring data integrity with respect to institutional deliveries and ANC's registered within the first trimester.

The Indian Himalayan Region is highly vulnerable to climate change, which is creating serious public health challenges, and Uttarakhand is a key part of this fragile mountain ecosystem. Since 1990, the state's average temperature has increased nearly twice as fast as the global average, driving more frequent natural disasters, retreating glaciers and unpredictable monsoon rainfall (Shekhar et al., 2023). These environmental changes act directly on health, water security and disaster risk, and indirectly through their effect on rural livelihoods, subsistence agriculture and pastoralism (Negi et al., 2017; State Agriculture Department, 2023).

Incidents such as the 2013 Kedarnath floods and the 2023 Haridwar rain-induced floods illustrate the need for better disaster preparedness and long-term health-system response strategies (SDMA, 2023; Rawat et al., 2019). Uttarakhand's health vulnerabilities stem jointly from its geography and the socioeconomic conditions of its communities: the variation in altitude from 200 to 7,000 metres creates microclimates that respond differently to warming, and rising temperatures have allowed disease vectors to invade highland areas previously unaffected.

1.5°C	300 m	1,350%	48%
Temperature rise since 1990 (Shekhar et al., 2023)	Malaria elevation shifts since 2000 (District Malaria Office, 2022)	Dengue case rise in Dehradun, 2016–2022 (The Hindu, 2022)	Districts rated highly vulnerable (IIT Mandi, 2020)

Headline indicators of climate impact on the Uttarakhand health sector.

Since 2000, malaria has moved up 300 metres in elevation, and dengue cases in Dehradun surged 1,350 per cent from 2016 to 2022 as expanding urban heat islands lengthened the vector season (The Hindu, 2022; District Malaria Office, 2022). Changes in water sources from melting glaciers and erratic rainfall have increased diarrhoeal disease by 30 per cent during monsoon seasons in Uttarkashi and Chamoli (Raina, 2009). Nearly half of Uttarakhand’s districts are rated highly vulnerable, and women, children, the elderly and indigenous communities such as the Bhotiyas face disproportionate hardship from limited healthcare access and heavy reliance on natural resources (Barua et al., 2020; Lancet Planetary Health, 2024).

The mental-health dimension of these changes is serious but under-researched: anxiety-disorder prevalence has risen from 3.2 per cent in 2011 to 11.7 per cent in 2022 in at-risk districts, alongside a 247 per cent rise in farmer suicides over the same period (National Mental Health Survey; State Agriculture Department, 2023). Traditional coping systems are disappearing rapidly, of 287 traditional healers (vaidyas) recorded in 1991, only 92 still practise, taking with them knowledge of plants such as *Swertia chirayita* for treating fever (UNESCO, 2023; Singh et al., 2025). Only 3–5 per cent of health-sector budgets for climate adaptation are directed to preparedness, leaving persistent gaps in inter-sectoral cooperation (Department of Environment & Climate Change, 2021; World Bank, 2023).

Community-based responses show that solutions are available: the Naula spring-revival programme cut waterborne disease by 61 per cent (Central Ground Water Board, 2022), even as institutional failures persist, the 2022 dengue outbreak response was delayed 42 days (District Malaria Office, 2022). With climate models projecting a 2°C temperature rise and 70 per cent glacial-mass loss by 2050 under current trajectories (IMD, 2023), Uttarakhand’s experience offers lessons for mountain health systems facing similar climate-health pressure.

The public health system follows the Government of India’s three-tier primary healthcare model, supplemented by district hospitals and referral centres.

Health facility type	Number of facilities	Number of beds
Sub-Centres (SC)	20	1,779
Primary Health Centres (PHC)	584	2,442
Community Health Centres (CHC)	81	1,819
District Hospitals	13 (one per district)	1,707
Medical Colleges (Govt.)	4	2,366
AYUSH Facilities	850	N/A
Air Ambulance Helipads (linked to SDRF)	5	N/A

Innovative Responses

The sector has seen targeted innovations with documented impact. The Apollo TeleHealth Services project connects 22 remote health sub-centres with specialists, reducing climate-related referral delays by 73 per cent (Journal of Telemedicine, 2023). MAMTA mobile clinics, equipped with portable ultrasound and automated weather alerts, reduced climate-related obstetric complications by 58 per cent in disaster-prone areas (UNFPA, 2023). An environment cell has been established within the Health Department to draft a state climate-health action plan and serve as the implementing entity for climate-health activities.

9.2 Defining Features and Their Climate Implication

Climate sensitivity in this context refers to the degree to which Uttarakhand's health sector is susceptible to harm from climate variability and change, given its inherent characteristics, independent of its capacity to adapt. A highly sensitive health system is one in which even a moderate shift in temperature, precipitation or extreme-event frequency produces significant disruption to service delivery, disease burden or population health outcomes.

Defining Feature	Description	Climate Implication
Large altitudinal gradient	Elevation ranges from approximately 200 m in the foothills to over 7,000 m in the Greater Himalaya, creating highly diverse climatic zones.	Rising temperatures enable vectors, pathogens, and climate-sensitive diseases to expand into higher elevations, exposing previously unaffected populations.
Rapid warming and changing climate	Rising temperatures, changing precipitation patterns, and increasing frequency of extreme weather events are altering local climatic conditions.	Changes in disease ecology, longer transmission seasons, increased heat stress, and changing patterns of climate-sensitive illnesses.
Mountainous terrain and dispersed settlements	Remote habitations, steep terrain, and seasonal road closures reduce physical accessibility to health facilities.	Extreme rainfall, landslides, and floods disrupt emergency referrals, supply chains, and continuity of essential healthcare services during disasters.
High exposure to hydro-meteorological hazards	Floods, landslides, cloudbursts, forest fires, heatwaves, and GLOFs occur with increasing frequency and intensity.	Increased injuries, mortality, displacement, disease outbreaks, and damage to health infrastructure during hazard events.
Predominantly rural and climate-dependent population	A large proportion of the population depends on agriculture, livestock, and natural resources for livelihoods.	Climate-induced livelihood losses contribute to malnutrition, mental health stress, reduced healthcare access, and increased vulnerability among rural communities.
Limited climate preparedness within the health system	Climate adaptation is not yet fully integrated into disease surveillance, emergency preparedness, infrastructure planning, or health service delivery.	Reduced institutional capacity to anticipate, prepare for, and respond to emerging climate-related health risks and increasing healthcare demand.

Direct and Indirect Health Impacts

The assessment of health impacts follows the IPCC risk framework, where climate hazards interact with exposure and vulnerability to determine health outcomes. Consistent with this framework, health impacts are classified as direct and indirect, reflecting whether climate hazards affect human health immediately or through intermediary environmental, ecological, or socio-economic pathways.

Direct health impacts

Direct health impacts occur as an immediate consequence of climate hazards and extreme weather events. These include mortality, injuries, and acute illnesses resulting from floods, landslides, cloudbursts, heatwaves, forest fires, and other climate-induced disasters.

In Uttarakhand, extreme rainfall events, flash floods, and landslides frequently result in fatalities, traumatic injuries, displacement, and disruption of emergency medical services. Heat stress has emerged as an additional concern, particularly in the lower-altitude districts where increasing temperatures elevate the risk of heat exhaustion, heatstroke, dehydration, and cardiovascular complications among outdoor workers, elderly populations, and other vulnerable groups.

Indirect health impacts

Indirect health impacts arise through changes in environmental, ecological, and socio-economic systems that influence disease transmission and population health over time.

Changing temperature and rainfall patterns alter the ecology of vectors and pathogens, facilitating the expansion of mosquito-borne diseases such as dengue and malaria into higher elevations where these diseases were previously uncommon. Variations in precipitation, flooding, and contamination of drinking water sources increase the incidence of waterborne diseases, including diarrhoeal diseases, cholera, hepatitis, and typhoid. Forest fires and deteriorating air quality exacerbate respiratory illnesses, while climate-induced crop losses and livelihood disruptions contribute to malnutrition, food insecurity, psychological stress, and adverse mental health outcomes.

It is important to distinguish between two forms of indirect impacts:

Indirect health impacts of climate change refer to health outcomes mediated through environmental and ecological pathways, such as vector expansion, declining water quality, reduced food security, and air pollution.

Indirect economic impacts through health refer to downstream economic consequences arising from climate-related illness, disability, reduced labour productivity, increased healthcare expenditure, and premature mortality.

Together, these direct and indirect pathways demonstrate that climate change affects not only the occurrence of disease but also the functioning of the healthcare system and the long-term resilience of communities, making climate-sensitive health planning an essential component of adaptation in Uttarakhand.

Population Vulnerability

Uttarakhand's population faces serious climate-health risk from the interaction of geography, economy and infrastructure. Remote villages on steep slopes can be isolated for months when monsoon road closures occur, while valley communities experience flooding. A 2020 IIT Mandi study (Barua et al., 2020) found 48% of districts "highly vulnerable" to climate impacts, with Rudraprayag, Chamoli and Bageshwar most affected through GLOF and landslide risk. About 70% of rural residents depend on agriculture and livestock, both of which suffer from changing weather; mandua (finger millet) yields have dropped 30–40% in hill areas over 15 years, with attendant nutrition consequences. Women face additional burden: PSI (2022) found hill women spend 4–6 more hours per day collecting water during drought, increasing physical strain and reducing capacity for childcare.

9.3 Key Climate Hazards for the Health Sector

Five hazards dominate climate risk to the Uttarakhand health sector. For each, this section sets out the physical mechanism, the stage of the health-system chain at which it acts, the populations and geographies affected, the documented historical record, and the residual data gap. Together the record establishes that climate risk to the sector is not prospective but an ongoing, measurable burden with two distinct signatures: rare, catastrophic events that destroy

health infrastructure and produce mass casualties, and a chronic, escalating burden of vector-borne and waterborne disease that never triggers a disaster declaration yet accounts for the larger share of cumulative illness.

The health-system chain runs from surveillance and early warning, through clinical diagnosis and treatment capacity, the physical infrastructure and cold-chain and supply lines that support them, and outbreak and emergency response, to longer-term rehabilitation and mental-health care. Several hazards are not independent extreme weather is the proximate trigger for waterborne outbreaks, and both feed the mental-health burden, so they are treated jointly in the composite index to avoid double-counting.

Hazard

Hazard	Mechanism of health impact	Health-system stage affected	Populations primarily affected	Principal districts
Vector-borne disease expansion	Warming extends vector survival season and elevational range; malaria and dengue establish at new altitudes	Surveillance; clinical diagnosis; treatment capacity	Whole population; emergent hill communities; fever patients	Statewide; emergent in Bageshwar, Chamoli, Champawat, Rudraprayag
Waterborne disease (flood & contamination-linked)	Flood/GLOF contaminates springs and groundwater with pathogens and heavy metals	Water-health nexus; outbreak response	Rural spring-dependent communities; children	Nainital, Chamoli, Uttarkashi; Rishiganga–Dhauliganga valleys
Extreme weather (flood / cloudburst / landslide)	Destroys health facilities, roads and cold chain; mass casualties; disrupted immunisation	Infrastructure; access; supply chain	Disaster-zone residents; project workers; facility-dependent patients	Uttarkashi, Chamoli, Rudraprayag, Pithoragarh, Bageshwar
Heat-related morbidity	Heat stroke and exhaustion in Terai labourers, pilgrims and the elderly	Emergency care; pilgrimage-season surge capacity	Terai labourers; elderly Char Dham pilgrims; outdoor workers	Haridwar, Udham Singh Nagar, lower Dehradun
Disaster-linked mental-health burden	Displacement, livelihood loss and repeated trauma drive anxiety, PTSD, depression, suicide	Mental-health service access (acutely under-resourced)	Disaster survivors; farmers; women; displaced and Bhotiya communities	Statewide; acute in disaster-affected and out-migration districts

Vector-Borne Diseases and Climatic Variability

Warming extends both the survival season and the elevational range of disease vectors. Minimum temperatures have risen 1.8°C over thirty years, creating conditions conducive to year-round vector survival (Meteorological Department, 2023). Malaria transmission has shifted approximately 300 m higher in elevation since 2000, with *Anopheles stephensi* now found in Tehri villages up to 1,800 m; in Almora the malaria season has lengthened from seven to nine months, with December cases confirmed for the first time in 2023. The overall signal is unambiguous: mosquito-borne disease incidence rose 137 per cent between 2010 and 2022,

and previously unaffected high-altitude regions now experience regular outbreaks (Varshney et al., 2024).

Why this hazard is uniquely severe for the health sector

- **It is the clearest climate signal in the State's disease record.** Range and season expansion track warming directly and are measurable at the district and elevation-band level.
- **Range expansion outruns health-system readiness.** A 2024 Pauri Garhwal study found 22 per cent of fever patients had a vector-borne disease, yet 60 per cent of doctors were untrained to identify these conditions, the burden arrives in places with no prior experience of it.
- **It is a chronic, escalating burden, not a single event.** It never triggers a disaster declaration, yet its cumulative illness load exceeds that of the catastrophic hazards in this register.

Documented impact

The State's notified-disease record over 2017-2025 captures both the control-programme gains and the emerging dengue and chikungunya threat.

Year	Malaria cases	Malaria deaths	Dengue cases	Dengue deaths	JE cases	Chikungunya
2017	508	0	849	0	0	0
2018	409	0	591	2	22	11
2019	296	0	10,622	8	9	1
2020	15	0	76	1	4	0
2021	13	0	738	2	1	1
2022	21	0	2,337	0	7	0
2023	24	0	4,320	17	0	0
2024	30	0	494	0	3	552
2025	41	0	993	0	0	71

Three trends carry particular significance. First, malaria cases declined 95.73 per cent from a 2016 baseline of 961 to just 41 in 2025, a documented control-programme success. Second, dengue exhibits explosive inter-annual variability, peaking at 10,622 cases in 2019 and 4,320 in 2023, the latter with a case-fatality ratio of 0.39 per cent, a fatality risk not previously associated with the disease in the State. Third, chikungunya, absent for several years, re-emerged with 552 cases in 2024, consistent with climate-enabled reactivation of Aedes populations at new altitudes (Varshney et al., 2024).

Dengue: the clearest climate signal in the state’s disease data

Dengue has become a major crisis, particularly in Dehradun, where average city temperature has risen 2.1°C since 1990. Cases surged from 127 (2016) to 1,843 (2022), with the peak season shifting to October–November as monsoon withdrawal delays. The 2022 outbreak

exposed system failures: hospitals operated at 180 per cent capacity, platelet shortages occurred, and transport delays caused preventable deaths (AIIMS Dehradun, 2023).

District	2019	2020	2021	2022	2023	2024	2025	Elevation zone
Dehradun	4,991	0	126	1,434	1,201	37	785	Plains, mid hill
Haridwar	708	69	561	285	690	79	110	Plains
Nainital	3,272	4	31	236	829	60	1	Plains, mid hill
Pauri Garhwal	145	1	0	196	971	302	91	Mid hill (1,200 m+)
Tehri Garhwal	115	0	0	42	25	0	0	Mid hill (1,200 m+)
U.S. Nagar	1,284	1	20	144	194	12	6	Plains, Terai
Almora	34	1	0	0	35	0	0	Mid hill (1,400 m+)
Bageshwar	3	0	0	0	147	0	0	Mid hill (1,400 m+)
Chamoli	4	0	0	0	75	0	0	High hill (1,800 m+)
Champawat	2	0	0	0	115	4	0	Mid hill (1,400 m+)
Rudraprayag	62	0	0	0	38	0	0	High hill (1,800 m+)
Pithoragarh	2	0	0	0	0	0	0	High hill (2,000 m+)
Uttarkashi	0	0	0	0	0	0	0	High hill (1,900 m+)
TOTAL	10,622	76	738	2,337	4,320	494	993	

District-wise dengue case burden, 2019–2025 (NVBDCP/DMHFW). Bold rows = first-time or climate-linked emergence in hill districts.

The 2025 entomological data confirm active *Aedes* breeding well beyond historical transmission zones: Pauri Garhwal recorded a House Index of 0.26 and a Breteau Index of 0.52, both above precautionary thresholds, validating vector establishment in mid-hill terrain. Other vectors are following. Japanese encephalitis virus has been found in mosquito pools at 1,200 m in Pauri Garhwal, far above its usual range, and scrub typhus, once rare above 1,500 m, now causes seasonal outbreaks in Chamoli with a 38 per cent case-fatality rate observed in 2022, driven by delayed diagnosis (Lancet Regional Health, 2023).

Emerging diseases

Japanese encephalitis virus has been found in mosquito pools at 1,200 m in Pauri Garhwal, far above its usual range. Scrub typhus, once rare above 1,500 m, now causes seasonal outbreaks in Chamoli with a 38 per cent case-fatality rate observed in 2022 due to delayed diagnosis (Lancet Regional Health, 2023). A 2024 Pauri Garhwal study found 22 per cent of fever patients had vector-borne disease, yet 60 per cent of doctors were untrained to identify these conditions.

Waterborne Diseases and Hydrological Change

Flood and glacial-lake outburst events contaminate springs and groundwater with pathogens and, distinctively, with heavy metals. The resource base is simultaneously being destabilised: glaciers are retreating at an average of about 20 m per year, disrupting the meltwater flow to the mountain springs that are the primary water source for around 70 per cent of rural communities (Raina, 2009), while total monsoon rainfall has fallen 12 per cent since 2000 even as extreme-rainfall events have risen 35 per cent (IMD, 2023).

Uniquely severe

It converts a water-supply loss into a chronic contamination crisis. After the 2021 Rishiganga disaster destroyed 17 water-supply schemes, subsequent groundwater tested at 3-8 times WHO arsenic limits, driving rising arsenical keratosis and bladder cancer in affected villages (CGWB, 2022), a multi-decade cancer liability generated by a single event.

It is predictable and recurrent. Flood-borne contamination follows the monsoon each year, yet is repeatedly treated as an emergency rather than an anticipated hazard.

It strikes where detection capacity is weakest. Most contamination goes unmeasured until an outbreak occurs.

Documented impact

The 2021 Nainital floods contaminated 73 springs with *E. coli*, causing 3,200 acute diarrhoea cases within two weeks; in Chamoli, paediatric hepatitis E rose 30 per cent during the monsoon, linked to sewage overflow into the Alaknanda. Beyond the acute events, microplastics have been found in 91 per cent of Bhagirathi River samples during glacial-melt periods, and the decline of the traditional naulas and dharas is itself associated with a 73 per cent higher incidence of diarrhoeal disease in villages that have lost these systems (UNESCO, 2023), tying the health burden directly to the spring-depletion hazard in the water sector.

Extreme Weather Events and Health Outcomes

Floods, cloudbursts and landslides destroy health facilities, roads and the cold chain simultaneously, produce mass casualties, and disrupt routine immunisation. The frequency is climbing sharply: hydro-meteorological disasters rose 240 per cent between 2000 and 2022 (SDMA, 2023), and cloudburst events rose from 11 in 2000-2010 to 83 in 2011-2023, causing 1,926 deaths.

Why this hazard is uniquely severe

The loss is total and simultaneous. A single event can destroy the facility, sever the access road and break the cold chain at once, then cascade into immunisation gaps and outbreaks, the 2021 Pithoragarh cloudburst destroyed four primary health centres, disrupted vaccine supply to 17 villages, and triggered a six-week diphtheria outbreak from broken immunisation continuity (cross-reference Section 10.4).

It produces a compound physical-and-mental burden. Casualty and injury load is followed by long-tail respiratory, dermatological and psychological illness.

The system is not built to withstand it. A 2022 PHFI assessment found 72 per cent of hill-district hospitals lack disaster-resilient infrastructure and only 11 per cent can maintain emergency oxygen beyond 48 hours.

Documented impact

The 2013 Kedarnath floods caused 5,748 deaths and 3,872 hospitalisations; post-event PM2.5 reached 480 µg/m³, driving a 320 per cent surge in asthma admissions, and 42 per cent of survivors developed chronic skin infections from contact with contaminated floodwater. A follow-up at AIIMS Dehradun (2016-2019) found 68 per cent of affected children showed signs of PTSD. The pattern is consistent: the physical disaster is only the first wave of the health burden.

Mental Health and Psychosocial Dimensions

Displacement, livelihood loss and repeated exposure to trauma drive anxiety, PTSD, depression and suicide. This is the sector's most chronic and least-visible pathway, acting on a service line that is acutely under-resourced across the State.

Why this hazard is uniquely severe

It is a large and fast-growing burden. Anxiety prevalence rose from 3.2 per cent (2011) to 11.7 per cent (2022) in the most climate-affected districts, farmer suicides rose 247 per cent since 2015 from climate-related crop failure, and women account for 73 per cent of climate-related depression cases.

It concentrates in specific populations. A 2022 World Bank study of 12,000 displaced families found 89 per cent of those affected by Char Dham road landslides reported sleep disorders and 62 per cent had somatic symptom disorders, while indigenous Bhotiya communities show four times the alcoholism rate following displacement from retreating glacial areas.

Service access is minimal. 78 per cent of psychiatric resources sit in Dehradun and Nainital, only three district hospitals have psychiatric units, and the National Mental Health Programme budget (₹2.7 crore per year) covers just 11 per cent of assessed need.

Documented impact and a working model

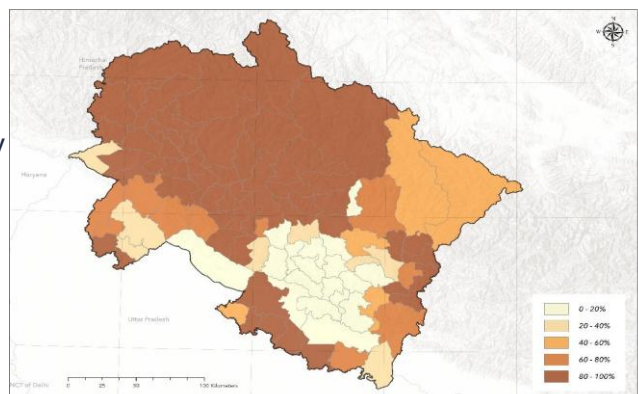
The burden is already measurable in the figures above. Against it stands one documented, low-cost, community-led response: the climate-smart AYUSH programme in Uttarkashi reduced anxiety symptoms 41 per cent using meditation and yoga, a scalable model the framework can build on. The residual data gap is that mental-health impact is largely unmeasured outside study snapshots; there is no routine post-disaster psychosocial surveillance, so the burden is almost certainly understated.

Emerging Heat-Related Health issues

Rising heat drives heat stroke and exhaustion among Terai agricultural labourers, pilgrims and the elderly. The IMD has recorded a statistically significant increase in days exceeding 40°C in Haridwar and Udham Singh Nagar since 2000, and heat-stroke admissions at Haridwar district hospital rose approximately 34 per cent over 2010-2023.

Why this hazard is distinctive for Uttarakhand

It collides with the pilgrimage calendar. The Char Dham Yatra season (April-June) now coincides with peak heat stress at the lower-altitude pilgrimage entry points, exposing elderly pilgrims to dangerous heat before they reach higher, cooler altitudes, a surge-capacity problem unique to the State's pilgrimage economy (cross-reference the roads and tourism sectors).



It is an occupational hazard in the plains. Heat-related morbidity is rising among Terai agricultural labourers, with the NCRB recording increasing heat-related deaths since 2015.

Chronology of Past Impacts on the Health Sector

Year	Event	Documented health-sector consequence	Lesson carried into the framework
2013	Kedarnath floods	5,748 deaths; 3,872 hospitalisations; 320% asthma-admission surge; 68% child PTSD signs	Compound disasters produce cascading physical and mental-health burden
2016-2022	Dengue surge, Dehradun	Cases 127 → 1,843; hospitals at 180% capacity in 2022 outbreak	Urban heat islands accelerate vector-borne disease independent of statewide trend
2018-2022	Malaria altitude shift	P. vivax +400% in Almora; malaria season 7→9 months	Vector range expansion is measurable and elevation-specific
2019 / 2023	Dengue emergence in hill districts	Bageshwar, Chamoli, Champawat, Rudrapur: first-ever indigenous transmission	Clearer single climate signal in the state disease record
2021 (Feb)	Rishiganga/Chamoli GLOF	17 water schemes destroyed; arsenic 3–8× WHO limits; rising arsenical keratosis, bladder cancer	GLOF converts a water-supply loss into a chronic contamination health crisis
2021	Nainital floods	73 springs contaminated; 3,200 diarrhoea cases in two weeks; Chamoli paediatric Hepatitis E +30%	Flood-borne contamination is a predictable, recurrent health hazard
2021	Pithoragarh cloudburst	4 PHCs destroyed; 17-village vaccine disruption; 6-week diphtheria outbreak	Infrastructure loss cascades into immunisation and outbreak risk
2022	Dengue outbreak, coordination failure	42-day delay between case detection and response	Parallel early-warning systems without integration cost critical response time
2011–2022	Rising mental-health burden	Anxiety 3.2%→11.7%; farmer suicides +247%	Mental health is a major, under-resourced climate-health pathway

Chronology of documented climate-hazard impacts on the Uttarakhand health sector.

9.4 Exposure Assessment

Exposure is assessed by intersecting hazard-susceptibility layers with the geo-referenced health-facility network and population. Four hazards are carried into the spatial overlay, vector-borne disease risk, waterborne/flood-linked disease risk, extreme weather (landslide/cloudburst/flood affecting facility access), and heat-related morbidity risk, because susceptibility layers or documented incidence for these are available at a resolution consistent with block-level reporting. Mental-health burden is treated on the sensitivity side (Section 11.6) because it is driven by cumulative disaster exposure rather than a discrete spatial hazard layer.

Data Layers

Hazard layer	Source and basis	Classification	Resolution
Vector-borne disease risk	NVBDCP/DMHFW case surveillance; entomological indices (HI/CI/BI); elevation & minimum-temperature thresholds (Mordecai et al., 2017)	5 classes (Very Low – Very High)	District, disaggregated to block by elevation
Waterborne/flood-linked disease risk	IDSP outbreak record; spring-dependence; NRSC flood hazard atlas	5 classes	Block
Extreme weather (facility/access)	SDMA disaster record; GSI/BMTPC landslide susceptibility; road-network hazard overlay	5 classes	30 m raster / block
Heat-related morbidity risk	IMD temperature series; THI thresholds; Char Dham pilgrimage-season overlay	5 classes	Block
Health assets	DMHFW facility register (PHC/CHC/District Hospital); cold-chain inventory; ASHA deployment	Point vector	Facility-level

Data layers used in the exposure overlay.

Exposure Metrics

Asset/population class	Exposure metric	Unit reported
PHC/CHC/District Hospital network	Facilities within high/very-high extreme-weather or landslide-access classes ÷ total facilities in block	% of facilities exposed
Population at vector-borne-disease risk	Population in blocks crossing the effective vector-establishment elevation/temperature threshold ÷ total block population	% of population exposed
Population on climate-exposed water sources	Population served by flood/GLOF-exposed springs or schemes ÷ total block population	% of population exposed
Heat-exposed population (Terai/pilgrimage)	Population/pilgrim-season inflow in blocks exceeding THI or Tmax thresholds ÷ total	% of population/season exposed

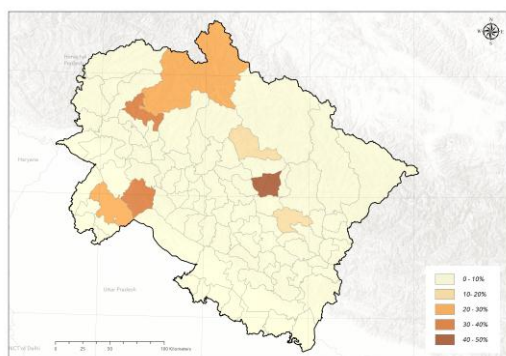
Exposure metrics by asset/population class. “Exposed” means intersecting the high or very-high class of the hazard layer.

Classification Bands

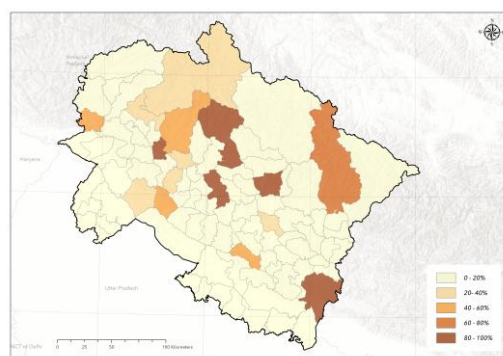
Class	Range (% of asset/population exposed)	Interpretation
Very Low	0 – 10 %	Marginal exposure; standard practice adequate
Low	>10 – 25 %	Localised exposure; targeted facility/surveillance strengthening

Moderate	>25 – 50 %	Material exposure; block-level preparedness required
High	>50 – 75 %	Systemic exposure; redundancy, cold-chain and surveillance priority
Very High	>75 – 100 %	Critical exposure; the health system is effectively co-located with the hazard

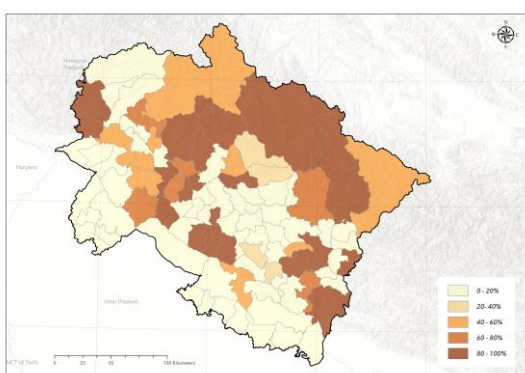
Exposure classification bands and the score used in the composite.



Hospitals exposed to flood



Hospitals exposed to Extreme Rainfall



Hospitals exposed to Landslide

Composite Exposure Formula

The block composite is the weighted mean of the four hazard-specific exposure scores. Weights reflect the relative contribution of each hazard to documented health-sector burden in the State, with extreme weather weighted highest because it directly destroys facilities and severs access, followed closely by vector-borne disease given its scale and rate of geographic expansion:

$$E(\text{block}) = 0.30 \cdot S(\text{Extreme Rainfall}) + 0.35 \cdot S(\text{Floods}) + 0.35 \cdot S(\text{Landslides})$$

Block-Wise Exposure of Health Infrastructure and Population

District	Block	Extreme Rainfall	Floods	Landslide	Composite exposure
Uttarkashi	Bhatwari	Very High	Moderate	High	High

	Dunda	High	Moderate	Moderate	Moderate
	Chinyalisaur	High	Moderate	Moderate	Moderate
	Naugaon	High	Moderate	Moderate	Moderate
	Purola	Moderate	Low	Moderate	Low
	Mori	High	Low	High	Moderate
Chamoli	Dasholi	Very High	Very High	High	Very High
	Joshimath	Very High	Very High	Very High	Very High
	Ghat	Very High	Very High	High	Very High
	Karnaprayag	High	High	High	High
	Dewal	High	Moderate	Moderate	Moderate
	Tharali	High	Moderate	High	Moderate
	Narayanbagar	High	Moderate	Moderate	Moderate
	Pokhri	High	Moderate	Moderate	Moderate
	Gairsain	Moderate	Low	Moderate	Low
Rudraprayag	Ukhimath	Very High	Very High	High	Very High
	Agastyamuni	High	High	High	High
	Jakholi	High	Moderate	High	Moderate
Tehri Garhwal	Chamba	High	Moderate	Moderate	Moderate
	Jaunpur	High	Low	Moderate	Moderate
	Kirtinagar	High	Moderate	Moderate	Moderate
	Narendranagar	High	Moderate	Moderate	Moderate
	Pratapnagar	High	Moderate	High	Moderate
	Devprayag	High	Moderate	Moderate	Moderate
	Jakhnidhar	High	Low	Moderate	Moderate
	Bhilangana	Very High	Moderate	High	High
	Thauldhar	High	Low	Moderate	Moderate
Dehradun	Chakrata	High	Low	Moderate	Moderate
	Kalsi	High	Moderate	Moderate	Moderate
	Vikasnagar	Moderate	High	High	High
	Sahaspur	Low	High	Moderate	Moderate
	Raipur	Moderate	High	High	High
	Doiwala	Low	High	High	High
Pauri Garhwal	Kot	High	Moderate	Moderate	Moderate
	Kaljikhali	High	Moderate	Moderate	Moderate
	Pauri	High	High	Moderate	High
	Pabau	High	Moderate	Moderate	Moderate
	Thalisain	High	Moderate	Moderate	Moderate
	Bironkhal	High	Moderate	Moderate	Moderate
	Dwarikhali	High	Low	Moderate	Moderate
	Dugadda	High	High	Moderate	High
	Jaharikhali	High	Moderate	Moderate	Moderate
	Rikhnikhali	Moderate	Low	Moderate	Low
	Yamkeshwar	High	Moderate	Moderate	Moderate
	Ekeswar	Moderate	Low	Moderate	Low
	Khirsu	Moderate	Low	Moderate	Low
	Nainidanda	High	Moderate	Moderate	Moderate
	Pokhra	High	Moderate	Moderate	Moderate
Nainital	Betalghat	High	High	Moderate	High
	Bhimtal	High	High	Moderate	High

	Dhari	High	Moderate	Moderate	Moderate
	Kotabag	Moderate	High	Moderate	High
	Okhalkanda	High	Moderate	Moderate	Moderate
	Ramgarh	High	Moderate	Moderate	Moderate
	Ramnagar	Low	High	Moderate	Moderate
	Haldwani	Low	Very High	High	High
Almora	Hawalbagh	High	High	Moderate	High
	Takula	High	Moderate	Moderate	Moderate
	Lamgara	High	Moderate	Moderate	Moderate
	Dhauladevi	High	Moderate	Moderate	Moderate
	Bhaisiyachhana	High	Moderate	Moderate	Moderate
	Bhikiyasain	High	Moderate	Moderate	Moderate
	Sult	High	Moderate	Moderate	Moderate
	Chaukhutia	High	Moderate	Moderate	Moderate
	Dwarahat	High	Moderate	Moderate	Moderate
	Syaldey	Moderate	Low	Moderate	Low
	Tarikheth	High	Moderate	Moderate	Moderate
Bageshwar	Bageshwar	Very High	Very High	High	Very High
	Kapkot	Very High	Very High	High	Very High
	Garur	High	High	High	High
Pithoragarh	Pithoragarh	High	Moderate	High	Moderate
	Munakot	High	Moderate	High	Moderate
	Gangolihat	High	Low	Moderate	Moderate
	Berinag	High	Moderate	High	Moderate
	Didihat	High	Moderate	High	Moderate
	Kanalichhina	High	Moderate	High	Moderate
	Dharchula	Very High	Low	Very High	High
	Munsyari	Very High	Low	Very High	High
Champawat	Champawat	High	Very High	Moderate	High
	Lohaghat	High	High	Moderate	High
	Pati	High	High	High	High
	Barakot	Moderate	Moderate	High	Moderate
Haridwar	Bahadrabad	Moderate	Moderate	High	High
	Roorkee	Moderate	Moderate	High	High
	Narsan	Low	Low	Moderate	Low
	Bhagwanpur	Low	Low	Moderate	Low
	Laksar	High	Moderate	Very High	High
	Khanpur	Moderate	Moderate	High	High
Udham Singh Nagar	Kashipur	Moderate	Moderate	Moderate	Moderate
	Jasipur	Moderate	Low	Moderate	Moderate
	Bazpur	Moderate	Low	Moderate	Moderate
	Gadarpur	High	Moderate	High	High
	Rudrapur	Moderate	Moderate	Moderate	Moderate
	Sitarganj	High	Moderate	High	High
	Khatima	High	Moderate	High	High

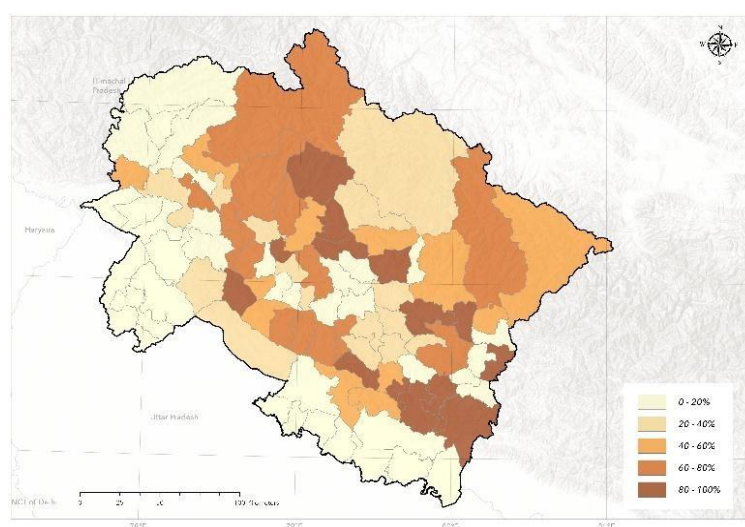
Priority Blocks

The Very High composite blocks combine Very High extreme-weather exposure with Very High vector-borne disease exposure and High waterborne-disease burden, concentrated in the high-Himalayan valleys of Chamoli, Rudraprayag and Bageshwar with documented GLOF, flood and dengue-emergence history. Dharchula and Munsyari carry the highest extreme-weather and waterborne-disease exposure in the state but currently lower documented vector-borne burden, placing them in the High rather than Very High composite class; this should be revisited once entomological surveillance is extended to these remote border blocks, as vector establishment may currently be under-detected rather than genuinely absent.

Block	District	Exposure driver	Health systems at stake
Joshimath	Chamoli	Extreme weather, vector-borne & waterborne disease all Very High	NH-7 corridor referral chain; valley PHCs/CHC
Dasholi	Chamoli	Extreme weather & vector-borne disease Very High; waterborne High	Alaknanda-corridor facilities
Ghat	Chamoli	Extreme weather & vector-borne disease Very High; waterborne High	District referral access
Ukhimath	Rudraprayag	Extreme weather & vector-borne disease Very High; waterborne High	Kedarnath-route PHCs; pilgrimage-season surge
Bageshwar	Bageshwar	Extreme weather & vector-borne disease Very High (dengue emergence signature district)	District hospital surge capacity
Kapkot	Bageshwar	Extreme weather & vector-borne disease Very High; waterborne High	Remote Pindari-catchment facilities
Dharchula / Munsyari	Pithoragarh	Extreme weather & waterborne disease Very High; remote, low redundancy (High composite)	Border-area facilities with minimal backup

Very-High and highest-High composite-exposure blocks and their health systems at stake.

Accessibility Disruption Mapping



Accessibility disruption, the severing of physical connections between patients and care, is treated as a cross-cutting impact across all pathways above rather than a discrete hazard, because it is the mechanism through which landslide, flood and cloudburst events convert into health outcomes: restoration time in hill terrain is frequently governed by road clearance rather than by

clinical capacity itself.

District-wise health-facility exposure

Aggregating the five layers to district level yields the following exposure profile. The hill districts return High composite exposure driven by the compounding of landslide, flood and access disruption; the plains districts return Low composite exposure on the facility hazards but carry the State's Very High heat exposure and its largest dependent populations - a distinction the composite alone does not capture and that is treated under population and service exposure below.

District	Flood	Landslide	Forest fire	Heat	Access disruption	Composite
Uttarkashi	Moderate	Very High	Moderate	Low	Very High	High
Chamoli	High	Very High	Moderate	Low	Very High	High
Rudrapur	High	Very High	Moderate	Low	High	High
Tehri Garhwal	Moderate	High	High	Moderate	High	High
Dehradun	High	Moderate	High	High	Low	Moderate
Pauri Garhwal	Low	High	Very High	Moderate	High	High
Haridwar	High	Very Low	Very Low	Very High	Low	Low
Nainital	Moderate	High	Very High	Moderate	Moderate	High
Almora	Low	High	Very High	Moderate	High	High
Bageshwar	High	High	High	Low	Very High	High
Pithoragarh	Moderate	High	High	Low	Very High	High
Champawat	Low	High	Very High	Moderate	High	High
Udham Singh Nagar	High	Very Low	Very Low	Very High	Low	Low

9.5 Climate Variability Impacting the Health Sector

Climate change impacts on human health in Uttarakhand are categorised into direct and indirect pathways, whether the effect of a climate risk is immediate or unfolds over time. This section examines the major health-impact domains with supporting surveillance data, then identifies where variability is already binding.

Observed Climate Variability and Trends

Climate variable	Observed change	Consequence for the health sector
Minimum temperature	+1.8°C over 30 years (Meteorological Dept., 2023)	Year-round vector survival conditions; extended transmission season
Mosquito-borne disease incidence	+137% (2010–2022) (Varshney et al., 2024)	Regular outbreaks in previously unaffected high-altitude regions
Malaria elevation range	+300 m since 2000	Anopheles established at 1,800 m in Tehri; P. vivax +400% in Almora (2018–2022)
Dengue geographic range	First transmission in 4 hill districts, 2023	Aedes established well above historical 1,000 m limit
Monsoon rainfall	–12% total; extreme events +35% since 2000	Less reliable water source; more flood-borne contamination events
Hydro-met disaster frequency	+240% between 2000 and 2022 (SDMA, 2023)	Recurrent facility destruction, mass casualty, immunisation disruption
Days >40°C (Terai)	Statistically significant increase since 2000 (IMD)	Rising heat-stroke admissions; pilgrimage-season risk
Anxiety/PTSD prevalence	3.2% (2011) → 11.7% (2022) in vulnerable districts	Chronic, under-resourced mental-health burden

Observed climate variability in Uttarakhand and its health-sector consequence.

Climate Variable → Health Service Chain Transmission Matrix

The matrix traces each climate variable through every stage of the health-service chain. Reading down a column shows what a single stage is exposed to; reading across a row shows how a single variable propagates.

Climate variable	Environmental determinant	Disease burden	Service delivery	Access and outcome
Rising air temperature and heat extremes	Higher indoor and outdoor thermal load; reduced water availability for hygiene; food spoilage	Heat exhaustion and heat stroke; excess cardiovascular, cerebrovascular and renal events; adverse pregnancy outcomes; heat rash and dehydration in infants	Cold-chain excursion risk for vaccines and insulin; pharmaceutical instability; heat stress on staff in unconditioned facilities; surge in outpatient load	Occupational income loss during treatment; presentation delayed to evening hours; excess mortality concentrated in the elderly and the chronically ill
Extreme rainfall and flooding	Faecal contamination of springs, handpumps and piped supply; standing water; damage to sanitation infrastructure	Diarrhoeal disease, cholera, typhoid, hepatitis A and E; leptospirosis; snakebite in the Tarai; drowning and trauma	Facility inundation; loss of power, water and oxygen supply; damage to drug and vaccine stores; loss of records	Referral roads severed at the moment of peak demand; maternal and trauma emergencies unable to reach the referral centre within any clinically meaningful window

Landslides and slope failure	Physical destruction of settlement and service infrastructure; displacement into temporary shelter	Crush and blunt trauma; wound infection; post-traumatic stress; disease transmission in congregate shelter	Loss of facility buildings and staff quarters; interruption of supply and referral routes for weeks	Isolation of entire villages; total loss of access rather than degraded access; care-seeking abandoned
Declining monsoon rainfall and drought	Spring and stream yield decline; longer water-collection distances; reduced water for hygiene	Water-washed disease (skin, eye and enteric infection); dehydration; musculoskeletal injury from head-loading	Facility water insecurity compromising infection prevention, dialysis and labour-room hygiene	Collection burden borne overwhelmingly by women and girls, displacing school attendance, antenatal visits and immunisation
Rising temperature at altitude (vector pathway)	Upward extension of thermally suitable altitude for <i>Aedes aegypti</i> , <i>Aedes albopictus</i> and <i>Anopheles</i> spp.	Introduction of dengue, chikungunya and malaria into hill populations with no acquired immunity; scrub typhus following post-monsoon mite activity	No established vector surveillance, entomological capacity or larvicidal programme above the historical transmission limit	Late clinical recognition and misdiagnosis in newly affected districts; higher case-fatality on first exposure
Forest-fire severity and air quality	Acute PM _{2.5} , PM ₁₀ and black-carbon loading during the fire season; winter inversion in valley towns	Asthma and COPD exacerbation; acute respiratory infection in children; ischaemic events; low birth weight on chronic exposure	Surge in respiratory outpatient and emergency load; oxygen and nebulisation demand; occupational exposure of fire crews	Chronic burden accumulating outside any surveillance system; no exposure-linked health advisory mechanism in place
Agricultural and nutritional pathway	Crop failure, livestock loss and dietary diversity contraction (see agriculture chapter)	Undernutrition, anaemia and micronutrient deficiency; increased infection susceptibility; impaired child growth	Nutrition rehabilitation and supplementation demand rising against a fixed delivery platform	Intergenerational transmission of vulnerability; nutritional status as a multiplier on every other pathway above
Compound and sequenced hazards	Simultaneous water, thermal, air and physical stress within a single season	Overlapping communicable and non-communicable burden rather than sequential episodes	Response capacity consumed by the first event and unavailable for the second	Non-linear escalation of mortality; catastrophic out-of-pocket expenditure; distress migration as a health outcome

Transmission of climate variability through the health-service chain. Blank cells indicate no material pathway.

Impact Pathway for the Health Sector

Climate Hazard	Climate Driver	Climate Stressor	Direct Impact	Functional Impact	Consequences	Confidence
Rising Temperature	Increasing average and maximum temperatures and more frequent heatwaves	Heat stress, warmer nights and prolonged exposure to high temperatures	Increased cases of heat-related illnesses, dehydration and exacerbation of cardiovascular and respiratory diseases	Increased demand for emergency care, outpatient services and hospital admissions	Higher heat-related morbidity and mortality, particularly among elderly, children, outdoor workers and vulnerable populations	High
Changing Rainfall Pattern	Erratic monsoon onset and withdrawal and uneven rainfall distribution	Seasonal water scarcity, declining water quality and sanitation challenges	Increased incidence of diarrhoeal diseases, dehydration and nutritional stress	Increased burden on primary healthcare facilities and public health programmes	Higher disease burden, especially among children and rural communities	High
Extreme Rainfall / Cloudbursts	High-intensity precipitation events and cloudbursts	Flooding of settlements, contamination of drinking water sources and localized infrastructure damage	Outbreaks of water-borne diseases and injuries associated with extreme weather events	Increased emergency response requirements and surge in healthcare demand	Increased morbidity, localized mortality and pressure on district health systems	High
Flooding / Flash Floods	River overflow and extreme runoff	Damage to health facilities, roads, utilities and communication networks	Disruption of hospitals, Primary Health Centres (PHCs), ambulance services and medical supply chains	Reduced accessibility and continuity of essential health services	Delayed treatment, increased mortality risk and interruption of maternal, child and emergency healthcare services	High
Landslides & Slope Instability	Intense or prolonged rainfall and unstable mountain slopes	Physical isolation of remote settlements and damage to	Increased trauma cases and disruption of referral transport systems	Reduced access to healthcare facilities and delayed	Increased preventable deaths, prolonged recovery and widening health inequities in	High

		transport infrastructure		emergency medical response	remote mountain communities	
Vector-borne Disease Expansion	Rising temperatures and changing rainfall regimes	Expansion of suitable habitats and longer breeding seasons for disease vectors	Increased transmission of dengue, malaria, scrub typhus and other vector-borne diseases into higher elevations	Increased disease surveillance, diagnostics and vector-control requirements	Expansion of disease burden into previously unaffected areas and increased public health expenditure	High
Forest Fires	Warmer, drier pre-monsoon conditions	Smoke, particulate matter (PM ₂₋₅) and reduced air quality	Increased respiratory illness, asthma exacerbation and cardiovascular complications	Increased hospital admissions and respiratory healthcare demand	Higher respiratory morbidity among vulnerable populations and increased healthcare costs	Medium-High
Cold Waves & Temperature Extremes	Greater seasonal temperature variability and cold spells at higher elevations	Cold stress and prolonged exposure to low temperatures	Increased hypothermia and worsening of cardiovascular and respiratory illnesses	Increased seasonal healthcare demand and pressure on health services	Increased winter morbidity and mortality among elderly and vulnerable populations	Medium
Climate-induced Ecosystem Change	Altered temperature and rainfall regimes affecting ecosystems and wildlife	Changing distribution of disease vectors, reservoirs and zoonotic pathogens	Increased risk of zoonotic disease emergence and altered infectious disease patterns	Enhanced surveillance, laboratory capacity and One Health coordination requirements	Increased public health risk and greater complexity of disease management	Medium
Compound Climate Hazards (Heat + Extreme Rainfall + Floods + Landslides)	Increasing frequency and convergence of multiple climate extremes	Simultaneous disruption of healthcare infrastructure, workforce, supply chains and emergency response	Reduced continuity of essential health services during disaster events	System-wide stress on the health system and reduced emergency response capacity	Increased mortality, prolonged service disruption, mental health impacts and reduced health system resilience	High

Indigenous Knowledge Systems and Traditional Healthcare

Traditional healthcare offers effective, ecologically grounded and rapidly eroding solutions. A 2023 UNESCO survey documented 217 medicinal plants used by traditional healers for climate-linked health conditions; Singh et al. (2021) found *Swertia chirayita* and *Aconitum heterophyllum* reduce insect-borne fever symptoms by 73 per cent, *Berberis aristata* and neem perform comparably to antibiotics for post-flood skin infections, and *Ocimum sanctum* and *Rhododendron arboreum* improved climate-related respiratory conditions by 68 per cent. The People's Biodiversity Register records active vaidyas falling from 287 (1991) to 92 today; of 217 medicinal plant species, 34 per cent face extinction risk (19% overharvesting, 42% habitat loss, 27% climate-driven habitat shift). Only 6 per cent of the state health budget supports traditional medicine. The draft Uttarakhand Traditional Healthcare Policy (2023) proposes climate-adapted cultivation of medicinal plants, digital documentation of traditional knowledge, and integration into the National Health Mission.

Health Infrastructure and Service Delivery: Increasing Disruption of Essential Health Services

Health infrastructure represents the backbone of climate-resilient healthcare delivery. Uttarakhand's health system is characterized by a network of Sub-Centres, Primary Health Centres (PHCs), Community Health Centres (CHCs), district hospitals, and tertiary referral institutions distributed across highly inaccessible mountainous terrain. While this decentralized network improves geographical coverage, it is also highly exposed to climate-induced disruptions.

Extreme rainfall, floods, landslides, and road failures increasingly interrupt access to health facilities, delay emergency referrals, and disrupt the transport of medicines, vaccines, blood products, oxygen supplies, and essential medical equipment. During major disasters, healthcare facilities located along river valleys or unstable slopes may themselves become inaccessible or sustain physical damage, while electricity and communication failures further reduce service continuity. Helicopter evacuations often become the only means of referral for remote mountain communities, significantly increasing response costs and treatment delays.

Projected increases in extreme rainfall intensity, compound flood-landslide events, and prolonged road closures are expected to increase healthcare service disruptions, particularly in upper Himalayan districts where single-road connectivity and long travel times already constrain access to emergency care.

Vector-Borne Diseases: Expansion into Higher Elevations

Climate warming is fundamentally altering the ecology of disease vectors across Uttarakhand. Temperature, humidity, and rainfall strongly influence mosquito survival, breeding cycles, biting rates, and pathogen development. Historically, diseases such as dengue and malaria were largely confined to the Terai and lower Himalayan foothills. Rising temperatures and warmer winters are gradually shifting the climatic suitability for *Aedes* and *Anopheles* mosquitoes toward mid-altitude settlements.

Increasing minimum temperatures extend vector survival through winter months, while longer monsoon periods create persistent breeding habitats in urban settlements, construction sites, water storage containers, and peri-urban environments. Intense rainfall events initially flush

breeding habitats but are followed by extensive water stagnation, increasing mosquito proliferation.

Urban centres such as Dehradun, Haridwar, Haldwani, Kotdwar and Srinagar are expected to remain major hotspots because of high population density, rapid urbanisation, and inadequate stormwater drainage. Simultaneously, warming is enabling climate-sensitive vectors to colonize higher elevations that historically experienced temperatures below transmission thresholds, increasing disease risk for populations with relatively low prior exposure.

Future climate projections indicate that both the spatial extent and duration of transmission seasons are likely to increase under continued warming, requiring disease surveillance systems to expand beyond traditionally endemic districts.

Water-Borne Diseases: Increasing Risk from Flooding and Water Contamination

Safe drinking water forms one of the most climate-sensitive determinants of public health in Uttarakhand. The state relies heavily on springs, small streams, gravity-fed water supply systems, and localized surface water sources that are increasingly affected by changing rainfall patterns and extreme weather.

Extreme rainfall and flash floods damage drinking-water infrastructure, contaminate springs and piped water systems, and increase the transport of sediments, sewage, pathogens, and animal waste into water supplies. Simultaneously, droughts reduce spring discharge and concentrate contaminants within shrinking water sources, further compromising water quality.

These processes increase the likelihood of diarrhoeal diseases, gastroenteritis, cholera, hepatitis, typhoid, and other enteric infections, particularly in rural settlements with limited treatment facilities. Temporary shelters established following floods and landslides often experience overcrowding and inadequate sanitation, further increasing outbreak potential.

Climate change therefore affects both the availability and quality of drinking water, making integrated water-health surveillance increasingly important for protecting public health.

Heat Stress and Non-Communicable Diseases

Although Uttarakhand has traditionally experienced a temperate mountain climate, increasing temperatures are creating new public health challenges, particularly across the Terai and lower hill districts.

Heat exposure is increasing the incidence of heat exhaustion, heatstroke, dehydration, acute kidney injury, cardiovascular disease, respiratory illnesses, and complications among individuals with diabetes and hypertension. Outdoor workers, agricultural labourers, construction workers, street vendors, elderly populations, children, and pregnant women are particularly vulnerable because of prolonged occupational exposure and limited cooling opportunities.

Urban centres such as Haridwar, Dehradun, Haldwani, Kashipur and Rudrapur are expected to experience increasing urban heat island effects due to expanding built-up areas, reduced vegetation cover, and growing energy demand. Rising nighttime temperatures also reduce physiological recovery during heat events, increasing cumulative health risks.

Projected increases in both average temperatures and the frequency of extreme heat events suggest that heat-related illnesses will become an increasingly important component of the state's future disease burden.

Nutrition, Food Security and Maternal: Child Health

Climate change influences health not only through disease but also through food systems and nutrition. Declining agricultural productivity resulting from droughts, erratic rainfall, pest outbreaks, and crop failures directly affects food availability and household incomes, particularly among rural farming communities.

Reduced dietary diversity, declining livestock productivity, and increasing food prices disproportionately affect women, children, elderly populations, and economically vulnerable households. Nutritional deficiencies increase susceptibility to infectious diseases, reduce childhood growth, and compromise maternal health outcomes.

Extreme weather events also disrupt antenatal services, routine immunization programmes, and nutritional supplementation delivered through Anganwadi Centres and frontline health workers. Damage to roads and communication infrastructure delays outreach services precisely when vulnerable populations require them most.

Climate-related food insecurity therefore represents an important indirect pathway through which climate change affects long-term public health and human development.

Mental Health and Psychosocial Impacts

The mental health consequences of climate change remain one of the least documented yet potentially significant dimensions of vulnerability in Uttarakhand.

Floods, landslides, forest fires, and displacement expose affected populations to trauma, bereavement, livelihood loss, housing insecurity, and prolonged uncertainty. Families experiencing repeated climate disasters often face chronic stress, anxiety, depression, post-traumatic stress disorder (PTSD), and declining psychological well-being.

Mountain communities dependent on agriculture and natural resources are particularly vulnerable because repeated climate shocks compound existing economic insecurity and increase distress migration. Frontline responders and healthcare workers also experience increasing occupational stress during prolonged emergency response operations.

As climate extremes become more frequent, integrating psychosocial support and community-based mental health services into disaster preparedness and primary healthcare systems will become increasingly important.

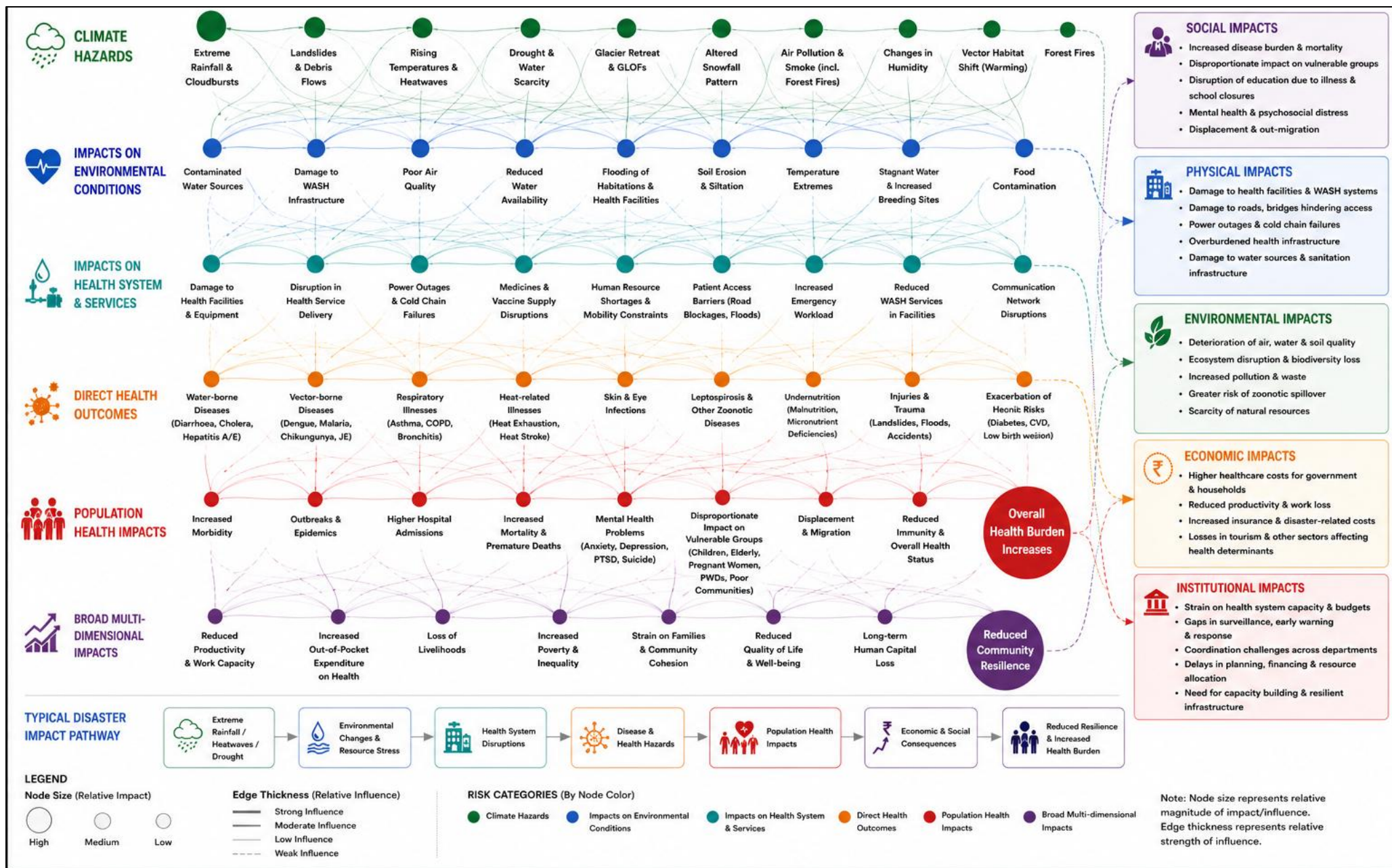
Health Workforce and Emergency Response Capacity

Climate change is increasing pressure not only on disease burden but also on the capacity of the health system itself. Extreme weather events generate sudden surges in healthcare demand while simultaneously reducing the availability of health personnel, transport, communication, electricity, and essential medical supplies.

Repeated disasters increase workloads for doctors, nurses, ASHA workers, ANMs, ambulance services, and emergency response teams. Access to remote villages becomes increasingly

difficult during landslides, floods, and snowfall, reducing continuity of routine healthcare delivery alongside emergency care.

Future climate scenarios indicate increasing requirements for climate-informed disease surveillance, mobile health services, telemedicine, early warning systems, emergency stockpiles, resilient healthcare infrastructure, and integrated disaster-health preparedness. Strengthening institutional preparedness will therefore be as important as reducing population exposure in building climate resilience within Uttarakhand's health sector.



9.5 Potential Future Climate Change

The health sector is influenced by climate through multiple pathways involving heat exposure, air quality, water availability, vector ecology, food systems, and the frequency of extreme weather events. Unlike infrastructure sectors, health outcomes respond not only to changes in mean climatic conditions but also to increasing climate variability and the occurrence of extreme events. Rising temperatures, changing precipitation regimes, increasing humidity, and more frequent climate extremes are projected to alter disease transmission dynamics, heat-related morbidity and mortality, nutritional outcomes, mental health, and the functioning of healthcare systems.

This section summarises the projected far-future climate (2071–2100 relative to the 1950–2014 baseline) under SSP1-2.6 and SSP5-8.5, highlighting those variables that are expected to have the greatest implications for public health in Uttarakhand.

Variable (Far Future, 2071–2100)	SSP1-2.6	SSP5-8.5	Mechanism and Health-sector Relevance
Mean annual temperature ($\Delta^{\circ}\text{C}$)	() – indicative +1.0 to +1.8 $^{\circ}\text{C}$	() – indicative +3.5 to +5.5 $^{\circ}\text{C}$	Increases heat stress, heat-related morbidity and mortality, reduces labour productivity, and elevates cooling requirements for healthcare facilities.
Frequency of hot days and warm nights	Moderate increase	Large increase	Extends heat exposure duration, increases cardiovascular and respiratory illnesses, and elevates mortality among elderly and vulnerable populations.
Mean annual precipitation ($\Delta\%$)	+17–24%	+56–63%	Alters water availability, sanitation conditions, and pathogen transport while increasing flood-related health risks.
Extreme rainfall events	Moderate increase	Large increase	Increases flooding, contamination of drinking-water sources, disruption of sanitation systems, injuries, and outbreaks of water-borne diseases.
Consecutive dry days / drought occurrence	Slight increase	Moderate increase	Reduces drinking-water availability, increases hygiene-related illnesses, nutritional stress, and mental health impacts associated with livelihood losses.
Relative humidity	Slight increase	Moderate increase	Creates favourable environmental conditions for mosquito breeding, vector survival, and transmission of climate-sensitive diseases.
Vector transmission suitability	Moderate expansion	Significant expansion	Expands the geographical and seasonal suitability for malaria, dengue, chikungunya, scrub typhus, and other vector-borne diseases into higher elevations.
Forest-fire frequency and duration	Moderate increase	Large increase	Increases smoke exposure, respiratory illnesses, cardiovascular disease, and air quality-related hospital admissions.
Flood frequency and intensity	Moderate increase	High increase	Increases injuries, displacement, disease outbreaks, healthcare disruption, and mental health impacts.
Landslide frequency	Moderate increase	Significant increase	Damages healthcare infrastructure, isolates remote communities, delays emergency response, and interrupts access to essential health services.
Climate-sensitive disease transmission season	Longer transmission season	Substantially longer season	Extends exposure periods for vector-borne, water-borne, and food-borne diseases, increasing disease burden and surveillance requirements.

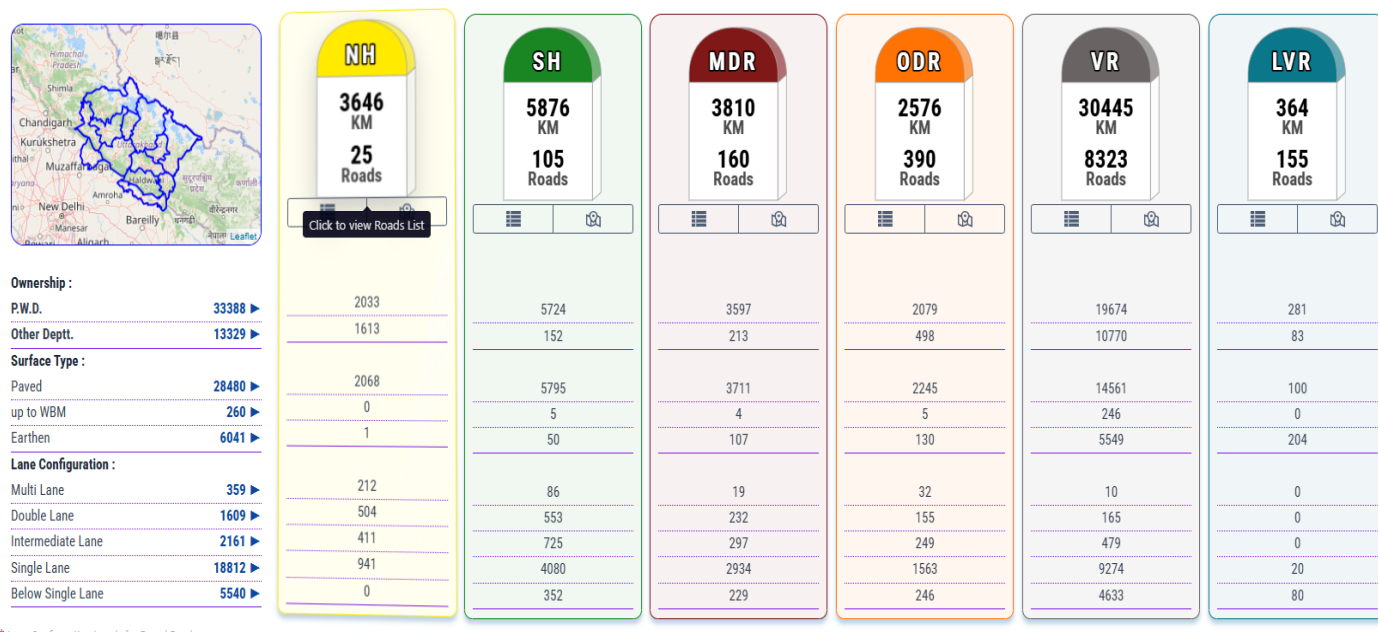
4. Roads & Linear Infrastructure

10.1 Overview

Roads in Uttarakhand are not merely transporting infrastructure; they are the lifeline of the state. Uttarakhand spans approximately 53,483 km² of which around 86 percent is mountainous, and 65 percent is under forest cover (Forest Survey of India, 2021). Within this terrain, road networks function as the principal medium for the movement of people, goods, healthcare, emergency response, defence logistics along the international border with China, and pilgrim and tourist mobility for the Char Dham Yatra. The dependence is so absolute that, in the words of the SAPCC's transport sub-strategy, "any disruption to the road network in the state translates directly into a humanitarian and economic emergency".

Because alternative modes are largely absent the state has almost no navigable waterways and only nascent rail penetration into the hills, this dependence has no practical substitute.

The network has expanded rapidly since the formation of the state. According to the Public Works Department, the length of metalled roads under the PWD rose from **8,080 km in 2000 to 17,100 km in 2010 and 26,701 km as on 31 March 2025**, more than a three-fold increase in a quarter of a century (Economic Survey of Uttarakhand, 2025–26). The current PWD-maintained network, as recorded on the PWD Uttarakhand Infrastructure Management Portal (<https://mis.pwduk.in/im/roadDashboard>), totals 48,266.6 km across all categories,



comprising National Highway sections (3,166 km), State (Provincial) Highways (5,810 km), Major District Roads (3,795 km), Other District Roads (2,570 km), Village Roads (29,176 km) and Border Roads (2,954 km).

Road / structure category	Length or count
Total PWD road network, all categories	48,266.6 km
National Highways (PWD)	3,166 km
State (Provincial) Highways	5,810 km
Major District Roads	3,795 km
Other District Roads	2,570 km
Border Roads	2,954 km
Village Roads	29,176 km
Other categories (LVR, BT, NUC)	797 km
Villages connected by road	13,916
Char Dham All-Weather Road (sanctioned)	736.994 km / ₹10,431 cr
Bridges inventoried	3,587 (≈120.7 km deck)
Bridges ≥ 40 years old	≈818 (23.5%)

Uttarakhand road-sector profile (compiled from the Economic Survey of Uttarakhand 2025-26, PWD IM Portal).

Beyond these state-implemented categories, the Border Roads Organisation maintains strategic alignments along the upper Bhagirathi, Alaknanda and Kali sub-basins, and the Ministry of Road Transport and Highways reports a larger declared National Highway length (of the order of 3,600 km) that reflects a broader definitional scope than the PWD-maintained figure. The two figures are not contradictory: they count different maintenance mandates, and both are cited here to avoid over-stating any single number.

Rural connectivity has been driven principally by the Pradhan Mantri Gram Sadak Yojana. As on December 2025, **13,916 of the state's Census-2011 villages were connected by road**, and road-construction works worth ₹3,457.70 crore were approved across schemes for the financial year 2025–26 (Economic Survey of Uttarakhand, 2025-26). The single largest recent investment is the **Char Dham All-Weather Road Project**: 47 works covering 736.994 km sanctioned at an estimated ₹10,431 crore, of which roughly 596.55 km had been widened to two lanes and 573.83 km resurfaced by late 2025.

Bridges are the most failure-sensitive elements of a mountain road network because a single lost span severs the entire corridor. The state bridge inventory analysed for this assessment records **3,587 structures totalling roughly 120.7 km of deck length**, including 2,324 high-level motor bridges and 1,169 pedestrian bridges (PWD Uttarakhand Infrastructure Management Portal, <https://mis.pwduk.in/im/roadDashboard>).

The stock is concentrated in the districts that also carry the densest hill road network, Pauri (471), Chamoli (422), Almora (373), Uttarkashi (361) and Tehri (306). Two attributes of this stock matter for climate risk. First, it is ageing: the oldest recorded bridge dates to 1886, 234 structures pre-date 1970, and **about 818 bridges (23.5 percent of those with a recorded construction year) are now 40 years or older**, approaching or exceeding typical design lives. Second, a large share was built to legacy loading standards, 888 structures to IRC Class A and 1,231 to Class B, with only 38 designed for the heaviest Class 70R/AA loads, leaving limited reserve capacity to absorb the combined stress of heavier traffic and more energetic flood and scour regimes.

This expansion has occurred in one of the world's most geologically and climatically hazardous terrains. The Uttarakhand Himalaya is characterised by young, tectonically active and fragile rock formations, deeply incised valleys, and a climate regime that is warming faster than the global average while generating an increasing frequency of extreme hydro-meteorological events (Tiwari et al., 2022). The National Crime Records Bureau documented 12,319 landslides in the state between 1988 and 2023, an average of more than 350 per year, and this figure understates the true incidence because it records only events associated with human casualties. The result is a structural paradox: the network that allows mountain communities to participate in the modern economy is simultaneously the most exposed and most fragile element of the state's physical infrastructure, and climate change is intensifying both its hazard exposure and the social dependency upon it.

Figure 2: Road Features



10.2 Defining Features and Their Climate Implication

The roads and linear-infrastructure sector in Uttarakhand is deeply embedded in, and continuously shaped by, the regional climate and geomorphological system. In a mountainous state where road transport provides the primary and frequently the only means of access to dispersed hill settlements, markets, health facilities and administrative centres, the performance of the network is governed by the same hydro-meteorological and slope processes that define the Himalayan environment. Roads, bridges, culverts, retaining structures and the linear ribbon settlements that develop alongside them traverse steep, geologically young and seismically active terrain, in which alignment, stability and serviceability are strongly conditioned by precipitation intensity, slope hydrology, river behaviour and geological processes. Consequently, projected changes in temperature, precipitation regimes, and the frequency and intensity of extreme weather events are expected to affect multiple components of the sector through interconnected physical pathways.

Roads in Uttarakhand intersect the natural hydrology of the region in three ways. First, they cross hundreds of streams and rivulets, each of which delivers a flash-flood pulse during high-intensity rainfall events. Second, they act as horizontal barriers on hillslopes, concentrating runoff at culvert points whose discharge capacity is often exceeded. Third, road cuts and muck dumping into stream channels alter sediment transport and base-level conditions, contributing to channel instability downstream (Asher & Bhandari, 2021). Road construction and maintenance in the hills are dominated by slope-stability processes. Hill roads are typically formed by cutting into steep valley sides, creating cut slopes and unsupported hill-and-valley faces whose stability depends on soil moisture, pore-water pressure and the integrity of the surrounding rock and debris mass. High-intensity and prolonged rainfall raises pore pressures and triggers cut-slope failures, debris slides and slumping, while inadequate cross-drainage concentrates runoff onto road formations and adjoining hillsides. Climate-induced increases in rainfall intensity and in the frequency of cloudbursts therefore act directly on the most vulnerable element of the hill-road system the slope on which it rests.

Linear transport corridors are closely coupled with river and geomorphic processes. Valley-floor and riverside roads, together with the bridges and causeways that carry them across Himalayan rivers, are exposed to flash floods, bank erosion, channel aggradation and debris flows; high-intensity runoff and sediment transport scour foundations, undermine abutments and bury carriageways. In glacier-fed catchments, glacial lake outburst floods and large sediment pulses pose an episodic but potentially catastrophic threat to bridges and low-lying road sections. Because a single damaged bridge or blocked reach can sever

access to entire valleys, the connectivity of the network is highly sensitive to these localised, high-energy events.

The reliability of the network is further governed by mass-movement processes along its length. Landslides, rockfalls, debris flows and, at higher elevations, avalanches repeatedly block and damage roads during and after the monsoon, isolating settlements for periods ranging from hours to weeks and imposing recurrent clearance and reconstruction costs. The linear, largely single-carriageway character of most hill roads means that the system carries little redundancy: the failure of one slope or one culvert can interrupt the only route serving a cluster of villages. Increasing climatic variability and more frequent extreme-rainfall events are expected to raise both the frequency and the duration of such disruptions.

Climate also shapes how, and how safely, the network is used, together with the settlements that depend upon it. Pilgrimage and tourism traffic concentrated along corridors such as the Char Dham routes and peaking in the pre-monsoon and monsoon-shoulder seasons coincides with the period of greatest hydro-meteorological hazard, amplifying the exposure of people and vehicles to weather-driven disruption. The linear, or “ribbon”, townships and settlements that have developed along road and river corridors, often on cut slopes, debris fans and floodplains, concentrate population and assets in precisely the zones most exposed to landslides, subsidence and flooding, so that road-sector hazards translate directly into settlement risk.

The most distinctive sensitivity axis is the population now resident along these roads. Census of India (2011) data and subsequent ethnographic work (Sati, 2020; Mamgain & Reddy, 2017; Rana & Barodia, 2024) show a clear pattern of upland-to-roadside settlement transfer in Uttarakhand. As traditional hill villages depopulate, 1,055 villages are recorded as fully uninhabited and another 405 with fewer than ten residents (Census, 2011; JIER, 2024), new informal and semi-formal settlements have aggregated along motorable road heads, where market access, education, healthcare and remittance-based livelihood opportunities are concentrated. The result is a population shift not merely in absolute terms but in terms of climate-risk exposure: people have moved out of relatively stable upland sites onto road-adjacent slopes that are themselves the most landslide-prone, drainage-disrupted and engineering-stressed locations in the catchment.

Defining Feature	Description	Relevance for Climate Sensitivity
Access-critical, low-redundancy network	Road transport is the primary and frequently the only means of access to dispersed hill settlements, with largely single-	The failure of a single slope, culvert or bridge can sever access to entire valleys, so network connectivity is acutely sensitive to localised, climate-driven disruptions.

Defining Feature	Description	Relevance for Climate Sensitivity
	carriageway, linear alignments and few alternative routes.	
Cut-slope hill-road construction	Hill roads are built by excavating steep valley sides, creating cut slopes and hill-and-valley faces of limited inherent stability.	Slope stability is governed by rainfall intensity, soil moisture and pore-water pressure; increasing extreme rainfall and cloudbursts directly raise the risk of cut-slope failure and road damage.
Bridges, causeways and river crossings	Linear corridors depend on bridges, causeways and culverts crossing steep, hydrologically dynamic Himalayan rivers.	Foundations and abutments are exposed to flash floods, scour, bank erosion, sediment aggradation and glacial lake outburst floods (GLOFs), making critical crossings sensitive to high-energy hydro-geomorphic events.
Alignment through landslide- and hazard-prone terrain	Corridors traverse geologically young, seismically active slopes subject to landslides, rockfall, debris flows and, at higher elevations, avalanches.	Road serviceability is repeatedly disrupted by mass-movement processes whose frequency and intensity are influenced by precipitation regimes and increasing climatic variability.
Dependence on drainage and cross-drainage	Road performance relies on culverts, side drains and cross-drainage works to manage hillside runoff across the formation.	Under high-intensity rainfall, drainage capacity is exceeded and concentrated runoff triggers erosion, washouts and slope failure, linking road integrity directly to precipitation extremes.
Climate-coincident demand (pilgrimage and tourism)	Traffic on key corridors, notably the Char Dham routes, peaks in seasons that overlap with the monsoon and its associated hazards.	Peak human and vehicular exposure coincides with peak hazard, so weather-driven disruptions carry elevated safety and economic consequences.
Linear (ribbon) settlements along corridors	Townships and settlements develop linearly along road and river corridors, frequently on cut slopes, debris fans and floodplains.	Population and assets are concentrated in the zones most exposed to landslides, subsidence and flooding, so road-corridor hazards translate directly into settlement and livelihood risk.

10.3. Key Climate Hazards Relevant to the Road Sector

Six climate-related hazards dominate risk to the Uttarakhand road sector. For each, this section sets out the physical mechanism, the element of the road network at which it acts, the assets and geographies affected, the documented historical record, and the residual data gap. Taken together, the record establishes that climate risk to the sector is not a prospective concern but an ongoing, measurable loss stream with two distinct signatures: rare, corridor-severing events that command national attention, and a chronic, high-frequency attrition of the lower-order rural network that does not but which accounts for the larger share of cumulative restoration cost.

A second structural feature distinguishes the road sector from other infrastructure sectors in this assessment. The road network is not only exposed to the landslide hazard; through hill-cutting, unmanaged spoil disposal and the loss of slope support, it partly manufactures that hazard. Eighty-one per cent of landslides recorded along roughly 800 km of pilgrimage-route highway occurred within 100 metres of the road alignment (Voisplanet, 2025). Hazard treatment for this sector therefore cannot be confined to protecting the asset from the environment; it must also address the asset's contribution to the hazard.

Hazard Register

Hazard	Mechanism of damage	Network element affected	Assets primarily affected	Principal districts / corridors
Landslide and slope failure	Planar, wedge and toppling failure of cuttings in quartzite, phyllite and metasedimentary lithozones; debris burial of the carriageway; loss of formation width	Carriageway; cut-slopes; hill-side alignment	Cut- and fill-slopes, breast and retaining walls, carriageway, formation	NH-7, NH-58, NH-109 corridors; acute in Rudraprayag, Chamoli, Tehri, Pauri, Uttarkashi
Extreme rainfall and cloudburst	Saturation and de-stabilisation of cut and fill slopes; hydraulic capacity of culverts and cross-drainage exceeded; embankment washout; stripping of pavement layers	Drainage; pavement; embankment	Culverts, cross-drainage works, embankments, bituminous surfacing	State-wide; concentrated in Uttarkashi, Chamoli, Rudraprayag, Pithoragarh
Riverine and flash flooding	Scour of bridge foundations and abutments; inundation of valley-floor alignments; concentration of discharge at stream crossings; channel migration	Bridges and cross-drainage; valley-floor alignment	Motor bridges, bridle bridges, causeways, approach embankments	Valley-floor alignments state-wide; Alaknanda, Mandakini and Bhagirathi corridors
Glacial-lake outburst flood (GLOF)	Instantaneous debris-laden surge destroys bridges, barrages and valley-floor carriageway far	Bridges; strategic and access corridors	Bridges, valley-floor roads,	Upper Bhagirathi, Alaknanda, Dhauliganga and

	downstream in minutes; post-event aggradation buries the alignment		hydropower-access and Border Roads network	Kali basins; Chamoli, Uttarkashi, Pithoragarh
Slow-onset land subsidence	Progressive settlement and cracking of formation on loose paraglacial material, with discrete accelerations triggered by extreme-rainfall episodes; slow-onset and effectively irreversible	Formation; nodal town alignments	National-highway formation, retaining structures, urban road nodes	Joshimath (Chamoli) on NH-7; other hill-cut and tunnelled reaches
Rising temperature and freeze-thaw cycling	Ageing and embrittlement of bituminous surfacing, cracking and loss of the waterproof seal, leading to water ingress and potholing; freeze-thaw fracturing of pavement and rock	Pavement asset; rock cut faces	Bituminous surfacing, sub-base, exposed rock faces	State-wide (heat); freeze-thaw acute in high-altitude blocks

Landslides and slope failures

Landslides are the single most consequential road hazard in the State. Cuttings made through quartzite, phyllite and metasedimentary lithozones, common across NH-7, NH-58 and NH-109 are prone to planar, wedge and toppling failures. Kinematic survey of 35 cut-slopes along approximately 17 km of NH-58 between Dhari Devi and Rudraprayag identified up to 18 as structurally vulnerable (Sati, Bhandari and Gahalaut, 2022). The controlling variables are slope angle, rainfall amount and lithology the same three that emerge from the most recent state-wide inventory.

Why this hazard is structurally severe for the road sector

The hazard is co-located with the asset by construction. Hill roads are built by cutting into the slope; the cut face is itself the failure surface, so exposure is not incidental but designed in.

Loss of service is disproportionate to loss of asset. A single blockage of a few tens of metres severs an entire corridor, because the Himalayan network is radial and largely unredundant- there is no parallel route to absorb the traffic.

Restoration is governed by access, not by the repair itself. Debris must be cleared before crews and plant can reach the site, and clearance capacity is the binding constraint during a multi-site monsoon event.

The sector generates part of its own hazard. Slope-cutting, unmanaged spoil disposal and toe erosion de-stabilise slopes above and below the alignment, so new construction raises the hazard baseline for existing assets.

Failure pathways

Failure pathway	What is lost	Why restoration is slow
Cut-slope failure above the alignment	Carriageway buried; corridor closed; traffic held at both faces	Debris volume must be removed before any assessment of the pavement beneath it can begin
Fill-slope or formation failure below the alignment	Loss of formation width; partial or total loss of carriageway	Requires re-construction of the retaining structure and, on unstable ground, re-alignment
Debris flow across a stream crossing	Culvert or causeway choked; localised washout	Clearance must precede repair, and the choked crossing re-fails in the next high-intensity event
Slide burying the approach to a settlement	No asset loss, but the settlement becomes unreachable	Restoration time is governed by clearance sequencing across competing sites
Village-road slope failure	Village-level connectivity cut; access to health and market services lost	Dispersed, low-priority assets are cleared last and dominate the restoration backlog

Landslide failure pathways and their consequences for the road network.

Documented impact

The most authoritative recent measurement of road-associated landslide density comes from *Mey et al. (2024)*. Following an anomalously wet September–October 2022 monsoon, a GPS-based inventory along the 247 km Rishikesh–Joshimath (NH-7) corridor documented 309 landslides totally or partially blocking the road of an average of 1.25 landslides per road kilometre. Modelling the inventory as a network-attached point pattern, the authors identified slope angle, rainfall amount and lithology as the principal controls, and concluded that climate change and rising exposure along this pilgrimage route will likely exacerbate landslide risk in future. A complementary survey of approximately 800 km of pilgrimage-route highway found 811 landslides, of which 81 per cent occurred within 100 metres of the road alignment, a direct signature of road construction as a landslide driver (*Voisplanet, 2025*).

Extreme and high-intensity rainfall, including cloudbursts

Extreme rainfall is the proximate trigger for the majority of road failures, and it damages the network through several mechanisms acting simultaneously: it saturates and de-stabilises cut- and fill-slopes; it overwhelms culverts and cross-drainage designed to a historical intensity–duration–frequency envelope; it scours bridge foundations; and it washes out embankments and pavement layers. Because extreme rainfall is also the trigger for the landslide hazard above, the two are not independent and must not be double counted in the composite index.

Documented impact

IMD records indicate that Uttarakhand has reported more than 7,750 extreme-rainfall events and cloudbursts since 2015, concentrated in the most recent years of the series (IMD, 2023). In October 2021 the State recorded 178.4 mm of rain in the first 18 days of the month 485 percent above the climatological normal triggering landslides and floods that killed at least 45 people and disrupted the Char Dham Yatra. The August 2025 Dharali (Uttarkashi) cloudburst produced a 24-hour rainfall episode reported at 369 per cent above the seasonal normal at the relevant station.

The August 2025 Dharali (Uttarkashi) cloudburst produced a 24-hour rainfall episode reported at 369 percent above the seasonal normal at the relevant station. High-intensity rainfall damages roads through several mechanisms simultaneously: it saturates and destabilises cut- and fill-slopes, overwhelms culverts and cross-drainage, scours bridge foundations, and washes out embankments and pavement layers.

The trend that matters

For the road sector the significant signal is not annual rainfall total but intensity concentration. Cross-drainage capacity, embankment side-slopes and pavement drainage were all dimensioned against a rainfall regime in which a given seasonal volume arrived over a larger number of lower-intensity days. The re-organisation of the same or a smaller total into fewer, more violent pulses shifts a large fraction of the network from within to outside its design envelope without any change in the asset itself - a form of stranded design capacity that no maintenance regime can recover

Riverine and flash floods

Because hill roads and their bridges are routinely aligned along valley floors and at stream crossings, they are directly exposed to riverine and flash-flood pulses. Each of the hundreds of streams and rivulets that a hill road intersects delivers a flash-flood surge during high-intensity events, concentrating discharge at culvert and bridge points whose hydraulic capacity is frequently exceeded.

Flood exposure in the State is bimodal, and the two modes require different treatment. In the hills, flooding is flashy, sediment-laden and channelised: it scours bridge foundations, undermines abutments and removes approach embankments. In the Bhabar–Tarai plains - Haridwar, Udham Singh Nagar and the southern blocks of Nainital and Dehradun - flooding is areal and prolonged: it inundates the carriageway over wide areas, affecting far larger populations but with lower per-asset destructive force. Hill flooding is a capital-destruction hazard; plains flooding is a service-continuity hazard. A single flood-exposure metric conceals this distinction.

Flood damage to bridges is disproportionately serious given the age and legacy loading of the stock (Section 12.1): scour around ageing foundations is a leading cause of span loss, and a

lost span is a corridor-level failure with a reconstruction timeline measured in years rather than weeks. Channel migration is a distinct and under-assessed sub-hazard: a river that shifts its thalweg after a high-magnitude flood can leave an alignment that was 200 m from the bank sitting on the bank, converting a transient event into a permanent change in exposure

Glacial-lake outburst floods (GLOFs)

Warming-driven glacier retreat is enlarging and multiplying moraine-dammed lakes in the upper Bhagirathi, Alaknanda, Dhauliganga and Kali basins. GLOF in the February 2021 Chamoli disaster, can destroy bridges, barrages and valley-floor roads far downstream within minutes, and disproportionately threatens the strategic Border Roads network and hydropower-access roads. This is a low-frequency, very-high-consequence hazard whose probability is rising with temperature.

Why this hazard requires separate treatment

- The failure mode is total rather than partial: bridges and valley-floor formation are removed, not damaged, so there is no repair pathway only reconstruction.
- The loss extends beyond the asset to the site: post-event sediment and debris aggradation can bury an alignment and impose clearance liabilities out of all proportion to the value of the road destroyed.
- The event has no useful precedent in the operational record, so a design basis cannot be derived from historical frequency- the central methodological difficulty in this hazard class.
- The exposed corridors are strategically significant: border-access and hydropower-access roads have no alternative alignment, so the connectivity consequence is disproportionate to the length of road lost.

The implication for this assessment is that GLOF exposure cannot be treated as a static layer. The number and size of glacial lakes at the termini of retreating glaciers continues to increase, so the hazard itself is growing under the same warming that drives the other hazards in this register. For the road sector this argues for treating GLOF exposure as a siting and alignment constraint particularly for bridge replacement and new valley-floor construction rather than as an insurable operating risk.

Slow-onset land subsidence

The Joshimath subsidence event of January 2023, in which more than 800 buildings cracked at the head of the Alaknanda valley on a critical NH-7 node, illustrates a slow-onset hazard that is modulated by climate. Geodetic analysis (Sreejith et al., 2024) demonstrated that Joshimath landslide kinematics are episodic, with discrete acceleration events triggered by extreme-rainfall episodes such as that of 19 October 2021. The cascading consequence for

the road sector was severe traffic restrictions on NH-7 and a construction ban and the analysis explicitly attributed part of the de-stabilisation to road widening, hill cutting and tunnelling above and below the town.

Subsidence is included in this register for two reasons. First, it acts on an asset class that is least amenable to retrofit: once the formation itself is moving, surface treatment achieves nothing, and the only responses are ground stabilisation or re-alignment. Second, it is invisible to hazard frameworks built exclusively around extreme events, because the damage accumulates between events even though its accelerations are event-triggered.

The cascading consequence for the road sector was severe NH-7 traffic restrictions and a construction ban and the analysis explicitly attributed part of the destabilisation to road widening, hill cutting and tunnelling above and below the town.

Rising temperature and freeze–thaw cycling (pavement hazard)

Distinct from the mass-movement hazards above, rising temperature is a chronic stressor acting on the pavement asset itself. Higher sustained temperatures accelerate the ageing and embrittlement of bituminous surfaces, which crack and lose their waterproofing seal; surface water then enters the pavement, causing potholing and rapid loss of condition (Shao, 2017). At higher elevations, more frequent freeze–thaw cycling fractures pavement and rock alike the latter feeding directly back into the rockfall and cut-slope hazard.

Extreme	Effect on the road asset	Where it binds in Uttarakhand
Heat	Ageing and embrittlement of bitumen; cracking and loss of the waterproof seal; rutting under load at high pavement temperature; accelerated loss of surface condition	Bhabar–Tarai belt and valley floors: Haridwar, Udham Singh Nagar, Dehradun valley, Haldwani
Freeze–thaw cycling	Fracture of pavement layers by ice expansion in cracks; frost heave of the sub-base; parallel fracturing of exposed rock cut faces, increasing rockfall frequency	High-altitude blocks and passes: Bhatwari, Mori, Joshimath, Dharchula, Munsyari, Kapkot, Chakrata
Warm winters (absence of cold)	Altered snow-clearance and closure regimes; changed timing of the freeze–thaw window; longer effective exposure season for high-altitude alignments	All high-altitude alignments and snow-bound passes

Documented past impacts: the restoration-cost evidence base

The scale of past hazard impact on the road sector is captured in a state-wide damaged-infrastructure dataset compiled from PWD divisional reporting. It records **6,363 damage spots across 13 districts and 62 divisions, with a total assessed restoration cost of ₹60,950.63 lakh (≈₹609.5 crore). These spots are distributed across roughly 1,649 assets, of which roads account for the overwhelming majority (1,601 road works), alongside 26**

motor bridges, 18 bridle bridges and 4 buildings. The damage is highly concentrated geographically and by road category.

District	Restoration cost (₹ lakh)	Share of state total
Dehradun	10,097	16.6%
Uttarkashi	8,632	14.2%
Tehri	7,670	12.6%
Pauri	7,571	12.4%
Chamoli	4,329	7.1%
Nainital	3,816	6.3%
Rudraprayag	3,798	6.2%
Haridwar	3,207	5.3%
Almora	3,044	5.0%
Bageshwar	2,955	4.8%
Pithoragarh	2,510	4.1%
Champawat	2,115	3.5%
Udham Singh Nagar	1,198	2.0%
State total	60,951	100%

District-wise road-sector restoration cost (compiled from the state Damaged-Infrastructure dataset). Figures rounded; a residual “unknown” division (~₹8.6 lakh) is omitted.

Impact is spatially concentrated. The four highest-damage districts: Dehradun, Uttarkashi, Tehri and Pauri- together account for more than half (55.8 per cent) of all restoration cost, confirming that damage is concentrated in the western Garhwal ranges and along the Char Dham approach corridors.

Damage is dominated by lower-order roads. Analysed by road type, Village Roads attract by far the largest restoration cost (of the order of ₹30,000 lakh), followed by State Highways (approximately ₹15,000 lakh) and Major District Roads (approximately ₹5,700 lakh). Village and other district roads are the least-engineered, most numerous and most poorly drained category, and they are precisely the roads on which recent ribbon settlement has concentrated (Section 12.5). The headline hazard events occur on the national highways; the cumulative cost accrues on the village roads.

The restoration backlog is a measurable signature of limited adaptive capacity. The status field of the same dataset shows that the great majority of damage spots remain at the “Identified” stage (of the order of 5,000 spots), with only a small fraction “Sanctioned”, “Under construction” or “Completed”. Damage is being identified far faster than it can be funded and repaired. These compounds risk directly: an unrepaired site enters the following monsoon in a degraded condition, so the backlog is not merely a delivery problem but a mechanism by which one season’s hazard raises the next season’s vulnerability.

Routinisation of corridor closures

The event record confirms that monsoon closure of the pilgrimage-route national highways (NH-7, NH-34, NH-94, NH-109) has become an annual rather than exceptional phenomenon.

In July 2023 the Char Dham pilgrimage highways and approximately 150 rural roads were simultaneously blocked; in July 2025 more than 100 hill roads were closed concurrently, stranding pilgrims near Sonprayag and Gaurikund. The state’s own 2025 estimate places recent-disaster damage at 946 roads and 15 bridges severely affected. The November 2023 collapse of the under-construction Silkyara Bend–Barkot tunnel on NH-134, trapping 41 workers for 16 days, further underscores that even climate-adaptation infrastructure (all-weather tunnels intended to bypass monsoon closures) is itself subject to compound geological risk in fragile Himalayan lithology.

Indicator	Baseline	Recent value	Direction
Char Dham route closures	Episodic	Annual, multi-NH	Routinised
Roads damaged in major events	Smaller pre-2013 incidents	946 roads + 15 bridges (2024–25)	Multi-fold rise
Road-associated landslide density (NH-7)	-	1.25 per road-km (2022)	Very high
Restoration backlog	—	~5,000 spots at “Identified” stage	Backlog growing
Forest felled, Char Dham project	—	≈690 ha / >55,000 trees	Major loss

Year	Event	Documented road-sector consequence	Lesson carried into the framework
2021 (Feb)	Rishi Ganga–Tapovan rock–ice avalanche and flash flood, Chamoli	Destruction of bridges and valley-floor road infrastructure downstream along the Rishi Ganga, Dhauliganga and Alaknanda; loss of hydropower-access and Border Roads connectivity	No-precedent, total-loss events cannot be appraised from the historical record; GLOF must be treated as an alignment constraint
2021 (Oct)	Extreme post-monsoon rainfall- 178.4 mm in 18 days, 485% above normal	Landslides and floods across the network; at least 45 fatalities; Char Dham Yatra disrupted; triggered a discrete acceleration of the Joshimath slide	The hazard season is lengthening beyond the conventional monsoon window
2022 (Sep–Oct)	Anomalously wet late monsoon; GPS landslide inventory, NH-7	309 landslides totally or partially blocking the 247 km Rishikesh–Joshimath corridor - 1.25 per road-km	Road-associated landslide density is now measurable and can anchor corridor-level prioritisation
2023 (Jan)	Joshimath land subsidence	More than 800 buildings cracked at a critical NH-7 node; traffic restrictions and a construction ban imposed	Slow-onset hazards need separate treatment from extremes, and construction activity is itself a driver
2023 (Jul)	Monsoon closures, Char Dham corridors	Pilgrimage highways and approximately 150 rural roads simultaneously blocked	Simultaneity across sites, not severity at one site, is the binding constraint on restoration capacity
2023 (Nov)	Silkyara Bend–Barkot tunnel collapse, NH-134	Under-construction all-weather tunnel collapsed; 41 workers trapped for 16 days	Adaptation infrastructure carries its own compound geological risk and must be appraised for residual risk
2025 (Jul)	Monsoon closures, Kedarnath approach	More than 100 hill roads closed concurrently; pilgrims stranded near Sonprayag and Gaurikund	Annual routinisation of closure has converted a hazard into an operating condition
2025 (Aug)	Dharali cloudburst, Uttarkashi	24-hour rainfall reported at 369% above the seasonal normal at the relevant station, with associated corridor damage	Cloudburst intensity continues to exceed the drainage design envelope

10.4 Hazard Exposure Analysis

Exposure is assessed by intersecting hazard-susceptibility layers with the geo-referenced road asset inventory. Four hazards are carried into the spatial overlay- landslide, flood and extreme rainfall because susceptibility layers for these are available at a resolution consistent with block-level reporting. GLOF is treated separately as a corridor- and basin-level siting constraint applied to bridges and valley-floor alignments; rising temperature is handled on the sensitivity side because its effect is mediated by binder grade and surfacing type rather than by location; and land subsidence is treated as a site-specific constraint at identified nodes.

The State Damaged-Infrastructure dataset- 6,363 spots and ₹609.5 crore of assessed restoration cost is attributed to district and division but not to a located chainage. It can therefore be used to validate the overlay at district level, and to weight the hazards, but it cannot yet be used to calibrate exposure at asset level. Chainage-locating that record is the single highest-value data improvement available to this assessment.

Exposure Metrics

Asset class	Exposure metric	Unit reported
National highways (incl. Char Dham corridors)	Length within high and very-high hazard classes ÷ total NH length in the block	% of NH length exposed
State highways and Major District Roads	Length within high and very-high classes ÷ total SH + MDR length in the block	% of length exposed
Other District Roads and Village Roads	Length within high and very-high classes ÷ total ODR + VR length in the block	% of length exposed
Motor bridges and causeways	Structures within high and very-high flood or landslide classes ÷ total structures in the block	% of structures exposed
Bridle bridges and footbridges	Structures within high and very-high classes ÷ total in the block	% of structures exposed
Cross-drainage works	Crossings on links within high and very-high extreme-rainfall classes ÷ total crossings in the block	% of crossings exposed
High-altitude alignments	Length above the freeze–thaw threshold elevation within high and very-high classes ÷ total length in the block	% of length exposed

Single-access habitations	Habitations whose sole serving road link intersects a high or very-high class ÷ total habitations in the block	% of habitations exposed
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The final metric has no counterpart in the energy sector and is the road-specific addition to the framework. Because the Himalayan Road network is radial and largely unredundant, a single blockage severs a corridor outright; the connectivity consequence of exposure therefore depends less on how much road is exposed than on whether any alternative alignment exists. Reporting single-access dependency alongside length-based exposure prevents a block with a short but critical exposed link from scoring as low-exposure.

Classification Bands

Class	Range (% of asset exposed)	Interpretation
Very Low	0 – 10 %	Marginal exposure; routine maintenance and standard drainage practice adequate
Low	>10 – 25 %	Localised exposure; targeted slope protection and drainage upgrade at identified spots
Moderate	>25 – 50 %	Material exposure; corridor-level risk treatment and a programmed slope-stabilisation schedule required
High	>50 – 75 %	Systemic exposure; re-alignment, engineered slope protection or redundant-route provision required
Very High	>75 – 100 %	Critical exposure; the network is effectively co-located with the hazard, and closure must be planned for rather than prevented

Composite Exposure Formula

The block composite is the weighted mean of the four hazard-specific exposure scores. Weights reflect the relative contribution of each hazard to documented road-sector damage in the State (Section 12.4.8), with landslide weighted highest because it is both the most frequent cause of network failure and the principal control on restoration time - debris clearance, not the repair itself, governs how long a corridor stays closed:

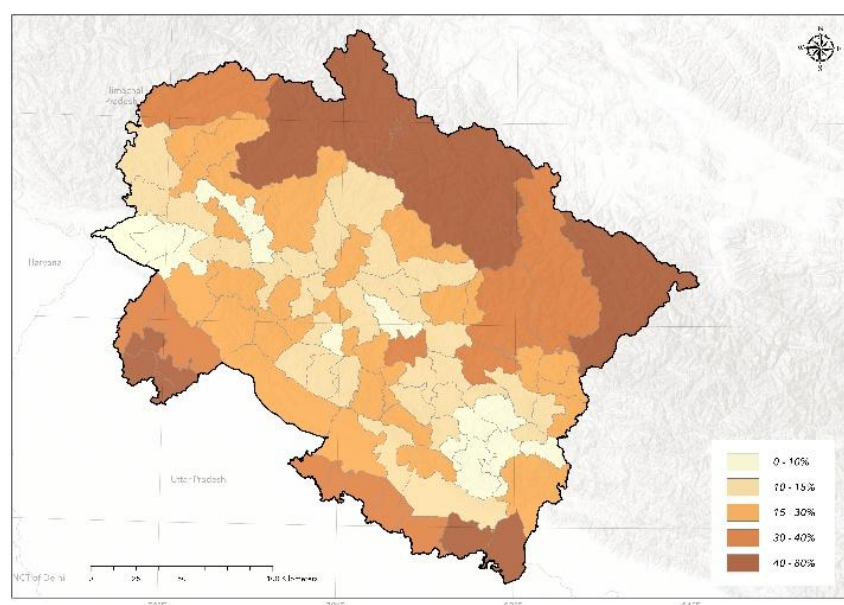
$$E(\text{block}) = 0.35 \cdot S(\text{landslide}) + 0.35 \cdot S(\text{extreme rainfall}) + 0.30 \cdot S(\text{flood})$$

Basis for the weights

Hazard	Weight	Justification
Landslide	0.35	Highest measured incidence (1.25 landslides per road-km on NH-7 in 2022) and the dominant control on closure duration.

		Weighted above the energy-sector value of 0.30 because the road asset is physically continuous - a single point failure disables the whole link, whereas a transmission network can sometimes be re-routed.
Extreme rainfall	0.25	The proximate trigger for the majority of road failures and the direct cause of culvert, embankment and pavement-layer damage. Held at the energy-sector value.
Flood	0.25	Raised above the energy-sector value of 0.20 because bridge scour is the sector's principal capital-destruction pathway: a lost span has a reconstruction timeline measured in years, against weeks for a cleared slide.
Freeze-thaw / snow-and-ice	0.15	Spatially restricted to high-altitude blocks and chronic rather than catastrophic in effect, but it degrades the pavement and fractures rock cut faces, feeding back into the landslide pathway.

Road Infrastructure Exposed to Floods in Hazards



The map illustrates the spatial distribution of road infrastructure exposed to flood hazards across Uttarakhand, categorized into exposure classes ranging from 0–10% (low exposure) to 40–80% (very high exposure). Higher exposure is concentrated in parts of the northern, eastern, and southwestern regions, while several central districts exhibit relatively lower exposure

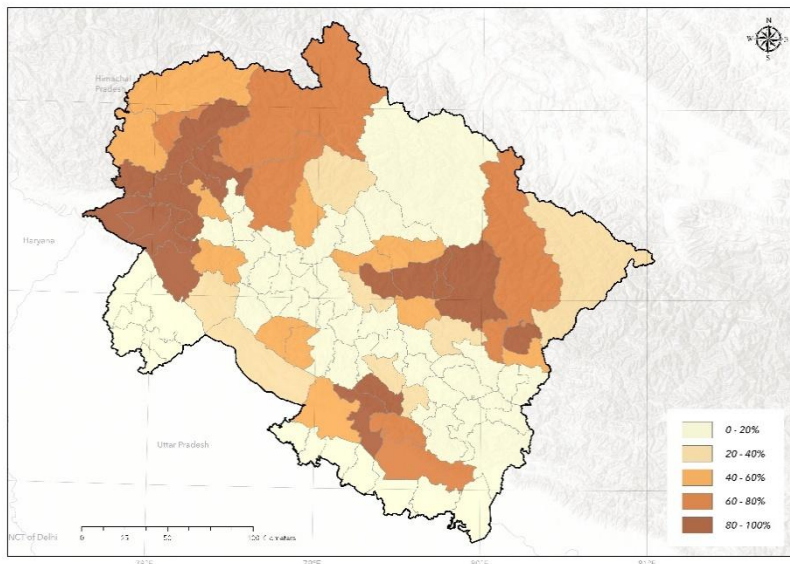
levels.

The observed pattern reflects the influence of river networks, valley systems, topography, intense rainfall events, and flood-prone catchments. Road infrastructure in Uttarakhand is particularly vulnerable because of the state's mountainous terrain and dependence on narrow transport corridors along river valleys and unstable slopes. Climate change is expected to intensify extreme rainfall events and alter hydrological patterns, increasing flood

frequency and severity. Flooding can damage road surfaces, bridges, culverts, and retaining structures, leading to prolonged transportation disruptions.

Road Infrastructure Exposed to Landslides in Uttarakhand

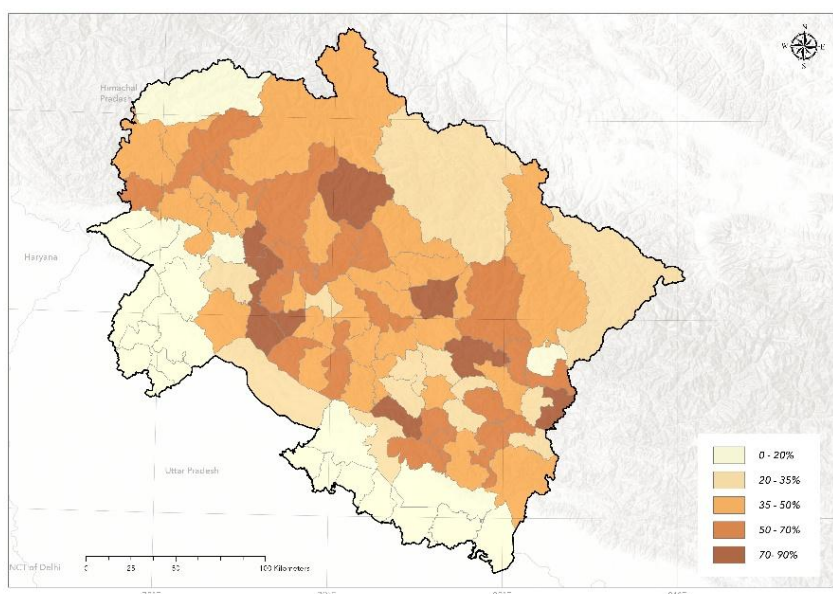
The map presents the spatial distribution of **road infrastructure exposed to landslide hazards across Uttarakhand**, categorized into exposure classes ranging from **0–20% (low exposure)** to **70–90% (very high exposure)**. The analysis indicates that a significant proportion of road networks in the **central and mountainous districts** fall under moderate to high exposure categories, reflecting the influence of steep terrain, geological fragility, slope instability, and intense rainfall patterns.



Landslide impacts on road infrastructure can have substantial social and economic consequences. Damage to roads and bridges disrupts transportation, isolates remote settlements, delays emergency response and relief operations, and restricts access to healthcare, education, and markets. In hilly regions, prolonged road closures can severely affect tourism activities and

livelihoods dependent on connectivity.

Road Infrastructure Exposed to Extreme Rainfall in Uttarakhand



The map illustrates the spatial distribution of road infrastructure exposed to extreme rainfall across Uttarakhand, categorized into exposure classes ranging from 0–20% (low exposure) to 80–100% (very high exposure). The analysis indicates that several northern, western, and central districts exhibit moderate to very high exposure levels, while

lower exposure is observed in some southern and valley regions. The spatial pattern reflects the influence of topography, rainfall intensity, drainage characteristics, and the concentration of transport networks in mountainous terrain.

Extreme rainfall events in Uttarakhand have become more frequent and intense under changing climatic conditions, increasing risks to road infrastructure. Heavy rainfall can trigger flash floods, slope failures, soil erosion, embankment instability, and road washouts, resulting in severe damage to transport corridors. Roads located along steep slopes and river valleys are particularly vulnerable to repeated disruptions.

10.4.1 Composite reading of block-wise exposure

Combining the three surfaces, the highest-priority exposure profile is found in the western Garhwal and Char Dham-corridor blocks (Uttarkashi, Tehri, Rudraprayag, Chamoli and upper Pauri), where high landslide exposure, high extreme-rainfall exposure and valley-floor flood exposure coincide on corridors of national strategic importance. A second cluster of high flood-and-rainfall exposure lies in the north-western blocks (parts of Dehradun district including the Chakrata–Kalsi belt) and along the eastern high-Himalayan margin. The table below summarises the dominant exposure band per hazard for the blocks:

District	Block	Landslide	Flood	Extreme rainfall	Composite exposure
Uttarkashi	Bhatwari	Very High	High	Very High	Very High
	Dunda	High	Moderate	High	High
	Chinyalisaur	High	Moderate	High	High
	Naugaon	High	Moderate	High	High
	Purola	Moderate	Moderate	Moderate	Moderate

District	Block	Landslide	Flood	Extreme rainfall	Composite exposure
	Mori	High	Moderate	High	High
Chamoli	Dasholi	Very High	High	Very High	Very High
	Joshimath	Very High	Very High	Very High	Very High
	Ghat	Very High	Moderate	High	High
	Karnaprayag	High	High	High	High
	Dewal	High	Moderate	High	High
	Tharali	High	High	High	High
	Narayanbagar	High	Moderate	High	High
	Pokhri	High	Low	Moderate	Moderate
	Gairsain	Moderate	Low	Moderate	Moderate
Rudraprayag	Ukhimath	Very High	High	Very High	Very High
	Agastyamuni	Very High	High	High	Very High
	Jakholi	High	Moderate	High	High
Tehri Garhwal	Chamba	High	Low	Moderate	Moderate
	Jaunpur	High	Low	Moderate	Moderate
	Kirtinagar	High	Moderate	Moderate	High
	Narendranagar	High	Moderate	Moderate	High
	Pratapnagar	High	Moderate	High	High
	Bhilangana	Very High	High	Very High	Very High
	Jakhnidhar	High	Moderate	Moderate	High
	Thauldhar	High	Low	Moderate	Moderate
	Devprayag	High	High	Moderate	High
Dehradun	Chakrata	High	Moderate	Very High	High
	Kalsi	High	Moderate	High	High
	Vikasnagar	Moderate	High	Moderate	Moderate
	Sahaspur	Low	High	Moderate	Moderate
	Raipur	Moderate	High	Moderate	Moderate
	Doiwala	Low	High	Low	Moderate
Pauri Garhwal	Kot	High	Low	Moderate	Moderate
	Kaljikhali	High	Low	Moderate	Moderate
	Pauri	High	Low	Moderate	Moderate
	Pabau	High	Low	Moderate	Moderate
	Thalisain	High	Moderate	Moderate	High
	Bironkhal	High	Low	Moderate	Moderate
	Dwarikhali	High	Moderate	Moderate	High

District	Block	Landslide	Flood	Extreme rainfall	Composite exposure
	Ekeshwar	High	Low	Moderate	Moderate
	Jaharikhal	High	Low	Moderate	High
	Nainidanda	High	Moderate	Moderate	High
	Pokhra	High	Low	Moderate	Moderate
	Rikhnikhal	High	Low	Moderate	Moderate
	Yamkeshwar	High	High	Moderate	High
	Dugadda	Very High	Moderate	High	High
	Khirsu	High	Low	Moderate	Moderate
Nainital	Betalghat	High	Moderate	Moderate	High
	Bhimtal	High	Moderate	Moderate	High
	Dhari	High	Moderate	Moderate	High
	Haldwani	Low	High	Low	Moderate
	Kotabag	Moderate	High	Moderate	High
	Okhalkanda	High	Moderate	Moderate	High
	Ramgarh	High	Low	Moderate	Moderate
	Ramnagar	Moderate	High	Moderate	High
Almora	Hawalbagh	High	Low	Moderate	Moderate
	Bhaisiyachhana	High	Low	Moderate	Moderate
	Bhikiyasain	High	Moderate	Moderate	High
	Chaukhutia	High	Moderate	Moderate	High
	Dhauladevi	High	Low	Moderate	Moderate
	Dwarahat	High	Low	Moderate	Moderate
	Lamgara	High	Low	Moderate	Moderate
	Sult	High	Moderate	Moderate	High
	Syaldey	High	Moderate	Moderate	High
	Takula	High	Low	Moderate	Moderate
	Tarikhhet	High	Low	Moderate	Moderate
Bageshwar	Bageshwar	High	High	High	High
	Garur	High	Moderate	High	High
	Kapkot	Very High	High	Very High	Very High
Champawat	Champawat	High	Moderate	Moderate	High
	Lohaghat	High	Moderate	Moderate	High
	Pati	High	Moderate	Moderate	High
	Barakot	High	High	Moderate	High
Pithoragarh	Berinag	High	Moderate	Moderate	High

District	Block	Landslide	Flood	Extreme rainfall	Composite exposure
	Didihat	High	Moderate	High	High
	Dharchula	Very High	Very High	Very High	Very High
	Gangolihat	High	Moderate	Moderate	High
	Kanalichhina	High	Moderate	Moderate	High
	Munakot	High	Moderate	Moderate	High
	Munsyari	Very High	High	Very High	Very High
	Vin (Berinag)	High	Moderate	Moderate	High
Haridwar	Bahadrad	Low	High	Low	Low
	Bhagwanpur	Low	Moderate	Low	Low
	Khanpur	Low	High	Low	Low
	Laksar	Low	Very High	Low	Moderate
	Narsan	Low	Moderate	Low	Low
	Roorkee	Low	High	Low	Low
Udham Singh Nagar	Bazpur	Low	High	Low	Low
	Gadarpur	Low	High	Low	Low
	Jasipur	Low	Moderate	Low	Low
	Kashipur	Low	Moderate	Low	Low
	Khatima	Low	Very High	Low	Moderate
	Rudrapur	Low	High	Low	Low
	Sitarganj	Low	Very High	Low	Moderate

10.5 Climate Variability Impacting Roads and Linear Settlements

This section synthesises the observed signals of climate variability already affecting Uttarakhand's roads and introduces the distinctive road–settlement feedback that shapes the sector's vulnerability. It first draws together the documented rainfall, landslide and corridor-disruption record, then turns to the ribbon-development dynamic through which roads reshape settlement and, in turn, amplify climate risk.

Rising Frequency and Intensity of Extreme Rainfall Events

IMD records indicate that Uttarakhand has reported more than 7,750 extreme rainfall events and cloudbursts since 2015, with a clear majority concentrated in the most recent years of the series (IMD, 2023). In October 2021, the state recorded 178.4 mm of rainfall in the first 18 days of the month, 485% above the climatological normal, triggering landslides and floods disrupted the Char Dham Yatra. The 2024 monsoon season witnessed similar episodes in Pithoragarh, Champawat, Bageshwar and Nainital districts, with orange and red rainfall alerts and resulting school closures recorded by the State Meteorological Department. The most recent confirmed event in the August 2025 Dharali (Uttarkashi) cloudburst, recorded a 24-hour rainfall episode that was 369% above the seasonal normal at the relevant IMD station.

Climate signal	Observed change / record	Consequence for the road system
Extreme rainfall & cloudburst frequency	>7,750 extreme rainfall/cloudburst events since 2015, concentrated in recent years (IMD, 2023; HT, 2021)	More frequent landslide and washout triggering; rising annual count of corridor closures
Rainfall intensity & concentration	Oct 2021: 178.4 mm in 18 days, 485% above normal; Aug 2025 Dharali 24-h total 369% above seasonal normal	Cloudburst-driven debris flows; carriageway, culvert and bridge loss; sudden corridor severance
Road-associated landslide density	309 landslides over 247 km of NH-7 - 1.25 per km- after the 2022 wet monsoon (Mey et al., 2024)	Chronic, spatially pervasive blockage independent of headline disasters
Proximity of failures to the alignment	811 landslides over ~800 km of pilgrimage highways, 81% within 100 m of the road (Geotech. & Geol. Eng.)	Failures concentrate exactly where traffic and roadside settlement are located
Rockfall exposure	Majority of engineered NH-7 slopes rated moderate-to-high rockfall risk; traffic a primary modifier (Sur et al., 2020; Singh et al., 2025)	Direct casualty and vehicle risk; hazard rating rises with traffic volume
Recurrence of corridor closure	July 2023: ~150 rural roads blocked; July 2025: >100 hill roads closed with pilgrims stranded (Newsonair, 2025)	Shift from exceptional to routine loss of connectivity
Off-season damaging events	Joshimath subsidence (Jan 2023) and Silkyara collapse (Nov 2023) fall outside the monsoon window	Maintenance and restoration planning built around a June–September season is mis-calibrated
Cumulative reconstruction burden	State 2025 estimate: 946 roads and 15 bridges severely damaged	Large, recurring rebuild cost absorbing sectoral budgets

Climate Variable → Road-System Transmission Matrix

The matrix below traces each climate variable through the four stages of the road system. Reading down a column shows what a single stage is exposed to; reading across a row shows how a single variable propagates from the terrain substrate to the settlements that depend on the corridor. Blank cells indicate no material pathway.

Climate variable	Slope & terrain substrate	Built assets (pavement, bridges, tunnels, retaining works)	Network & connectivity	Users & dependent settlements
Increased rainfall intensity	Rapid saturation, rising pore pressure, loss of cohesion; landslide and debris-flow triggering	Carriageway washout, culvert and bridge scour, retaining-wall failure, portal debris	Corridor severance with no alternate path; downstream nodes isolated	Stranded travellers and pilgrims; loss of provisioning, medical and market access
Rainfall concentration (fewer, more intense pulses)	Repeated rapid wet–dry cycling of already-fragile slopes	Drainage designed to lower-intensity storms is overtopped	Clustered, simultaneous closures across multiple corridors	Compound, network-wide disruption in a single event
Rising air temperature	Enhanced freeze–thaw weathering and permafrost/ice–cement degradation at altitude → rockfall	Bitumen softening and rutting; thermal stress on bridge joints	—	Heat exposure of stranded travellers (secondary)
GLOF / mass movement	Catastrophic channel and valley reconfiguration; aggradation	Total loss of bridges, roadbed and tunnels	Long-duration severance and loss of access	Prolonged regional isolation

Density of Road-Associated Landslides

The most consequential peer-reviewed finding on Uttarakhand's roads is from Mey, Guntu, Plakias, Silva de Almeida, and Schwanghart (2024), published in *Natural Hazards and Earth System Sciences* (Vol. 24, 3207–3223). Following an anomalously wet September–October 2022 monsoon, the authors conducted a GPS-based inventory of mass-movement features along the 247-km Rishikesh-Joshimath NH-7 corridor and documented 309 landslides totally or partially blocking the road - an average of 1.25 landslides per road kilometre. Modelling the inventory as a network-attached point pattern, they identified slope angle, rainfall amount and lithology as the principal controlling variables. The authors conclude unambiguously that "climate change and increasing exposure along this pilgrimage route will likely exacerbate landslide risk along the NH-7 in the future" (Mey et al., 2024).

A complementary, peer-reviewed survey published in *Geotechnical and Geological Engineering* (cited in Voisplanet, 2025) covering approximately 800 km of pilgrimage-route highways identified 811 landslides, of which 81% occurred within 100 metres of the road

alignment. A subsequent rockfall hazard assessment along NH-7 by Sur et al. (2020) and Singh et al. (2025), using the Rockfall Hazard Rating System (RHRS) and numerical simulation, classifies the majority of engineered slopes as exposed to moderate-to-high rockfall risk, with traffic exposure being a primary modifier in the rating.

Recurrent Disruption of Char Dham and National Highway Corridors

Documented in continuous news reporting and official Akashvani/State Emergency Operations Centre records, monsoon-season closure of the four pilgrimage-route national highways (NH-7, NH-34, NH-94, NH-109) has become an annual rather than exceptional phenomenon. In July 2023 alone, the Char Dham pilgrimage highways and approximately 150 rural roads were simultaneously blocked by landslides. In July 2024, the Gangotri NH near Bishanpur in Uttarkashi was blocked overnight by a single heavy landslide event (Newsonair, 2024). In July 2025, more than 100 hill roads were closed concurrently, with pilgrims stranded near Sonprayag and Gaurikund (Newsonair, 2025). The combined economic impact has not been comprehensively quantified, but the state's own 2025 estimate places damages from recent disasters at 946 roads and 15 bridges severely damaged.

Ribbon Development and the Linear Settlement Phenomenon

Road construction in Uttarakhand has not merely served pre-existing settlement patterns, it has generated new ones. The logic of this process is straightforward: a new road or a road improvement reduces transport costs for agricultural produce, improves access to schools and hospitals, and enables commercial activity at previously unviable locations. Households and enterprises rationally respond by locating as close to the road as possible, maximising accessibility. Over time, this produces the characteristic form of mountain linear settlement, a continuous or near-continuous strip of structures along both sides of a road, often extending for several kilometres without the two-dimensional spread that would characterise a market town on flat terrain.

This phenomenon is well-documented in the Uttarakhand context. Research from the Urban-Global Environmental Change (UGEC) Viewpoints (2016) on Nainital and surrounding areas documents the acceleration of urban growth in road-accessible mountain zones, with ribbon settlement emerging as the dominant spatial form. Research published on governance and urban risk in Uttarakhand Himalayan towns (ScienceDirect, 2019) identifies Almora as exemplifying this pattern: a historical ridge-top settlement that has extended as ribbon development along approach roads into the valley below, occupying terrain with high landslide susceptibility. Sati and Kumar (2019), writing in *Mountain Research and Development*, characterise mountain communities in Uttarakhand as being "caught between a rock and a hard place" simultaneously exposed to natural hazards on steep terrain and economically dependent on market access that is only available by situating settlements in road corridors that traverse that terrain.

The Joshimath case provides the most extensively documented and climatically significant example. Joshimath (population approximately 23,000; elevation 1,890 metres), the gateway town for Badrinath, Auli ski resort, Hemkund Sahib, and the Valley of Flowers, has experienced a near-doubling of its built-up area over 17 years, driven primarily by the tourism and pilgrimage economy accessing these destinations along the only road corridor through the upper Alaknanda Valley (Pandey, 2023; Down to Earth, 2023). The Char Dham road widening beneath the town, which required blasting and heavy excavation at the toe of the slope on which Joshimath sits, is documented as a contributing factor to the acceleration of subsidence, illustrating the direct interaction between road-driven settlement, road investment, and geotechnical destabilisation (Down to Earth, 2023).

Climate Hazard	Climate Driver	Climate Stressor	Direct Impact	Functional Impact	Consequences	Confidence
Rising Temperature	Increasing average and summer temperatures and more frequent heat extremes	Thermal expansion, pavement softening and accelerated material deterioration	Rutting, cracking and reduced service life of flexible pavements; thermal stress on bridges and road structures	Increased maintenance requirements and reduced structural performance of transport infrastructure	Higher maintenance costs, reduced asset lifespan and disruption to transport services	Medium–High
Changing Rainfall Pattern	Erratic monsoon onset and withdrawal and uneven rainfall distribution	Reduced drainage efficiency, fluctuating soil moisture and prolonged wet conditions	Accelerated pavement deterioration, shoulder erosion and drainage failure	Reduced road serviceability and increased maintenance frequency	Increased travel delays, higher maintenance expenditure and reduced transport reliability	Medium–High
Extreme Rainfall / Cloudbursts	High-intensity precipitation events and cloudbursts	Surface runoff exceeding drainage capacity, debris flow and slope failures	Washout of roads, culverts and retaining structures; bridge scour and embankment erosion	Closure of highways, district roads and rural roads and interruption of traffic movement	Transport disruption, economic losses and delayed emergency response	High
Flooding / Flash Floods	River overflow and extreme runoff	Inundation of roads, bridges and low-lying transport corridors	Damage to bridges, culverts, road embankments and river-crossing infrastructure	Loss of transport connectivity and prolonged restoration of road services	Isolation of communities, disruption of supply chains and increased reconstruction costs	High
Landslides & Slope Instability	Intense or prolonged rainfall and unstable mountain slopes	Rockfalls, debris slides and slope failures along road corridors	Damage to carriageways, retaining walls, tunnels and protective structures	Reduced accessibility and prolonged closure of strategic transport corridors (e.g., Char Dham routes)	Isolation of settlements, disruption of tourism and trade and increased maintenance burden	High
Glacier Retreat & Permafrost Degradation	Rising temperatures, declining glacier mass and changing freeze–thaw cycles	Increased slope instability, glacial sediment movement and changing hydrological regimes	Damage to high-altitude roads, bridges and protective infrastructure	Reduced reliability of transport infrastructure in glacier-fed catchments	Increased maintenance costs, higher infrastructure risk and reduced connectivity to remote regions	Medium

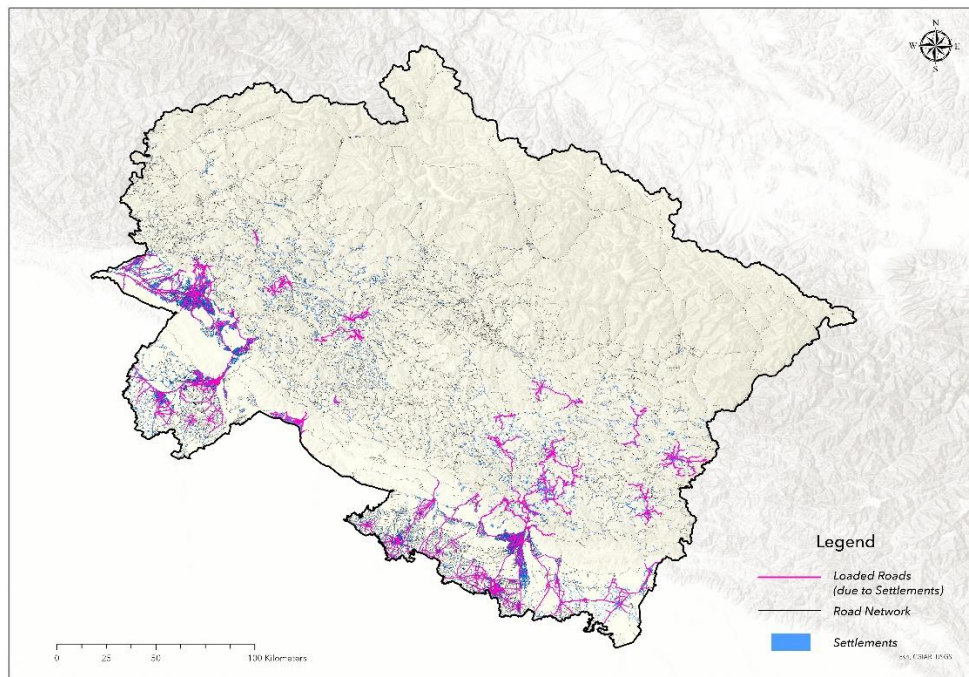
River Morphological Change & Sedimentation	Intense rainfall, erosion and changing river dynamics	Channel migration, aggradation and increased bridge scour	Undermining of bridge foundations, culvert blockage and embankment instability	Reduced structural safety of transport infrastructure	Increased bridge failures, traffic restrictions and expensive rehabilitation works	Medium–High
Forest Fires	Warmer, drier pre-monsoon conditions	Fire damage, smoke and reduced visibility along road corridors	Damage to roadside safety infrastructure, power supply to transport systems and reduced driver visibility	Increased accident risk and temporary road closures	Disruption of transport operations and increased emergency management costs	Medium
Freeze–Thaw Variability	Reduced winter snow cover and fluctuating temperatures around freezing point	Repeated freeze–thaw weathering of pavement and slope materials at higher elevations	Pavement cracking, pothole formation and rockfall initiation	Reduced pavement quality and increased slope maintenance requirements	Higher maintenance costs and increased road safety risks	Medium
Compound Climate Hazards (Extreme Rainfall + Floods + Landslides)	Increasing frequency and convergence of multiple climate extremes	Simultaneous failure of slopes, bridges, drainage systems and transport infrastructure	Widespread damage across road networks and prolonged closure of major transport corridors	System-wide disruption of passenger movement, freight transport and emergency access	Long-term connectivity loss, severe economic impacts, disruption of tourism and reduced transport system resilience	High

How Linear Settlement Amplifies Climate Vulnerability...

The co-location of road infrastructure and linear settlement creates a compound vulnerability system that is qualitatively different from either hazard in isolation. Four mechanisms link linear settlement to amplified climate risk, and each is documented in the Uttarakhand record.

1. **Cumulative slope destabilisation from distributed construction.** Building on steep hillsides above and below the alignment requires excavation that replicates, at smaller scale, the destabilisation dynamics of road-cuts: reduced cohesion, disrupted drainage and elevated pore-water pressure. Septic systems, domestic water channels and unlined drains concentrate additional water on slopes already weakened by cutting. In aggregate, a ribbon of several hundred structures along a single kilometre applies a distributed but cumulative load-and-drainage burden that standard geological hazard assessments conducted once, at the time of road construction do not capture.
2. **Occupation of marginal, high-susceptibility terrain.** Because accessibility is maximised by proximity to the road, ribbon development forces construction onto the very steep slopes a flat-terrain town would avoid. The housing stock is therefore disproportionately located in zones of high landslide and rockfall susceptibility, as the Almora and Joshimath cases illustrate, so exposure is not incidental but structural to the settlement form.
3. **Concentration of exposure and dependence on a single, non-redundant corridor.** As settlement density along a corridor rises, so does the number of people whose daily provisioning, medical access and economic activity depend on that road's operational status. Because the corridor is typically the only route radial and series-connected, directly analogous to the hill power grid, a single severance isolates the entire ribbon at once, with no alternative path for supply, evacuation or emergency response. Restoration time is then governed not by the repair itself but by physical access, which the same hazard has usually cut, so disruption in inhabited corridors is measured in days rather than hours.
4. **Feedback between road investment and geotechnical destabilisation.** The road access that drives ribbon development also drives its upgrading: widening, hill-cutting and tunnelling undertaken to serve the settlement and its pilgrimage traffic destabilise the very slopes the settlement occupies. The Char Dham widening beneath Joshimath, blasting and excavation at the toe of the slope on which the town

sits and the Silkyara tunnelling illustrate a positive feedback in which the economic driver of ribbon development recursively increases the hazard it is exposed to.



The Joshimath Land Subsidence Event (2023) and Cascading Highway Stress

On 2 January 2023, residents of Joshimath, a town at the head of the Alaknanda valley and a critical node on NH-7, reported widespread cracking of more than 800 buildings, with documented vertical displacement progressing over subsequent weeks. Geodetic analysis by Sreejith and colleagues (2024), published in *Geophysical Research Letters*, demonstrated that Joshimath landslide kinematics are episodic, with discrete acceleration events triggered by extreme rainfall episodes, most notably the 19 October 2021 event. The cascading consequence for the road sector was severe: NH-7 traffic restrictions and a state-imposed construction ban affected the upstream Tapovan-Vishnugad hydropower project access. The event also drew explicit attention to the fact that road widening, hill cutting and tunnelling along NH-7 above and below Joshimath had contributed to the destabilisation (Sreejith et al., 2024).

Tunnel Failures and Strategic Vulnerability

On 12 November 2023, a section of the under-construction Silkyara Bend–Barkot tunnel on NH-134 collapsed, trapping 41 tunnel workers for 16 days before successful rescue on 28 November (Wikipedia, NH-134; PIB, 2023). While the collapse mechanism was primarily geological, its occurrence at a strategic all-weather Char Dham connector, designed precisely to reduce climate-driven monsoon closures, underscores that climate adaptation through tunnelling is itself subject to compound risk in fragile Himalayan lithology.



10.6 Potential Future Climate Changes and Their Implications for Roads

Climate projections from CMIP6-based downscaling exercises consistently point to intensified hazard regimes for Uttarakhand's roads. The most policy-relevant projections are summarised below.

Projected Change in Road-Relevant Variables

Variable (far future, 2071–2100)	SSP1-2.6	SSP5-8.5	Mechanism and road-sector relevance
Mean annual temperature ($\Delta^{\circ}\text{C}$)	() - indicative +1.0 to +1.8 $^{\circ}\text{C}$	() - indicative +3.5 to +5.5 $^{\circ}\text{C}$ [confirm mean-annual value from projections chapter]	Drives freeze–thaw weathering of rock slopes and pavement, permafrost degradation and pavement thermal distress; elevation-dependent warming means high-altitude alignments - where the most hazard-exposed length lies - warm faster than the State mean
Extreme-heat days (TXx)	() - moderate increase	+3.9 to +4.8 $^{\circ}\text{C}$ (State ensemble hottest-day change)	Asphalt rutting, bleeding and shoving under traffic; thermal stress on rigid pavements and bridge expansion joints
Mean annual precipitation ($\Delta\%$)	+17 to +24 % (State ensemble, district means)	+56 to +63 % (State ensemble, district means)	Total load on drainage rises, but the increase is concentrated in high-intensity events, so the annual total understates the design challenge
Extreme-rainfall day frequency	() - moderate increase	() - large increase	The dominant trigger of shallow landslides, debris flows and embankment washouts; each event overtops cross-drainage sized to a stationary return period that no longer holds - the single most consequential road driver
Short-duration rainfall intensity (sub-daily maxima)	() - moderate increase	() - large increase	Sub-daily bursts exceed culvert and side-drain capacity, causing overtopping and scour even when daily totals are moderate
Snowfall fraction / seasonal snow cover	() - moderate decline	() - steep decline	Fewer sustained snow-closure days at mid-elevation, but a shift toward rain-on-snow and wet-snow avalanche cycles; net effect on closure days is ambiguous and elevation-dependent
Zero-degree isotherm / freezing-level rise	() - rises	() - rises substantially	Pushes the rain (rather than snow) domain upslope, saturating and destabilising cut slopes and colluvium at elevations previously protected by a frozen or snow-bound regime - expands the landslide-prone area
Freeze–thaw cycle regime	() - band migrates upslope	() - pronounced upslope migration	High-altitude pavements and rock slopes see more thaw cycles as winters warm into the cycling range, accelerating fatigue and rockfall; lower elevations see fewer
High-altitude permafrost / active-layer depth	() - indicative degradation	() - substantial degradation	Thaw of ice-cemented slopes and embankment fill causes settlement, loss of bearing capacity and foundation instability for high-altitude roads and bridges; exposure layer is non-stationary

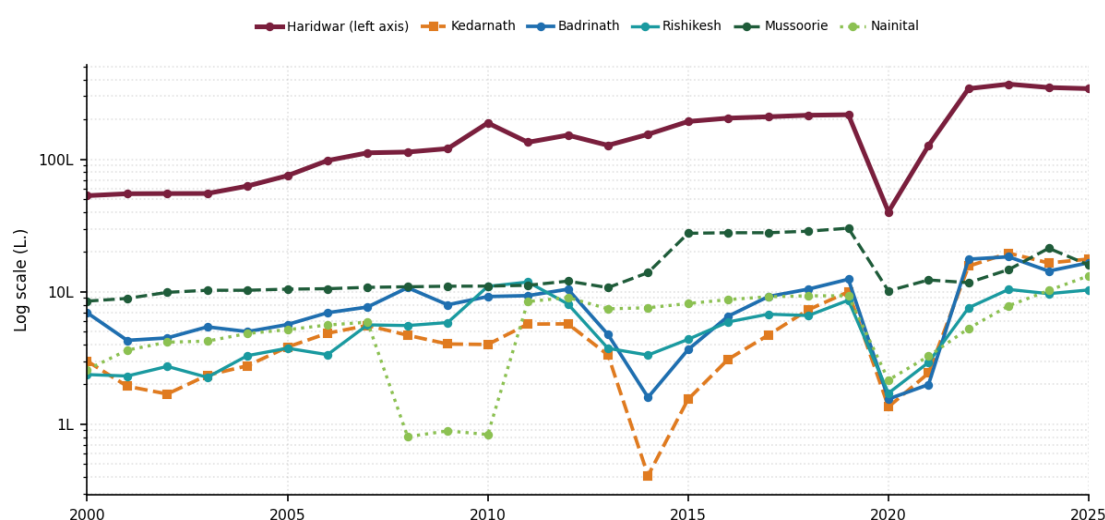
Landslide- and debris-flow-triggering days	() - moderate increase	() - large increase	The critical result: the hazard that most directly and most frequently closes the network intensifies, lengthens its season and migrates upslope into terrain previously treated as stable - connectivity is the binding constraint on a network asset
Flood magnitude at river crossings	() - higher, modestly variable	() - higher volume, high interannual variance	Increased bridge scour, pier undermining and approach-embankment erosion; design floods exceeded, and the exposure of valley-floor alignments grows
GLOF frequency and lake area	() - elevated	() - substantially elevated	Outburst floods can destroy valley-floor road and bridge assets in a single event; hazard itself grows and the exposed network is non-stationary

5. Tourism

Tourism constitutes a critical pillar of Uttarakhand's economy, contributing significantly to employment generation, local livelihoods, and regional economic development. The state's diverse physiographic and ecological characteristics support a broad spectrum of tourism activities, attracting substantial numbers of domestic and international visitors throughout the year. Nature-based tourism is concentrated in the Himalayan landscapes, protected areas, and hill destinations, including Corbett National Park and Tiger Reserve, Nainital, Mussoorie, Almora, Kausani, Ranikhet, Lansdowne, Auli, Chakrata, and Bhimtal. These destinations offer opportunities for wildlife tourism, landscape-based recreation, and mountain tourism.

Uttarakhand is also one of India's most prominent pilgrimage destinations, reflecting its cultural and spiritual significance. The state hosts several nationally and internationally recognized religious sites, including the Char Dham shrines, Haridwar, Rishikesh, Jageshwar, Devprayag, and Hemkund Sahib, which collectively attract millions of pilgrims annually and constitute a major component of the state's tourism economy.

The state's rich natural capital, encompassing snow-clad mountain ranges, glaciers, alpine meadows, river systems, forests, and high-altitude lakes, provides the foundation for a rapidly expanding adventure tourism sector. Activities such as trekking, mountaineering, skiing, rock climbing, camping, paragliding, and white-water rafting have emerged as important tourism products, supported local enterprises and enhanced livelihood opportunities in mountain communities. Furthermore, Uttarakhand is home to several internationally recognized natural heritage sites, including the UNESCO World Heritage Site comprising the Valley of Flowers National Park and the Nanda Devi National Park, which contribute to the state's global tourism profile and biodiversity conservation objectives.



Long-run visitor arrivals at major Uttarakhand destinations, 2000-2025 (log scale, lakhs)

Source: Uttarakhand Tourism Department

The state attracts an estimated 37-39 million tourists annually at the pre-COVID peak (2018-19), rising to over 55 million in recent years (Uttarakhand Tourism Department, tourist statistics 2000-2025), generating approximately ₹12,000-15,000 crore in direct and indirect tourism

revenues and supporting livelihoods across the hotel, transport, guiding, retail, and food sectors (Government of Uttarakhand, 2022). Tourism contributes approximately 5-6% of state GDP and is identified as a priority growth sector in successive state economic policy frameworks.

The sector is structured across six distinct but climatically interdependent sub-categories, each occupying a specific ecological and altitudinal niche within Uttarakhand's Himalayan geography.

Sub-sector	Scale indicator	Peak season	Dominant climate drivers
Pilgrimage and religious tourism (Char Dham, Hemkund Sahib)	4.5-5.6 million pilgrims/yr; ₹7,500 crore/yr	May-June; Sept-Oct	Snowpack persistence; monsoon landslide/flood risk; heat at approach towns
Adventure tourism (trekking, rafting, skiing)	~700,000 adventure tourists/yr across segments	Apr-Jun; Sept-Nov; Jan-Mar (skiing)	Snowpack depth; monsoon river discharge; weather-window stability
Eco-tourism and wildlife tourism	~360,000 visitors/yr to Corbett alone (2024)	Nov-June (parks closed in monsoon)	Habitat and species phenology; flood damage to park infrastructure
Heritage and cultural tourism	20-38 million visitors/yr (Haridwar/Rishikesh)	Year-round; Kumbh cycle every 12 yrs	River flow and water quality; monsoon flood risk to ghats
Health and wellness tourism	Hundreds of thousands of international/ domestic visitors	Year-round; peak Oct-Mar	Air quality; forest-fire smoke; river water quality
Hill-station and leisure tourism	Millions of domestic plains visitors/yr	Apr-June (summer escape)	Cool-refuge differential vs plains; landslide risk to hill-town infrastructure

Religious and Pilgrimage Tourism

The Char Dham Yatra, comprising the four shrines of Kedarnath, Badrinath, Gangotri, and Yamunotri, constitutes the largest single pilgrimage circuit in the Indian Himalaya, drawing 4.5-5.6 million pilgrims annually in recent years (Badrinath-Kedarnath Temple Committee, 2023). The yatra operates within a narrow seasonal window (May-June and September-October), concentrated in high-altitude valley corridors at elevations of 1,900-3,880 metres. Hemkund Sahib (4,329 m), attracting Sikh pilgrims, and the Panch Kedar circuit complete the state's high-altitude religious tourism portfolio. The economic multiplier of Char Dham is substantial: pilgrim expenditure on transport, accommodation, food, and ritual services circulates across mountain communities along the route corridors.

Adventure Tourism

Trekking and mountaineering form the backbone of Uttarakhand's international tourism brand, with the state hosting routes to some of the most celebrated high-altitude destinations in Asia, Roopkund (5,029 m), Valley of Flowers (UNESCO World Heritage), Kedarkantha, Har Ki Dun, Milam Glacier, and Pindari Glacier. River rafting and kayaking on the Ganga (Rishikesh-Devprayag segment), Alaknanda, and Tons rivers attract an estimated 500,000 adventure tourists annually (Adventure Tourism Association of India, 2022). Skiing at Auli (2,519-3,049 m, Chamoli district) and snow sports at Munsiyari represent Uttarakhand's winter adventure economy, with Auli serving as the site of the National Winter Games.

Eco-Tourism and Wildlife Tourism

Jim Corbett National Park, India's oldest national park and a premier tiger reserve, anchors the state's wildlife tourism economy, receiving approximately 360,000 visitors annually (Uttarakhand Tourism Department, 2024). Rajaji Tiger Reserve, Nanda Devi National Park (UNESCO World Heritage Site), Govind Pashu Vihar, and Binsar Wildlife Sanctuary complete the protected area network. The Valley of Flowers National Park, an alpine meadow UNESCO site at 3,352-3,658 m, attracts high-value ecotourists during its narrow seasonal window (July-September) when its globally significant wildflower assemblage is in bloom.

Heritage and Cultural Tourism

Uttarakhand's heritage tourism encompasses ancient temple complexes of Haridwar and Rishikesh, which together receive more than 20 million visitors annually (rising above 35 million in recent years) for religious bathing, the Kumbh Mela cycle (every 12 years), and yoga and wellness retreats, alongside the colonial hill-station heritage of Mussoorie, Nainital, Lansdowne, and Almora. The Garhwali and Kumaoni folk traditions, crafts, and cuisine anchor cultural tourism programming across the hill districts.

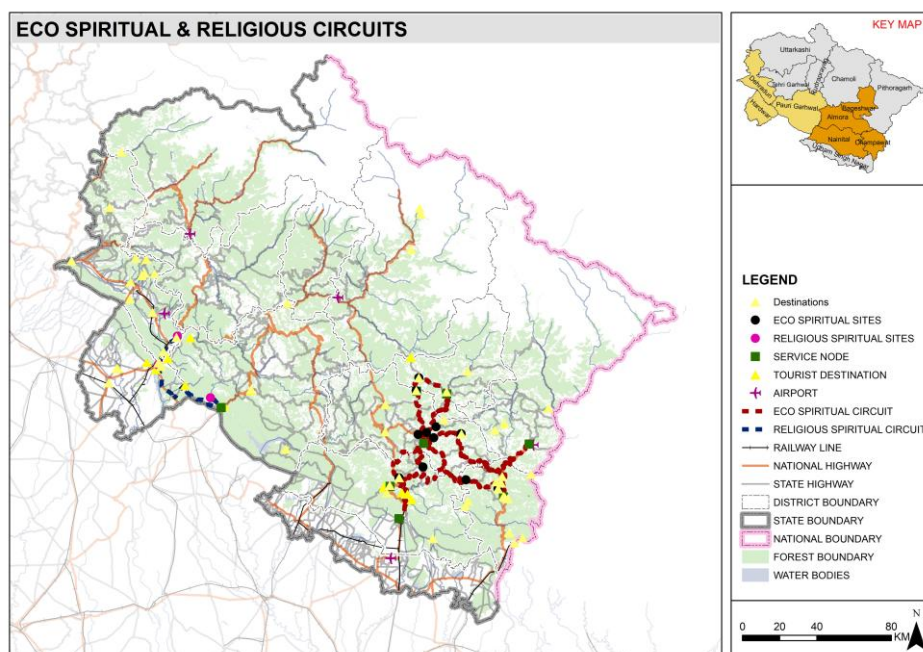
Health and Wellness Tourism

Rishikesh has established an internationally recognised identity as the "Yoga Capital of the World," hosting thousands of yoga ashrams and wellness retreat centres that attract a growing international clientele. Ayurvedic health tourism and forest bathing (shinrin-yoku) programmes in the Himalayan temperate forest zones are emerging sub-segments. The sector depends critically on clean air, temperate climate conditions, and the spiritual and ecological character of mountain environments, attributes that are directly threatened by climate change.

In recent years, the tourism sector has undergone increasing diversification, with growing emphasis on ecotourism, rural tourism, cultural tourism, agri-tourism, wellness tourism, and recreational freshwater angling. These emerging tourism segments promote community participation, diversify rural income sources, and support the sustainable utilization of natural and cultural resources. The continued growth and diversification of the tourism sector underscore its strategic importance for the state's socio-economic development while simultaneously increasing its dependence on the integrity of climatic, ecological, and environmental systems.

Institutional Structure

The sector is coordinated by the Uttarakhand Tourism Development Board and the Tourism Department, with execution split across several bodies whose mandates any adaptation measure in this chapter must be assignable to: Garhwal Mandal Vikas Nigam (GMVN) and Kumaon Mandal Vikas Nigam (KMVN) for accommodation and circuit infrastructure; the Badrinath-Kedarnath Temple Committee (BKTC) for the Char Dham shrines; the Forest Department and park management authorities for Corbett, Rajaji, Nanda Devi and Valley of Flowers; the State Disaster Management Authority (SDMA) for Yatra-route safety and evacuation; and the Public Works Department (PWD) for the road network on which every sub-sector ultimately depends. The vulnerability index for the sector is structured so that each indicator maps to one of these owners.



Uttarakhand tourism destinations, circuits and access network.

11.2 Defining Drivers and Their Climate Implication

This section sets out the climate stressors and drivers acting on each tourism sub-sector: the physical climate mechanisms (temperature, precipitation, snowpack, discharge, extreme events) through which climate change alters conditions for tourism activity.

Tourism	Relevance for Climate Sensitivity	Primary Climate Drivers	Climate Stressors
Nature- and pilgrimage-dependent tourism	Tourism is strongly dependent on the condition and accessibility of mountain landscapes, glaciers, rivers, forests, biodiversity, and religious destinations.	Rising temperature, changing precipitation patterns, glacier retreat, extreme rainfall, GLOFs	Changes in environmental quality, seasonal accessibility, hydrological regimes, and hazard exposure.
Highly seasonal tourism	Tourism activities are concentrated within distinct pilgrimage, summer, winter, wildlife, and trekking seasons.	Rising temperature, changing snowfall patterns, monsoon variability, extreme weather events	Changes in seasonal climatic conditions, snowfall duration, rainfall distribution, and weather reliability.
Mountain accessibility dependence	Tourism destinations rely primarily on mountain transport corridors traversing steep and geologically fragile terrain.	Extreme rainfall, flash floods, landslides, avalanches, GLOFs	Increased exposure of transport corridors to slope instability, flooding, snow accumulation, and geomorphic hazards.
Snow- and glacier-dependent tourism	Winter tourism and high-altitude recreation depend on persistent snow cover and stable cryospheric conditions.	Rising temperature, reduced snowfall, earlier snowmelt, glacier retreat	Reduced snow accumulation, shorter snow persistence, glacier recession
Climate-sensitive tourism infrastructure	Tourism infrastructure is concentrated in river valleys, hill settlements, and other environmentally sensitive locations.	Extreme rainfall, floods, landslides, heat stress, water scarcity	Increased exposure to hydrometeorological hazards, thermal stress, and variability in water availability.

Tourism-dependent local economy	Tourism supports a substantial share of employment and local economic activity across multiple service sectors.	Climate variability, extreme weather events, multiple climate hazards	Increased variability in tourism operating conditions and seasonal climatic suitability.
Dependence on ecosystem services	Tourism relies on healthy forests, biodiversity, rivers, wetlands, and scenic landscapes to sustain recreational and cultural experiences.	Rising temperature, changing precipitation, forest fires, ecosystem change	Changes in ecosystem structure, ecological processes, habitat condition, and environmental quality.
High disaster-risk perception	Tourism demand is influenced by perceptions of destination safety and environmental stability.	Extreme rainfall, floods, landslides, GLOFs, forest fires	Increased occurrence and visibility of hydro-meteorological and geophysical hazard conditions.

The tourism sector in Uttarakhand encompasses diverse sub-sectors, including pilgrimage and religious tourism, adventure and snow-based tourism, ecotourism and wildlife tourism, and hill-station and wellness tourism, each exhibiting distinct sensitivities to climatic variability and long-term climate change. Pilgrimage tourism, particularly the Char Dham Yatra and other high-altitude religious circuits, is intrinsically linked to seasonal climatic conditions that govern shrine accessibility, road connectivity, and visitor safety. The operational period of these pilgrimages is influenced by winter snow accumulation, the timing of snowmelt, monsoon rainfall, and the occurrence of landslides, flash floods, and glacial lake outburst floods (GLOFs), all of which affect accessibility along narrow Himalayan valleys.

Activities such as skiing, winter trekking, mountaineering, and snow-based recreation rely on the duration, depth, and reliability of seasonal snow cover, while high-altitude trekking routes are influenced by snowmelt dynamics, slope stability, and the frequency of extreme weather events. River-based adventure tourism, including white-water rafting, is closely associated with seasonal river discharge, which is regulated by snowmelt and monsoonal precipitation and is increasingly influenced by climatic variability.

Ecotourism and wildlife tourism depend on the ecological integrity of forests, protected areas, and riverine ecosystems that support biodiversity and wildlife populations. Climatic changes affecting temperature regimes, precipitation patterns, habitat conditions, and ecosystem processes have the potential to alter species distribution, seasonal behaviour, vegetation dynamics, and habitat suitability, thereby influencing the ecological attributes that underpin nature-based tourism. In addition, protected area accessibility and visitor services may be affected by extreme weather events and associated disruptions to transport infrastructure.

Hill-station and wellness tourism derive their comparative advantage from the temperate climatic conditions, scenic landscapes, and environmental quality of the Himalayan region. Climatic warming, changing precipitation patterns, declining snow cover, increased frequency of forest fires, and seasonal water stress have the potential to modify the environmental conditions that support these tourism products. As climate change continues to alter the physical and ecological characteristics of mountain environments, the relative attractiveness, seasonality, and operational conditions of multiple tourism sub-sectors are expected to evolve, highlighting the need to integrate climate considerations into tourism planning and destination management across Uttarakhand.

Eight hazards dominate climate risk to the tourism sector in Uttarakhand. For each, this section sets out the physical mechanism, the tourism-chain stage at which it acts, the sub-sectors and geographies affected, the documented historical record, and the residual data gap. Together the record establishes that climate risk to the sector is not a prospective concern but an ongoing, measurable loss stream - with two distinct signatures: rare, catastrophic, life-and-capital-destroying events (the 2013 Kedarnath disaster), and a chronic, high-frequency attrition of route access, pilgrim health and destination reputation that never dominates the national news cycle but accounts for the larger share of cumulative economic and human loss.

Hazard Register

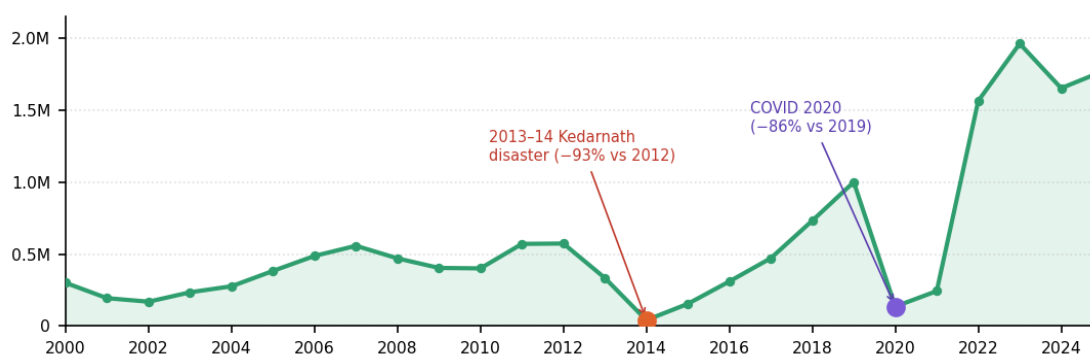
Hazard	Mechanism of damage	Tourism-chain stage affected	Sub-sectors primarily affected	Principal districts / circuits
GLOF	Instantaneous debris-laden surge destroys bridges, shrine approaches and riverside infrastructure; severs access for weeks	Access; resource (glacier-proximate sites)	Pilgrimage; adventure trekking; hydropower-adjacent tourism	Chamoli, Uttarkashi, Pithoragarh, Bageshwar
Landslide and slope failure	Road burial and bridge loss; strands pilgrims and tourists; blocks evacuation	Access; visitor safety	All circuits — Char Dham, hill-station, adventure	Rudraprayag, Chamoli, Tehri, Pauri, Uttarkashi, Pithoragarh
Extreme rainfall / cloudburst	Flash flooding of valley-floor accommodation and shrine approach towns; triggers landslides	Access; accommodation; visitor safety	Char Dham Yatra; hill-station; wellness (Rishikesh/Haridwar)	Uttarkashi, Chamoli, Rudraprayag, Nainital, Pithoragarh
Riverine and flash flooding	Inundation of riverfront ghats and rafting infrastructure; channel migration	Resource; accommodation	Wellness and pilgrimage (Haridwar/Rishikesh); river rafting	Ganga corridor: Haridwar, Dehradun, Tehri
Forest fire	Destroys wildlife habitat and hill-station forest cover; smoke degrades air quality and visibility	Resource; visitor experience	Wildlife/eco-tourism; hill-station and wellness	Pauri Garhwal, Nainital, Almora, Champawat, Tehri
Temperature extremes and altitude stress	Heat stress at approach towns compounds with unacclimatised rapid ascent; cardiac and altitude mortality	Visitor health and safety	Char Dham pilgrimage (elderly-dominant)	Haridwar, Rishikesh, Dehradun approach; Kedarnath, Badrinath shrines
Snowpack declines and winter-	Erosion of natural seasonal snow reservoir; shortens	Resource; visitor experience	Winter adventure tourism (skiing)	Auli (Chamoli), Munsiyari (Pithoragarh)

Hazard	Mechanism of damage	Tourism-chain stage affected	Sub-sectors primarily affected	Principal districts / circuits
season compression	commercially viable ski window			
Phenological shift	Earlier or less predictable flowering/bloom timing decouples fixed booking calendars from actual peak conditions	Resource; visitor demand	Eco-tourism (Valley of Flowers)	Chamoli (Bhyundar valley)

Extreme Rainfall, Landslides, and Flood Events

Extreme hydrometeorological events are the most sudden and economically destructive climate hazards for Uttarakhand's tourism system. Cloudbursts - rainfall exceeding 100 mm per hour at a point - and the flash floods and landslides they trigger destroy road access, damage or eliminate accommodation, strand pilgrims and tourists in hazard-exposed locations, and generate the catastrophic events that dominate tourism's risk profile. The State Disaster Management Authority documents a 240% increase in hydro-meteorological disasters between 2000 and 2022, with cloudburst events rising from 11 (2000-2010) to 83 (2011-2023) (SDMA, 2023). ISRO recorded 12,319 landslide events across Uttarakhand between 1988 and 2023, with the 2024 monsoon season generating 1,521 landslides in a single 17-day period - a rate with no documented precedent (PreventionWeb, 2024).

- The 2013 Kedarnath disaster - triggered by cloudbursts delivering over 340 mm in 24 hours - resulted in **5,748 confirmed deaths**, destruction of 1,520 road segments and 154 bridges, and a **30% decline in total state tourist arrivals** for the subsequent year. The World Bank (2013) estimated total economic losses at **USD 3.8 billion** - exceeding the entire state budget for FY 2013-14.
- In August 2023, 420 mm of rainfall fell in 24 hours at select Uttarakhand catchments; the Gaurikund landslide buried three vehicles killing four pilgrims from Madhya Pradesh; widespread Char Dham route closures stranded thousands (Deccan Herald, 2023). Infrastructure losses across the North India region during August 2023 floods were estimated at USD 1.2 billion.
- In 2024, **2,946 landslides** were recorded - 1,521 in the first 17 monsoon days - repeatedly closing all four Char Dham corridors simultaneously, with 132 critical landslide zones mapped on Yatra routes (SLRMC, 2024; Prevention Web, 2024).



Kedarnath annual pilgrim arrivals, 2000-2025 (Uttarakhand Tourism Department). The 2013-14 disaster cut arrivals by ~93% versus the 2012 peak; COVID cut 2020 arrivals by ~86% versus 2019.

Note: Mey et al. (2024, Natural Hazards and Earth System Sciences): 300 landslides surveyed along the NH-7 (Rishikesh-Joshimath) corridor following the September-October 2022 rainfall event - more than one road-blocking landslide per kilometre of highway. Identifies slope angle, rainfall intensity, and lithology as primary controls - all worsening under climate projections. Scientific Reports (2025, Char Dham Carrying Capacity): Documents pilgrim growth from ~1 million/yr (early 2000s) to ~5 million+ in 2022-23 and quantifies trail carrying capacity limits. SDMA (2023): 240% increase in hydrometeorological disasters 2000-2022.*

Temperature Rise, Heat Stress, and Altitude Sickness - The Pilgrim Health Crisis

Climate warming at valley-floor entry towns and the absence of acclimatisation protocols for rapid high-altitude ascent constitute the most lethal but most under-acknowledged tourism-climate risk in Uttarakhand. Dehradun recorded 43°C in May 2024, unprecedented for a station at 640 m (IMD, 2024). Haridwar and Rishikesh, the primary pilgrimage entry points at 290 m and 372 m respectively, sit in districts recording a statistically significant increase in days exceeding 40°C since 2000 (IMD, 2023). The Char Dham shrines open in late April and May - precisely the period of maximum pre-monsoon heat stress at approach towns and maximum pilgrim volumes. Pilgrims, predominantly elderly (over 60% aged above 55 years), arrive after long rail or road journeys in heat, then ascend to 3,880 m (Kedarnath) or 3,300 m (Badrinath) by helicopter or foot often without any acclimatisation.

- Documented annual Char Dham pilgrim deaths: 112 (2017), 102 (2018), 90+ (2019), 300+ (2022), 230+ (2023), 200+ estimated (2024). The 2022 season saw 120 deaths in just the first 27 days, prompting Union Health Ministry intervention - an acute death rate documented in a medical journal paper by Government Doon Medical College physicians as 'not at par with previous years' records' (Deccan Herald, 2022).
- The primary cause in the majority of cases is cardiac arrest at altitude directly triggered by the combination of unacclimatised rapid ascent (helicopter to 3,880 m without any altitude preparation), exertion in cold, low-oxygen conditions, and pre-existing cardiovascular or hypertensive comorbidities common in the elderly pilgrim demographic (Gulf News, 2022; Drishti IAS, 2024).
- The helicopter access model - which enables pilgrims to bypass the 16-18 km trek from Gaurikund to Kedarnath - paradoxically increases death risk by eliminating the gradual acclimatisation that the walk provides. The case of Vimla Ben (64, Gujarat) who died of

cardiac arrest immediately upon helicopter landing at Kedarnath is emblematic of this pattern (Hindustan Times, 2022).

GLOF Risk

The state's cryospheric assets - 968 glaciers covering 2,857 km² - underpin multiple tourism sub-sectors: the spiritual and visual draw of glacier-proximate shrines, the snowpack sustaining ski tourism at Auli, the perennial river flows sustaining white-water rafting on the Ganga and Alaknanda, and the ecological integrity of high-altitude trekking environments. Climate change is rapidly degrading all of these simultaneously. The 2024 winter was described as snowless at several mid-altitude tourism sites, forcing the Auli ski resort to rely on artificial snow-making at commercial scale for the first time (Carbon Impacts, 2024).

- The February 2021 Chamoli GLOF - a 27-million m³ rock-ice avalanche at ~25 m/s (Shugar et al., 2021, Science) - killed over 200 people, destroyed two hydropower projects and 11 bridges, and severely disrupted the Niti Valley tourism circuit accessing the Nanda Devi Biosphere Reserve. SLRMC (2025) has inventoried 426 glacial lakes in Uttarakhand, of which 25 are classified as Potentially Dangerous Glacial Lakes (PDGLs) each posing direct GLOF risk to downstream tourism corridors.
- The October 2022 Draupadi Ka Danda avalanche killed 27 mountaineers from the Nehru Institute of Mountaineering, India's worst mountaineering disaster on record, during a season with anomalously low snow consolidation attributable to warming-induced thermal extremes (Wikipedia, 2022). The June 2024 Sahastra Tal blizzard killed 9 trekkers from the Karnataka Mountaineering Association during an unseasonable late-spring snowstorm (Wikipedia, 2024).

Note: Shugar et al. (2021, Science, 373(6552), 300-306): Comprehensive analysis of 2021 Chamoli GLOF causation and consequences. SLRMC (2025): 25 PDGLs classified out of 426 glacial lakes; 25 PDGLs with downstream tourism exposure. Scientific Reports (2025, Char Dham carrying capacity): Documents glacial and environmental stress on Char Dham tourism ecosystem. Mina et al. (2023, Science of Total Environment, 862, 160679): Forest fire escalation linked to warming, 41,600 incidents in 2019; smoke and fire disrupting tourism landscapes. Wikipedia (2022, 2024): Draupadi Ka Danda and Sahastra Tal fatal events documented.*

11.3 Observed Climate Impacts on Tourism

Tourism is among the most climate-exposed pillars of Uttarakhand's economy, and also one of the most economically important. The state's appeal rests almost entirely on climate- and landscape-dependent assets: the seasonal opening of the high Char Dham shrines, the cool summer air of its hill stations, the snow of its winter-sport slopes, the white water of its rivers, and the forests, meadows and wildlife of its protected areas. Each of these is the product of a particular climatic regime, and each is now being reshaped by climate change.

The sector's economic weight makes this exposure consequential. Services dominate the state economy, and tourism is among their principal drivers; the Char Dham Yatra alone is the single largest revenue earner, drawing millions of pilgrims each season and supporting a dense ecosystem of hotels, transport operators, guides, porters and small traders across the hills (India Tourism Statistics, 2023). Because much of this activity is highly seasonal and concentrated along a few routes and destinations, it is unusually sensitive to

disruption: a blocked road, a cancelled snow season or a fouled river translates quickly into lost visitor-days and lost livelihoods.

The Setting: A Landscape-Dependent Economy Along Fragile Routes

Uttarakhand's tourism sector is best understood as a set of distinct, climate-sensitive segments. Religious and pilgrimage tourism above all the Char Dham circuit of Yamunotri, Gangotri, Kedarnath and Badrinath, together with Haridwar and Rishikesh is the largest by volume and revenue. Around it sit hill-station leisure tourism (Nainital, Mussoorie and others), adventure and trekking tourism, river and riverfront tourism centred on the rafting economy of Rishikesh, wildlife and ecotourism in the protected areas, and a small but high-value winter and snow-tourism segment at destinations such as Auli. Each segment sells a different facet of the same climate-shaped landscape.

The economic stakes are substantial. The service sector accounts for more than half of the state's gross domestic product, and tourism is one of its main engines of income and employment; the state recorded of the order of fifty-three to fifty-eight million tourist and pilgrim arrivals annually in recent years, with the Char Dham Yatra as the dominant revenue earner (India Tourism Statistics, 2023). For large numbers of hill households, hotel and guesthouse operators, drivers, guides, mule handlers, camp operators and traders tourism is the difference between staying in the hills and migrating out of them.

Two structural features turn this economic base into acute climate exposure. First, the product is the climate and landscape itself: the snow, the cool air, the flowing river, the flowering meadow. When the climatic conditions that create these assets shift, the product degrades directly, in a way that a factory or an office does not. Second, the delivery of that product depends on a fragile and highly concentrated network, narrow, landslide-prone mountain roads, single-access destinations, and stretched local water and sanitation systems so that even where the attraction survives, the ability to reach it, stay at it and be served safely can fail. Tourism in Uttarakhand is therefore vulnerable at both ends of the chain: in the asset and in the access to it.

This aligns with the IPCC Sixth Assessment Report's framing of mountain regions, which concludes that climate-change impacts in mountains have increased in recent decades with serious and observable consequences for people and economies explicitly including mountain tourism and that mountains sit at the intersection of accelerated warming and large dependent populations (Adler et al., 2022). The pathway makes explicit the mechanism by which a change in the climate becomes a loss of visitors, income or life.

The confidence ratings in Table 1 are deliberately mixed. The physical-hazard pathways extreme rainfall, floods, landslides and the compound pathway are rated High, because both the hazard trend and its disruptive effect on a route-dependent tourism system are well established. The thermal, snow, fire and water-scarcity pathways are rated Medium–High: the underlying climate signal is firm, but the tourism outcome also depends on visitor behaviour and demand response, which adds uncertainty. The ecosystem-change pathway is rated Medium, reflecting the greater complexity of ecological and phenological responses.

Climate Hazard	Climate Driver	Climate Stressor	Direct Impact	Functional Impact	Consequences	Confidence
Rising Temperature	Increasing average and summer temperatures and a rising snowline	Declining thermal comfort at lower hill stations, reduced snow reliability and heat stress on visitors	Shorter, less reliable snow-tourism season and reduced comfort advantage of established hill stations	Erosion of the cool-retreat and winter-sport value proposition and shifting seasonality of demand	Changing visitor demand and revenue patterns, with pressure on higher-altitude destinations	Medium-High
Changing Rainfall Pattern	Erratic monsoon onset and withdrawal and uneven rainfall distribution	Unpredictable weather windows and unreliable travel and trekking conditions	Disrupted trip planning, cancellations and uncertain timing of peak seasons	Reduced predictability of tourism operations, staffing and booking cycles	Revenue volatility and weakened stability of the seasonal tourism economy	Medium-High
Extreme Rainfall / Cloudbursts	High-intensity precipitation events and cloudbursts	Sudden road washouts, slope failures and hazardous conditions along tourist and pilgrimage routes	Blockage of Char Dham and hill routes, stranding of tourists and damage to trails	Interruption of visitor movement and mid-season closure of destinations	Stranded and endangered visitors, cancellations and reputational damage to destinations	High
Flooding / Flash Floods	River overflow and extreme runoff	Inundation of riverfront and valley-floor tourism assets and infrastructure	Damage to hotels, guesthouses, riverside camps, roads and rafting/riverfront tourism	Loss of tourism infrastructure and disruption of destination operations	Capital loss, livelihood loss for tourism-dependent households and slow recovery	High
Landslides & Slope Instability	Intense or prolonged rainfall and slope instability	Blockage of and damage to roads, trekking trails and hillside settlements	Severed access to destinations and prolonged route and trail closures	Reduced accessibility and reliability of the tourism transport network	Lost visitor days, isolation of tourism-dependent economies and rising maintenance costs	High
Glacier Retreat & Snow Cover Decline	Rising temperatures and declining glacier mass and seasonal snow cover	Degradation of snow- and glacier-dependent scenery and activities and emerging glacial-lake hazards	Diminished ski and snow tourism (e.g., Auli), altered high-altitude trekking and GLOF risk on trekking routes	Erosion of the adventure- and scenery-based tourism product	Decline of high-value niche segments and heightened high-altitude safety risk	Medium-High
Forest Fires	Warmer, drier pre-monsoon conditions	Increased frequency and intensity of forest fires (notably chir pine) and smoke haze	Reduced scenic visibility, closure of forest and wildlife areas and health discomfort	Disruption of nature-, wildlife- and scenery-based tourism during the peak spring season	Diminished visitor experience, cancellations and damage to ecotourism assets	Medium-High
Water Scarcity at Destinations	Rising temperatures, declining spring discharge and erratic rainfall	Seasonal water shortages aggravated by concentrated peak-season tourist influx	Strain on drinking-water and sanitation supply at hill stations and pilgrimage centres	Exceedance of destination carrying capacity and service breakdown at peak load	Host-community–tourist competition for water and constraints on sustainable tourism growth	Medium-High
Climate-induced Ecosystem Change	Altered temperature and rainfall regimes and shifting phenology	Changes in wildlife distribution, alpine-meadow bloom timing and forest health	Reduced predictability of wildlife sightings and shifting flowering windows (e.g., Valley of Flowers)	Degradation of nature-based and ecotourism attractions	Diminished destination appeal and reduced long-term ecotourism value	Medium
Compound Climate Hazards (Heat + Extreme Rainfall + Landslides)	Increasing frequency and convergence of multiple climate extremes	Simultaneous disruption of access, safety and infrastructure during peak tourist season	Large-scale destination closure, mass stranding and casualties (e.g., 2013 Kedarnath disaster)	Systemic breakdown of the tourism value chain and collapse of visitor confidence	Severe revenue and livelihood loss, long-term reputational damage and reduced destination resilience	High

Warming, Snow and the Shifting Seasons

Two pathways, rising temperature and glacier retreat with snow-cover decline, act on the seasonal and scenic foundations of the sector. Rising average and summer temperatures, together with a rising snowline, erode the “cool-retreat” advantage that has long defined the lower hill stations and impose heat stress on visitors during the peak summer season. Because Uttarakhand is warming faster than the global average (Banerjee et al., 2021), this pressure is intensifying, gradually shifting the seasonality and altitude of comfortable leisure demand.

The snow and glacier pathway strikes the winter and adventure segments more directly. Reduced and less reliable snowfall shortens the snow-tourism season and degrades the scenery that underpins high-altitude tourism; the winter-sports economy at Auli has been repeatedly disrupted, with the national ski championship postponed or cancelled several times over the past decade for want of adequate snow, and operators reporting waves of cancellations in low-snow winters as the western disturbances that bring winter snow become weaker or more erratic (as reported, 2024–2025). Over the longer term, the loss of one-third to two-thirds of Himalayan glacier volume projected this century (Wester et al., 2019) will degrade glacier-dependent scenery and, through the growth and instability of glacial lakes, raise the risk of glacial-lake outburst floods along high trekking routes — the same class of high-mountain cryospheric hazard that produced the 2021 Chamoli disaster (Shugar et al., 2021). A likely secondary effect is the migration of demand towards higher, still-snowbound or still-scenic altitudes, concentrating tourist pressure on the most fragile zones.

Rainfall, Extreme Events and the Problem of Access

If warming reshapes the product, it is the route-dependence of the sector that makes extreme events so damaging. Four pathways, changing rainfall, extreme rainfall and cloudbursts, flooding and flash floods, and landslides converge on the network by which tourists reach and move between destinations.

Changing rainfall patterns act first through unpredictability. An erratic monsoon, with variable onset and withdrawal and uneven distribution, produces unreliable weather windows for travel and trekking, disrupting trip planning, triggering cancellations and blurring the timing of peak seasons. The functional effect is a loss of predictability in operations, staffing and booking cycles; the consequence is revenue volatility that undermines the stability of what is already a highly seasonal economy.

Extreme rainfall and cloudbursts turn this unpredictability into acute hazard. High-intensity events wash out roads, destabilise slopes and create dangerous conditions along precisely the pilgrimage and hill routes on which the sector depends, blocking the Char Dham and hill-station corridors, stranding visitors and damaging trails. The June 2013 Kedarnath disaster in which exceptional monsoon rainfall combined with the breach of a moraine-dammed glacial lake to devastate one of the state’s most important pilgrimage sites at the height of the yatra season remains the defining illustration of how catastrophic this pathway can be. Flooding and flash floods extend the damage to riverfront and valley-floor assets, inundating hotels, guesthouses, riverside camps and rafting infrastructure; the 2021 Chamoli event destroyed downstream infrastructure outright (Shugar et al., 2021), and river-valley tourism assets sit directly in the path of such flows.

Landslides complete this group and are especially disabling in a system built on single-access routes. Intense or prolonged rainfall and slope instability block and damage roads and trekking trails, severing access to destinations for days or weeks and isolating tourism-dependent economies from their market. The functional impact is a persistent loss of reliability in the tourism transport network; the consequence is lost visitor-days and a rising

maintenance burden pressures compounded by the unplanned construction on fragile slopes that has accompanied tourism growth, as the Joshimath subsidence crisis brought into sharp relief.

Fire, Water and the Quality of the Destination Experience

Two further pathways degrade the experience at the destination itself, rather than the route to it. The first is forest fire. Warmer, drier pre-monsoon conditions increase the frequency and intensity of the surface fires that are endemic to Uttarakhand's extensive chir-pine forests; analysis of two decades of fire data shows that pre-monsoon temperature, the proportion of a division under chir pine, and low pre-monsoon and winter precipitation together explain much of the variation in fire incidence, with the hottest, driest pre-monsoon years, such as 2016 and 2019 being the worst for fire (Singh et al., 2024). For tourism, the timing is unfortunate: the fire season peaks in March-May, the same window as the spring tourist and early-yatra season, and the resulting smoke haze reduces scenic visibility, forces the closure of forest and wildlife areas, and causes health discomfort diminishing the visitor experience precisely when demand is high.

The second is water scarcity at the destination. The peak tourist season coincides with the lean pre-monsoon months, when spring discharge and surface-water availability are already at their lowest, and the concentrated influx of visitors sharply raises demand at exactly that moment; a hotel guest typically consumes several times as much water per day as a resident. The result is severe seasonal strain on the drinking-water and sanitation systems of hill stations and pilgrimage centres. Nainital's Naini Lake, the town's principal source, has fallen to multi-year lows, and the wider pattern of tourism-driven water stress in Himalayan towns is now well documented (Observer Research Foundation, 2026); the near "day-zero" crisis in Shimla is a cautionary regional parallel. This pathway pits host communities against visitors in direct competition for a shrinking resource, and it has already prompted regulatory scrutiny of destination carrying capacity, most notably in the post-Joshimath review of Mussoorie (National Green Tribunal, 2023).

Ecosystem Change and Nature-Based Tourism

A slower pathway works through the ecosystems that underpin nature-based and wildlife tourism. Altered temperature and rainfall regimes and shifting phenology change the distribution of wildlife, the timing of alpine-meadow blooms and the health of forests, reducing the predictability of the natural attractions on which this segment depends from wildlife sightings in the protected areas to the flowering window of the Valley of Flowers. The functional impact is a gradual degradation of nature-based and ecotourism attractions; the consequence is diminished destination appeal and a slow erosion of long-term ecotourism value. This pathway is assessed at Medium confidence, reflecting the greater complexity and site-specificity of ecological and phenological responses, but its direction is consistent with the broader evidence of climate-driven ecological shifts across mountain regions (Adler et al., 2022).

Compound Hazards and the Tourism Value Chain

The final pathway addresses the convergence of hazards. When heat, extreme rainfall and landslides occur together or in close succession during the peak tourist season, they disrupt access, safety and infrastructure simultaneously, so that the whole tourism value chain can fail at once. The direct impact is large-scale destination closure, mass stranding and, in the worst cases, casualties, the 2013 Kedarnath catastrophe being the archetype; the functional impact is a systemic breakdown of the value chain and a collapse of visitor confidence; and the consequence is severe revenue and livelihood loss, lasting reputational damage to affected destinations, and a reduction in overall destination resilience.

Regulation has become part of this pathway's landscape. In response to mounting environmental and safety pressures, the authorities have restricted riverside camping along the Ganga between Kaudiyala and Rishikesh (National Green Tribunal, 2015), commissioned carrying-capacity studies of hill destinations following the Joshimath crisis (National Green Tribunal, 2023), and faced growing calls to reimagine Himalayan tourism as a water-sensitive and climate-resilient pathway rather than an unconstrained growth model (Observer Research Foundation, 2026). These measures are not merely external constraints; they are a signal that the sector's own expansion is generating the very conditions crowding, water stress, slope destabilisation that heighten its exposure to climate hazards.

That shared structure points to clear priorities. Diversifying the tourism calendar and destination base spreading load across the year and across sites reduces the sector's dependence on any single climatic window and eases peak-season carrying-capacity stress. Climate-resilient access infrastructure and slope stabilisation along pilgrimage and hill routes address the extreme-rainfall, flood and landslide pathways together. Carrying-capacity-based regulation and destination-level water and waste management, of the kind now being trialled for Mussoorie, address the water-scarcity and compound pathways. Early-warning and route-safety systems including monitoring of glacial-lake and cloudburst hazards on trekking and pilgrimage routes directly reduce the mortality and stranding risk that defines the sector's most serious consequences. And product-level adaptation planning for shrinking snow and scenery segments, and scheduling nature-based activity around the fire season preserves value as the underlying conditions change. The value of the pathway approach is that, by making each mechanism explicit, it shows not only where tourism losses accumulate but where a single measure can protect the sector across several pathways at once.

The sector-specific and district-level assessments that follow build on this framework, quantifying exposure and sensitivity for Uttarakhand's principal tourism destinations and translating these impact chains into prioritised, context-specific adaptation responses.

11.5 Future Climate Projections and Tourism Implications

The table below presents projected changes in key climate metrics relevant to Uttarakhand's tourism and visitor economy, across two Global Warming Levels (GWL) relative to the 1950-2014 baseline.

Tourism-Relevant Metric	Baseline (1980-2014)	+1°C	+2.0°C
Extreme heat days/yr - valley-floor entry towns (days >40°C at Haridwar, Udham Singh Nagar)	Baseline: 2.1 days/yr at Haridwar (1950-2014 mean). Observed trend: significant increase since 2000 (IMD, 2023)	3-5 days/yr; heat stress risk for arriving pilgrims increases; April-May Yatra season increasingly heat-stressed at entry points	7-10 days/yr; dangerous heat exposure window overlaps with peak Yatra season; elderly pilgrim cardiac risk materially elevated
Extreme rainfall events/yr (≥100 mm/day) - landslide-triggering threshold on Yatra routes	~18 events/yr statewide; +35% increase observed 2000-2024; Char Dham routes blocked multiple times/season (IMD, 2023)	~22 events/yr; Yatra route closures become near-systematic in monsoon months; peak pilgrim volumes coinciding with peak hazard	~27 events/yr; 50-yr rainfall events become 20-yr events; multi-week simultaneous route closures increasingly common (Krishnan et al., 2020)
Pilgrim deaths per Char Dham season (cardiac arrest and altitude sickness)	Baseline (2017-19): 90-112 deaths/season. Post-COVID surge: 300+ (2022), 230+ (2023), 200+ est. (2024). Upward trend (Deccan Herald, 2022)	Near-current trajectory unless acclimatisation protocols, health screening, and cap enforcement are implemented	Mortality elevated further as heat at approach towns worsens cardiovascular loading before altitude exposure; without intervention, 350-400+/season plausible
Natural snowpack at Auli ski resort (2,519 m amsl) - commercial ski season viability	Historical: reliable Jan-Mar natural snow. 2024: near-zero natural snowfall; first commercial artificial snow-making (CarbonImpacts, 2024)	Marginal snow seasons increasing; artificial snow-making required in majority of years; ski season compressed	Natural ski season unreliable; artificial snow costs rise; investment viability of ski infrastructure in question below 3,000 m amsl
GLOF risk to downstream tourism infrastructure - glacial lake surface area expansion	426 glacial lakes; 25 PDGLs; 2021 Chamoli GLOF: 200+ dead, 11 bridges destroyed (SLRMC, 2025; Shugar et al., 2021)	Lake inventory expanding; PDGL count likely growing; downstream tourism corridor risk increasing	+60-90% glacial lake area expansion; GLOF risk to Char Dham and trekking corridor bridges materially higher (Jain et al., 2025)
Valley of Flowers phenological alignment - tourist booking window vs actual peak bloom	Historical: peak bloom reliably July-September. Rhododendron arboreum flowering advancing 25-30 days (Kumar et al.,	Moderate seasonal uncertainty; some years visitor arrival misaligns with bloom; reputation risk begins to emerge	Significant phenological disruption; bloom window shifts and compresses; heritage site tourism value

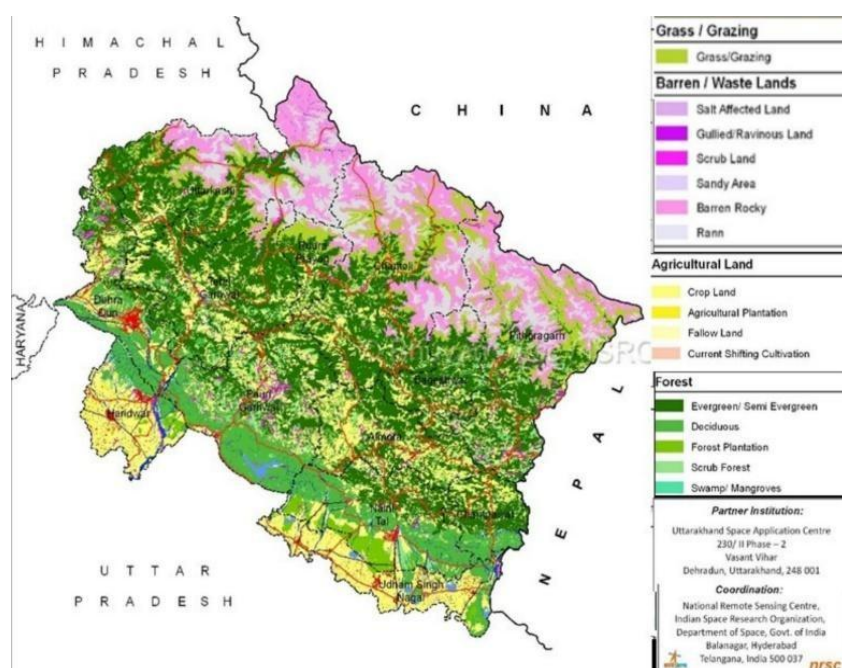
Tourism-Relevant Metric	Baseline (1980-2014)	+1°C	+2.0°C
	2010). Misalignment risk growing		decreasing (Kumar et al., 2010)
Tourism sector annual revenue disruption from climate events (INR crore)	2013 event: ₹12,000-25,000 crore over 3 years (PHD Research Bureau, 2013). Annual partial disruption losses: growing	Increasing annual disruption as extreme event frequency grows; cumulative losses across operators in affected districts rising	Structural seasonal disruption as monsoon extreme events and Yatra season increasingly overlap; compound economic

6. Forest and Biodiversity

12.1 Overview of the Forest and Biodiversity Sector in Uttarakhand

Uttarakhand's forests are among the most ecologically significant in South Asia. The state's 53,483 km² geographic area supports a forest cover of approximately 24,303 km² (45.44%) of its land area, comprising 37 distinct forest types across eight forest type groups, from tropical moist deciduous Sal forests in the Terai-Bhabar belt to dry alpine scrub above 3,500 m (Champion C Seth, 1968; MoEFCC, 2019). As a recognised global biodiversity hotspot, one of 36 designated worldwide, Uttarakhand harbours over 4,500 species of flowering plants (93 endemic), 743 species of birds (60% of India's avifauna), 93 species of mammals (25% of India's known species), 72 species of reptiles, 439 species of butterflies, and over 700 species of medicinal plants used in drug, pharmaceutical, and traditional Ayurvedic and Unani systems (Uttarakhand Biodiversity Board, 2013; MoEFCC, 2014; ENVIS Centre-UEPPCB, 2013).

Uttarakhand's forests acts as the headwater catchment for the Ganga, Yamuna, Alaknanda, Bhagirathi, and Kali river systems, the lifelines of one of the world's most densely populated plains. The Indian Himalayan Region (IHR), of which Uttarakhand forms the central core, is estimated to sequester approximately 65 million tonnes of carbon annually (Tolangay C Moktan, 2020). At the national level, Uttarakhand holds the second-largest growing stock of forest trees (400.02 million m³), behind only Arunachal Pradesh (FSI, 2023). Six national parks, seven wildlife sanctuaries, and four conservation reserves cover 7,818 km² (14.6% of the geographic area), including two UNESCO World Heritage Sites, the Nanda Devi Biosphere Reserve and Valley of Flowers National Park (WII, 2017).



Community forest governance adds a further critical layer: 12,167 Van Panchayats manage 13.41% of the state's forest area, making Uttarakhand host to one of India's most extensive and historically significant community forest management institutions, a system institutionalised since 1931 after sustained resistance against colonial forest appropriation (UFD, 2025; Chauhan C Todaria, 2025).

Forest condition in Uttarakhand is deteriorating on two fronts that matter directly for climate exposure: canopy density and species composition. Dense forest cover declined from 36.82 per cent in 2000 to 26.06 per cent

by 2015, even as open forest and scrub expanded, a pattern consistent with fire-driven canopy opening and climate-induced compositional shift (Dabas, 2016). Chir pine (*Pinus roxburghii*), which covers approximately 28 per cent of the State's forest area, dominates the mid-altitude fire economy, and its continuing expansion at the expense of banj oak (*Quercus leucotrichophora*) is the single most consequential compositional trend in the State's forest resource base.

Segment	Extent / status	Share of forest estate	Dominant climate dependency
Very dense / dense forest	Declining — 36.82% (2000) → 26.06% (2015) [indicative]	Majority canopy cover	Fire; drought stress; regeneration failure
Moderately dense forest	Transitional class; expanding at margins	Substantial share	Compositional shift (oak→pine); phenology
Open forest / scrub	Expanding since 2000	Growing share	Fire-driven canopy opening; invasive species
Chir pine (<i>Pinus roxburghii</i>) belt	~28% of forest area, 1,000–2,000 m	Large and expanding	Highly flammable litter; drought-tolerant, fire-adapted
Banj oak (<i>Quercus leucotrichophora</i>) belt	Contracting under pine replacement	Declining share	Spring recharge; moisture retention; regeneration
Alpine / sub-alpine (bugyal) meadows	Above ~3,500 m	Small but ecologically critical	Upward species compression; invasive colonisation

Against this backdrop of extraordinary ecological wealth, climate change is operating as a structural destabiliser, altering thermal regimes, precipitation patterns, species distributions, fire behaviour, and the hydrological systems that the forests both depend on and regulate. Since 1990, Uttarakhand's average temperature has risen by approximately 1.5°C, roughly twice the global mean driven by Elevation-Dependent Warming (EDW) characteristic of the Himalayan altitudinal gradient (IMD, 2023; Shekhar et al., 2023; Pepin et al., 2015). Forest fire incidents have escalated from 922 in 2002 to 41,600 in 2019, making Uttarakhand the most fire-afflicted state in India by FSI fire alert counts in 2024 (Mina et al., 2023; ScienceDirect, 2025). Dense forest cover declined from 36.82% in 2000 to 26.06% by 2015, even as open forest and scrub expanded a pattern consistent with fire-driven canopy opening and climate-induced compositional shifts (Dabas, 2016). More than 300 plant species are now classified as threatened or endangered, many under direct pressure from habitat loss, climate-driven range compression, and over-harvesting of commercially valuable medicinal species.

The protected area network, while extensive faces increasing climate-mediated pressure through habitat contraction at upper elevation limits, species range shifts that push wildlife into human-dominated landscapes, and the progressive drying of spring-shed catchments that once sustained both forest regeneration and community water security. The confluence of climate change, demographic out-migration, governance weakening of Van Panchayats, and increasing anthropogenic pressure from infrastructure development creates a compound threat to forest and biodiversity assets that no single-sector intervention can address.

Institutional Structure

Governance of the forest and biodiversity sector is distributed across the Uttarakhand Forest Department (UFD), which holds primary custodial and management responsibility; the Uttarakhand Biodiversity Board (UBB), responsible for species and genetic-resource conservation; the Van Panchayat Directorate, which oversees the State's community forest institutions; and the GB Pant National Institute of Himalayan Environment (GBPNHE), which provides ecosystem-service assessment and research support. Central institutions, the Wildlife Institute of India (WII) and the Forest Survey of India (FSI) supply the national-level monitoring, census and mapping baselines on which State-level planning depends. Every adaptation measure proposed in this chapter must be assignable to one of these owners; the vulnerability index is structured accordingly.

The National Mission for a Green India (GIM), one of eight missions under India's National Action Plan on Climate Change (NAPCC), allocated approximately Rs. 220 crore to Uttarakhand in 2023–24 for afforestation and forest health, though absorptive-capacity constraints limit utilisation (Outlook Business, 2024). The National Mission on Sustaining Himalayan Ecosystem (NMSHE), launched in 2010, specifically targets cryospheric and biodiversity conservation in Himalayan states.

Community Forest Governance: Van Panchayats

Community forest governance adds a further critical layer, and one without direct parallel in the energy sector's institutional structure: 12,167 Van Panchayats (VPs) manage 13.41 per cent of the State's forest area, approximately 7,164 km² making Uttarakhand host to one of India's most extensive and historically significant community forest management institutions, a system institutionalized since 1931 after sustained resistance against colonial forest appropriation (UFD, 2025; Chauhan C Todaria, 2025). These institutions are simultaneously the State's most granular climate-monitoring network, fire-prevention force and biodiversity custodian, yet they face an accelerating erosion of autonomy, budget and demographic base due to out-migration and policy reversal.

Biodiversity and Protected Area Network

Six national parks, seven wildlife sanctuaries and four conservation reserves cover 7,818 km² (14.6 per cent of the geographic area), including two UNESCO World Heritage Sites, the Nanda Devi Biosphere Reserve and the Valley of Flowers National Park (WII, 2017). The State harbours 2,276 leopards, 2,026 elephants and 560 tigers, among the highest densities of large carnivores and megafauna in the world, cohabiting with a predominantly rural, agriculture-dependent mountain population (Mongabay, 2025).

Ecosystem Services and Downstream Dependencies

Uttarakhand's forests are not passive recipients of climate change; they actively modulate it through carbon sequestration, evapotranspiration, albedo effects and hydrological regulation. The State's forests intercept monsoon rainfall, recharge the over 3,000 perennial springs (dhaaras and naulas) that supply more than 70 per cent of rural communities, and stabilise slopes that would otherwise fail under intense precipitation (Central Ground Water Board, 2022). A locally significant non-timber forest produce (NTFP) economy, Rhododendron arboreum (Buransh) squash, juice and flower products chief among them, depends directly on stable phenology and harvest windows (Singh et al., 2025).

12.2 Defining Features and Their Climate Implication

The forest and biodiversity sector in Uttarakhand is intrinsically linked to the state's climatic, hydrological, and geomorphological systems. As one of the most biologically diverse regions of the Indian Himalaya, Uttarakhand supports a wide range of ecosystems distributed across steep elevational gradients, from subtropical forests in the foothills to alpine meadows and permanent snowfields at high elevations. This exceptional ecological diversity is sustained by strong climatic gradients that regulate temperature, precipitation, snow cover, hydrological processes, and ecosystem productivity. Consequently, projected changes in temperature regimes, precipitation variability, cryospheric dynamics, and the increasing frequency and intensity of extreme climatic events are expected to influence forest structure, species composition, ecosystem functioning, and biodiversity across the state.

The state's forests are distributed along one of the world's steepest elevational gradients, creating distinct ecological zones within relatively short horizontal distances. Temperature and moisture largely determine the distribution of vegetation types, while snow accumulation and seasonal melt regulate high-altitude ecosystem processes. Climate warming is expected to shift suitable climatic niches upward, resulting in progressive changes in species distributions, altered community composition, and changes in ecosystem boundaries. However, the upward migration of species is constrained by the limited availability of habitat at higher elevations, a phenomenon commonly referred to as the "**conical mountain effect**" which may reduce habitat availability for alpine and sub-alpine species and increase the risk of local extirpation, particularly for endemic and range-restricted taxa.

Forest ecosystems also play a fundamental role in regulating watershed hydrology and sustaining downstream water resources. The forests of Uttarakhand form the upper catchments of the Ganga, Yamuna, and numerous tributary river systems while supporting thousands of springs that provide water for domestic use, agriculture, and ecosystem functioning. Forest condition influences infiltration, groundwater recharge, runoff generation, erosion, sediment transport, and slope stability. Climate-induced changes in forest health, vegetation cover, wildfire occurrence, and disturbance regimes may therefore have cascading impacts on watershed functioning and water security across multiple sectors beyond forestry.

The composition and functioning of Himalayan forests are strongly influenced by climatic variability. Mid-elevation forests dominated by **Chir pine (*Pinus roxburghii*)** are naturally fire-adapted ecosystems where wildfire occurrence is closely linked to temperature, humidity, soil moisture, and the duration of the dry season. Rising temperatures, prolonged dry periods, and increasing moisture deficits are expected to lengthen the fire season and increase the frequency, intensity, and spatial extent of forest fires. Recurrent fires may alter forest composition, reduce regeneration, accelerate land degradation, and affect ecosystem services including carbon sequestration, watershed protection, and biodiversity conservation.

Climate change is also expected to modify wildlife distributions and ecological interactions. Uttarakhand supports globally significant populations of elephants, tigers, leopards, Himalayan ungulates, and numerous endemic bird, reptile, and plant species. As climatic conditions shift, many species are projected to alter their ranges, phenology, and habitat use in response to changing temperature and moisture regimes. Such shifts may increase overlap between wildlife habitats and human settlements, contributing to rising incidences of human-wildlife conflict, particularly in the mid-hills and forest-agriculture interface where settlements are closely interspersed with forest landscapes.

The adaptive capacity of the forest sector extends beyond ecological processes and is closely associated with local governance institutions. Uttarakhand possesses one of India's largest community-based forest management systems through its extensive network of Van Panchayats, which play an important role in forest protection, resource management, fire prevention, and ecosystem restoration. The effectiveness of climate adaptation within the sector therefore depends not only on ecological resilience but also on institutional capacity, community participation, governance effectiveness, and sustained management of forest resources.

Defining feature	Description	Implied climate sensitivity
Biodiversity hotspot	One of 36 global hotspots; >4,500 flowering plant species, 93 endemic	Endemic and range-restricted species are disproportionately exposed to thermal-envelope loss
Altitudinally compressed	~7,500 vertical m of habitat gradient in a compact footprint	The 'conical mountain effect' — shrinking habitat area as species shift upslope
Hydrologically pivotal	Headwater catchment for Ganga, Yamuna and tributaries; recharges 3,000+ springs	Forest degradation propagates directly into water security for downstream sectors
Fire-adapted mid-hills	Chir pine covers ~28% of forest area at 1,000–2,000 m, coinciding with settlement density	Fire season length and severity are climate-modulated and rising
Community-governed	12,167 Van Panchayats manage 13.41% of forest area	Adaptive capacity is bound to institutional health, not just physical assets
Wildlife-dense, conflict-prone	Among world's highest large-carnivore and megafauna densities	Climate-driven range shifts translate directly into human safety and livelihood risk

Structural features of the forest and biodiversity sector and the climate sensitivity each implies.

12.3 Key Climate Hazards on the Forest and Biodiversity Sector

Climate sensitivity in this context refers to the degree to which forest and biodiversity systems in Uttarakhand are susceptible to harm from climate variability and change, given their inherent structural characteristics and ecological dependencies, independent of adaptive capacity. A highly sensitive forest ecosystem is one where even moderate shift in temperature, precipitation timing, or extreme event frequency produces measurable and potentially irreversible changes in species composition, regeneration dynamics, carbon stocks, or hydrological function.

The analytical framework applied here follows the IPCC AR6 risk model (Ara Begum et al., 2022) and is grounded in the conceptual advances of vulnerability science applied to mountain forest ecosystems by researchers including Thadani (2015), Mina et al. (2023). Four dimensions of climate sensitivity are particularly acute in Uttarakhand's forest and biodiversity sector.

Hazard Register

Hazard	Mechanism of damage	Ecological-chain stage affected	Assets primarily affected	Principal districts
Forest fire	Resinous chir-pine litter ignition; extended dry season	Resource base; carbon stock; biodiversity	Chir-pine belt; wildlife habitat	Pauri Garhwal, Almora, Nainital, Tehri Garhwal, Dehradun
Drought / moisture stress	Reduced pre-monsoon and winter rainfall; extreme heat years	Ecological function (regeneration); resource base	Oak forests; regenerating stands	Almora, Pauri, Tehri, Bageshwar
Extreme rainfall / cloudburst	Short-duration high-intensity precipitation	Ecological function; ecosystem services	Riparian forest; spring-shed catchments	Uttarkashi, Chamoli, Rudraprayag, Pithoragarh
Landslide / slope failure	Rainfall-triggered slope failure on fractured lithology	Resource base; ecosystem services	Forest cover on steep slopes; access routes	Uttarkashi, Chamoli, Rudraprayag, Pithoragarh, Bageshwar
Elevation-dependent warming	Amplified warming with altitude	Resource base; species pool	Alpine and sub-alpine species; endemic flora	Chamoli, Uttarkashi, Pithoragarh, Bageshwar (high-altitude blocks)
Phenological disruption	Advanced flowering / leafing decoupled from pollinator cues	Ecological function; ecosystem services (NTFP)	Rhododendron arboreum and keystone flowering species	Chakrata, Pauri, Tehri, Almora

Human-wildlife conflict amplification	Habitat contraction and prey shifts drive wildlife into settled areas	Human interface	Communities at forest edge; livestock	Pauri Garhwal, Almora, Pithoragarh, Rudraprayag, Tehri, U.S. Nagar, Haridwar, Nainital
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Forest Fire

Mechanism and dual pathway

Uttarakhand's mid-altitude belt is dominated by chir pine (*Pinus roxburghii*), whose resinous, highly flammable needle litter forms a near-ideal surface fuel bed through the dry pre-monsoon months. Fire acts on the sector through three pathways that are usually assessed separately and should not be. The carbon pathway: Bargali, Pandey, Bhatt and Sundriyal (2024) quantify carbon-stock loss across Sal, Pine and Mixed forest types along an elevation gradient and show that recurrent fires now exceed the regenerative sequestration capacity of these stands. The biodiversity pathway: recurrent burning reduces herbaceous diversity (Singh and Thadani (2022) report), a fall from a maximum of 37 species in unburnt plots to just 12 in high-fire-frequency plots and the fire season now overlaps the breeding and flowering windows of vulnerable native species, including 55 IUCN-listed threatened birds and mammals such as the Himalayan musk deer (*Moschus leucogaster*) and the snow leopard (*Panthera uncia*). The cross-sector pathway: smoke and black-carbon deposition on Himalayan glaciers accelerates cryospheric melt, the Wadia Institute of Himalayan Geology (2016) measured summer black-carbon concentrations near the Gangotri glacier at roughly 400 times winter values, with downstream water-security implications far beyond the fire perimeter.

Why this hazard leads the register

- **It is the fastest-growing hazard in the sector.** The 2024 season alone recorded fire alerts at a scale that overwhelms current suppression capacity, and the risk season is projected to lengthen by 15–45 days under continued warming (Barik & Roy, 2023).
- **It is self-reinforcing and drought-coupled.** Fire consumes the regeneration that would re-stock the stand and builds fuel-ladders that raise the next fire's severity; drought lowers the ignition threshold, so the two hazards escalate together.
- **It attacks the sector's own adaptation strategy.** The State's programme of Rare, Endangered and Threatened (RET) plant reintroduction and protected-area management depends on stable fire regimes, which fire directly undermines.

Documented impact

The 2024 fire season recorded 3,338 fire alerts during summer, the highest among Indian states claiming 12 lives, including six Forest Department personnel (Earth Journalism Network, 2025). Fire incidents have escalated from 922 in 2002 to 41,600 in 2019, making Uttarakhand the most fire-afflicted state in India by FSI fire-alert counts (Mina et al., 2023; ScienceDirect, 2025).

Fire season	Incidents	Area affected (ha)	Notable features / key data
2015–16	20,667	4,433	30% surge over 2015; 1,200 simultaneous fire locations on 26 April; IAF helicopter deployment; temperature 7°C above normal (Newsline, 2021)
2017–18	1,165 → 4,714	Not fully compiled	Incidents rose from 1,165 (2017) to 4,714 (2018); initiation of FSI SNPP-VIIRS satellite monitoring (Mina et al., 2023)
2018–19	41,600	~18,000 (est.)	Peak fire year; 45× increase from the 2002 baseline of 922; coincides with the driest pre-monsoon in a decade (Negi, 2019; Mina et al., 2023)
2020–21	989 (to 4 Apr 2021)	1,297	Season began Oct 2020 (3 months early); State sought MI-17 helicopter deployment; Uttarakhand High Court suo motu action initiated (Newsline, 2021)
2023–24	930+ (to 11 May 2024)	1,196	National-highest FSI alert count: 3,338; IAF MI-17 and Army deployment; 5 human deaths; 300 FIRs filed for arson (Deccan Herald, 2024)
2025–26	226 (to 27 Apr 2026)	144.22	Garhwal: 177 incidents (110.52 ha); Kumaon: 28 incidents; Badrinath FD: 57 incidents, highest by division (National Herald India, 2026)

Forest fire incidents and area affected in Uttarakhand, 2015–2026. From 2001 to 2025, the State lost 1.3 kha of tree cover to fire and 23 kha to all other drivers of loss combined; 2009 was the single worst fire year on record, with 220 ha lost (23% of that year's total tree-cover loss).

Drought and Moisture Stress

Extreme heat years measurably impair tree physiology before any visible canopy damage occurs. Singh and Singh (2010) found that a single extreme heat year reduced tree water potential in *Quercus floribunda* (tilonj oak) in Kumaon Himalaya to -5.5 MPa, against a normal range of -2.0 to -2.3 MPa, a threshold that suppresses leafing, creates drought stress, and pre-conditions trees to fire. This is a slow, cumulative hazard: it does not produce a single dramatic event, but it lowers the ignition threshold for the fire hazard above and suppresses regeneration success independently of fire.

The downstream hydrological consequence compounds the ecological one. Oak forests, with deep root systems and dense leaf litter, are critical for the recharge of Himalayan springs (naulas, dhaaras) on which over 70 per cent of rural water supply depends (Raina, 2009; Tiwari & Sunny, 2024). Where drought-stressed oak is progressively replaced by pine, which has a shallower root system, hydrophobic litter and a higher evapotranspirative demand spring drying accelerates. This correlation has been documented across Almora, Pauri and Tehri districts (Joshi et al., 2024), and Sugandha et al. (2025) explicitly identify the oak–pine transition as a ‘climate-anthropogenic compound stressor’ that propagates from forest into water, agriculture and ultimately livelihood domains.

Extreme Rainfall and Cloudbursts

Short-duration, high-intensity precipitation damages the sector chiefly through riparian forest and spring-shed catchments, and it is the proximate trigger for the landslide hazard that follows. Because the two are not independent, they are treated jointly in the composite index and must not be double-counted. The direct forest consequence is disruption of ecological function and of the water-regulation ecosystem service in headwater catchments, where intense pulses scour channels, strip riparian vegetation and mobilise sediment.

The trend is the same reorganisation of rainfall observed across the other sectors: less total monsoon rainfall delivered in fewer, more violent pulses (IMD, 2023). For the forest this means fewer of the gentle, high-infiltration rain events that sustain regeneration and recharge, and more of the erosive events that trigger slope failure and remove forest cover.

Landslide and Slope Failure

Uttarakhand's geology compounds the rainfall trigger: young, tectonically active, highly fractured lithologies with steep relief and thin, poorly cohesive colluvial cover. Where slopes carrying forest cover fail, the loss is not only standing timber and habitat but the very slope-stabilising function that intact forest performs, a hazard that partly manufactures its own trigger, because once a stand is thinned or burned the loss of root cohesion raises the probability of the next failure.

Landslide susceptibility mapping showed localised pockets of moderate-to-very-high forest exposure concentrated in the southern and central mountainous districts, corresponding to fragile slopes, steep terrain, intense-rainfall zones and geologically unstable landscapes. Forest degradation from landslides produces loss of vegetation cover, habitat fragmentation, reduced biodiversity and disruption of soil-stabilisation and water-regulation functions. The residual data gap is the coupling coefficient between forest condition and slope stability: the degree to which thinning or burning a specific stand raises local failure probability is qualitatively established but not yet quantified for the State's slope classes.

Elevation-Dependent Warming and Altitudinal Range Compression

Uttarakhand's forest biodiversity is structured across a steep altitudinal gradient from tropical moist deciduous Sal forest below 600 m to alpine scrub above 3,500 m, spanning over 7,500 vertical metres within a compact footprint. Since 1990 the State's average temperature has risen by roughly 1.5°C, about twice the global mean, driven by the Elevation-Dependent Warming (EDW) characteristic of the Himalayan gradient (IMD, 2023; Shekhar et al., 2023; Pepin et al., 2015). Because warming amplifies with altitude, climate-driven range shifts translate into rapid displacement over short horizontal distances, but with progressively less habitat available as elevation rises. The conical mountain effect (IPCC AR5; confirmed for the Himalaya by Thadani, 2015) predicts that high-altitude species meet shrinking habitat envelopes as isotherms move upward, ultimately facing local extinction at summit limits.

Uniquely severe

- **Its losses are terminal.** Local extinction at a summit limit has no reconstruction pathway; unlike a damaged asset, an extirpated endemic cannot be restored.
- **It is existential for the State's endemic flora.** For Uttarakhand's 93 endemic plant

species, many confined to narrow altitudinal bands, upslope isotherm shift is an existential threat. Extrapolating the upward shifts of 237.9 ± 219.8 m documented for endemic plants in the Sikkim Himalaya over 1850-2010 (Telwala et al., 2013) with 70 per cent of species shifting 100-400 m and extremes of 600-800 m implies that by 2050 the thermal envelopes of many endemics will have moved beyond the available habitat ceiling.

At least 34 high-value threatened medicinal plant species occupy the trans-Himalayan zone, several with global potentially-suitable habitat of less than 10 km^2 (Kumar et al., 2023). The IUCN critically endangered *Meizotropis pellita* (locally 'Patwa'), endemic to Uttarakhand, is restricted to under 10 km^2 of suitable habitat under combined pressure from deforestation, fire and climate-driven habitat contraction. The residual data gap is the coarse resolution of altitudinal-band occupancy data for most endemics, which makes the timing of individual summit-limit extinctions difficult to project with confidence.

Phenological Disruption and Pollinator Mismatch

Phenological advancement, the earlier onset of flowering, leafing or breeding is among the clearest 'fingerprints' of climate change on biological systems, and it is a hazard with no direct equivalent in the energy sector's register: it damages ecological function without destroying any physical asset. The State tree of Uttarakhand, *Rhododendron arboreum* (locally 'Buransh'), is the best-documented case in the Western Himalaya. Rana, Bhattacharyya and colleagues (2024), comparing real-time field observations with herbarium records dating to 1893 using Generalized Additive Models, document a flowering advancement of 88–97 days over the last century at Chakrata Forest Division. Mongabay India (2020) reports widespread December–January flowering in mid-altitude populations, a complete season-shift from the historical February–March bloom.

R. arboreum is a keystone species supporting passerine and insect pollinators across elevations; precocious flowering risks plant–pollinator desynchronisation, since most Himalayan pollinator emergences remain temperature-cued at ground level rather than at the canopy (Mamgain et al., 2017; Rawat et al., 2024). Memmott et al. (2007) establish that even moderate decoupling leads to pollinator decline and consequent plant reproductive failure. For Uttarakhand, the economic risk is direct: Buransh squash, juice and flower-based products underpin a locally significant NTFP economy, with women-led collectives in Pauri, Tehri and Almora reporting reduced harvest windows and yields since 2018 (Singh et al., 2025). The broader synthesis of western Himalayan forest ecology by Bargali et al. (2025), covering 614 Scopus-indexed studies over three decades, identifies further phenological anomalies: early flowering and fruiting in *Myrica esculenta*, disrupted seed-viability windows for *Shorea robusta* (sal), and early seed development in *Quercus floribunda* (tilonj oak).

Invasive Species Expansion Under Warming

Climate change has facilitated the proliferation of invasive species across Uttarakhand's forest landscapes, threatening native biodiversity through competitive displacement and altered fire dynamics. *Lantana camara* and *Ageratina adenophora* (Eupatorium) are the most widespread and ecologically damaging invaders; their broad altitudinal adaptability means their ranges are expanding upward as thermal barriers are removed by warming (Bargali et al., 2025). *L. camara* now occupies large areas of the Sal belt and is advancing into lower Himalayan mixed forests, where its dense thickets suppress native tree regeneration, create fire fuel-ladders and reduce habitat quality for native fauna. *Rumex* and *Polygonum* species are invading alpine meadow (bugyal) margins at elevations formerly restricted to native alpine flora, driven by warming-induced expansion of the growing season (Thadani, 2015). Invasive species control in Van Panchayat-managed forests depends entirely on resident community labour- precisely the resource that out-migration is depleting.

Human-Wildlife Conflict Amplification

Human-wildlife conflict (HWC) in Uttarakhand has intensified into one of the most consequential and under-addressed dimensions of the forest-sector climate-impact nexus. Climate change amplifies both the frequency and spatial spread of conflict through three pathways: habitat contraction at lower altitudes driving animals into agricultural margins; climate-stressed prey depletion forcing dietary shifts; and the vacuum of human presence in out-migrating villages that paradoxically increases wildlife access to remaining settled areas.

Between 2000 and 2020, approximately 500 people lost their lives to leopard attacks, while approximately 1,396 leopards died over the same period (Think Wildlife Foundation, 2024). In Narendranagar forest division alone, 740 livestock casualties, 126 human injuries and 36 human deaths from wildlife were recorded over 20 years (Academia, 2022). The geography of HWC hotspots - Pauri Garhwal, Almora, Pithoragarh, Rudraprayag, Tehri, Narendra Nagar, maps directly onto the districts of highest climate vulnerability and greatest out-migration: as villages hollow out and fields are abandoned, forest edges advance into former cultivated buffers, creating new predator-access corridors precisely where fewer residents remain to maintain vigilance.

Chronology of Past Impacts on the Forest and Biodiversity Sector

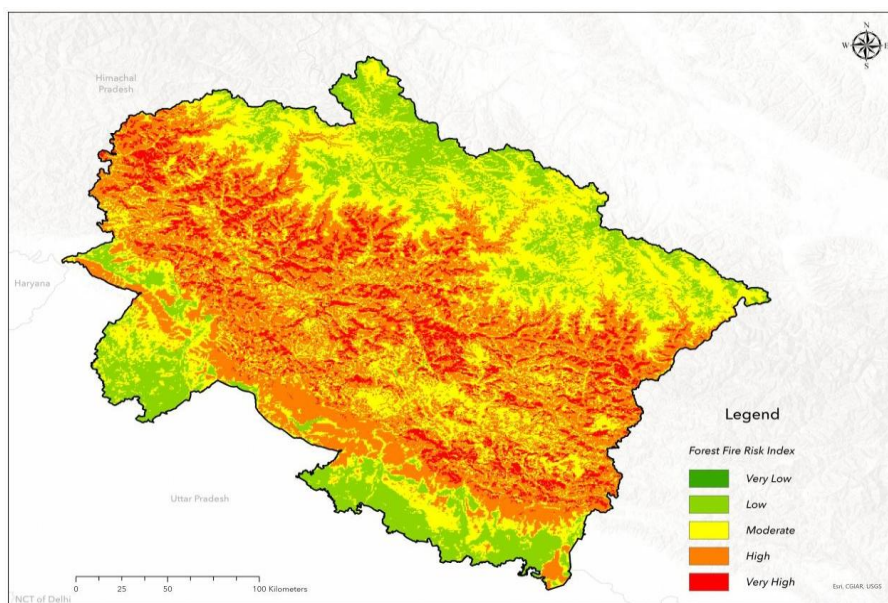
Year	Event	Documented consequence	Lesson carried into the framework
2000–15	Progressive canopy decline	Dense forest cover fell from 36.82% to 26.06%	Chronic compositional and canopy loss, not a single event, sets the baseline degradation rate
2002–19	Escalating fire incidence	Fire incidents rose from 922 to 41,600	Fire is a rising structural trend, not an anomaly

2010	Monsoon landslides and cloudbursts (Kumaon and Garhwal)	Extensive forest and slope damage; access disruption	Landslide impacts on forest are recurrent and monsoon-linked
2018–19	Peak fire year	41,600 incidents; ~18,000 ha affected; driest pre-monsoon in a decade	Drought and fire are coupled hazards
2020–21	Early-onset fire season	Season began three months early (Oct 2020); High Court suo motu action	The hazard season is lengthening beyond the conventional dry-season window
2021	HWC mortality continues	20 leopard-attack deaths recorded	The HWC toll is persistent, not declining, despite mitigation programmes
2023–24	National-highest fire season	3,338 FSI alerts; 5 deaths; 300 arson FIRs filed	Suppression capacity is outpaced by fire-season severity
2024 (Mar)	Panchayati Van Niyamawali 2023 shelved	Reform work replaced by token amendments; plan cycles extended 5→10 years	Adaptive governance capacity can be reversed by a single policy decision
2025 (Jul)	RET reintroduction programme launched	First systematic in-situ-plus-ex-situ programme for 14 threatened species	In-situ conservation alone is no longer sufficient
2025–26	Continuing fire seasons	226 incidents to 27 Apr 2026; 144.22 ha affected	The hazard remains active and is tracked in near-real time

12.4 Exposure Assessment

This section applies a structured hazard pathway approach to trace how specific climate-induced hazards cascade into impacts on the forest ecosystem, its biodiversity, and the communities that depend on it. Each pathway is traced from the hazard trigger through exposure mechanisms to ecological, economic, and social outcomes, consistent with the methodology applied to other sectors in this assessment series.

Forest Fire Susceptibility Mapping:

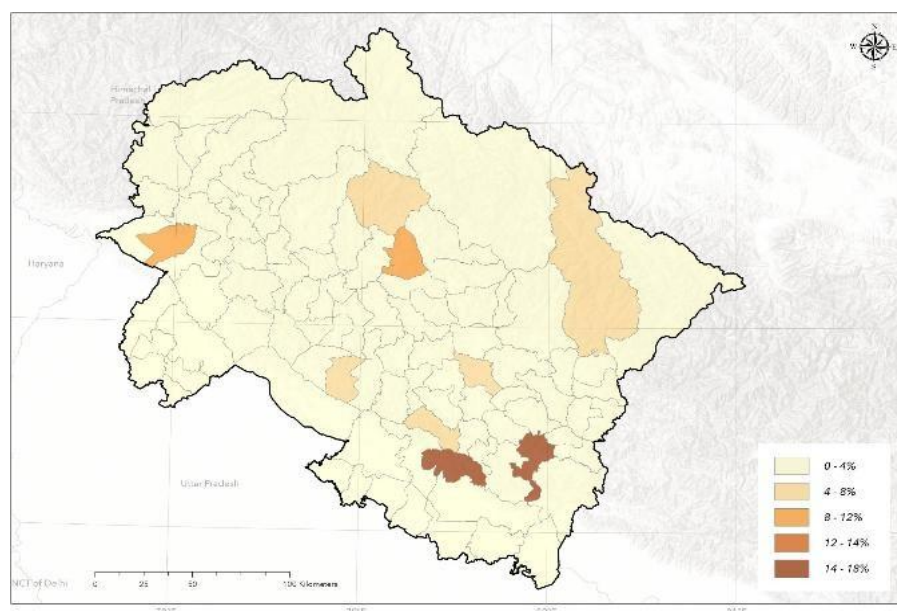


The Forest Fire Risk Index map of Uttarakhand indicates that large parts of the state, particularly the central and southern mountainous belts, fall under high to very high fire risk zones. Moderate to low-risk areas are primarily observed in the northern higher-altitude regions and some foothill areas. The distribution reflects the combined influence of temperature rise, prolonged dry

spells, vegetation type, topography, and changing precipitation patterns.

Climate change has intensified forest fire occurrence in Uttarakhand through increasing temperatures, reduced pre-monsoon rainfall, and extended dry periods. Forest fires pose substantial public health risks and ecosystem impacts.

Forest Area Exposed to Landslide

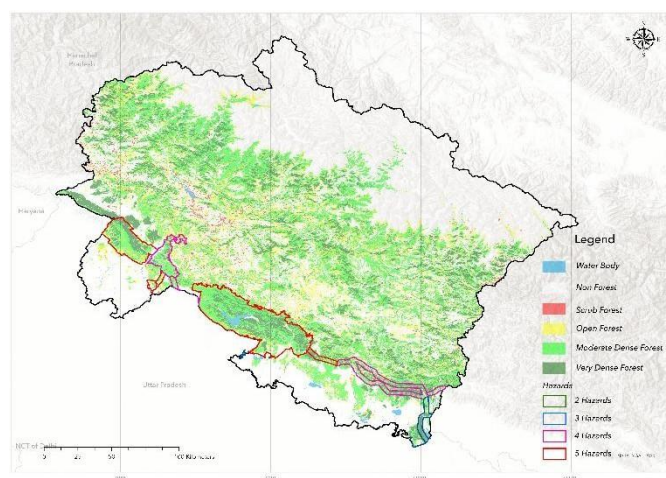


The map illustrates the spatial distribution of forest areas exposed to landslide risk across Uttarakhand, categorized into exposure classes ranging from 0-4% (very low exposure) to 14-18% (very high exposure).

Most forested regions in the state fall within the lower exposure category; however,

localized pockets in the southern and central mountainous districts exhibit moderate to very high exposure levels. These areas correspond to fragile slopes, steep terrain, intense rainfall zones, and geologically unstable landscapes. Landslides pose significant threats to forest ecosystems and their associated environmental services. It is expected to increase the occurrence of intense rainfall events and slope instability, thereby increasing landslide susceptibility in vulnerable forest regions. Forest degradation due to landslides can result in loss of vegetation cover, habitat fragmentation, reduced biodiversity, and disruption of ecosystem functions such as soil stabilization and water regulation.

Wildlife Corridors Exposed to Multi-Hazards



The map illustrates the spatial distribution of forest cover and wildlife corridors in relation to multi-hazard exposure zones across Uttarakhand. Forest classes, including open forest, moderately dense forest, dense forest, and very dense forest, are overlaid with wildlife corridors and hazard exposure categories ranging from two-hazard to five-hazard exposure zones. The analysis highlights the interaction between ecological connectivity and climate-related risks.

The mapped wildlife corridors traverse ecologically important landscapes and connect fragmented habitats across the state, enabling

movement of species and maintaining ecosystem functions. The spatial pattern indicates that several wildlife corridors overlap with high multi-hazard exposure zones, suggesting that critical ecological pathways are increasingly vulnerable to multiple climate-induced and environmental stressors.

12.5 Climate Variability impacting the Forest and Biodiversity Sector

The impacts of climate change on Uttarakhand's forests can be broadly categorised into direct impacts those arising immediately from thermal and precipitation changes or extreme events, and indirect impacts those operating through cascading ecological, hydrological, and social disruptions. The following sections trace both pathways with reference to peer-reviewed evidence and state-level observational data.

Observed Climate Variability and Trends

Climate variable	Observed change	Consequence for the forest and biodiversity sector
Air temperature	Warming, with elevation-dependent amplification at altitude (IMD, 2023)	Upward displacement of bioclimatic envelopes; treeline advance; removal of the thermal barrier that previously excluded invasive species from higher elevations
Winter precipitation and snowfall	Declining snowfall fraction; rising snowline	Shortened snow-cover duration reduces soil moisture through the establishment window; alpine growing season lengthens, opening niche space to colonisers
Total monsoon rainfall	Decline of approximately 12% since 2000 (IMD, 2023)	Lower spring-shed recharge; reduced soil moisture available to moisture-dependent climax species during regeneration
Extreme rainfall frequency	Increase of approximately 35% (IMD, 2023)	Slope failure removes cover together with the soil and seed bank on which recovery depends; habitat and corridor fragmentation
Pre-monsoon dryness	Extended dry spells and rising atmospheric moisture demand through the March–June window [figure to be inserted from IMD district series]	Fuel desiccation across chir pine litter; sharp reduction in the ignition threshold
Fire-season severity	3,338 fire alerts recorded in summer 2024 the highest of any Indian State	Carbon-stock loss exceeding the regenerative sequestration capacity of Sal, Pine and Mixed stands (Bargali et al., 2024); herbaceous diversity collapse
Phenological timing	Flowering advancement of 88–97 days over the last century in <i>Rhododendron arboreum</i> at Chakrata Forest Division (Rana, Bhattacharyya et al., 2024)	Plant–pollinator desynchronisation risk in a keystone species; contraction of the NTFP harvest window

Spring discharge	Approximately 30% decline across Kumaon and Garhwal districts since 2000 (Raina, 2009)	Simultaneously a forest-health indicator, a climate signal and a rural water-security crisis
Cumulative cover loss	1.3 kha of tree cover lost to fire and 23 kha to all other drivers, 2001–2025 (Global Forest Watch)	Progressive erosion of the carbon and habitat stock against which all adaptation gains must be measured

Climate Variable → Forest and Biodiversity Chain Transmission Matrix

The matrix below traces each climate variable through every stage of the forest value chain, applying the same discipline used for the energy sector. Reading down a column shows what a single stage of the chain is exposed to; reading across a row shows how a single variable propagates from the resource base to the household. The chain is defined as: resource base (the physical and climatic substrate), ecosystem structure and composition (the living stock), service flow (the regulating and provisioning functions the stock delivers), and dependent livelihoods (the human systems that draw on that flow). Blank cells indicate no material pathway.

Climate variable	Resource base	Ecosystem structure and composition	Service flow	Dependent livelihoods and cross-sector
Rising air temperature	Upward displacement of bioclimatic envelopes; contraction of alpine and sub-alpine climatic space	Treeline advance; upslope migration across taxonomic groups; establishment of invasives above former thermal limits	Altered carbon uptake trajectory; declining availability of high-altitude medicinal flora	Erosion of the medicinal-plant and herb-collection economy; loss of the alpine cultural and pilgrimage landscape
Declining snowfall fraction and rising snowline	Shorter snow-cover duration; earlier soil warming; loss of the slow-release moisture reservoir	Compositional change in alpine meadows (bugyals); niche opening for Rumex and Polygonum spp.	Reduced regenerative interval for alpine pasture; declining pasture quality	Falling productivity of migratory grazing systems; pressure on transhumant Bhotiya and Gujjar livelihoods
Reduced winter and pre-monsoon rainfall	Soil-moisture deficit through the germination and establishment window	Regeneration failure of moisture-dependent climax species; seedling mortality; competitive release of pine	Declining forest renewal capacity; loss of leaf-litter and humus production	Fodder, fuelwood and leaf-manure scarcity; increased collection time borne largely by women
Declining total monsoon rainfall	Reduced spring-shed recharge; lower sub-surface storage	Decline of Banj oak (Quercus leucotrichophora) on mid-elevation slopes	Reduced and increasingly seasonal discharge from naulas and dharas	Drinking-water insecurity for the ~70% of rural households dependent on springs; waterborne disease burden
Increased rainfall intensity	Loss of soil profile and seed bank on failed slopes; debris deposition burying regeneration	Fragmentation of contiguous habitat; severance of wildlife corridors	Loss of slope-stabilisation and sediment-regulation service	Downstream sediment loading on hydropower intakes and irrigation channels; road blockage and access loss
Pre-monsoon dryness and fire severity	Combustion of litter, humus and below-ground carbon; loss of fuelwood biomass stock	Herbaceous diversity decline from a maximum of 37 species in unburnt plots to 12 under high fire frequency; competitive advantage to chir pine	Carbon-stock loss exceeding sequestration capacity; black carbon deposition on glaciers at ~400 times winter values near Gangotri	Loss of grazing and NTFP resource; human and Forest Department casualties; downstream cryospheric and water-security consequence
Warming winters (flora)	Earlier accumulation of growing-degree days	Flowering advancement in Rhododendron arboreum; desynchronisation from ground-cued pollinator emergence	Reduced fruit and seed set; declining reproductive success in a keystone species	Contraction of the Buransh harvest window and yields reported by women-led collectives in Pauri, Tehri and Almora since 2018
Warming winters (fauna)	Reduced hibernation depth and duration in Himalayan black bear (WII, 2019)	Altered predator and megafauna distribution; leopard range expansion following prey upslope	—	Human–wildlife conflict extending to mid-altitude communities with no historical exposure; livestock and crop loss; night-mobility restriction

14.1.1

Climate Hazard	Climate Driver	Climate Stressor	Direct Impact	Functional Impact	Consequences	Confidence
Rising Temperature	Increasing average and maximum temperatures and warmer winters	Upward shift of vegetation zones, altered growing conditions and increased heat stress	Changes in forest composition, reduced regeneration of climate-sensitive species and altered species distribution	Declining ecosystem stability and reduced habitat suitability for native flora and fauna	Loss of biodiversity, reduced ecosystem resilience and increased vulnerability of endemic species	High
Changing Rainfall Pattern	Erratic monsoon onset and withdrawal and uneven rainfall distribution	Soil moisture variability, altered hydrological regimes and moisture stress	Reduced forest productivity, poor natural regeneration and changes in flowering and fruiting cycles	Declining forest health and disruption of ecosystem processes	Reduced ecosystem services, lower biodiversity and declining forest resilience	High
Drought	Prolonged rainfall deficit and increasing evapotranspiration	Moisture stress, drying of forest soils and reduced streamflow	Increased tree mortality, reduced regeneration and decline in understory vegetation	Degradation of forest ecosystems and reduced habitat quality	Loss of carbon sequestration, reduced ecosystem productivity and increased ecosystem vulnerability	High
Extreme Rainfall / Cloudbursts	High-intensity precipitation events and cloudbursts	Soil erosion, slope instability and high sediment movement	Damage to forest stands, riparian ecosystems and alpine vegetation	Disruption of forest ecosystem structure and regeneration processes	Habitat degradation, biodiversity loss and increased restoration requirements	High
Flooding / Flash Floods	River overflow and extreme runoff	Riverbank erosion and inundation of riparian forests	Damage to riverine vegetation, wetlands and wildlife habitats	Loss of habitat connectivity and degradation of aquatic–terrestrial ecosystems	Reduced ecological integrity and decline in ecosystem services	High
Forest Fires	Warmer, drier pre-monsoon conditions and prolonged dry	Increased frequency, intensity and duration of	Loss of forest cover, wildlife habitat and regeneration	Reduced forest productivity and degradation of	Biodiversity loss, increased carbon emissions, reduced	High

	spells	forest fires	potential	ecosystem functions	ecosystem services and heightened human–wildlife interactions	
Glacier Retreat & Alpine Ecosystem Change	Rising temperatures and declining snow cover	Shrinking alpine habitats and altered snowmelt regimes	Changes in alpine meadows, treeline shifts and habitat loss for high-altitude species	Reduced suitability of alpine ecosystems and altered species assemblages	Decline of endemic and climate-sensitive species and reduced ecological resilience	Medium–High
Climate-induced Phenological Change	Rising temperatures and changing seasonal patterns	Altered timing of flowering, fruiting, migration and breeding	Mismatch between species interactions (e.g., pollinators and flowering plants)	Disruption of ecological processes and food-web dynamics	Reduced reproductive success, biodiversity decline and ecosystem instability	Medium–High
Invasive Species Expansion	Rising temperatures and disturbed ecosystems	Expansion of invasive plant species and increased competition with native vegetation	Replacement of native species and degradation of forest habitats	Reduced ecosystem diversity and altered ecological functioning	Declining native biodiversity, increased management costs and reduced ecosystem resilience	Medium–High
Human–Wildlife Conflict	Climate-induced habitat degradation and shifting resource availability	Wildlife movement into agricultural and human-dominated landscapes	Increased crop raiding, livestock depredation and human–wildlife encounters	Reduced coexistence between communities and wildlife and increased management burden	Livelihood losses, increased conflict incidents and reduced support for biodiversity conservation	Medium
Compound Climate Hazards (Heat + Drought + Forest Fire + Extreme Rainfall)	Increasing frequency and convergence of multiple climate extremes	Simultaneous ecosystem degradation, habitat fragmentation and reduced	Large-scale forest degradation and loss of ecological connectivity	System-wide decline in ecosystem functioning and biodiversity resilience	Long-term biodiversity loss, reduced carbon storage, diminished ecosystem services and increased ecological risk	High

		regeneration				
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Intensifying Forest Fires and Carbon Stock Loss

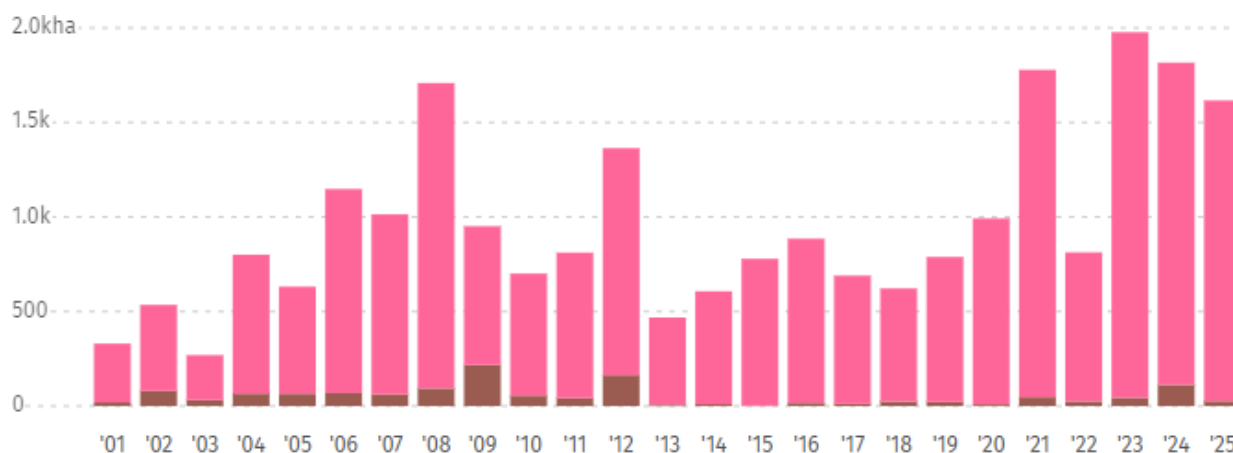
The 2024 fire season in Uttarakhand recorded 3,338 fire alerts during summer, the highest among Indian states, claiming 12 lives, including six Forest Department personnel (Earth Journalism Network, 2025; Anantam IAS, 2025). Bargali, Pandey, Bhatt, and Sundriyal (2024), in a landmark study published in *Frontiers in Forests and Global Change*, quantify carbon-stock loss across Sal, Pine and Mixed forest types along an elevation gradient and demonstrate that recurrent fires now exceed the regenerative carbon-sequestration capacity of these stands. Chir pine (*Pinus roxburghii*) forests, covering approximately 28% of the state's forest area, dominate this fire economy: the species accumulates highly flammable resinous needle-litter, and the altitudinal band it occupies (1,000–2,000 m) coincides with the densest concentration of human settlement and transmission infrastructure (Fulé et al., 2021; Bargali et al., 2024).

The consequences extend well beyond the fire perimeter. First, recurrent burning reduces herbaceous diversity, Singh and Thadani (2022) report a decline from a maximum of 37 species in unburnt plots to just 12 in high-fire-frequency plots in Uttarakhand. Second, smoke and black carbon deposition on Himalayan glaciers accelerate cryospheric melt; the Wadia Institute of Himalayan Geology (2016) measured black carbon concentrations near the Gangotri glacier in summer at approximately 400 times winter values, with implications for downstream water security. Third, the fire season now overlaps with the breeding and flowering windows of vulnerable native species including 55 IUCN listed threatened birds and the state's mammalian fauna such as the Himalayan musk deer (*Moschus leucogaster*) and the snow leopard (*Panthera uncia*) (Down to Earth, 2024).

Forest Fire Incidents and Area Affected in Uttarakhand (2016-2026)

Fire Season	Incidents	Area Affected (ha)	Notable Features / Key Data
2015-16	20,667	4,433	30% surge over 2015; 1,200 simultaneous fire locations on April 26; IAF helicopter deployment; temp. 7°C above normal (Newsline, 2021)
2017-18	~1,165 / 4,714	Not fully compiled	Incidents rose from 1,165 (2017) to 4,714 (2018); initiation of FSI SNPP-VIIRS satellite monitoring (Mina et al., 2023)
2018-19	41,600	~18,000 (est.)	Peak fire year; 41,600 incidents, 45x increase from 2002 baseline of 922; coincides with driest pre-monsoon in a decade (Negi, 2019; Mina et al., 2023)
2020-21	989 (to Apr 4, 2021)	1,297	Season began Oct 2020 (3 months early); state sought MI-17 helicopter deployment from Centre; Uttarakhand High Court suo motu action initiated (Newsline, 2021)
2023-24	930+	1,196	National highest FSI alert count: 3,338; IAF MI-17 C Army deployment; 5 human deaths; 721 incidents (Nov 2023–May 2024) destroyed 902.5 ha; Rs 3.24 crore reported economic loss in Himachal (comparable data from Tribune, 2024); 300 FIRs filed for arson (Deccan Herald, 2024)
2025-26	226	144.22	Garhwal: 177 incidents (110.52 ha); Kumaon: 28 incidents; Wildlife zones: 21 incidents; Badrinath FD: 57 incidents, highest by division; Rudraprayag: 21.82 ha, highest area loss; UHC monitoring continues (National Herald India, 2026)

From 2001 to 2025, Uttarakhand lost 1.3 kha of tree cover from fires and 23 kha from all other drivers of loss. The year with the most tree cover loss due to fires during this period was 2009 with 220 ha lost to fires, 23% of all tree cover loss for that year.



Species Distribution Shifts and Treeline Dynamics

Climate-induced upward altitudinal shifts in species distributions are among the most comprehensively documented ecological responses in Uttarakhand's forests. The upward movement phenomenon operates across taxonomic groups, but its most consequential manifestations occur in the dominant forest-forming species that define ecosystem services.

The single most consequential sensitivity in this sector is the replacement of moisture-loving Banj oak (*Quercus leucotrichophora*) by drought- and fire-tolerant chir pine across mid-elevation slopes, an ongoing pinification of the Himalayan mid-altitudes (Sharma et al., 2018; Kumar and Khanduri, 2024). Joshi, Sundriyal, Chandra and Arya (2024), in an assessment of Binsar Wildlife Sanctuary published in *Environmental Challenges*, recorded 271 vascular plant species across 208 genera and 74 families but flagged a significant correlation between rising disturbance levels and reduced regeneration of climax species.

What makes this transition the sector's equivalent of the peak-water problem is that it is self-reinforcing and effectively irreversible on management timescales. Pine encroachment raises fire frequency through litter flammability; fire in turn destroys oak seedlings, which are fire-sensitive, while chir pine regenerates readily on burnt ground; the resulting stand then produces still more flammable litter. The system moves through a threshold rather than along a gradient, and once past it, restoring oak is not a matter of reducing pressure but of active, sustained and expensive intervention against a fire regime that now favours the replacement species. A management plan calibrated to the composition observed at the time of its drafting is therefore calibrated to a state the forest is in the process of leaving, the

same class of assumption failure that stationary hydrology represents for hydropower appraisal.

The hydrological consequences are the most severe. Oak forests, with deep root systems and dense leaf litter, are critical to the recharge of Himalayan springs, the *naulas* and *dharas* on which over 70% of rural water supply depends (Raina, 2009; Tiwari and Sunny, 2024). Their replacement by pine shallower-rooted, with hydrophobic litter and high evapotranspirative demand has been correlated with the drying of perennial springs across Almora, Pauri and Tehri districts (Joshi et al., 2024). Sugandha et al. (2025) identify the oak–pine transition explicitly as a climate-anthropogenic compound stressor that propagates from forest into water, agriculture and ultimately livelihood domains.

Decline of High-Altitude Medicinal Plants and Endemic Flora

Uttarakhand's trans-Himalayan zone hosts at least 34 high-value threatened medicinal plant species, several with their entire global potentially suitable habitat measuring less than 10 km² (Kumar et al., 2023). The IUCN-listed critically endangered *Meizotropis pellita* (locally 'Patwa'), endemic to Uttarakhand, is restricted to less than 10 km² of suitable habitat due to combined pressures of deforestation, forest fires and climate-driven habitat contraction (Optimize IAS, 2022). *Picrorhiza kurroa*, *Saussurea obvallata* (the iconic 'Brahma Kamal'), *Dactylorhiza hatagirea* and *Sinopodophyllum hexandrum* face habitat constraints further intensified by warming and altered precipitation (Kumar et al., 2023; Sharma et al., 2024).

Recognising this crisis, the Uttarakhand Forest Department's Research Wing in July 2025 launched the country's first systematic Rare, Endangered and Threatened (RET) plant reintroduction programme covering 14 species, including *Gentiana kurroo* (Himalayan Gentian), *Nardostachys jatamansi* (Indian Spikenard) and *Lilium polyphyllum* (White Himalayan Lily) (Indian Masterminds, 2025; Usthadian Academy, 2025). The intervention is a tacit acknowledgement that in situ conservation alone is no longer sufficient under current climate trajectories, a substantial shift in conservation philosophy for the state.

Altered Forest Composition and Regeneration

Climate-driven changes are increasingly evident in the structural and compositional fabric of Uttarakhand's mid-altitude forests. Joshi, Sundriyal, Chandra, and Arya (2024), in a comprehensive assessment of Binsar Wildlife Sanctuary published in *Environmental Challenges*, recorded 271 vascular plant species across 208 genera and 74 families but flagged a significant correlation between rising disturbance levels and reduced regeneration of climax species. The most ecologically consequential trend is the replacement of moisture-loving Banj Oak (*Quercus leucotrichophora*) by drought- and fire-tolerant Chir Pine (*Pinus*

roxburghii) across mid-elevation slopes — an ongoing 'pinification' of the Himalayan mid-altitudes that simultaneously reduces water retention, increases fire risk, and depletes fodder, fuel and biodiversity (Sharma et al., 2018; Kumar C Khanduri, 2024).

The downstream hydrological consequences of this compositional shift are particularly severe. Oak forests, with their deep root systems and dense leaf litter, are critical for the recharge of Himalayan springs (naulas, dharas) on which over 70% of rural water supply depends (Raina, 2009; Tiwari C Sunny, 2024). Their replacement by pine, which has a shallower root system, hydrophobic litter, and a high evapotranspirative demand has been correlated with the drying of perennial springs across Almora, Pauri and Tehri districts (Joshi et al., 2024). Sugandha et al. (2025) explicitly identify this oak–pine transition as a 'climate-anthropogenic compound stressor' that propagates from forest into water, agriculture, and ultimately livelihood domains.

Phenological Disruption

Phenological advancement, the earlier onset of flowering, leafing or breeding is among the clearest 'fingerprints' of climate change on biological systems. The state tree of Uttarakhand, *Rhododendron arboreum* (locally 'Buransh'), is now perhaps the best-documented case in the Western Himalaya. Rana, Bhattacharyya, and colleagues (2024), comparing real-time field observations with herbarium records dating back to 1893 using Generalized Additive Models (GAM), document a flowering advancement of 88–97 days over the last century at Chakrata Forest Division, Uttarakhand. Mongabay India (2020) further reports widespread observations of December–January flowering in mid-altitude populations, a complete season-shift from the historical February–March bloom.

The downstream consequences are substantial. *R. arboreum* is a keystone species supporting passerine and insect pollinators across elevations; precocious flowering risks plant–pollinator desynchronisation, since most Himalayan pollinator emergences remain temperature-cued at ground level rather than at the canopy (Mamgain et al., 2017; Rawat et al., 2024). Memmott et al.'s (2007) seminal work establishes that even moderate decoupling leads to pollinator decline and consequent plant reproductive failure. For Uttarakhand, the economic risk is direct: Buransh squash, juice and flower-based products underpin a small but locally significant non-timber forest produce (NTFP) economy, with women-led collectives in Pauri, Tehri and Almora reporting reduced harvest windows and yields since 2018 (Singh et al., 2025).

Human-Wildlife Conflict: A Climate-Amplified Crisis

Human-wildlife conflict (HWC) in Uttarakhand has intensified into one of the most consequential and under-addressed dimensions of the forest-sector climate impact nexus. The state harbours 2,276 leopards (WII census, December 2021–July 2022), 2,026 elephants (2020 census, concentrated in Corbett Tiger Reserve and Rajaji National Park), and 560 tigers (2022

census), among the highest densities of large carnivores and megafauna in the world (Mongabay, 2025). Cohabitation of these densities with a predominantly rural, agriculture-dependent mountain population has always generated conflict; climate change is now amplifying both the frequency and spatial spread of that conflict through three pathways: habitat contraction at lower altitudes (driving animals into agricultural margins), climate-stressed prey depletion forcing dietary shifts, and the vacuum of human presence in out-migrating villages that paradoxically increases wildlife access to remaining settled areas.

The mortality data are stark. According to Uttarakhand Forest Department records, 244 people were killed by leopard attacks between 2000 and 2010; 234 between 2011 and 2021; and at least 20 in 2022 alone, demonstrating a persistent, not-declining toll despite successive mitigation programmes (Mongabay, 2023). Between 2000 and 2020, approximately 500 people lost their lives to leopard attacks, while approximately 1,396 leopards died in the same period, illustrating the bilateral cost (Think Wildlife Foundation, 2024). Elephant-human conflict, concentrated in the Terai corridor (Haridwar, U.S. Nagar, Nainital districts), kills 20-25 persons annually and causes substantial crop damage, forcing the Forest Department to maintain a 24×7 information centre and develop a digital compensation platform. In Narendranagar forest division alone, 740 livestock casualties, 126 human injuries, and 36 human deaths from wildlife were recorded over 20 years (Academia, 2022).

The geography of HWC hotspots, Pauri Garhwal, Almora, Pithoragarh, Rudraprayag, Tehri, Narendra Nagar, maps directly onto the districts of highest climate vulnerability and greatest out-migration. As villages hollow out and agricultural fields are abandoned, forest edges advance into what were previously cultivated buffers, creating new predator access corridors and amplifying the human-wildlife interface at precisely the locations where fewer residents remain to maintain vigilance. The attack by leopards, tigers, elephants, or other wildlife animals are not reported' (Mongabay, 2023). Climate change-induced altitudinal expansion of leopard range, responding to upward prey movement and habitat shifts, is now bringing conflict to mid-altitude communities that historically had no such exposure.

Species	Population (latest census)	Human Deaths (cumulative)	HWC Hotspot Districts	Climate Change Driver
Leopard	2,276	~500 (2000–2020); 20 in 2022 alone	Pauri, Almora, Pithoragarh, Rudraprayag, Tehri, Narendra Nagar	Range expansion to higher altitudes following prey; habitat fragmentation by fires C roads
Elephant	2,026	20–25 annually (2017 onwards)	U.S. Nagar, Haridwar, Nainital (Terai corridor)	Rainfall variability alters crop/food availability; corridor fragmentation by infrastructure

Tiger	560 (	Scattered; rising in Terai interface zones	Corbett-Rajaji landscape; Terai forest edges	Prey base movement driven by temperature and vegetation shifts
Black Bear	-	Injuries and property damage reported annually	All hill districts bordering reserve forests	Reduced hibernation depth / duration due to warmer winters; foraging in lower villages during traditional hibernation periods (WII, 2019)

Van Panchayat Community Forest Management, Climate Stress and Governance Erosion

Uttarakhand's Van Panchayat (VP) system is one of the most historically significant community forest governance institutions in South Asia, institutionalised in 1931 after sustained resistance against

colonial forest appropriation. The 12,167 VPs manage 13.41% of the state's forest area, approximately

7,164 km², across the hill districts, providing the most granular forest protection, fire early warning, spring conservation, and sustainable harvest regulation system available in the state's terrain (UFD, 2025; 101 Reporters, 2021). Each VP formulates its own management rules, from forest guard selection to defaulter penalties, calibrated to local ecology and livelihood patterns. Villages with intact VP management have measurably healthier forest cover and higher spring discharge than those relying exclusively on state forest department oversight.

Between 2020 and 2022, the state undertook significant VP governance reforms: land record anomalies were corrected, VP elections were revived (compliance rising from 6% to 69%), and a comprehensive Composite Management Plan (CMP) framework aligned with ecosystem services was developed over three years of multi-stakeholder consultation (Daily Pioneer, 2025). The 58-amendment Panchayati Van Niyamawali 2023 draft was positioned to be the most progressive VP governance instrument since the system's founding, integrating carbon market access, biodiversity value recognition, and climate-adaptive management planning.

The implications are severe: the policy reversal has frozen community-linked management frameworks at a moment when climate variability is accelerating beyond the predictive range of decade-old management plans. Forest fires burning more hectares each year, invasive species expanding range, and spring discharge declining, all require adaptive, responsive management cycles. The VP system's demographic base is simultaneously eroding through out-migration: as younger residents leave climate-stressed villages for plains employment, the resident guardians of 7,000+ km² of community forest are disappearing. This creates what Joshi et al. (Tropical Ecology, 2022), in a study of Van Panchayat and Soyam forests across 98 villages in six Uttarakhand districts, identified as the key driver of reduced regeneration: reduced winter rainfall combined with increased anthropogenic disturbance in VPs with weakened governance.

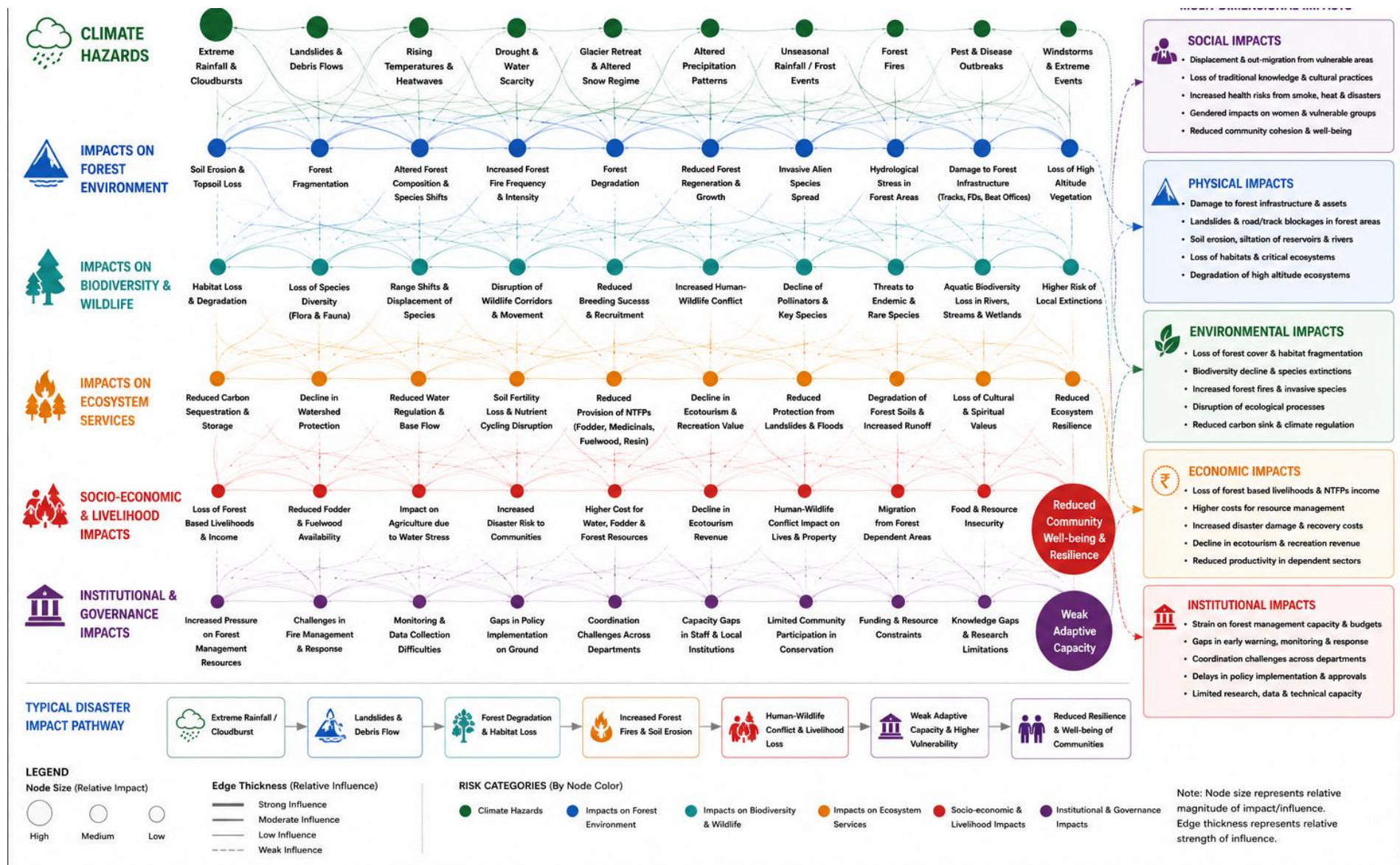
Invasive Species Expansion Under Warming

Climate change has facilitated the proliferation of invasive species across Uttarakhand's forest landscapes, threatening native biodiversity through competitive displacement and altered fire dynamics. *Lantana camara* and *Ageratina adenophora* (Eupatorium) are the most widespread and ecologically damaging invaders, displaying broad altitudinal adaptability, their ranges are expanding upward as thermal barriers are removed by warming (Bargali et al., ScienceDirect, 2025). *L. camara* now occupies large areas of the Sal belt and is advancing into the lower Himalayan mixed forests; its dense thickets suppress native tree regeneration, create fire fuel ladders, and reduce habitat quality for native fauna. *Ageratina adenophora*, introduced from Mexico, has colonised large areas of the sub-alpine zone. *Rumex* and *Polygonum* species are invading the alpine meadow (bugyal) margins, with documented occurrence at elevations formerly restricted to native alpine flora, driven by warming-induced expansion of the growing season (Thadani, 2015). The forest management implications are significant: invasive species control in VP-managed forests depends entirely on resident community labour, which is precisely the resource that out-migration is depleting.

Hydrological Stress: Springs, Streams, and Forest Hydrology

Uttarakhand's forests are the primary spring-shed recharge systems for the over 3,000 perennial springs (dhare and naulas) upon which 70% of rural mountain communities depend for drinking water (Central Ground Water Board, 2022). The progressive drying of these springs, at a rate of approximately 30% decline in discharge documented across Kumaon and Garhwal districts since 2000, is simultaneously a forest health indicator, a climate signal, and a rural livelihood crisis (Raina, 2009). The mechanism is partially climate-driven and partially land-use-driven: reduced snowpack and erratic precipitation reduce recharge, while pine expansion over oak reduces soil moisture retention and evapotranspiration patterns that historically sustained sub-surface flow. A People's Science Institute study of spring revival through

community-based interventions found that villages with restored traditional Naula spring chambers had 73% fewer waterborne disease cases than those relying exclusively on government water supply, demonstrating that forest-hydrological restoration has direct human health co-benefits (PSI, 2023).



13 Gender Inclusion

Uttarakhand's hill agriculture is, in practice, run by women. Field studies place women's share of farm operations at roughly two-thirds of total labour: a Farmers FIRST programme study across the mid-hills found women perform 64.6 per cent of farm operations and work 12–16 hours a day, with over one in ten women working more than 16 hours a day, while men retain control of only the highest-status, least time-intensive tasks (Ansari et al., 2025). This division of labour is not a static cultural feature; it is actively produced by out-migration. As men leave for urban wage employment, the AR6 finding that “labour migration in the context of climate change is also gendered” (IPCC, 2022) plays out at household level: women left behind absorb the full farm workload on top of existing domestic and care responsibilities, a pattern documented in a seven-village Kumaon Himalaya study, which finds that while male out-migration does increase women's day-to-day household and farm decision-making authority, it does not translate into comparable gains in financial control or asset ownership (Jatav et al., 2025).

The People's Science Institute's household survey found women spending four to six additional hours a day collecting water during drought (PSI, 2022), which identifies water-collection duties, together with childcare and firewood collection, as unpaid labour that intensifies specifically under climate stress. Across Southern Asia 71 per cent of women's total employment sits within agrifood systems, compared with 47 per cent of men's (FAO, 2023). FAO's global synthesis attributes a persistent 24 per cent productivity gap between male- and female-managed farms of equal size not to any difference in skill, but to unequal access to land, credit, inputs and extension services (FAO, 2023), a mechanism with direct force in Uttarakhand, where 71 per cent of rural women labourers report being unable to access agricultural training, against 60 per cent of rural men (UN Women & UNEP EmPower Initiative).

Considering the actual fact that Labour and land title have decoupled in Uttarakhand's hills, and that decoupling is itself a climate-vulnerability factor: it means the people doing the adaptive work of farming, choosing what to plant, when to irrigate, whether to invest in a drought-tolerant variety, are frequently not the people legally entitled to make those investment decisions or to access the credit that would fund them. Only 8.5 per cent of Uttarakhand's operational agricultural landholdings are legally registered to women, against 91.5 per cent to men (Ansari et al., 2025), a figure consistent with the 2010-11 Agricultural Census finding that women managed only 12.8 per cent of operational holdings statewide, cited by the state government itself when it proposed, in 2020, to amend the Uttar Pradesh Zamindari Abolition and Land Reforms Act (as adapted in Uttarakhand) to grant women co-ownership of agricultural land, on the

grounds that roughly 90 per cent of hill farm labour is performed by women while 89 per cent of Uttarakhand women hold no property rights over the household landholding they work (Matters India, 2020).

The same report finds that women constitute only around 16 per cent of PM-KISAN beneficiaries in Uttarakhand, against a national average nearer 20-30 per cent (Singh, 2026), meaning the state's principal direct-income-support scheme for farmers is structurally under-reaching the people actually farming. Where women have gained formal governance authority, the pattern is uneven but instructive.

Another major concern, Climate exposure and land quality are not evenly distributed by caste in Uttarakhand. Reporting from Dhudholi and Chinoni villages documents that 36 per cent of the state's farmed land is marginal (under one hectare) and 28 per cent is small (one to two hectares), that only 45 per cent of farmland is irrigated, and that almost no landholdings above four hectares are owned by Scheduled Caste families, meaning Dalit farming households are disproportionately confined to the smallest, least-irrigated, most climate-exposed plots (Thapa, 2023).

IPCC identifies children and older people as groups whose climate-related health effects “can be especially devastating” (IPCC, 2022, Sec. 10.3.3), and specifically flags rising mental-health impacts “particularly for children, adolescents, elderly, and those with underlying health conditions” (IPCC, 2022). In Uttarakhand's hills, out-migration has left agricultural and crop-guarding labour disproportionately to women, the elderly and children, the youngest and oldest residents of the state's most climate-exposed blocks are also its primary defenders against crop loss from human-wildlife conflict, an acute-hazard exposure layered directly onto age-related physical vulnerability.

8 Psychological Impacts of Climate-Induced Disasters: Recovery, Chronicity and Who Is Left Behind

The impact of disaster also occurs psychologically. Uttarakhand's own disaster record shows that psychological recovery is neither fast nor automatic, that it is systematically harder for women, children and the economically precarious, and that the state's mental-health response infrastructure is not currently built to match the duration of the need.

Unlike a flood, the loss of a season's crop or a decade-old orchard is a cumulative economic and psychological shock rather than a single traumatic event, and the

literature on agrarian distress in India treats it accordingly. A national econometric study attributes approximately 59,300 suicides in India since 1980 to the effect of above-threshold growing-season temperatures on crop yields, roughly 7 per cent of the total rise in the national suicide rate over that period, operating through the specific mechanism of financial hardship following crop failure, not heat exposure itself (Carleton, 2017). No equivalent Uttarakhand-specific quantitative study exists, but the qualitative record from the state's own hill farmers is consistent with the mechanism and gives it a human register the national statistic cannot: farmers in Dhudholi and Chinoni describe the compounding effect of repeated crop failure from unseasonal rain and flooding on top of the social isolation that follows it, with one farmer describing simply that “no one ever visits us” (Thapa, 2023).

For the grower who plants an orchard as a multi-decade asset and then watches it fail cultivar by cultivar as the chilling regime shifts, the decision to uproot is very plausibly also a chronic grief process, distinct from and longer-lasting than the acute trauma the disaster-psychology literature describes below, because there is no single day on which the loss occurred and therefore no single day from which recovery can be dated. Land abandonment driven by human-wildlife conflict compounds the same mechanism: where guarding a terrace against wild boar or macaque becomes physically unsustainable for the ageing, feminised labour force left in a village, the resulting abandonment is experienced by the household not as a discrete event but as a slow withdrawal from a livelihood and an identity, a pathway the broader eco-distress but has not yet quantified specifically for Uttarakhand's farming communities.